SM-X818U_REV0.5_SVC

2023.05.02

sheet01 : RF (1/3)

sheet02 : RF (2/3)

sheet03 : RF (3/3)

sheet04 : CONNECTIVITY - BT/WIFI(WCN6856), GPS

sheet05 : AP_SM8550(Secondary PCB), JTAG, UFS

sheet06 : PMIC(Secondary PCB), IF PMIC, D/C, OVP

sheet07 : AUDIO

sheet08 : CAMERA, DISPLAY

sheet09 : SENSOR, SD/SIM, KEY, I/F, eSIM

sheet10 : RF PMIC, 5G IFIC

sheet11 : WACOM PCB (Ext.) , ANT SUB PCB (Ext.)

sheet12 : USB PCB (Ext.)

sheet13 : WIFI TR PCB (Ext.) / TL PCB (Ext.), CLIP, SMR, SUS, SHIELDCAN

sheet14 : Sencondary PCB PAD (MAIN PCB)

mmW,NA : X818U
Global,EUR : X816
WIFI : X810

TITLE TabS7P
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 0 of 15
Date Drawing Date

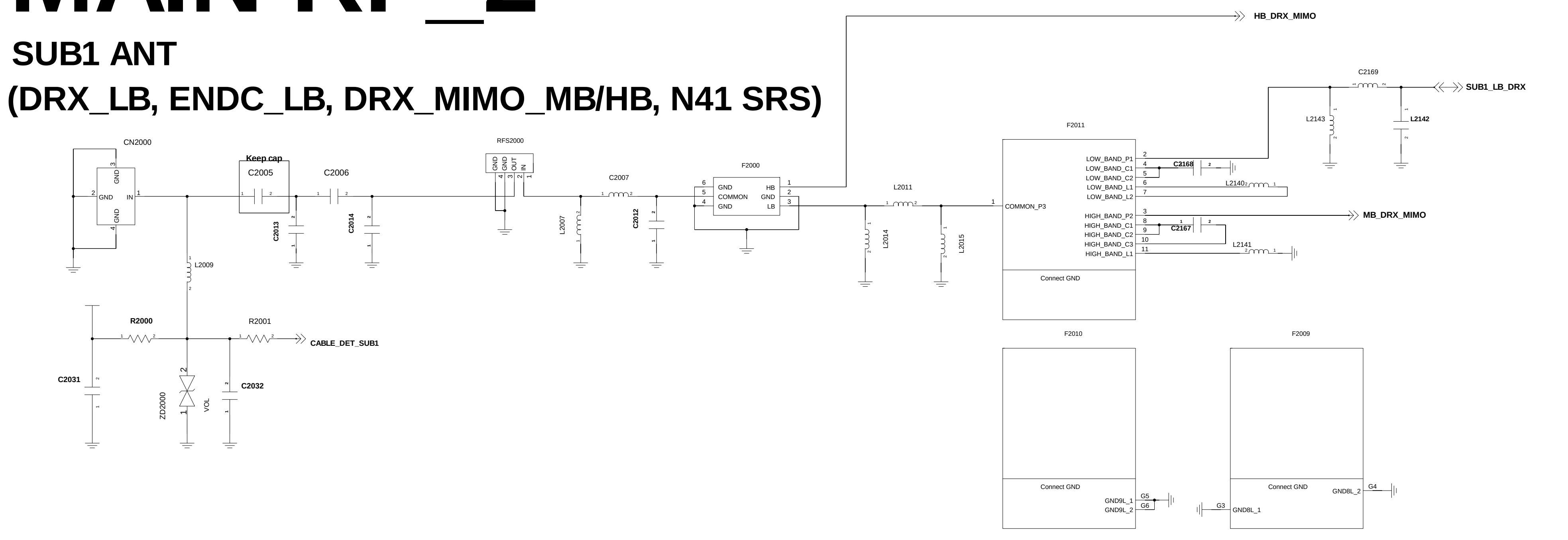
MAIN RF_1

MAIN1 ANT
(TRX_LB / MB / HB / N41)
need for WIFI BOM
U1001
RF_MAIN_1
SDM_RFFE0_DATA
SDM_RFFE0_CLK
RFS1000
F1001
U1000
MAIN1_HB_TRX
MAIN1_MB_TRX
MAIN1_LB_TRX
F1003
U1004
MAIN1_GRIP
WIFI0_GRIP
U1003
need for WIFI BOM
MAIN1 ANT XGND1
need for WIFI BOM
MAIN1 ANT XGND2
LB LPAMID
Global B5/8/12/13/17/20/26/28, G850/900
NA B5/8/12/13/14/20/26/28/29/71
U1002
Global only
TX_TR_OUT0_LB
TX_TR_OUT0_GSM_LB
PRX1_TR_IN_L
PRX2_TR_IN_L
B13/B14
F1005
TRX_B13_14
LNA_IN_AUX_B13_14
B28
F1004
TRX_B20_28
LNA_IN_AUX_B20_28
F1000
B71
TRX_B71
LNA_IN_AUX_B71
F1002
B29
LNA_IN_AUX_B29
NA only
OMH LPAMID
Global 1/2/3/4/7/25/32/38/40/41/66, G1800/1900, N41
NA 1/2/3/4/7/25/30/40/41/66/70, N41
TH1000
PA_THM0
THERMISTERS
U1017
TX_TR_OUT1_MB1
TX_TR_OUT1_HB1
TX_TR_OUT0_GSM_HB
Global only
PRX1_TR_IN_MH
PRX1_TR_IN_JMH
PRX2_TR_IN_MH
PRX2_TR_IN_JMH
Global only
F1007
PRX_B32
PRX_LNA_IN_B32
NA only
F1006
TRX_B30
TX_PA_OUT_B30
PRX_LNA_IN_B30
MAIN2 ANT
(N77/N78 TRX)
CN1000
Keep cap
RFS1001
U1007
CABLE_DET_MAIN2
U1009
CPL1_n77_78_OUT
SDR1_RFFE5_DATA
SDR1_RFFE5_CLK
0901 check!
SDR1_GRFC14
U1008
TX_TR_OUT0_UHB
TX_TR_OUT1_UHB
PRX1_TR_IN_UN
SDR1_RFFE2_DATA
SDR1_RFFE2_CLK
SUB3 ANT
(GPS_L1)
GPS_L1
CN1001
Keep cap
F1008
GPS_L1_RF_IN
CABLE_DET_GPS
U1011
U1012
LB/UHB APT Modulator(QET6105#0)
MHB/UHB APT Modulator(QET6105#1)
TITLE TabS7P
SIZE A1
DRAWN BY User Name
SHEET 1 of 15

MAIN RF_2

SUB1 ANT

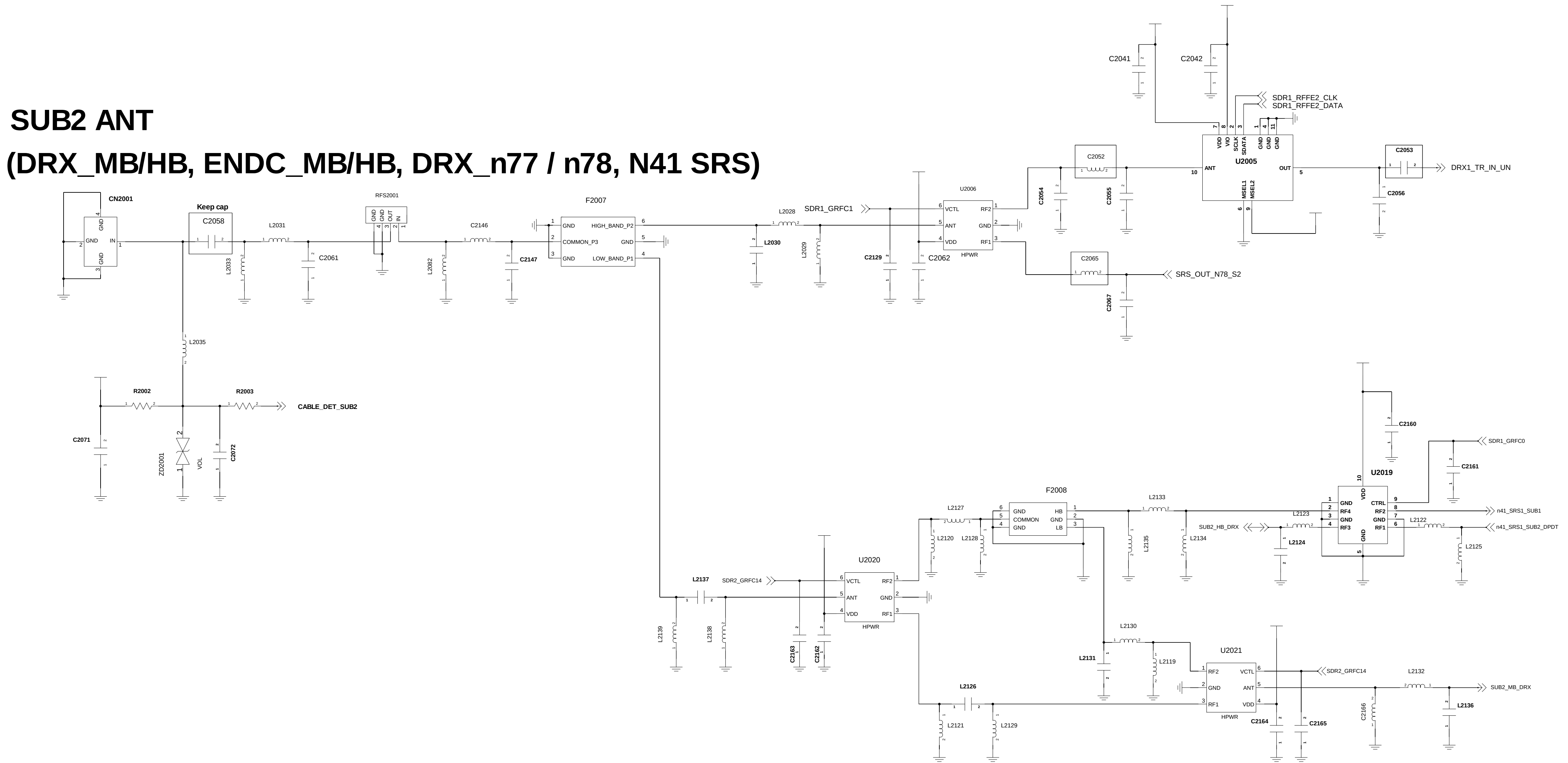
(DRX_LB, ENDC_LB, DRX_MIMO_MB/HB, N41 SRS)



CN2000
Keep cap
C2005
C2006
RFS2000
C2007
F2000
HB_DRX_MIMO
L2011
F2011
COMMON_P3
LOW_BAND_P1
HIGH_BAND_P2
Connect GND
C2169
SUB1_LB_DRX
L2143
L2142
MB_DRX_MIMO
C2168
C2167
L2140
L2141
L2009
R2000
R2001
CABLE_DET_SUB1
C2031
ZD2000
VOL
C2032
F2010
F2009
GND9L_1
GND9L_2
GND8L_1
GND8L_2

SUB2 ANT

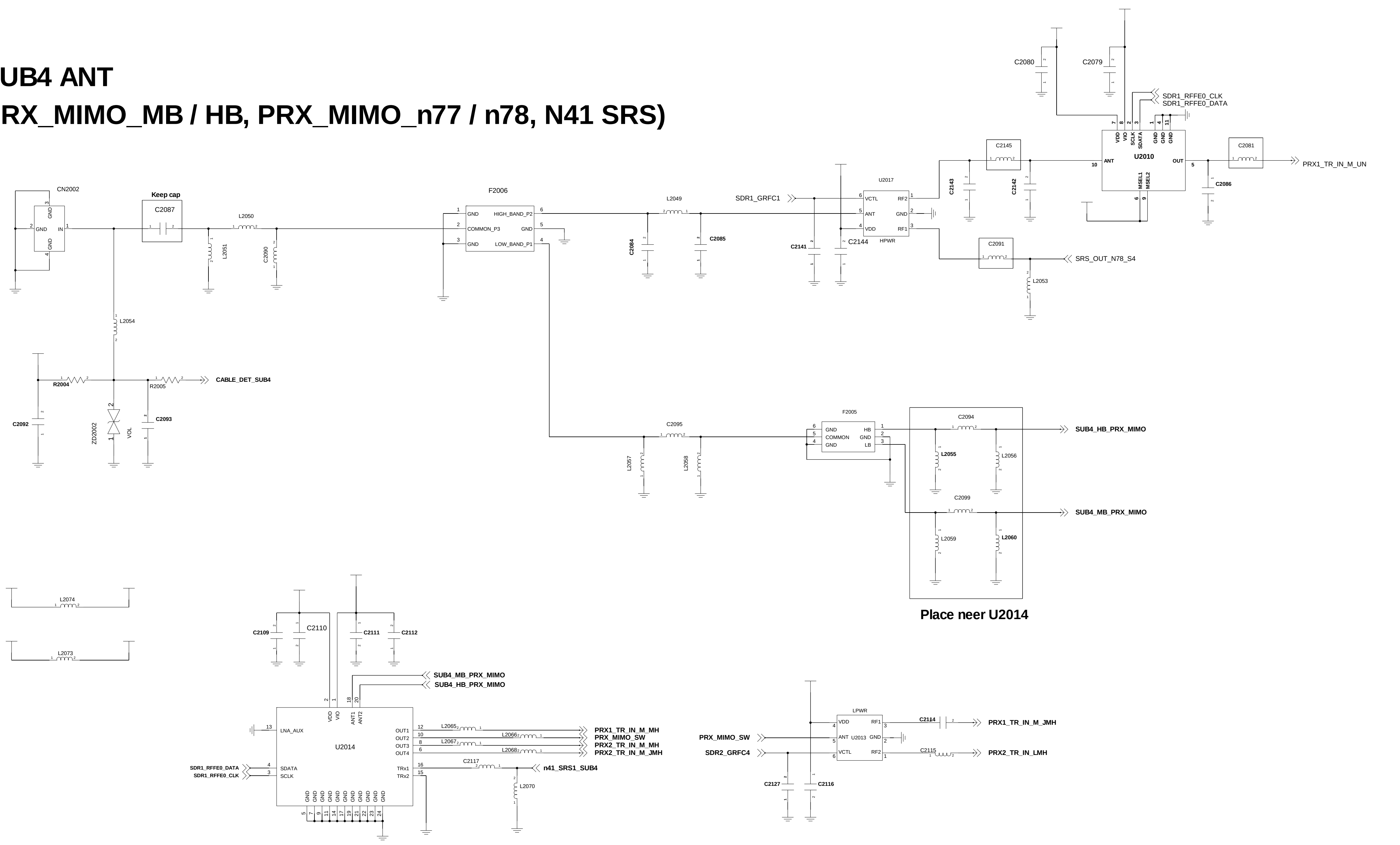
(DRX_MB/HB, ENDC_MB/HB, DRX_n77 / n78, N41 SRS)



C2041
C2042
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
U2005
DRX1_TR_IN_UN
C2053
C2056
C2052
C2054
C2055
U2006
SDR1_GRFC1
HPWR
C2065
SRS_OUT_N78_S2
C2067
CN2001
Keep cap
C2058
L2031
RFS2001
C2146
F2007
C2147
L2028
L2030
L2029
C2129
C2062
L2035
R2002
R2003
CABLE_DET_SUB2
C2071
ZD2001
VOL
C2072
C2160
SDR1_GRFC0
C2161
U2019
n41_SRS1_SUB1
n41_SRS1_SUB2_DPDT
F2008
L2133
SUB2_HB_DRX
L2123
L2124
L2122
L2125
L2127
L2120
L2128
L2135
L2134
U2020
L2137
SDR2_GRFC14
L2139
L2138
C2163
C2162
L2130
L2131
L2119
U2021
L2132
SUB2_MB_DRX
L2136
C2164
C2165
C2166
L2126
L2121
L2129

SUB4 ANT

(PRX_MIMO_MB / HB, PRX_MIMO_n77 / n78, N41 SRS)

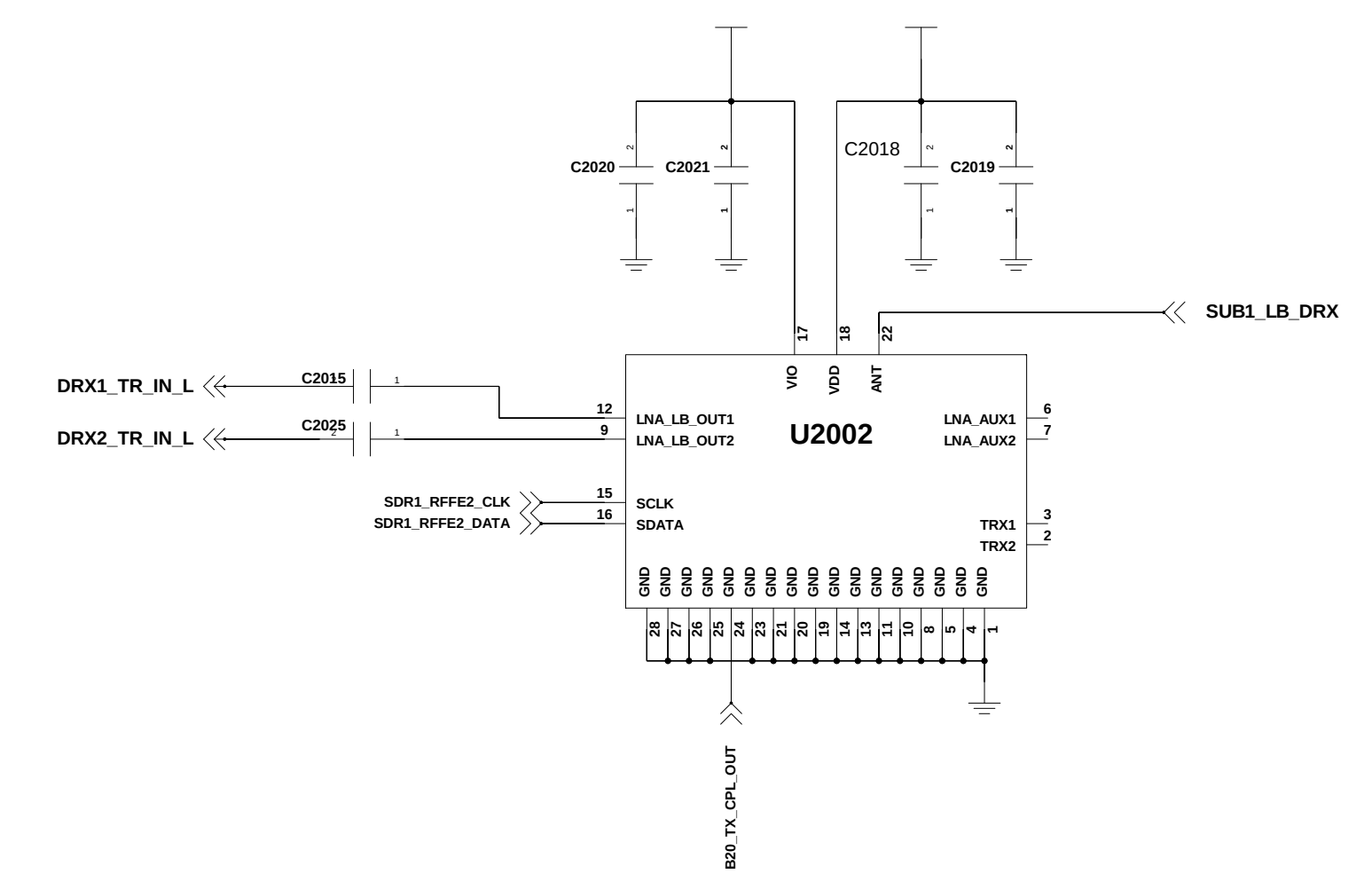


C2080
C2079
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
U2010
C2081
PRX1_TR_IN_M_UN
C2086
C2145
C2143
C2142
U2017
SDR1_GRFC1
HPWR
C2091
SRS_OUT_N78_S4
L2053
CN2002
Keep cap
C2087
L2050
L2051
C2090
F2006
L2049
C2084
C2085
C2141
C2144
L2054
R2004
R2005
CABLE_DET_SUB4
C2092
ZD2002
VOL
C2093
C2095
L2057
L2058
F2005
C2094
L2055
L2056
SUB4_HB_PRX_MIMO
C2099
L2059
L2060
SUB4_MB_PRX_MIMO
Place neer U2014
L2074
L2073
C2109
C2110
C2111
C2112
U2014
SDR1_RFFE0_DATA
SDR1_RFFE0_CLK
L2065
L2066
L2067
L2068
PRX1_TR_IN_M_MH
PRX_MIMO_SW
PRX2_TR_IN_M_MH
PRX2_TR_IN_M_JMH
C2117
n41_SRS1_SUB4
L2070
LPWR
U2013
PRX_MIMO_SW
SDR2_GRFC4
C2114
PRX1_TR_IN_M_JMH
C2115
PRX2_TR_IN_LMH
C2127
C2116

MHB MIMO_LNA OUT

DRX_LB

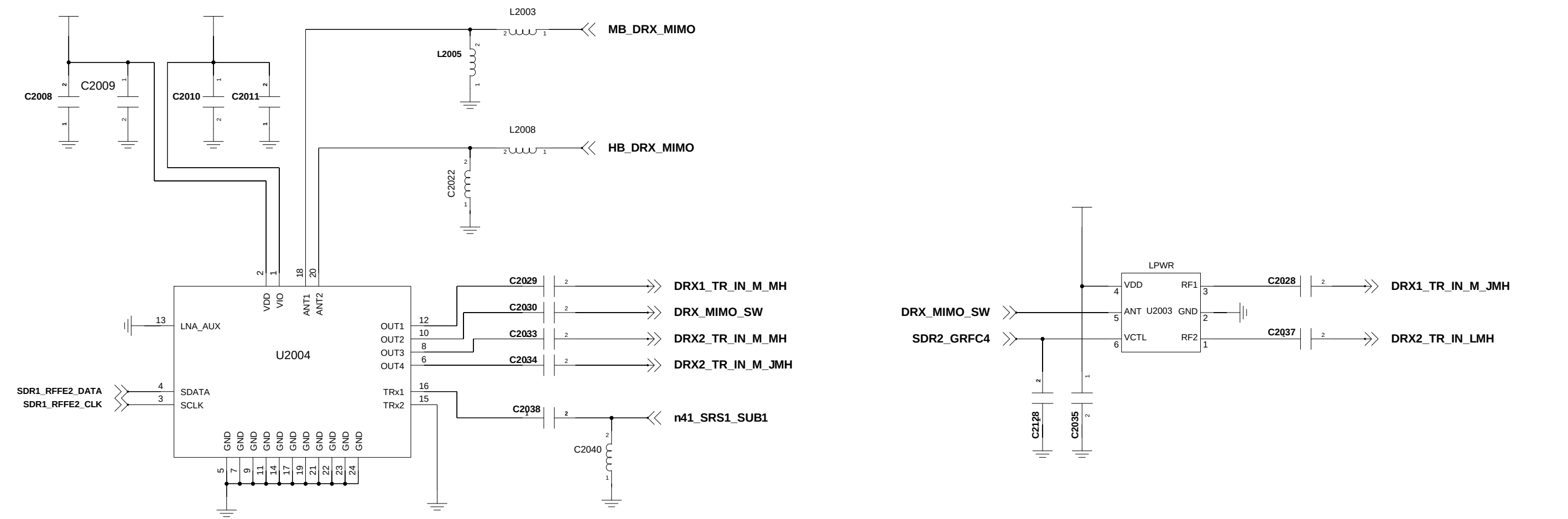
(B5/8/12/13/14/17/18/19/20/26/28/29/71_G850/900)



C2020
C2021
C2018
C2019
SUB1_LB_DRX
DRX1_TR_IN_L
DRX2_TR_IN_L
C2015
C2025
U2002
LNA_LB_OUT1
LNA_LB_OUT2
LNA_AUX1
LNA_AUX2
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
TRX1
TRX2
B20_TX_CPL_OUT

DRX_MHB_2

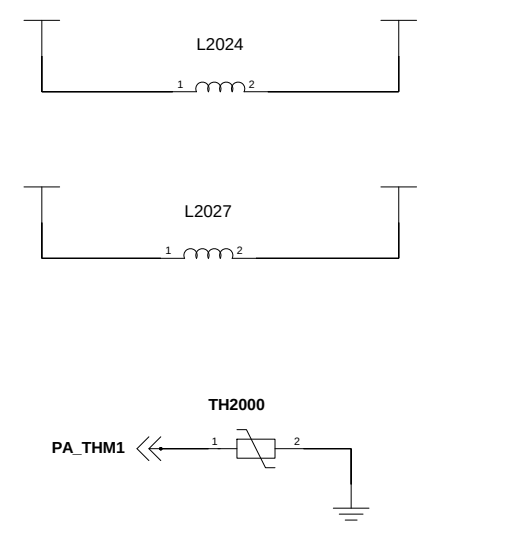
(B2/4/7/25/30/41/66)



C2008
C2009
C2010
C2011
L2003
MB_DRX_MIMO
L2005
L2008
HB_DRX_MIMO
C2022
U2004
LNA_AUX
SDR1_RFFE2_DATA
SDR1_RFFE2_CLK
C2029
DRX1_TR_IN_M_MH
C2030
DRX_MIMO_SW
C2033
DRX2_TR_IN_M_MH
C2034
DRX2_TR_IN_M_JMH
C2038
n41_SRS1_SUB1
C2040
LPWR
U2003
DRX_MIMO_SW
SDR2_GRFC4
C2028
DRX1_TR_IN_M_JMH
C2037
DRX2_TR_IN_LMH
C2128
C2035

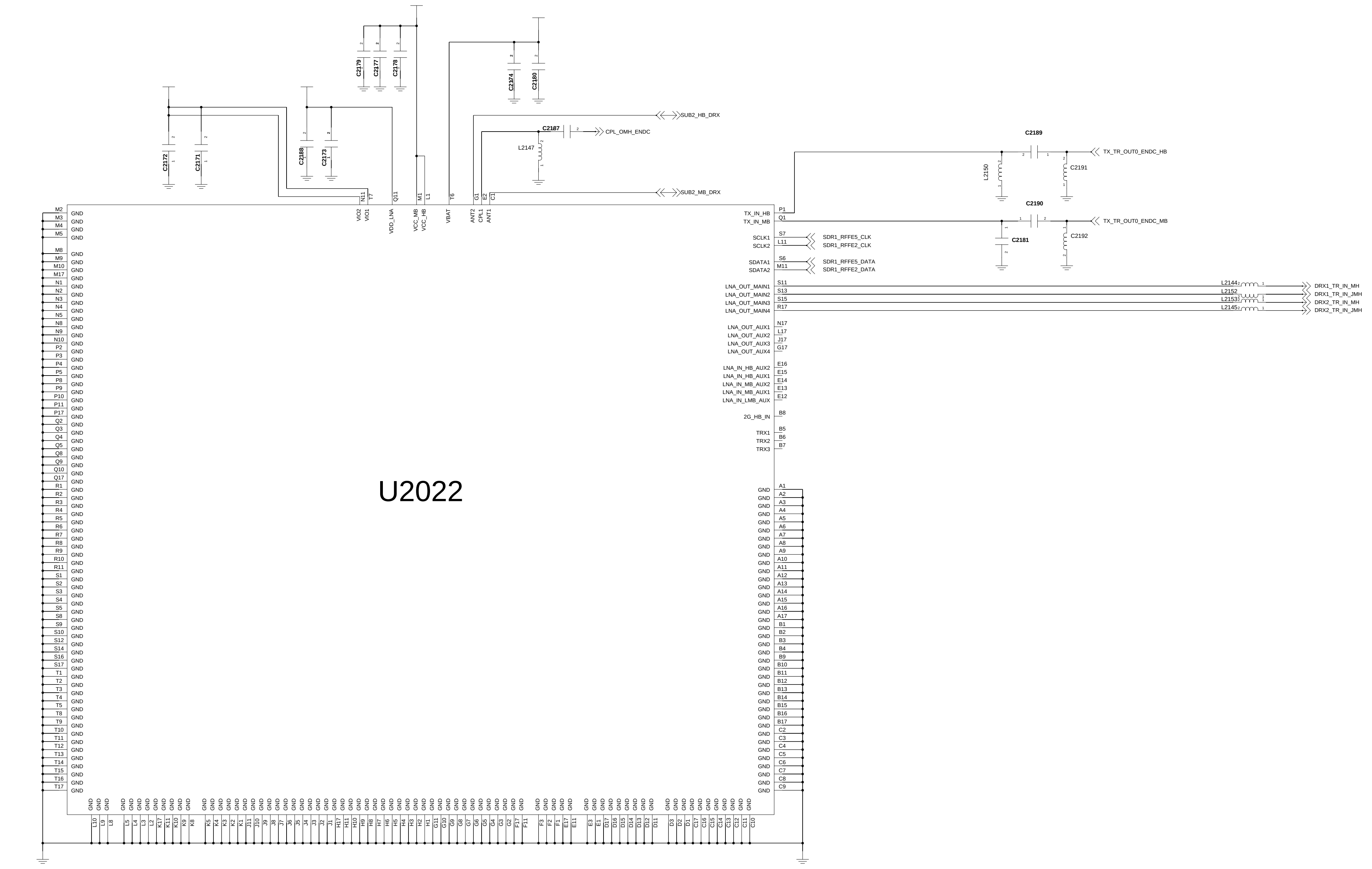
ENDC module

(B1/2/3/4/7/25/30/39/41/66/70_DCS/PCS)

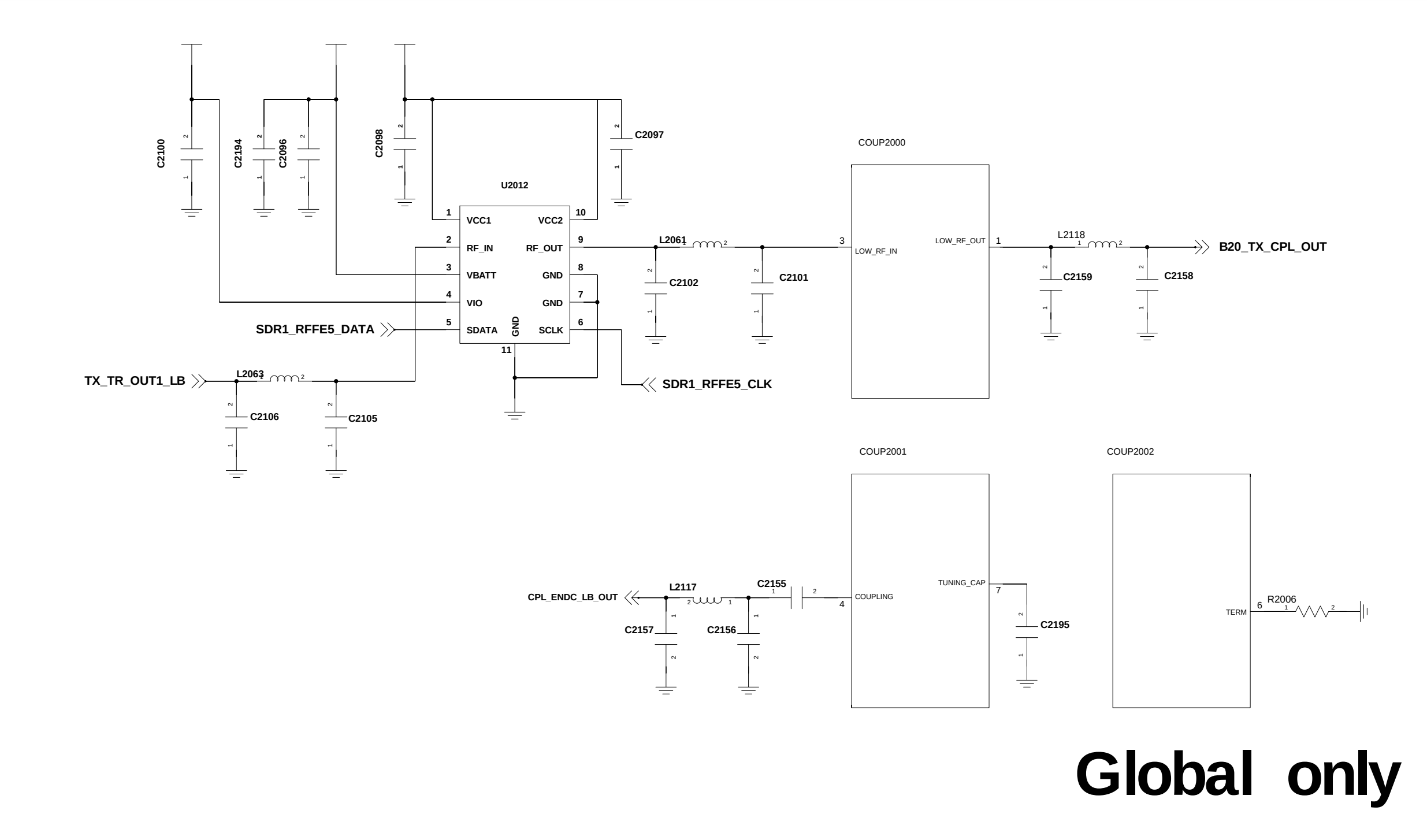


L2024
L2027
TH2000
PA_THM1

THERMISTERS

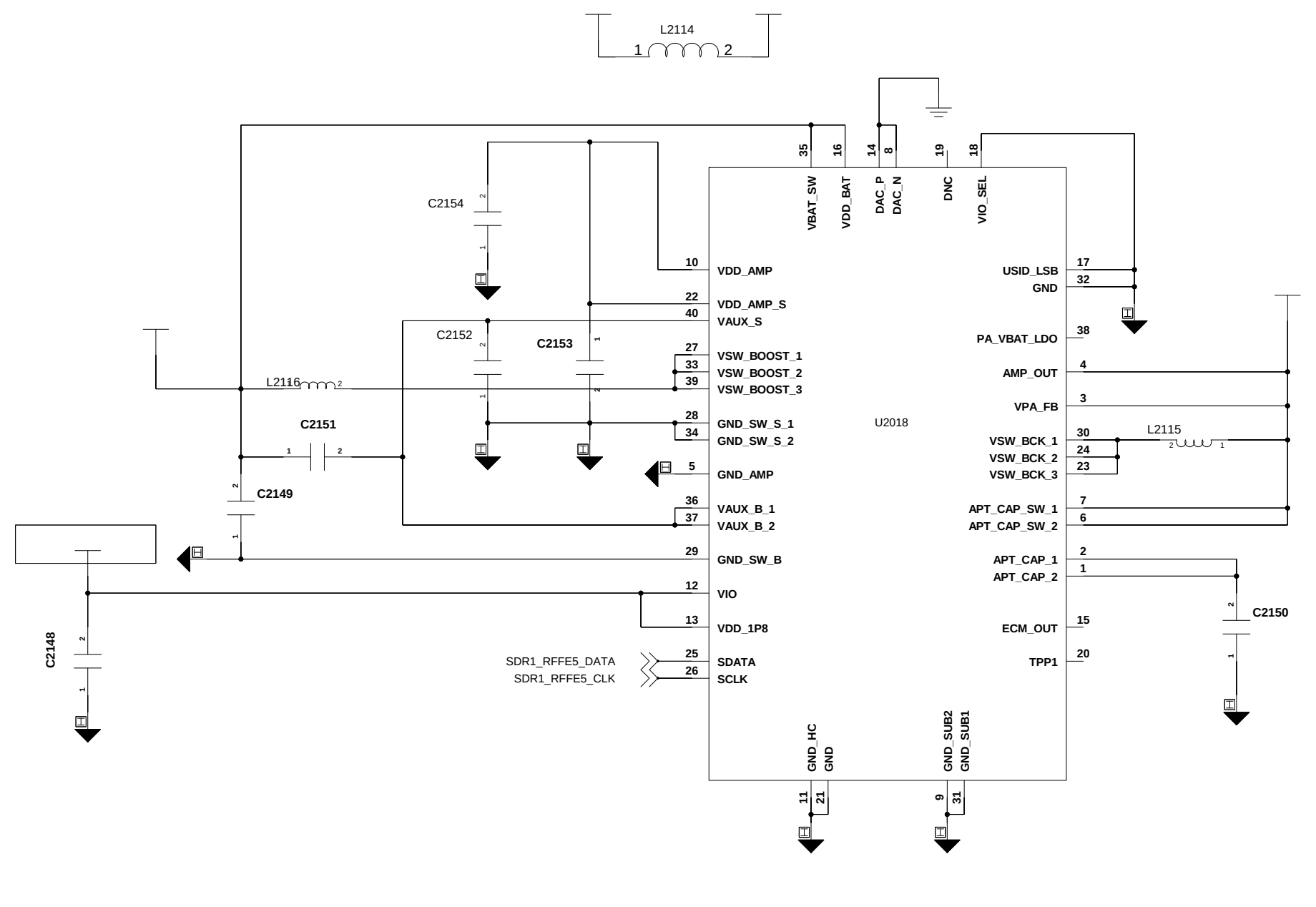


U2022
C2179
C2177
C2178
C2172
C2171
C2188
C2173
C2174
C2180
SUB2_HB_DRX
C2187
CPL_OMH_ENDC
L2147
SUB2_MB_DRX
C2189
TX_TR_OUT0_ENDC_HB
L2150
C2191
C2190
TX_TR_OUT0_ENDC_MB
C2181
C2192
TX_IN_HB
TX_IN_MB
SCLK1
SCLK2
SDATA1
SDATA2
SDR1_RFFE5_CLK
SDR1_RFFE2_CLK
SDR1_RFFE5_DATA
SDR1_RFFE2_DATA
LNA_OUT_MAIN1
LNA_OUT_MAIN2
LNA_OUT_MAIN3
LNA_OUT_MAIN4
L2144
L2152
L2153
L2145
DRX1_TR_IN_MH
DRX1_TR_IN_JMH
DRX2_TR_IN_MH
DRX2_TR_IN_JMH
LNA_OUT_AUX1
LNA_OUT_AUX2
LNA_OUT_AUX3
LNA_OUT_AUX4
LNA_IN_HB_AUX2
LNA_IN_HB_AUX1
LNA_IN_MB_AUX2
LNA_IN_MB_AUX1
LNA_IN_LMB_AUX
2G_HB_IN
TRX1
TRX2
TRX3
VIO2
VIO1
VDD_LNA
VCC_MB
VCC_HB
VBAT
ANT2
CPL1
ANT1
GND



C2100
C2194
C2096
C2098
C2097
U2012
VCC1
VCC2
RF_IN
RF_OUT
VBATT
VIO
SDATA
SCLK
GND
SDR1_RFFE5_DATA
SDR1_RFFE5_CLK
TX_TR_OUT1_LB
L2063
C2106
C2105
L2061
C2102
C2101
COUP2000
LOW_RF_IN
LOW_RF_OUT
L2118
C2159
C2158
B20_TX_CPL_OUT
COUP2001
COUPLING
TUNING_CAP
C2195
CPL_ENDC_LB_OUT
L2117
C2155
C2157
C2156
COUP2002
TERM
R2006

Global only

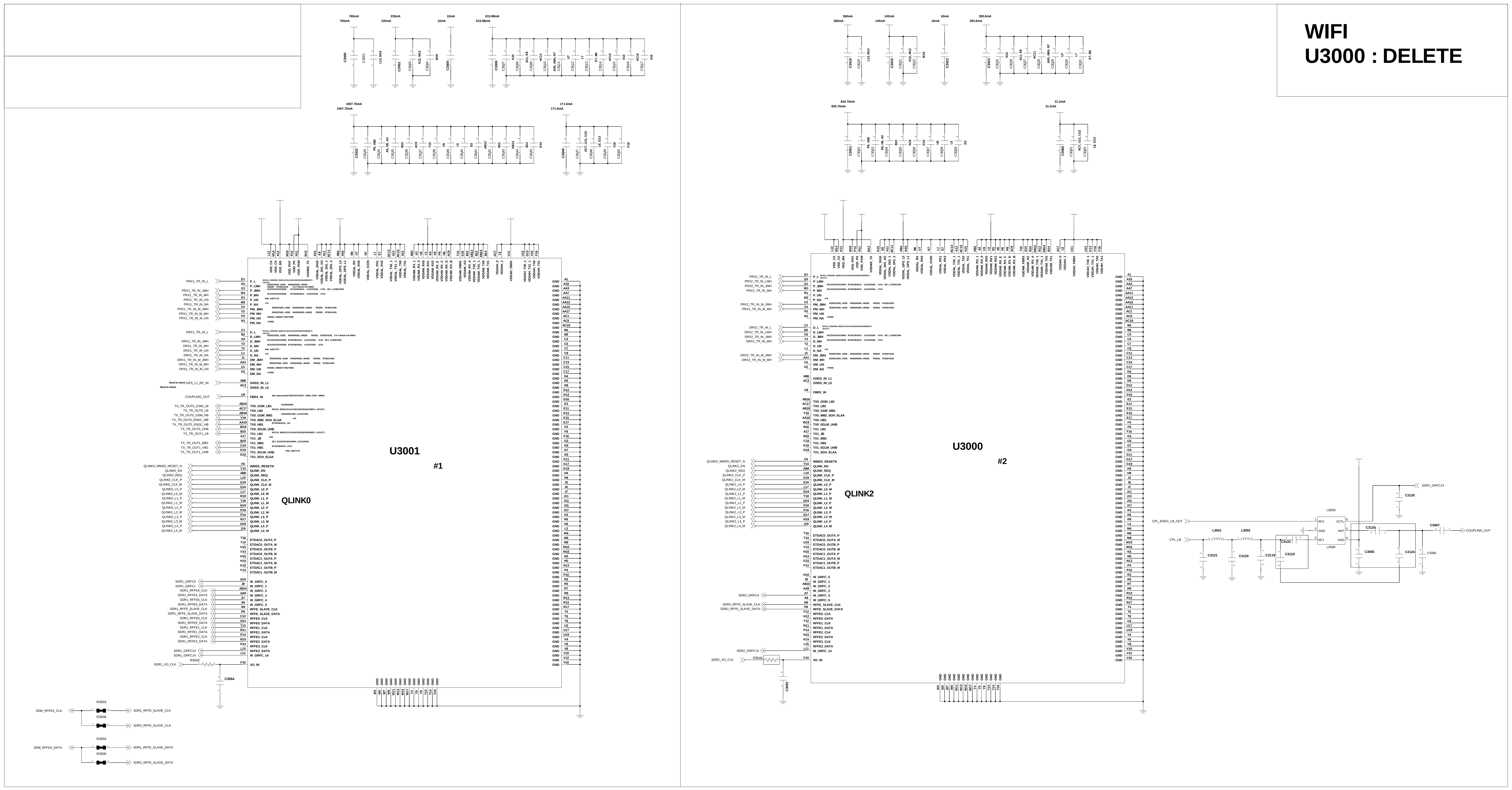


L2114
C2154
C2152
C2153
L2116
C2151
C2149
C2148
U2018
VBAT_SW
VDD_BAT
DAC_P
DAC_N
DNC
VIO_SEL
VDD_AMP
VDD_AMP_S
VAUX_S
VSW_BOOST_1
VSW_BOOST_2
VSW_BOOST_3
GND_SW_S_1
GND_SW_S_2
GND_AMP
VAUX_B_1
VAUX_B_2
GND_SW_B
VIO
VDD_1P8
SDATA
SCLK
SDR1_RFFE5_DATA
SDR1_RFFE5_CLK
GND_HC
GND_SUB2
GND_SUB1
USID_LSB
PA_VBAT_LDO
AMP_OUT
VPA_FB
VSW_BCK_1
VSW_BCK_2
VSW_BCK_3
L2115
APT_CAP_SW_1
APT_CAP_SW_2
APT_CAP_1
APT_CAP_2
ECM_OUT
TPP1
C2150

ENDC APT Modulator(QET5100)

TITLE TabS7P
SIZE A1 DWG NO REV
DRAWN BY User Name
SHEET 2 of 15 Date Drawing Date

MAIN RF_3

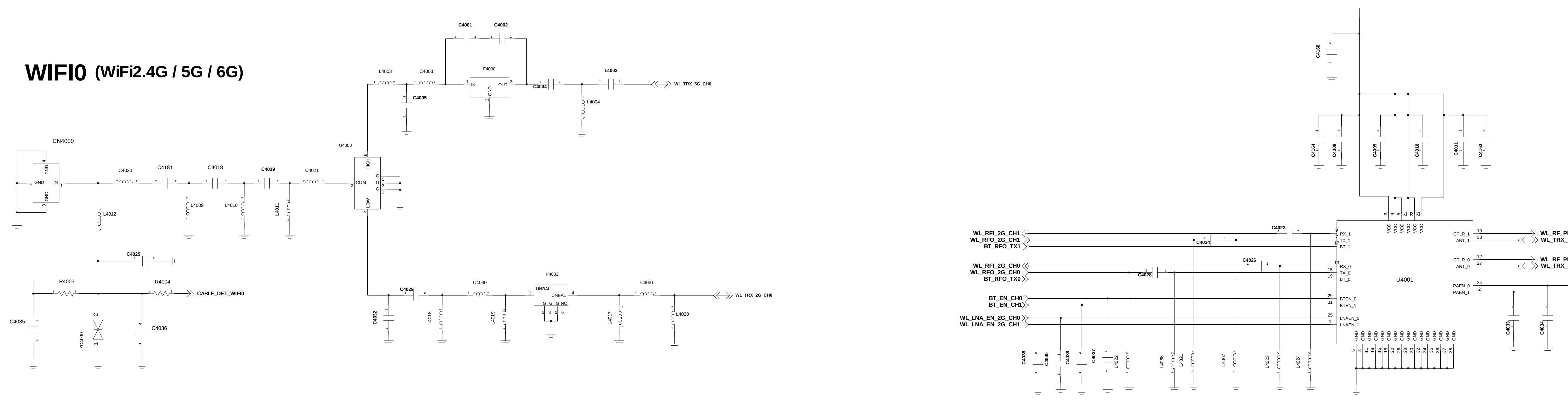

WIFI
U3000 : DELETE
U3001
#1
QLINK0
U3000
#2
QLINK2
765mA
235mA
10mA
615.98mA
1007.76mA
171.6mA
360mA
145mA
10mA
393.6mA
842.76mA
31.2mA
C3000
C3001
C3002
C3003
C3004
C3005
C3006
C3018
C3019
C3020
C3021
C3022
C3023
C3024
C3032
C3046
C3051
C3060
VDD_CX
VDD_MX
VDD_PHY
VDD_PX
VDD_PXM
VDDMX_TX
VDDAL_DIG0
VDDAL_DIG_XO
VDDAL_GPS_L5
VDDAL_GPS_L1
VDDAL_RX
VDDAL_ICON
VDDAL_TX0
VDDAL_TX1
VDDAM_RX_1
VDDAM_FBRX
VDDAM_XO
VDDAH_0
VDDAH_1
VDDAH_FBRX
VDDAH_TX0
VDDAH_TX1
PRX1_TR_IN_L
PRX1_TR_IN_JMH
PRX1_TR_IN_MH
PRX1_TR_IN_UN
PRX1_TR_IN_NA
PRX1_TR_IN_M_JMH
PRX1_TR_IN_M_MH
PRX1_TR_IN_M_UN
DRX1_TR_IN_L
DRX1_TR_IN_JMH
DRX1_TR_IN_MH
DRX1_TR_IN_UN
DRX1_TR_IN_NA
DRX1_TR_IN_M_JMH
DRX1_TR_IN_M_MH
DRX1_TR_IN_M_UN
Need to check.
GPS_L1_RF_IN
Need to check.
COUPLING_OUT
TX_TR_OUT0_GSM_LB
TX_TR_OUT0_LB
TX_TR_OUT0_GSM_HB
TX_TR_OUT0_ENDC_MB
TX_TR_OUT0_ENDC_HB
TX_TR_OUT0_UHB
TX_TR_OUT1_LB
TX_TR_OUT1_MB1
TX_TR_OUT1_HB1
TX_TR_OUT1_UHB
QLINK0_WMSS_RESET_N
QLINK0_EN
QLINK0_REQ
QLINK0_CLK_P
QLINK0_CLK_M
QLINK0_L0_P
QLINK0_L0_M
QLINK0_L1_P
QLINK0_L1_M
QLINK0_L2_P
QLINK0_L2_M
QLINK0_L3_P
QLINK0_L3_M
QLINK0_L4_P
QLINK0_L4_M
SDR1_GRFC0
SDR1_GRFC1
SDR1_RFFE4_CLK
SDR1_RFFE4_DATA
SDR1_RFFE5_CLK
SDR1_RFFE5_DATA
SDR1_RFFE_SLAVE_CLK
SDR1_RFFE_SLAVE_DATA
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
SDR1_RFFE1_CLK
SDR1_RFFE1_DATA
SDR1_RFFE2_CLK
SDR1_RFFE2_DATA
SDR1_GRFC13
SDR1_GRFC14
R3042
SDR1_XO_CLK
C3064
R3003
R3006
R3004
R3005
SDM_RFFE4_CLK
SDR1_RFFE_SLAVE_CLK
SDR2_RFFE_SLAVE_CLK
SDM_RFFE4_DATA
SDR1_RFFE_SLAVE_DATA
SDR2_RFFE_SLAVE_DATA
P_L
P_LMH
P_JMH
P_MH
P_UN
P_NA
PM_JMH
PM_MH
PM_UN
PM_NA
D_L
D_LMH
D_JMH
D_MH
D_UN
D_NA
DM_JMH
DM_MH
DM_UN
DM_NA
GNSS_IN_L1
GNSS_IN_L5
FBRX_IN
TX0_GSM_LB1
TX0_LB2
TX0_GSM_MB1
TX0_MB2_5GH_ELAA
TX0_HB1
TX0_5GLM_UHB
TX1_LB1
TX1_JB
TX1_MB1
TX1_HB1
TX1_5GLM_UHB
TX1_5GH_ELAA
WMSS_RESETN
QLINK_EN
QLINK_REQ
QLINK_CLK_P
QLINK_CLK_M
QLINK_L0_P
QLINK_L0_M
QLINK_L1_P
QLINK_L1_M
QLINK_L2_P
QLINK_L2_M
QLINK_L3_P
QLINK_L3_M
QLINK_L4_P
QLINK_L4_M
ETDAC0_OUTA_P
ETDAC0_OUTA_M
ETDAC0_OUTB_P
ETDAC0_OUTB_M
ETDAC1_OUTA_P
ETDAC1_OUTA_M
ETDAC1_OUTB_P
ETDAC1_OUTB_M
W_GRFC_0
W_GRFC_1
W_GRFC_2
W_GRFC_3
W_GRFC_4
W_GRFC_5
RFFE_SLAVE_CLK
RFFE_SLAVE_DATA
RFFE0_CLK
RFFE0_DATA
RFFE1_CLK
RFFE1_DATA
RFFE2_CLK
RFFE2_DATA
RFFE3_CLK
RFFE3_DATA
W_GRFC_14
XO_IN
GND
Max input power(CDMA/3G/4G/5G : 5dBm, GSM : 9dBm)
PRX2_TR_IN_L
PRX2_TR_IN_LMH
PRX2_TR_IN_JMH
PRX2_TR_IN_MH
PRX2_TR_IN_M_JMH
PRX2_TR_IN_M_MH
DRX2_TR_IN_L
DRX2_TR_IN_LMH
DRX2_TR_IN_JMH
DRX2_TR_IN_MH
DRX2_TR_IN_M_JMH
DRX2_TR_IN_M_MH
QLINK2_WMSS_RESET_N
QLINK2_EN
QLINK2_REQ
QLINK2_CLK_P
QLINK2_CLK_M
QLINK2_L0_P
QLINK2_L0_M
QLINK2_L1_P
QLINK2_L1_M
QLINK2_L2_P
QLINK2_L2_M
QLINK2_L3_P
QLINK2_L3_M
QLINK2_L4_P
QLINK2_L4_M
SDR2_GRFC4
SDR2_RFFE_SLAVE_CLK
SDR2_RFFE_SLAVE_DATA
SDR2_GRFC14
R3041
SDR2_XO_CLK
C3063
U3002
RF2
GND
RF1
VCTL
ANT
VDD
LPWR
CPL_ENDC_LB_OUT
CPL_LB
L3001
L3002
C3121
C3120
C3119
C3122
C3123
C3065
C3125
C3124
C3126
C3066
C3067
SDR1_GRFC13
COUPLING_OUT

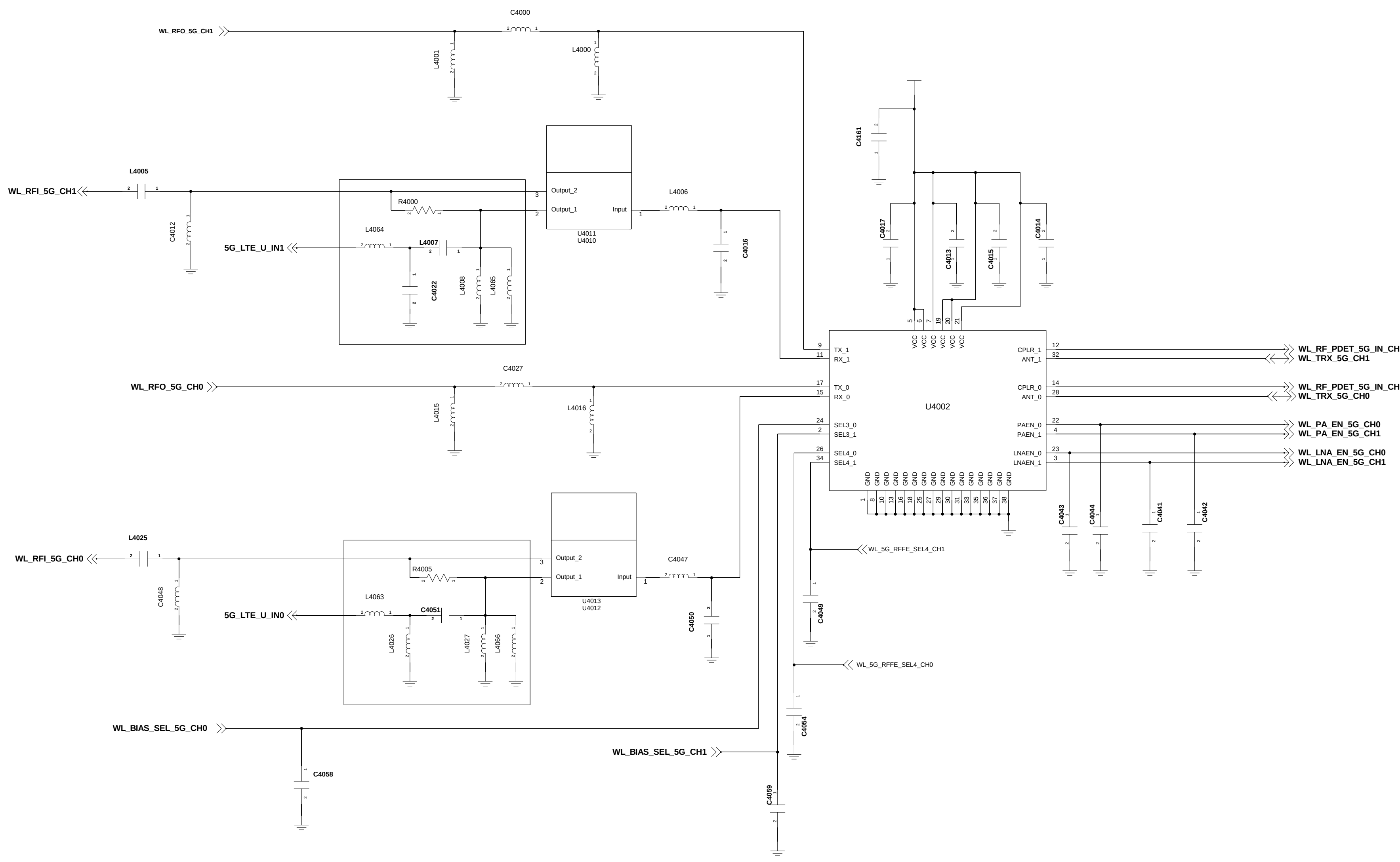
TRANSCEIVER(Dual SDR735)

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 3 of 15
Date

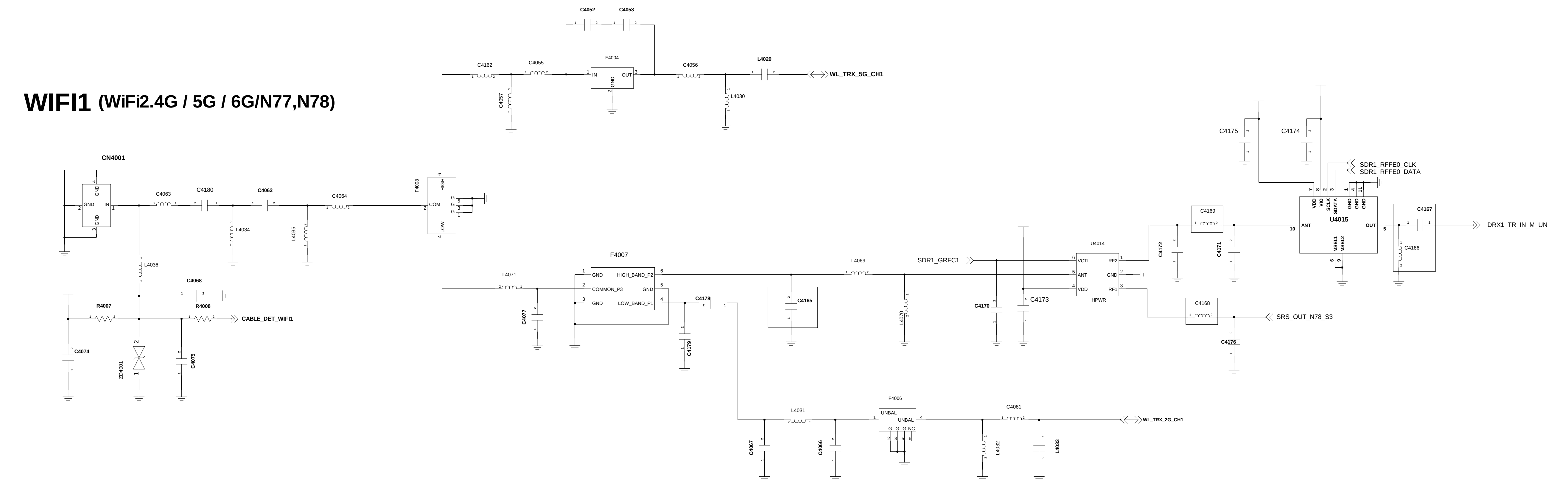
Connectivity

WIFI0 (WiFi2.4G / 5G / 6G)

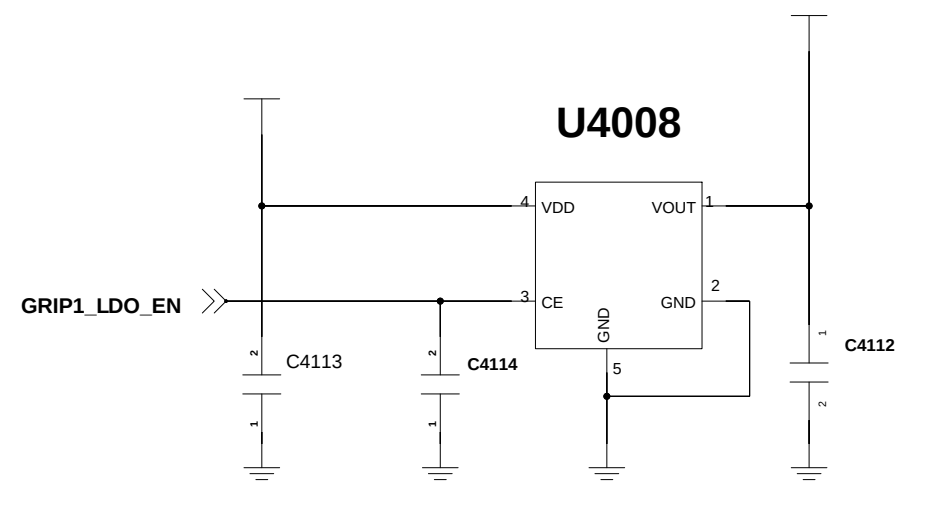

CN4000
GND
IN
C4020
C4181
C4018
C4019
C4021
L4009
L4010
L4011
L4012
U4000
HIGH
COM
LOW
L4003
C4003
C4001
C4002
F4000
IN
OUT
C4004
C4005
L4002
L4004
WL_TRX_5G_CH0
C4025
R4003
R4004
CABLE_DET_WIFI0
C4035
ZD4000
C4036
C4029
C4032
C4030
L4018
L4019
F4002
UNBAL
C4031
L4017
L4020
WL_TRX_2G_CH0
C4160
C4164
C4008
C4009
C4010
C4011
C4163
U4001
VCC
RX_1
TX_1
BT_1
RX_0
TX_0
BT_0
BTEN_0
BTEN_1
LNAEN_0
LNAEN_1
CPLR_1
ANT_1
CPLR_0
ANT_0
PAEN_0
PAEN_1
GND
WL_RFI_2G_CH1
WL_RFO_2G_CH1
BT_RFO_TX1
WL_RFI_2G_CH0
WL_RFO_2G_CH0
BT_RFO_TX0
BT_EN_CH0
BT_EN_CH1
WL_LNA_EN_2G_CH0
WL_LNA_EN_2G_CH1
C4023
C4024
C4026
C4028
C4038
C4040
C4039
C4037
L4022
L4068
L4021
L4067
L4023
L4024
WL_RF_PDET_2G_IN_CH1
WL_TRX_2G_CH1
WL_RF_PDET_2G_IN_CH0
WL_TRX_2G_CH0
WL_PA_EN_2G_CH0
WL_PA_EN_2G_CH1
C4033
C4034


WL_RFO_5G_CH1
C4000
L4001
L4000
L4005
WL_RFI_5G_CH1
C4012
L4064
L4007
R4000
5G_LTE_U_IN1
C4022
L4008
L4065
U4011
U4010
Output_2
Output_1
Input
L4006
C4016
C4161
C4017
C4013
C4015
C4014
U4002
VCC
TX_1
RX_1
TX_0
RX_0
SEL3_0
SEL3_1
SEL4_0
SEL4_1
CPLR_1
ANT_1
CPLR_0
ANT_0
PAEN_0
PAEN_1
LNAEN_0
LNAEN_1
GND
WL_RF_PDET_5G_IN_CH1
WL_TRX_5G_CH1
WL_RF_PDET_5G_IN_CH0
WL_TRX_5G_CH0
WL_PA_EN_5G_CH0
WL_PA_EN_5G_CH1
WL_LNA_EN_5G_CH0
WL_LNA_EN_5G_CH1
WL_RFO_5G_CH0
C4027
L4015
L4016
C4043
C4044
C4041
C4042
WL_5G_RFFE_SEL4_CH1
L4025
WL_RFI_5G_CH0
C4048
L4063
R4005
C4051
5G_LTE_U_IN0
L4026
L4027
L4066
U4013
U4012
C4047
C4050
C4049
WL_5G_RFFE_SEL4_CH0
C4054
WL_BIAS_SEL_5G_CH0
C4058
WL_BIAS_SEL_5G_CH1
C4059

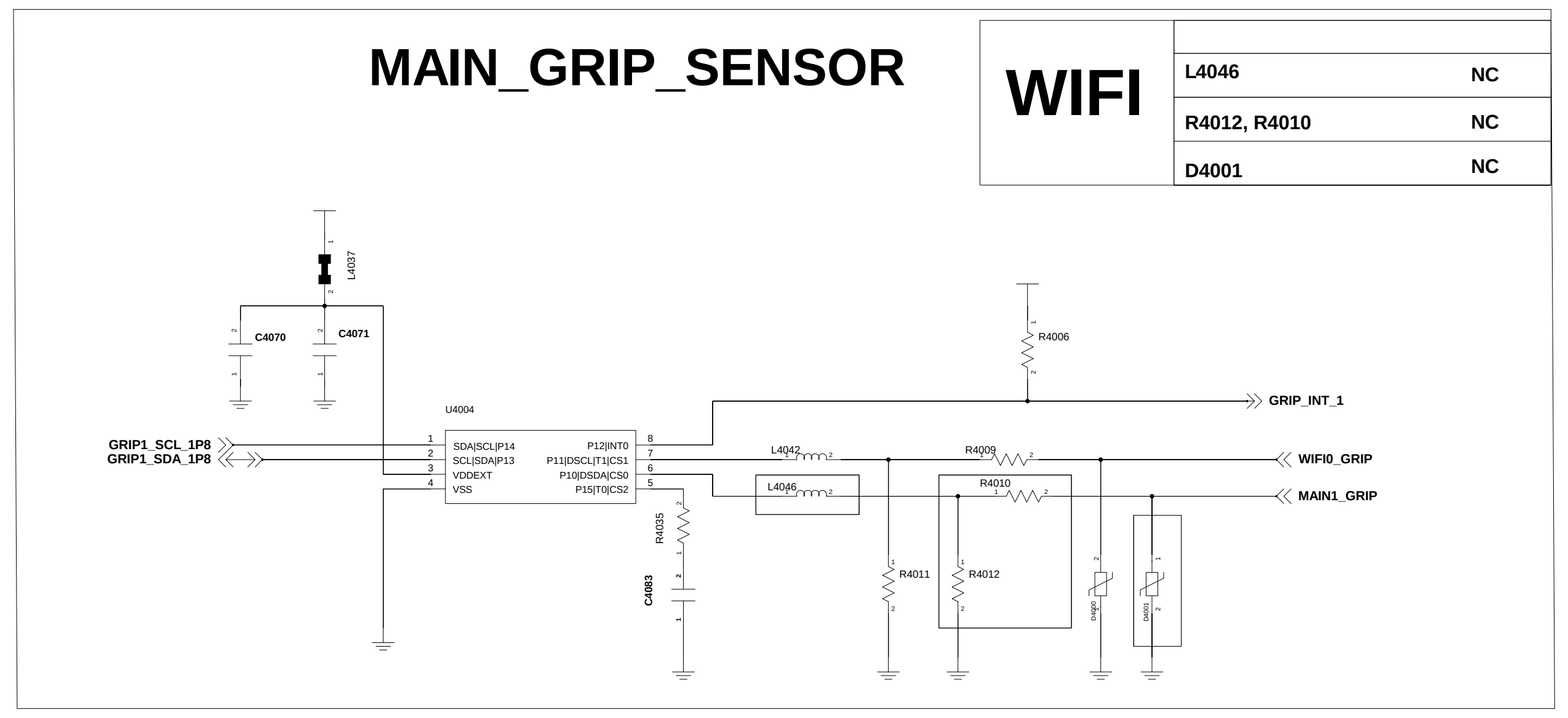
WIFI1 (WiFi2.4G / 5G / 6G/N77,N78)


CN4001
GND
IN
C4063
C4180
C4062
C4064
L4034
L4035
F4008
HIGH
COM
LOW
L4036
C4068
R4007
R4008
CABLE_DET_WIFI1
C4074
ZD4001
C4075
C4162
C4055
C4052
C4053
F4004
IN
OUT
C4056
L4029
C4057
L4030
WL_TRX_5G_CH1
L4071
F4007
GND
COMMON_P3
HIGH_BAND_P2
LOW_BAND_P1
C4077
C4178
C4179
C4165
L4069
L4070
SDR1_GRFC1
C4170
C4173
U4014
VCTL
ANT
VDD
RF2
RF1
HPWR
C4169
C4172
C4171
C4168
SRS_OUT_N78_S3
C4176
C4175
C4174
SDR1_RFFE0_CLK
SDR1_RFFE0_DATA
U4015
VDD
VIO
SCLK
SDATA
MSEL1
MSEL2
OUT
C4167
C4166
DRX1_TR_IN_M_UN
L4031
C4067
C4066
F4006
UNBAL
C4061
L4032
L4033
WL_TRX_2G_CH1

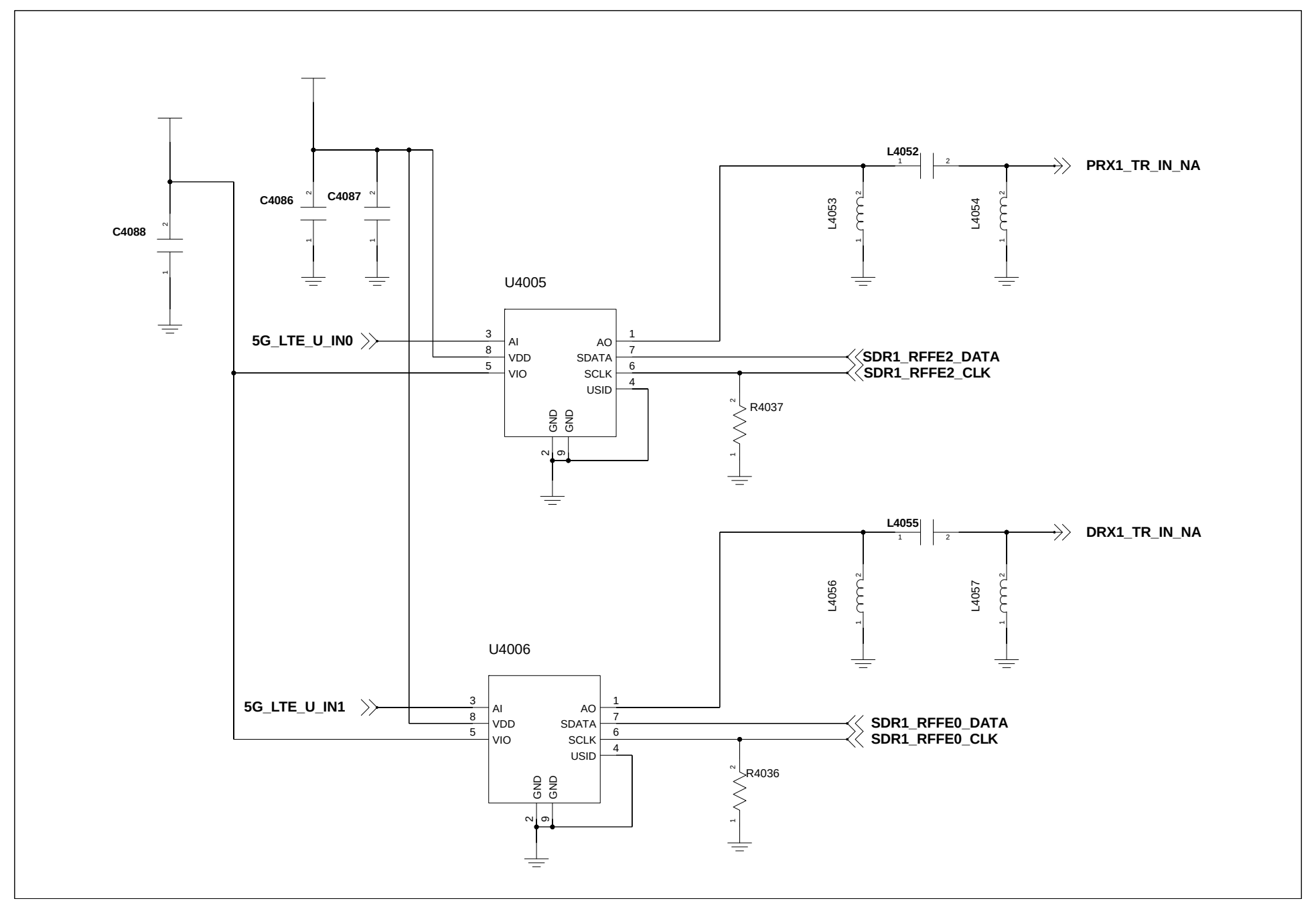
GRIP1 LDO


U4008
VDD
VOUT
CE
GND
GRIP1_LDO_EN
C4113
C4114
C4112

MAIN_GRIP_SENSOR

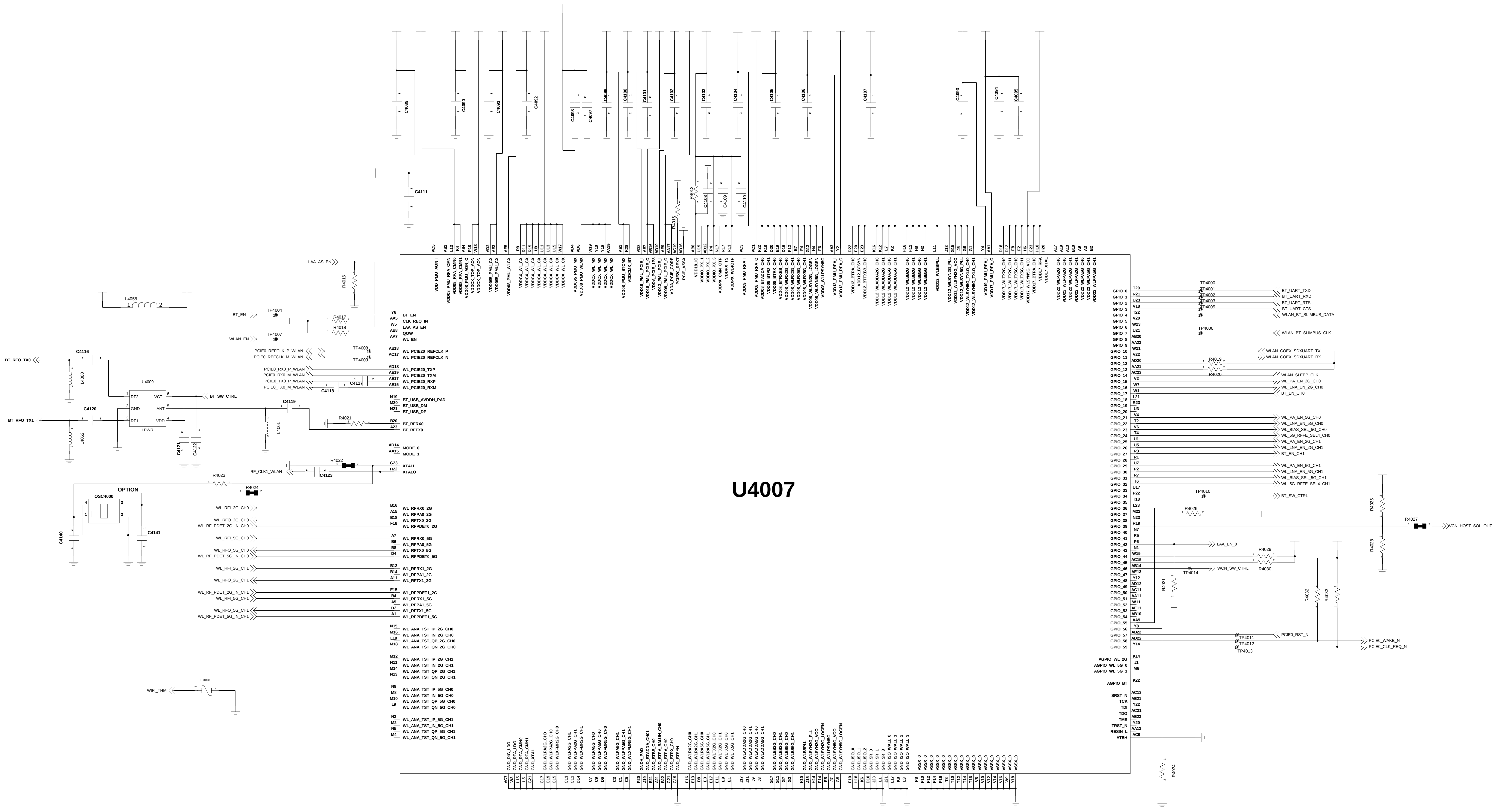

L4037
C4070
C4071
U4004
SDA|SCL|P14
SCL|SDA|P13
VDDEXT
VSS
P12|INT0
P11|DSCL|T1|CS1
P10|DSDA|CS0
P15|T0|CS2
GRIP1_SCL_1P8
GRIP1_SDA_1P8
R4006
GRIP_INT_1
L4042
L4046
R4009
R4010
WIFI0_GRIP
MAIN1_GRIP
R4035
C4083
R4011
R4012
D4000
D4001

WIFI	
L4046	NC
R4012, R4010	NC
D4001	NC

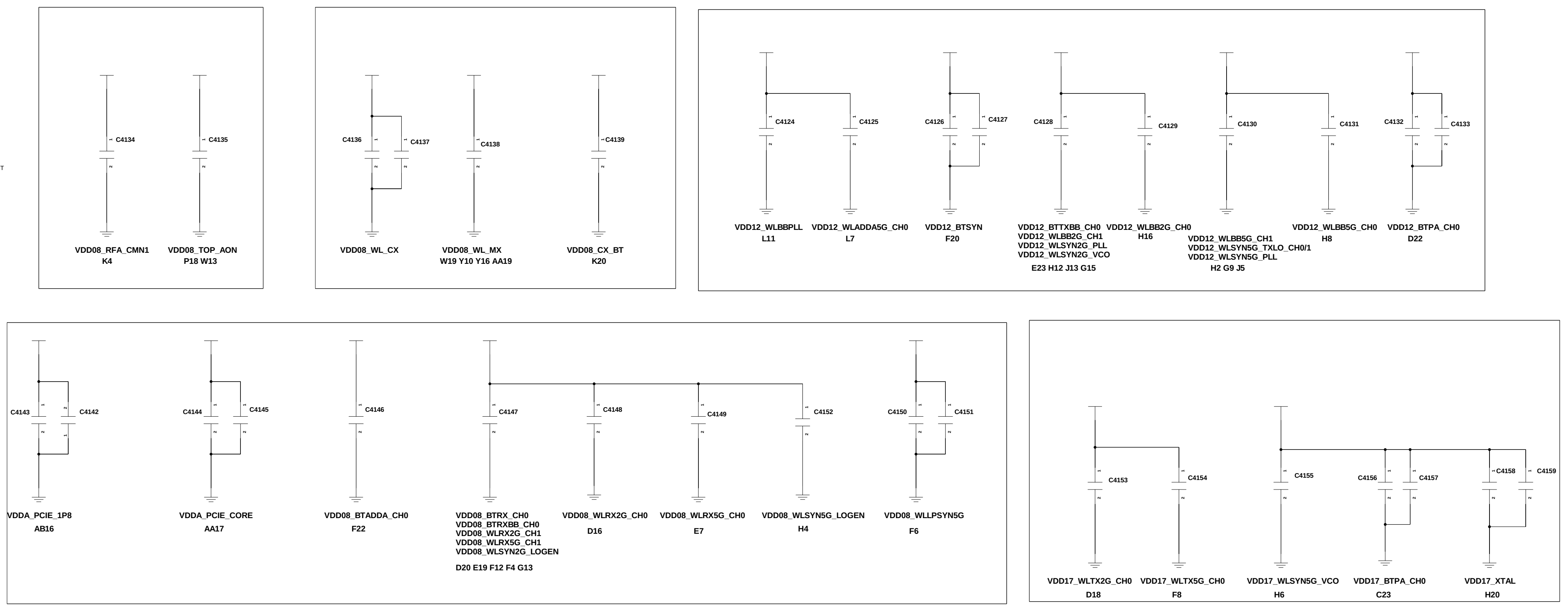

C4088
C4086
C4087
U4005
AI
VDD
VIO
AO
SDATA
SCLK
USID
GND
5G_LTE_U_IN0
SDR1_RFFE2_DATA
SDR1_RFFE2_CLK
R4037
L4052
L4053
L4054
PRX1_TR_IN_NA
L4055
L4056
L4057
DRX1_TR_IN_NA
U4006
5G_LTE_U_IN1
SDR1_RFFE0_DATA
SDR1_RFFE0_CLK
R4036

LAA LNA

LAA Option
mmW : O
SUB6, WIFI : X


U4007
C4089
C4090
C4091
C4092
C4098
C4097
C4099
C4100
C4101
C4102
C4103
C4104
C4105
C4106
C4107
C4093
C4094
C4095
C4111
R4013
R4015
C4108
C4109
C4110
LAA_AS_EN
R4016
L4058
BT_EN
TP4004
R4017
R4018
WLAN_EN
TP4007
BT_EN
CLK_REQ_IN
LAA_AS_EN
QOW
WL_EN
PCIE0_REFCLK_P_WLAN
PCIE0_REFCLK_M_WLAN
TP4008
TP4009
WL_PCIE20_REFCLK_P
WL_PCIE20_REFCLK_N
PCIE0_RX0_P_WLAN
PCIE0_RX0_M_WLAN
PCIE0_TX0_P_WLAN
PCIE0_TX0_M_WLAN
C4117
C4118
WL_PCIE20_TXP
WL_PCIE20_TXM
WL_PCIE20_RXP
WL_PCIE20_RXM
BT_RFO_TX0
C4116
L4060
U4009
RF2
GND
RF1
VCTL
ANT
VDD
LPWR
BT_SW_CTRL
BT_RFO_TX1
C4120
L4062
C4119
L4061
R4021
C4121
C4122
BT_USB_AVDDH_PAD
BT_USB_DM
BT_USB_DP
BT_RFRX0
BT_RFTX0
MODE_0
MODE_1
RF_CLK1_WLAN
R4022
C4123
XTALI
XTALO
R4023
R4024
OPTION
OSC4000
C4140
C4141
WL_RFI_2G_CH0
WL_RFO_2G_CH0
WL_RF_PDET_2G_IN_CH0
WL_RFI_5G_CH0
WL_RFO_5G_CH0
WL_RF_PDET_5G_IN_CH0
WL_RFI_2G_CH1
WL_RFO_2G_CH1
WL_RF_PDET_2G_IN_CH1
WL_RFI_5G_CH1
WL_RFO_5G_CH1
WL_RF_PDET_5G_IN_CH1
WL_RFRX0_2G
WL_RFPA0_2G
WL_RFTX0_2G
WL_RFPDET0_2G
WL_RFRX0_5G
WL_RFPA0_5G
WL_RFTX0_5G
WL_RFPDET0_5G
WL_RFRX1_2G
WL_RFPA1_2G
WL_RFTX1_2G
WL_RFPDET1_2G
WL_RFRX1_5G
WL_RFPA1_5G
WL_RFTX1_5G
WL_RFPDET1_5G
WL_ANA_TST_IP_2G_CH0
WL_ANA_TST_IN_2G_CH0
WL_ANA_TST_QP_2G_CH0
WL_ANA_TST_QN_2G_CH0
WL_ANA_TST_IP_2G_CH1
WL_ANA_TST_IN_2G_CH1
WL_ANA_TST_QP_2G_CH1
WL_ANA_TST_QN_2G_CH1
WL_ANA_TST_IP_5G_CH0
WL_ANA_TST_IN_5G_CH0
WL_ANA_TST_QP_5G_CH0
WL_ANA_TST_QN_5G_CH0
WL_ANA_TST_IP_5G_CH1
WL_ANA_TST_IN_5G_CH1
WL_ANA_TST_QP_5G_CH1
WL_ANA_TST_QN_5G_CH1
WIFI_THM
TH4000
GPIO_0 … GPIO_59
TP4000
TP4001
TP4002
TP4003
TP4005
BT_UART_TXD
BT_UART_RXD
BT_UART_RTS
BT_UART_CTS
WLAN_BT_SLIMBUS_DATA
TP4006
WLAN_BT_SLIMBUS_CLK
WLAN_COEX_SDXUART_TX
WLAN_COEX_SDXUART_RX
R4019
R4020
WLAN_SLEEP_CLK
WL_PA_EN_2G_CH0
WL_LNA_EN_2G_CH0
BT_EN_CH0
WL_PA_EN_5G_CH0
WL_LNA_EN_5G_CH0
WL_BIAS_SEL_5G_CH0
WL_5G_RFFE_SEL4_CH0
WL_PA_EN_2G_CH1
WL_LNA_EN_2G_CH1
BT_EN_CH1
WL_PA_EN_5G_CH1
WL_LNA_EN_5G_CH1
WL_BIAS_SEL_5G_CH1
WL_5G_RFFE_SEL4_CH1
TP4010
BT_SW_CTRL
R4026
R4025
R4027
R4028
WCN_HOST_SOL_OUT
LAA_EN_0
R4029
R4030
TP4014
WCN_SW_CTRL
R4031
R4032
R4033
TP4011
TP4012
TP4013
PCIE0_RST_N
PCIE0_WAKE_N
PCIE0_CLK_REQ_N
AGPIO_WL_2G
AGPIO_WL_5G_0
AGPIO_WL_5G_1
AGPIO_BT
SRST_N
TCK
TDI
TDO
TMS
TRST_N
RESIN_L
ATBH
R4034

WCN6856 Decap


C4134
C4135
VDD08_RFA_CMN1
K4
VDD08_TOP_AON
P18 W13
C4136
C4137
C4138
C4139
VDD08_WL_CX
VDD08_WL_MX
W19 Y10 Y16 AA19
VDD08_CX_BT
K20
C4124
C4125
C4126
C4127
C4128
C4129
C4130
C4131
C4132
C4133
VDD12_WLBBPLL
L11
VDD12_WLADDA5G_CH0
L7
VDD12_BTSYN
F20
VDD12_BTTXBB_CH0
VDD12_WLBB2G_CH1
VDD12_WLSYN2G_PLL
VDD12_WLSYN2G_VCO
E23 H12 J13 G15
VDD12_WLBB2G_CH0
H16
VDD12_WLBB5G_CH1
VDD12_WLSYN5G_TXLO_CH0/1
VDD12_WLSYN5G_PLL
H2 G9 J5
VDD12_WLBB5G_CH0
H8
VDD12_BTPA_CH0
D22
C4143
C4142
VDDA_PCIE_1P8
AB16
C4144
C4145
VDDA_PCIE_CORE
AA17
C4146
VDD08_BTADDA_CH0
F22
C4147
C4148
C4149
C4152
C4150
C4151
VDD08_BTRX_CH0
VDD08_BTRXBB_CH0
VDD08_WLRX2G_CH1
VDD08_WLRX5G_CH1
VDD08_WLSYN2G_LOGEN
D20 E19 F12 F4 G13
VDD08_WLRX2G_CH0
D16
VDD08_WLRX5G_CH0
E7
VDD08_WLSYN5G_LOGEN
H4
VDD08_WLLPSYN5G
F6
C4153
C4154
C4155
C4156
C4157
C4158
C4159
VDD17_WLTX2G_CH0
D18
VDD17_WLTX5G_CH0
F8
VDD17_WLSYN5G_VCO
H6
VDD17_BTPA_CH0
C23
VDD17_XTAL
H20

TITLE	TabS7P
SIZE	A1
DWG NO	
REV	
DRAWN BY	User Name
SHEET	4 of 15
Date	

SM8550

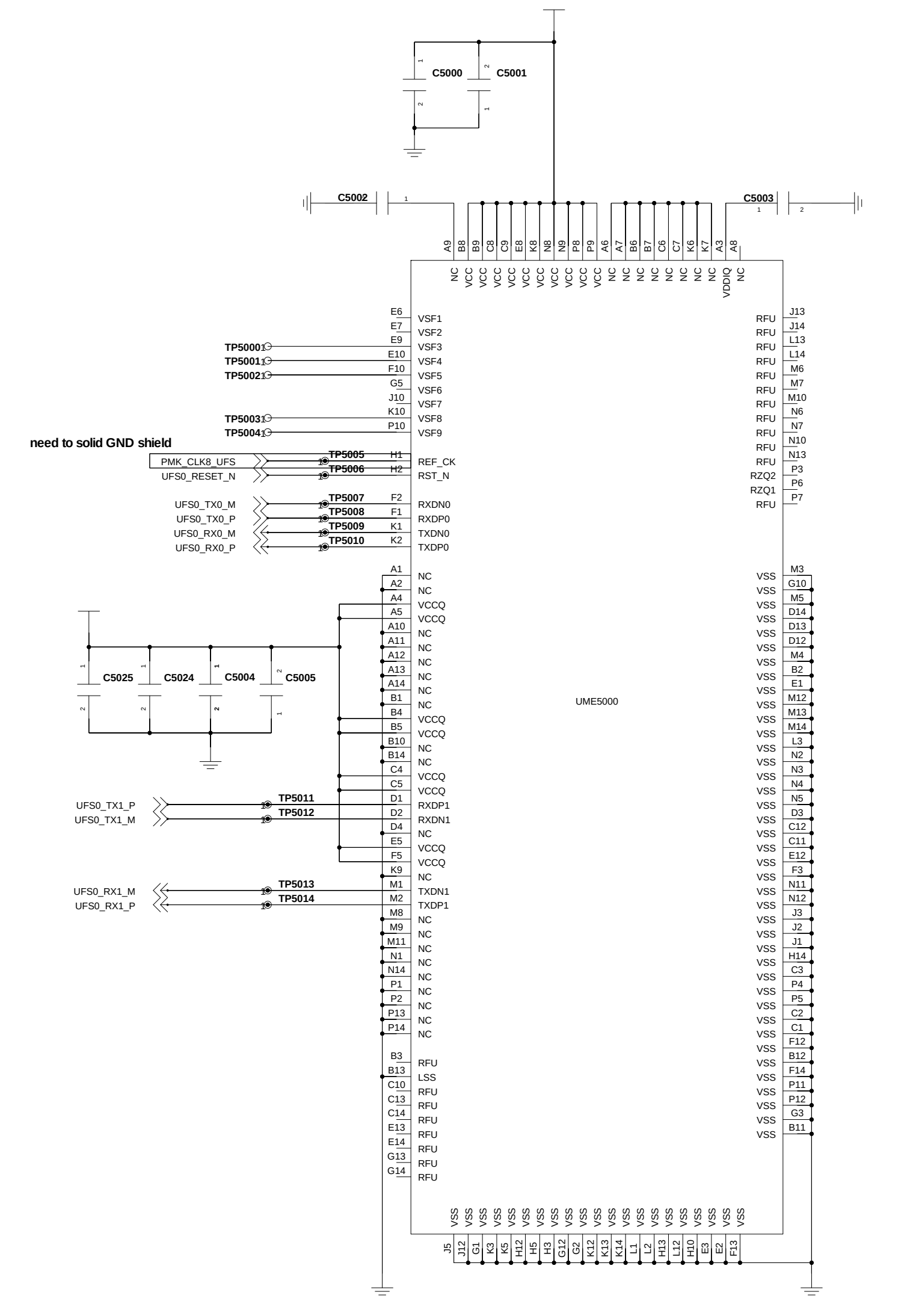
(INTERFACE/GPIO)

System

System

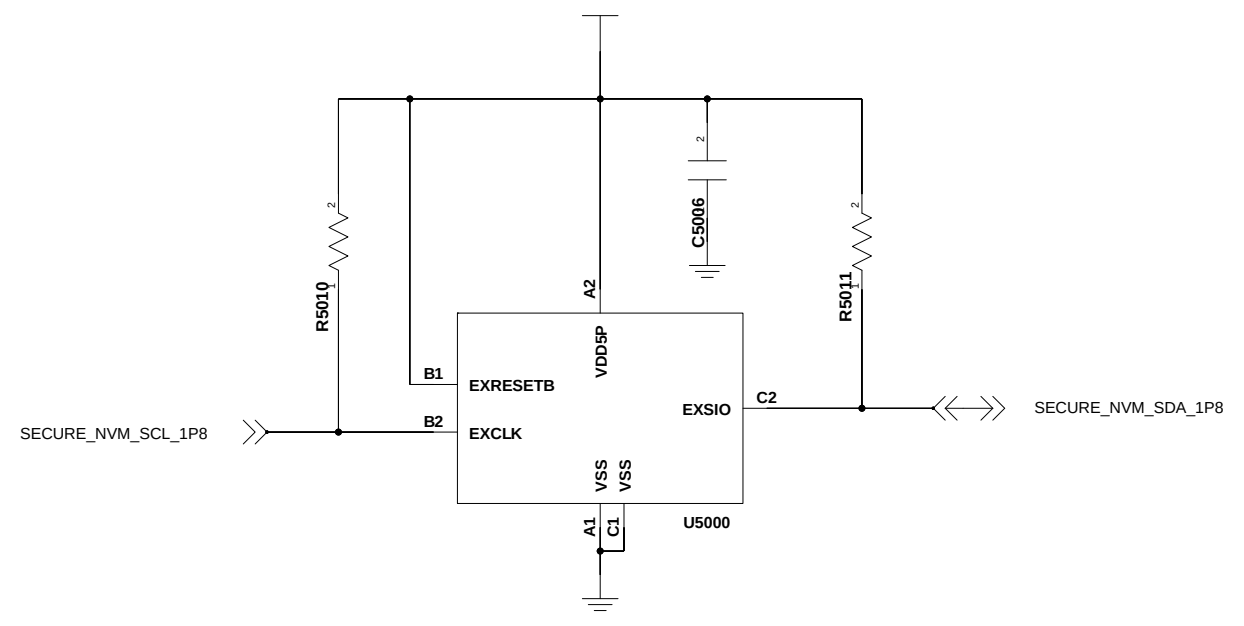
on SECONDARY PCB

need to solid GND shield



C5000
C5001
C5002
C5003
C5025
C5024
C5004
C5005
UME5000
TP5000
TP5001
TP5002
TP5003
TP5004
TP5005
TP5006
TP5007
TP5008
TP5009
TP5010
TP5011
TP5012
TP5013
TP5014
PMK_CLK8_UFS
UFS0_RESET_N
UFS0_TX0_M
UFS0_TX0_P
UFS0_RX0_M
UFS0_RX0_P
UFS0_TX1_P
UFS0_TX1_M
UFS0_RX1_M
UFS0_RX1_P
VSF1
VSF2
VSF3
VSF4
VSF5
VSF6
VSF7
VSF8
VSF9
REF_CK
RST_N
RXDN0
RXDP0
TXDN0
TXDP0
RXDP1
RXDN1
TXDN1
TXDP1
VCCQ
VCC
VDDIQ
NC
RFU
LSS
RZQ2
RZQ1
VSS

UFS 4.0 256G

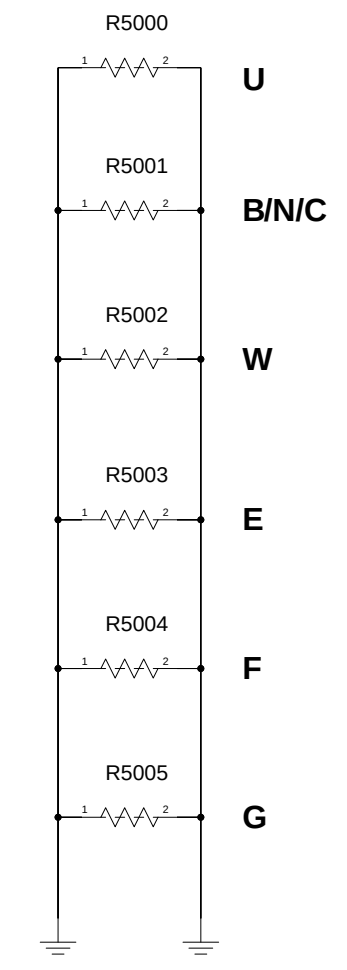


R5010
R5011
C5006
A2
VDDSP
B1
EXRESETB
B2
EXCLK
EXSIO
C2
VSS
VSS
A1
C1
U5000
SECURE_NVM_SCL_1P8
SECURE_NVM_SDA_1P8

SECURE NVM

REGION

ANT PCB	SHORT
X818U (U)	R5000
X816B/N/C (B/N/C)	R5001
X810 (W)	R5002
ETC (E)	R5003
ETC (F)	R5004
ETC (G)	R5005



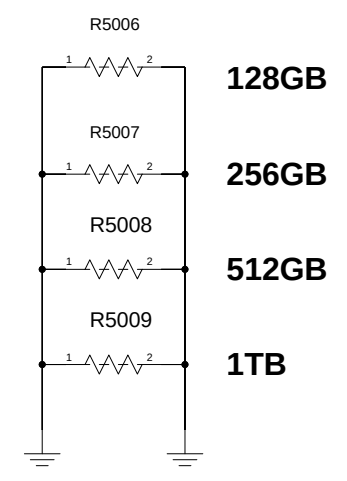
R5000
U
R5001
B/N/C
R5002
W
R5003
E
R5004
F
R5005
G

Memory_POP (CORE)

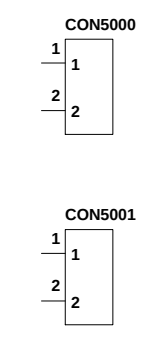
12GB	S9P

UFS Storage (MAIN)

256GB	S9P
512GB	S9P



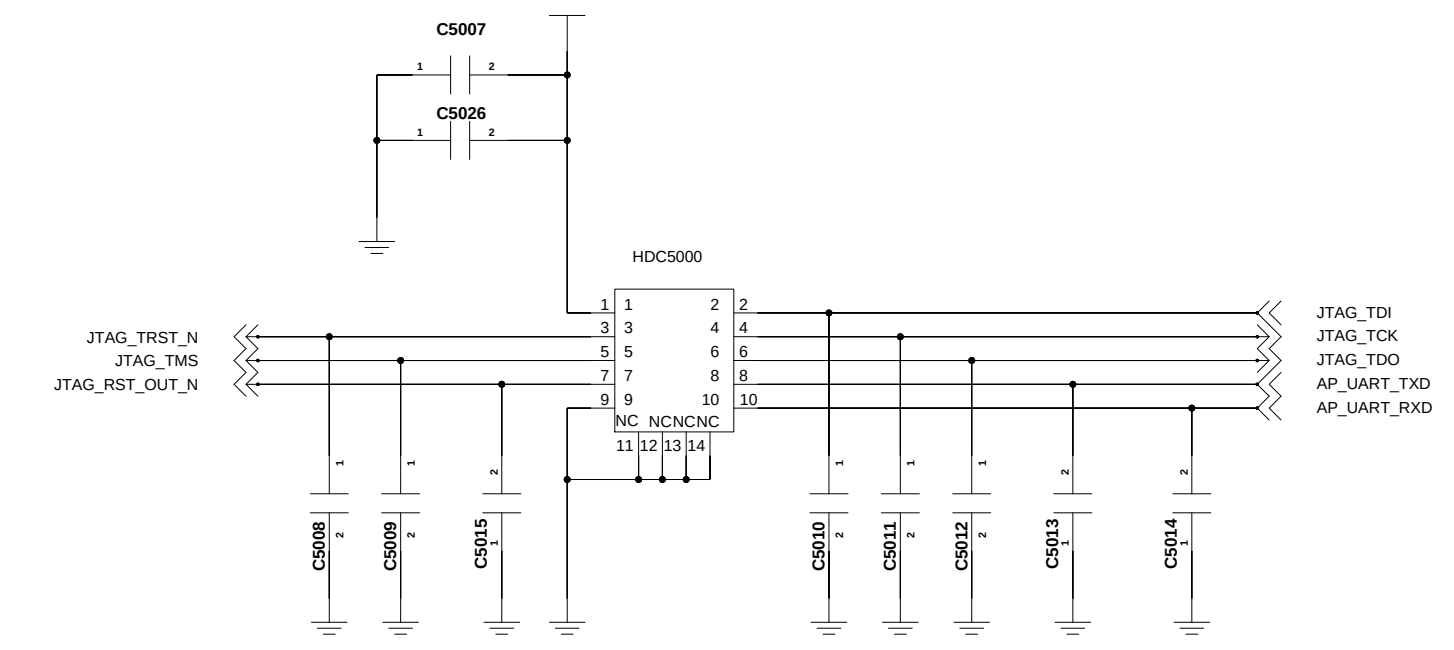
R5006
128GB
R5007
256GB
R5008
512GB
R5009
1TB



CON5000
CON5001
1
2

2D BARCODE

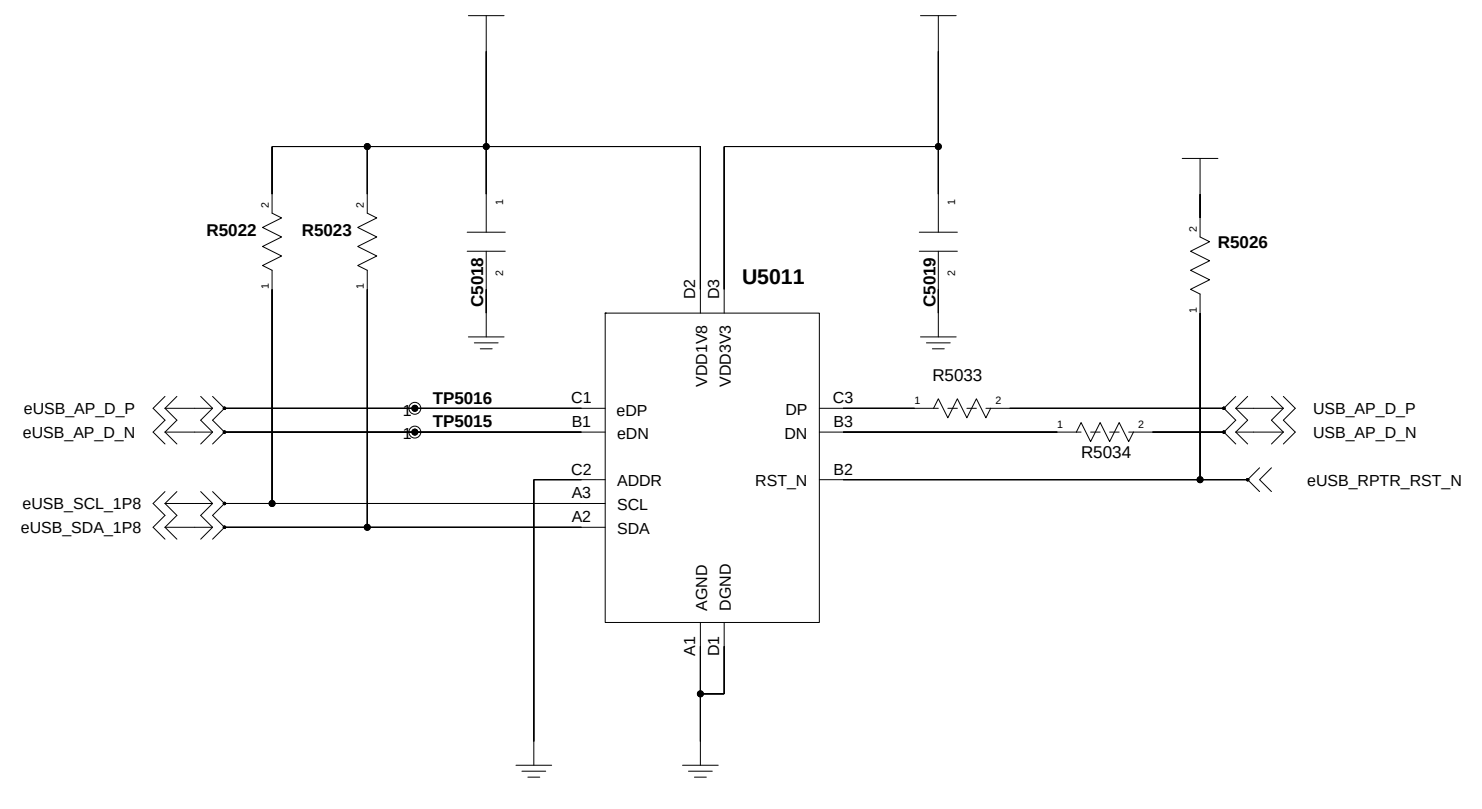
SMD NC



C5007
C5026
HDC5000
JTAG_TRST_N
JTAG_TMS
JTAG_RST_OUT_N
JTAG_TDI
JTAG_TCK
JTAG_TDO
AP_UART_TXD
AP_UART_RXD
C5008
C5009
C5015
C5010
C5011
C5012
C5013
C5014

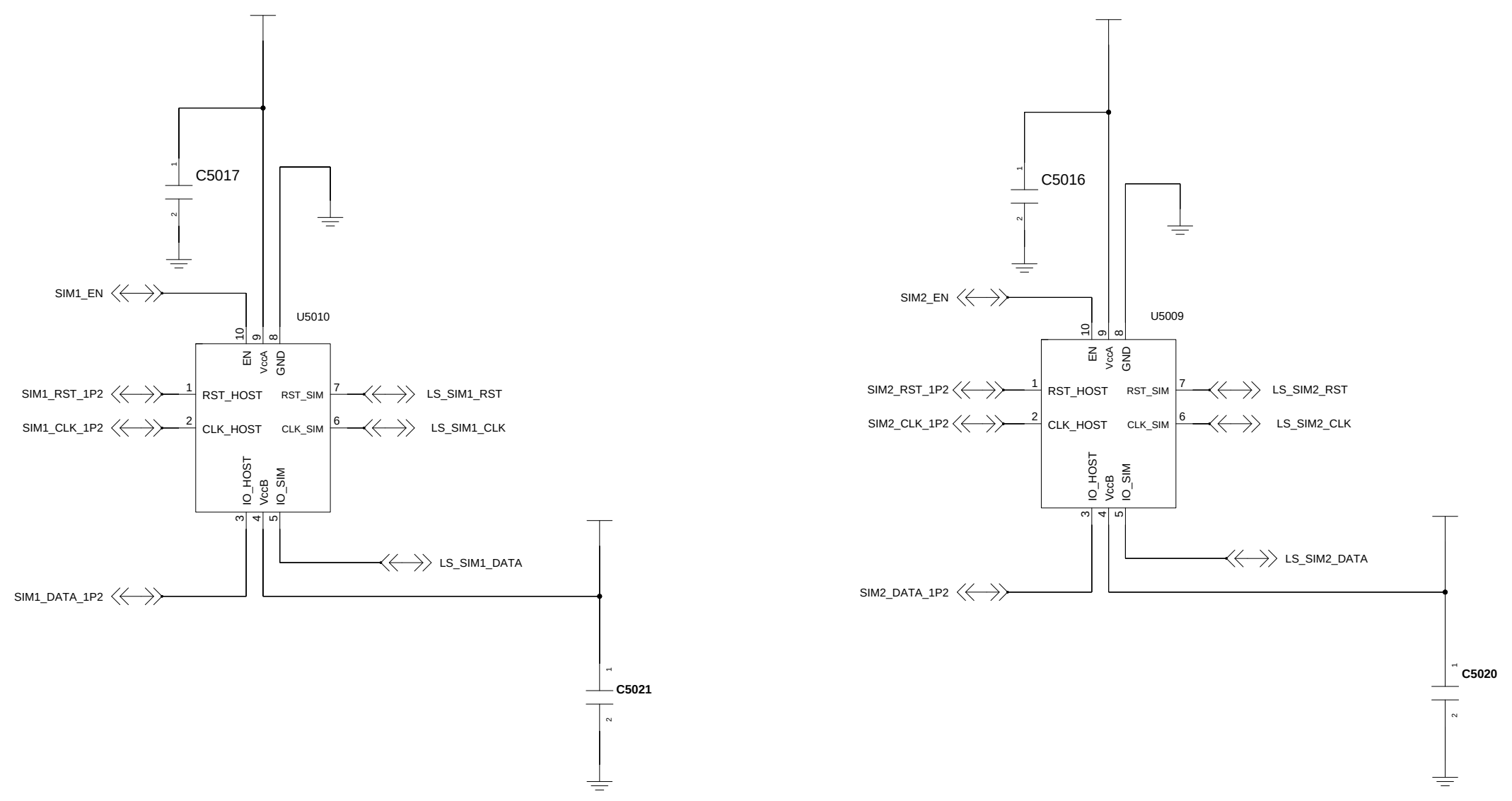
BOM NC

AP JTAG CONN



R5022
R5023
C5018
C5019
R5026
U5011
TP5016
TP5015
eUSB_AP_D_P
eUSB_AP_D_N
eUSB_SCL_1P8
eUSB_SDA_1P8
eDP
eDN
ADDR
SCL
SDA
VDD1V8
VDD3V3
DP
DN
RST_N
AGND
DGND
R5033
R5034
USB_AP_D_P
USB_AP_D_N
eUSB_RPTR_RST_N

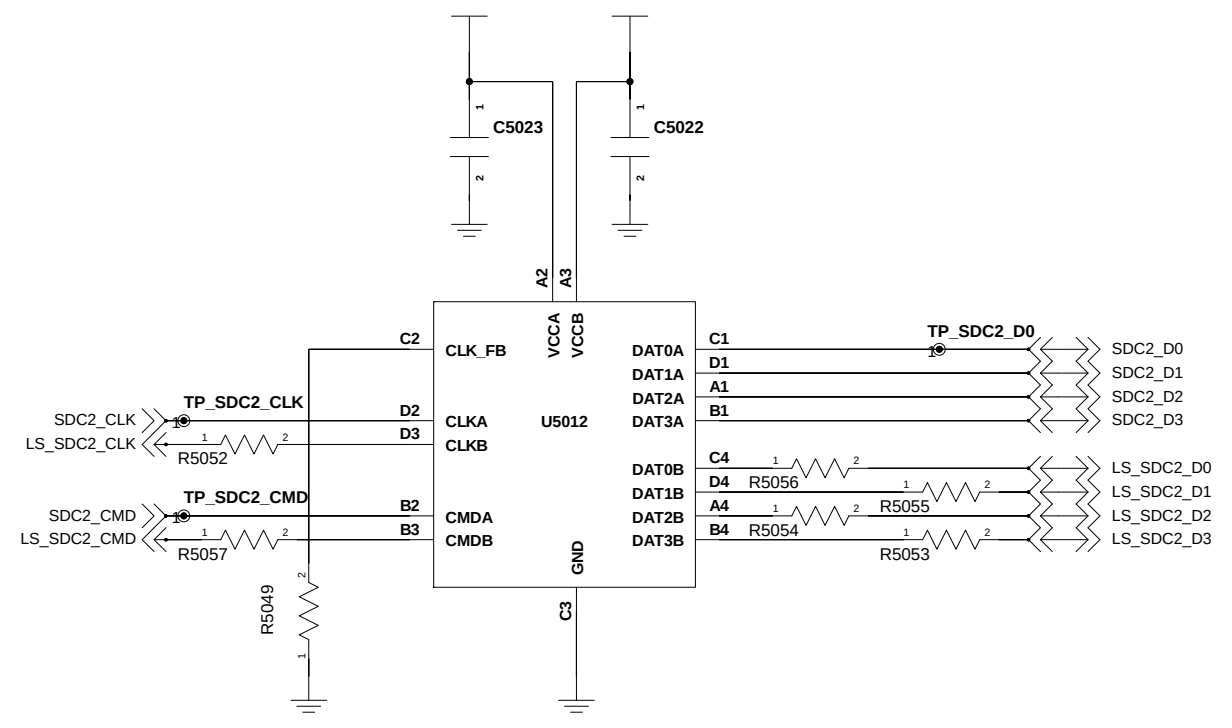
eUSB REPEATER



C5017
U5010
SIM1_EN
SIM1_RST_1P2
SIM1_CLK_1P2
SIM1_DATA_1P2
LS_SIM1_RST
LS_SIM1_CLK
LS_SIM1_DATA
C5021
C5016
U5009
SIM2_EN
SIM2_RST_1P2
SIM2_CLK_1P2
SIM2_DATA_1P2
LS_SIM2_RST
LS_SIM2_CLK
LS_SIM2_DATA
C5020
EN
VccA
GND
RST_HOST
CLK_HOST
RST_SIM
CLK_SIM
IO_HOST
VccB
IO_SIM

SIM Level Shifter

WIFI delete

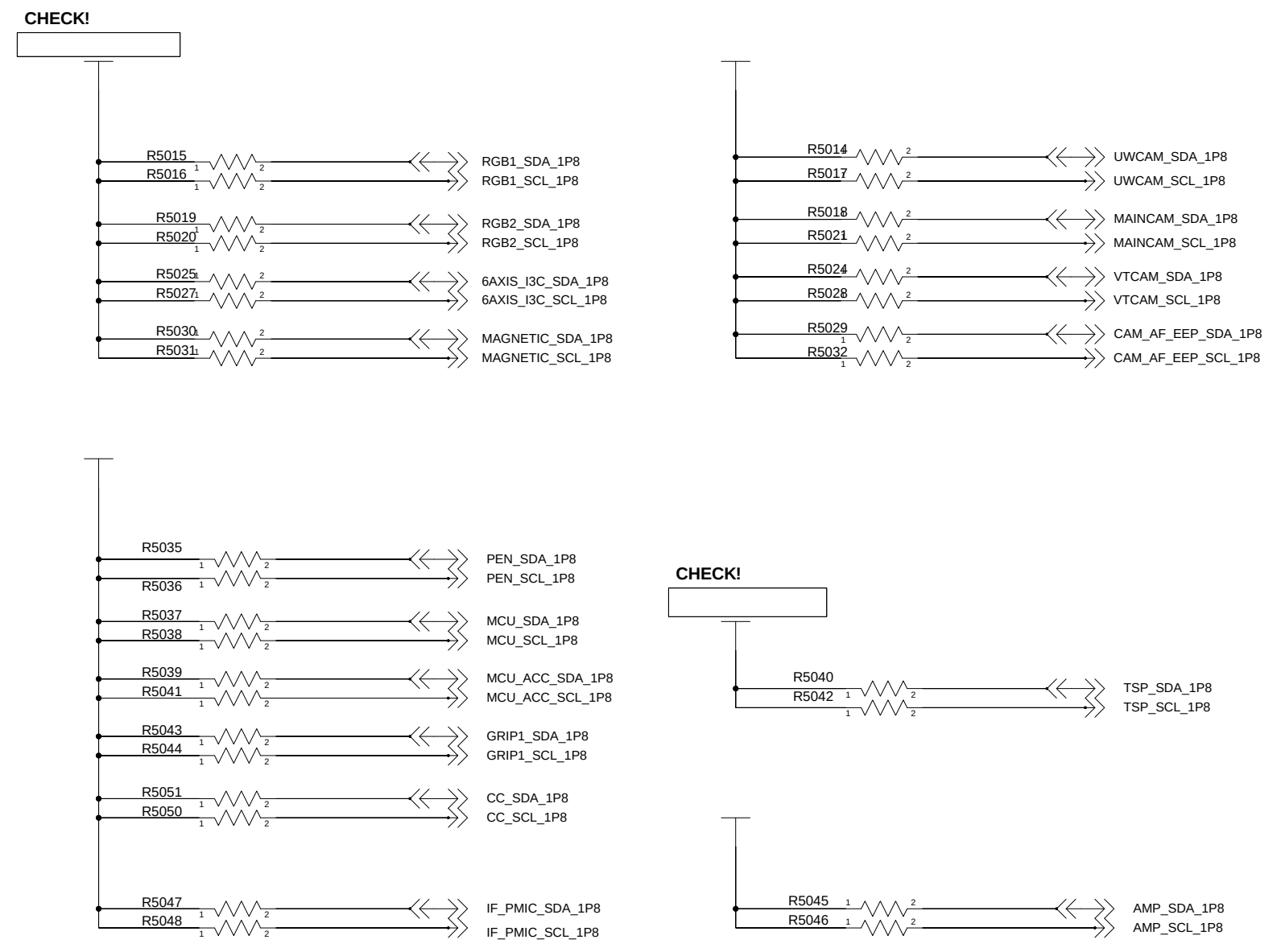


C5023
C5022
U5012
TP_SDC2_CLK
TP_SDC2_CMD
TP_SDC2_D0
SDC2_CLK
LS_SDC2_CLK
SDC2_CMD
LS_SDC2_CMD
R5052
R5057
R5049
R5056
R5055
R5054
R5053
CLK_FB
CLKA
CLKB
CMDA
CMDB
VCCA
VCCB
DAT0A
DAT1A
DAT2A
DAT3A
DAT0B
DAT1B
DAT2B
DAT3B
GND
SDC2_D0
SDC2_D1
SDC2_D2
SDC2_D3
LS_SDC2_D0
LS_SDC2_D1
LS_SDC2_D2
LS_SDC2_D3

SD Level Shifter

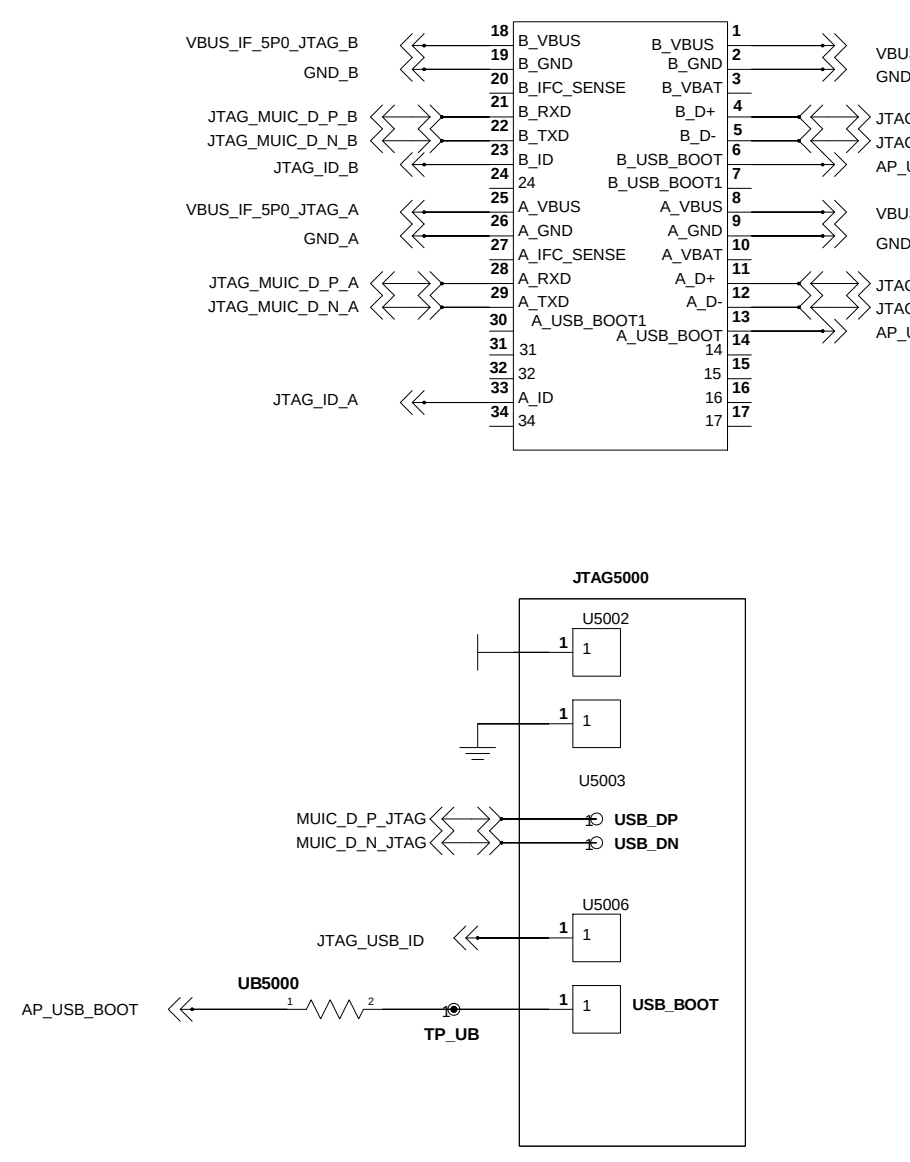
Place close to connector

CHECK!



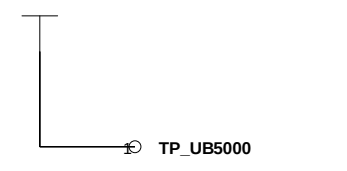
R5015
R5016
RGB1_SDA_1P8
RGB1_SCL_1P8
R5019
R5020
RGB2_SDA_1P8
RGB2_SCL_1P8
R5025
R5027
6AXIS_I3C_SDA_1P8
6AXIS_I3C_SCL_1P8
R5030
R5031
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
R5014
R5017
UWCAM_SDA_1P8
UWCAM_SCL_1P8
R5018
R5021
MAINCAM_SDA_1P8
MAINCAM_SCL_1P8
R5024
R5028
VTCAM_SDA_1P8
VTCAM_SCL_1P8
R5029
R5032
CAM_AF_EEP_SDA_1P8
CAM_AF_EEP_SCL_1P8
R5035
R5036
PEN_SDA_1P8
PEN_SCL_1P8
R5037
R5038
MCU_SDA_1P8
MCU_SCL_1P8
R5039
R5041
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
R5043
R5044
GRIP1_SDA_1P8
GRIP1_SCL_1P8
R5051
R5050
CC_SDA_1P8
CC_SCL_1P8
R5047
R5048
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
R5040
R5042
TSP_SDA_1P8
TSP_SCL_1P8
R5045
R5046
AMP_SDA_1P8
AMP_SCL_1P8

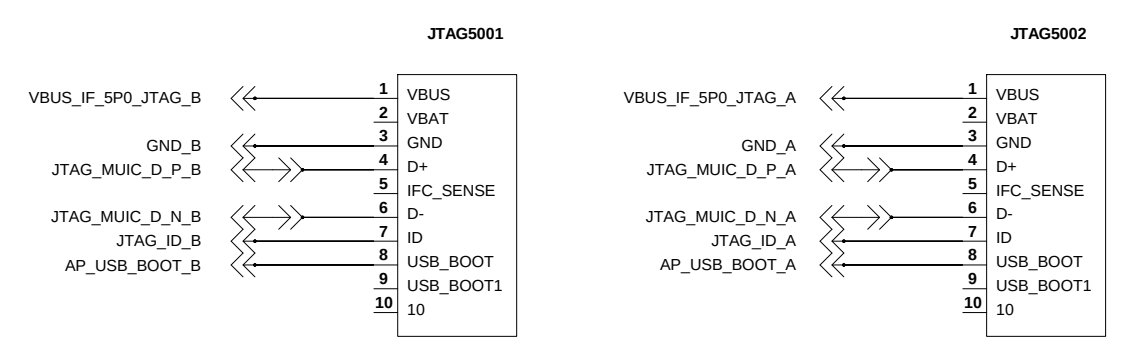
CHECK!



JTAG5000
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
JTAG_ID_A
AP_USB_BOOT_B
AP_USB_BOOT_A
B_VBUS
B_GND
B_IFC_SENSE
B_RXD
B_TXD
B_ID
B_VBAT
B_D+
B_D-
B_USB_BOOT
B_USB_BOOT1
A_VBUS
A_GND
A_IFC_SENSE
A_RXD
A_TXD
A_ID
A_VBAT
A_D+
A_D-
A_USB_BOOT
A_USB_BOOT1
U5002
U5003
U5006
MUIC_D_P_JTAG
MUIC_D_N_JTAG
USB_DP
USB_DN
JTAG_USB_ID
UB5000
AP_USB_BOOT
TP_UB
USB_BOOT

CAUTION: PLACEMENT
Near to TP AP_USB_BOOT





JTAG5001
VBUS_IF_5P0_JTAG_B
GND_B
JTAG_MUIC_D_P_B
JTAG_MUIC_D_N_B
JTAG_ID_B
AP_USB_BOOT_B
JTAG5002
VBUS_IF_5P0_JTAG_A
GND_A
JTAG_MUIC_D_P_A
JTAG_MUIC_D_N_A
JTAG_ID_A
AP_USB_BOOT_A
VBUS
VBAT
GND
D+
IFC_SENSE
D-
ID
USB_BOOT
USB_BOOT1

PCB array JTAG

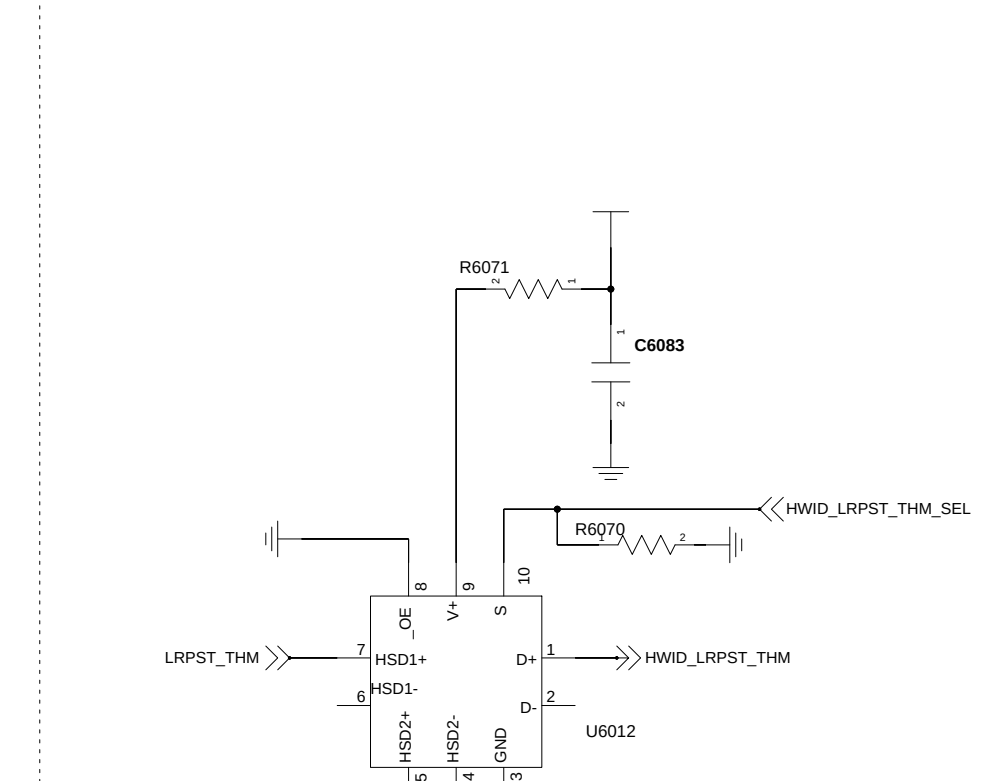
TITLE	TabS7P	
SIZE A1	DWG NO	REV
DRAWN BY User Name	SHEET 5 of 15	Date

AP PMIC set & CLK Generator

Power

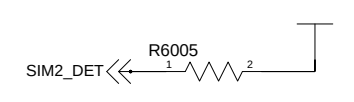
PMIC (PM8550/8550VS/8550VE)

on SECONDARY PCB

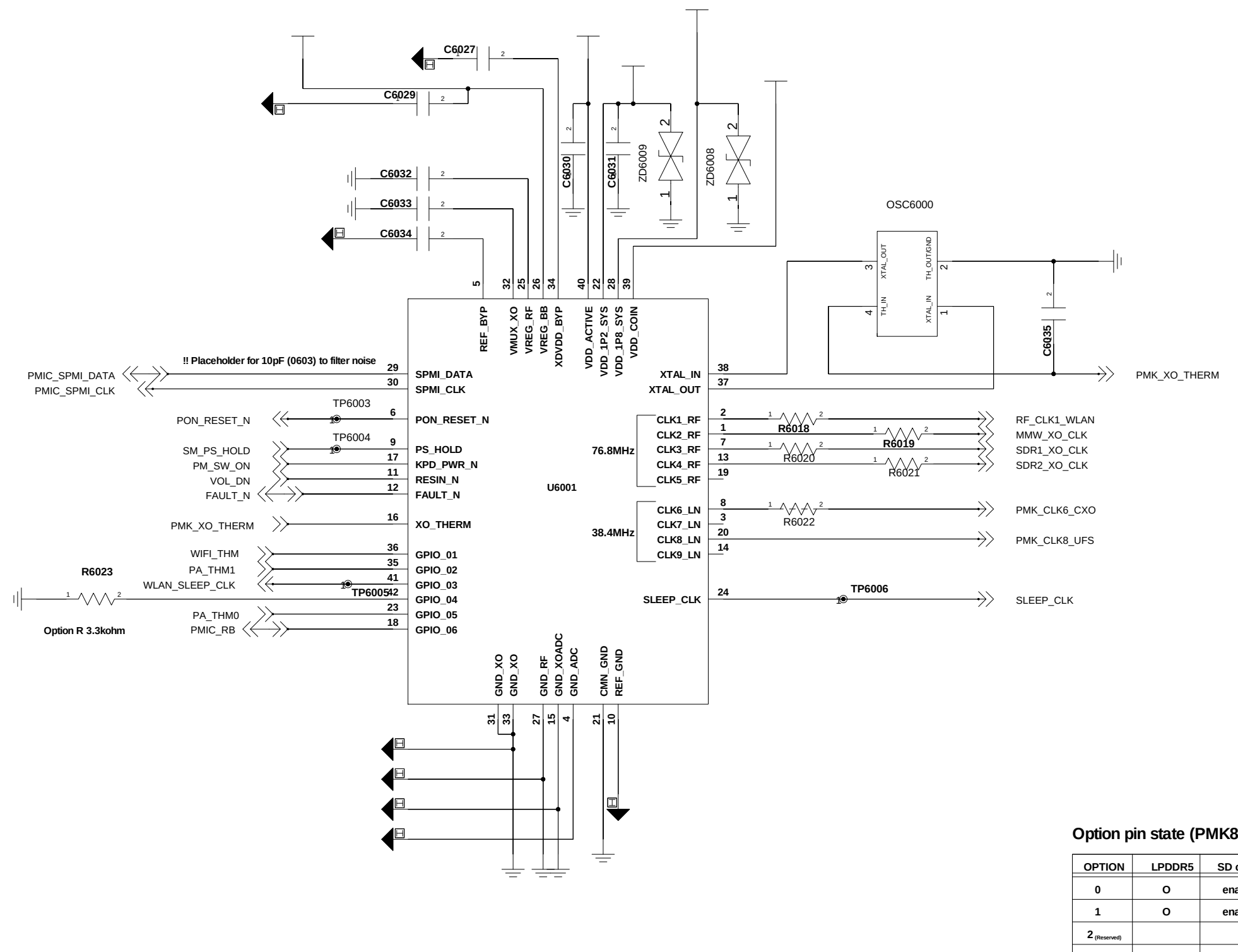


R6071
C6083
HWID_LRPST_THM_SEL
R6070
LRPST_THM
HWID_LRPST_THM
U6012
_OE
V+
S
HSD1+
HSD1-
HSD2+
HSD2-
GND
D+
D-
HW_ID

WIFI_MMW_THM_SEL	RESULT
H	LRPST_THM
L	HW_ID



R6005
SIM2_DET

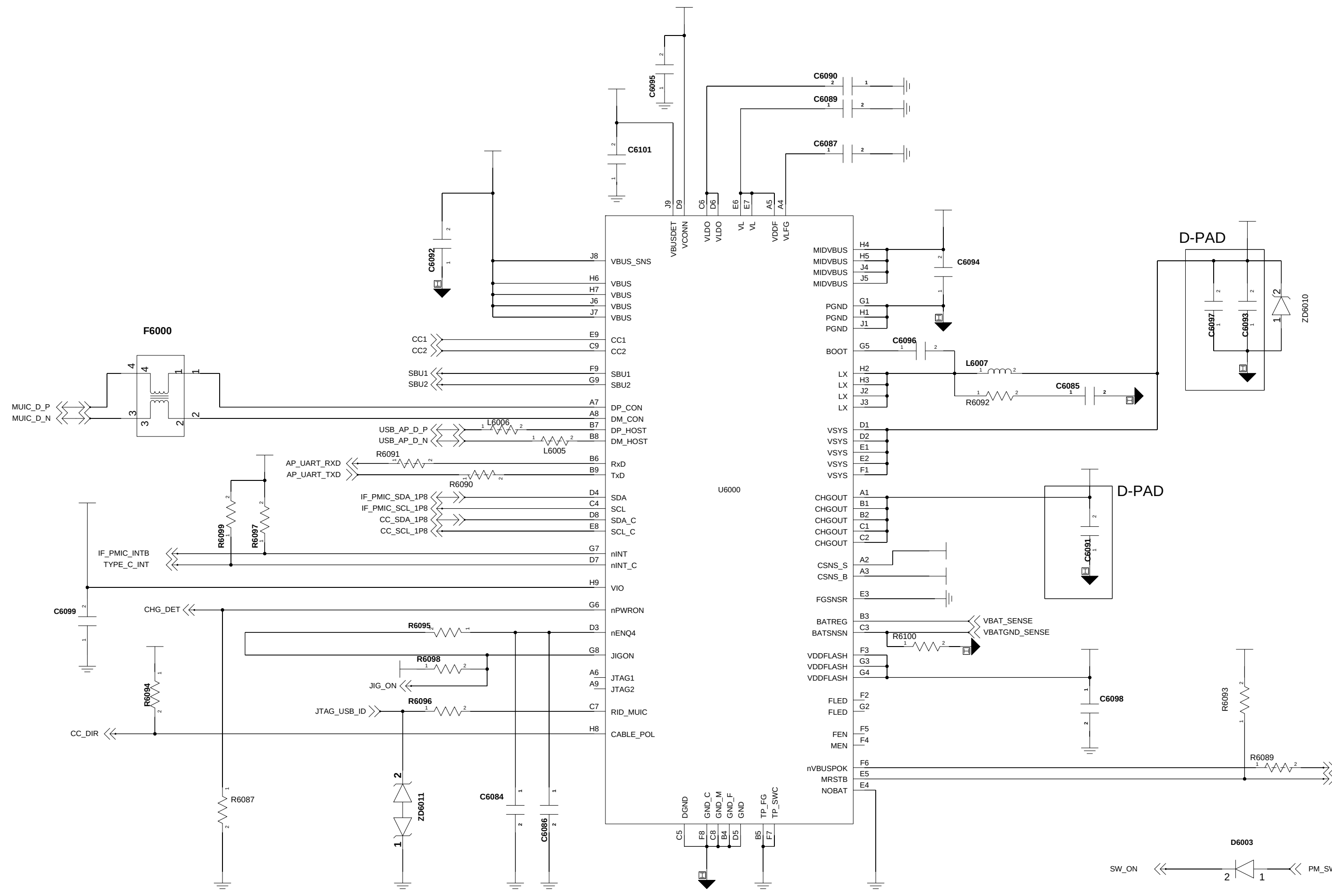


C6027
C6029
C6032
C6033
C6034
C6030
C6031
ZD6009
ZD6008
OSC6000
C6035
U6001
!! Placeholder for 10pF (0603) to filter noise
PMIC_SPMI_DATA
PMIC_SPMI_CLK
TP6003
PON_RESET_N
TP6004
SM_PS_HOLD
PM_SW_ON
VOL_DN
FAULT_N
PMK_XO_THERM
WIFI_THM
PA_THM1
WLAN_SLEEP_CLK
TP6005
R6023
Option R 3.3kohm
PA_THM0
PMIC_RB
REF_BYP
VMUX_XO
VREG_RF
VREG_BB
XDVDD_BYP
VDD_ACTIVE
VDD_1P2_SYS
VDD_1P8_SYS
VDD_COIN
SPMI_DATA
SPMI_CLK
PS_HOLD
KPD_PWR_N
RESIN_N
XO_THERM
GPIO_01
GPIO_02
GPIO_03
GPIO_04
GPIO_05
GPIO_06
XTAL_IN
XTAL_OUT
76.8MHz
CLK1_RF
CLK2_RF
CLK3_RF
CLK4_RF
CLK5_RF
38.4MHz
CLK6_LN
CLK7_LN
CLK8_LN
CLK9_LN
SLEEP_CLK
GND_XO
GND_RF
GND_XOADC
GND_ADC
CMN_GND
REF_GND
R6018
R6019
R6020
R6021
R6022
TP6006
RF_CLK1_WLAN
MMW_XO_CLK
SDR1_XO_CLK
SDR2_XO_CLK
PMK_CLK6_CXO
PMK_CLK8_UFS
SLEEP_CLK

Option pin state (PMK8550; 76.8MHz XO)

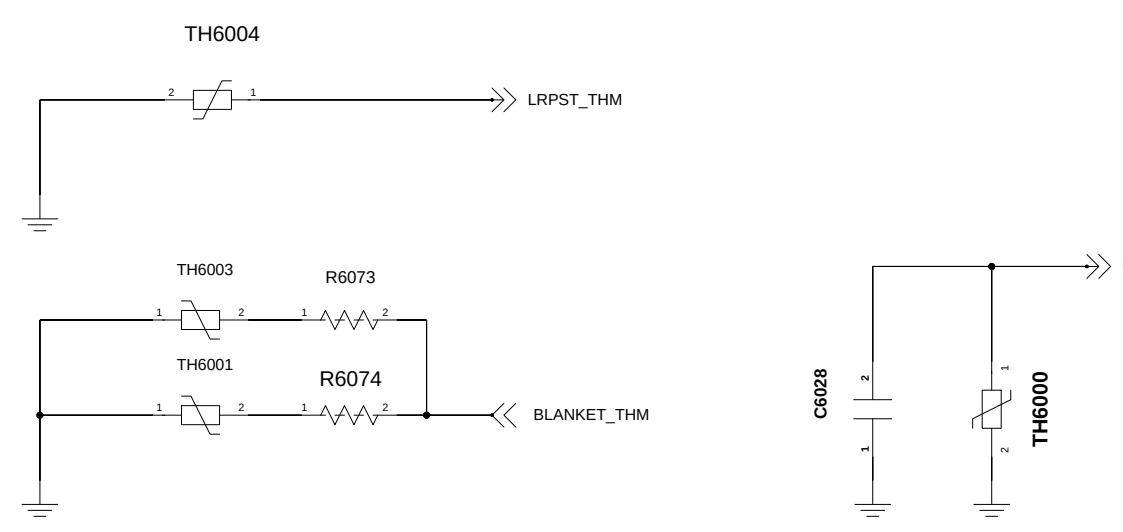
OPTION	LPDDR5	SD card LDO	GPIO05B Active	R value
0	O	enable	HIGH	
1	O	enable	LOW	
2 (Reserved)				
3 (Reserved)				
4	O	skip	HIGH	
5	O	skip	LOW	
6 (Reserved)				
7 (Reserved)				

PMK8550



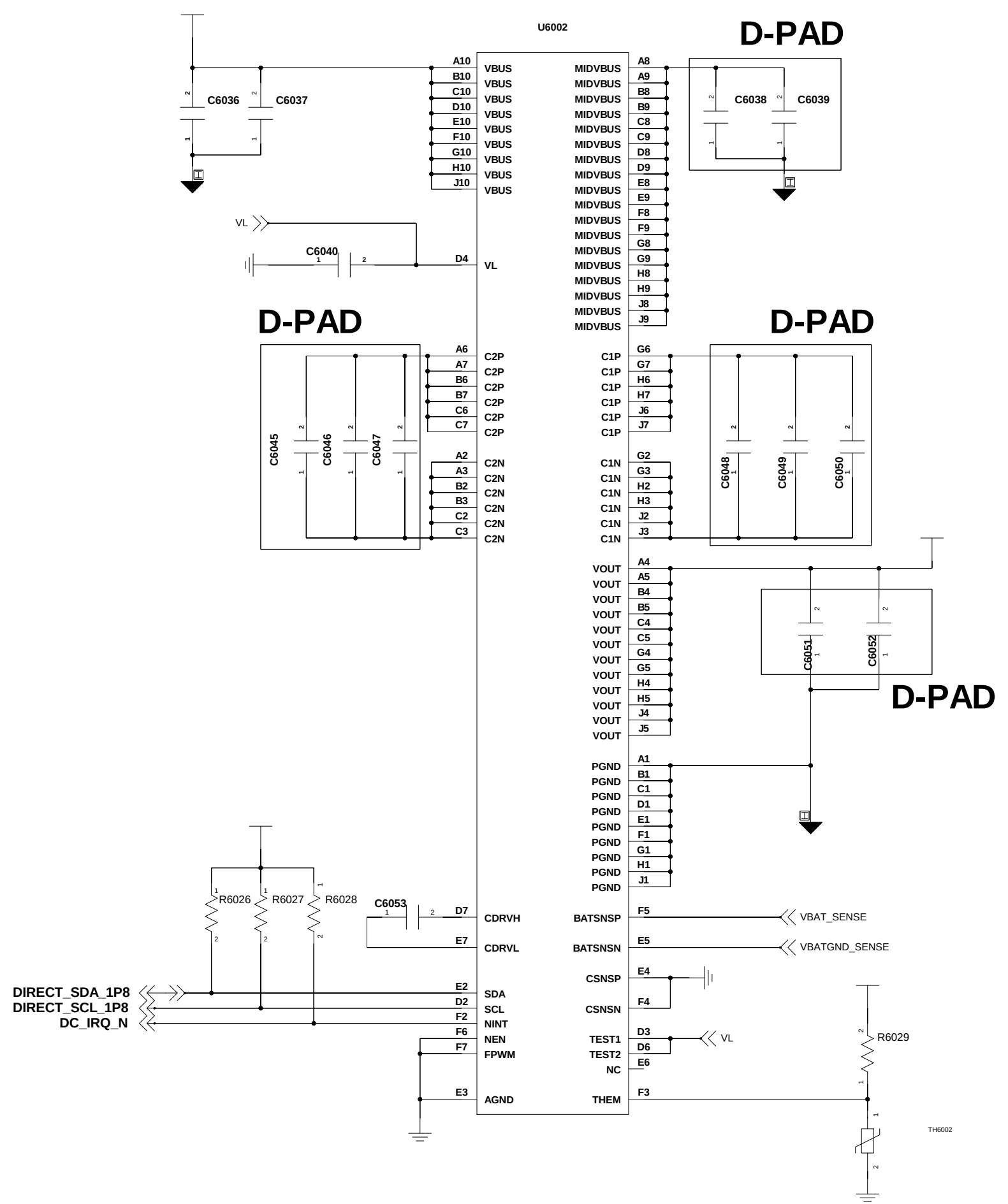
C6095
C6101
C6090
C6089
C6087
C6092
F6000
MUIC_D_P
MUIC_D_N
CC1
CC2
SBU1
SBU2
USB_AP_D_P
USB_AP_D_N
L6006
L6005
AP_UART_RXD
AP_UART_TXD
R6091
R6090
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
CC_SDA_1P8
CC_SCL_1P8
R6099
R6097
IF_PMIC_INTB
TYPE_C_INT
C6099
CHG_DET
R6095
R6098
JIG_ON
R6096
JTAG_USB_ID
R6094
CC_DIR
R6087
ZD6011
C6084
C6086
U6000
VBUS_SNS
VBUS
DP_CON
DM_CON
DP_HOST
DM_HOST
RxD
TxD
SDA
SCL
SDA_C
SCL_C
nINT
nINT_C
VIO
nPWRON
nENQ4
JIGON
JTAG1
JTAG2
RID_MUIC
CABLE_POL
MIDVBUS
PGND
BOOT
LX
VSYS
CHGOUT
CSNS_S
CSNS_B
FGSNSR
BATREG
BATSNSN
VDDFLASH
FLED
FEN
MEN
nVBUSPOK
MRSTB
NOBAT
C6094
C6096
L6007
R6092
C6085
D-PAD
C6097
C6093
ZD6010
C6091
VBAT_SENSE
VBATGND_SENSE
R6100
C6098
R6093
R6089
TSP_TA
SW_ON
D6003
PM_SW_ON

IF PMIC



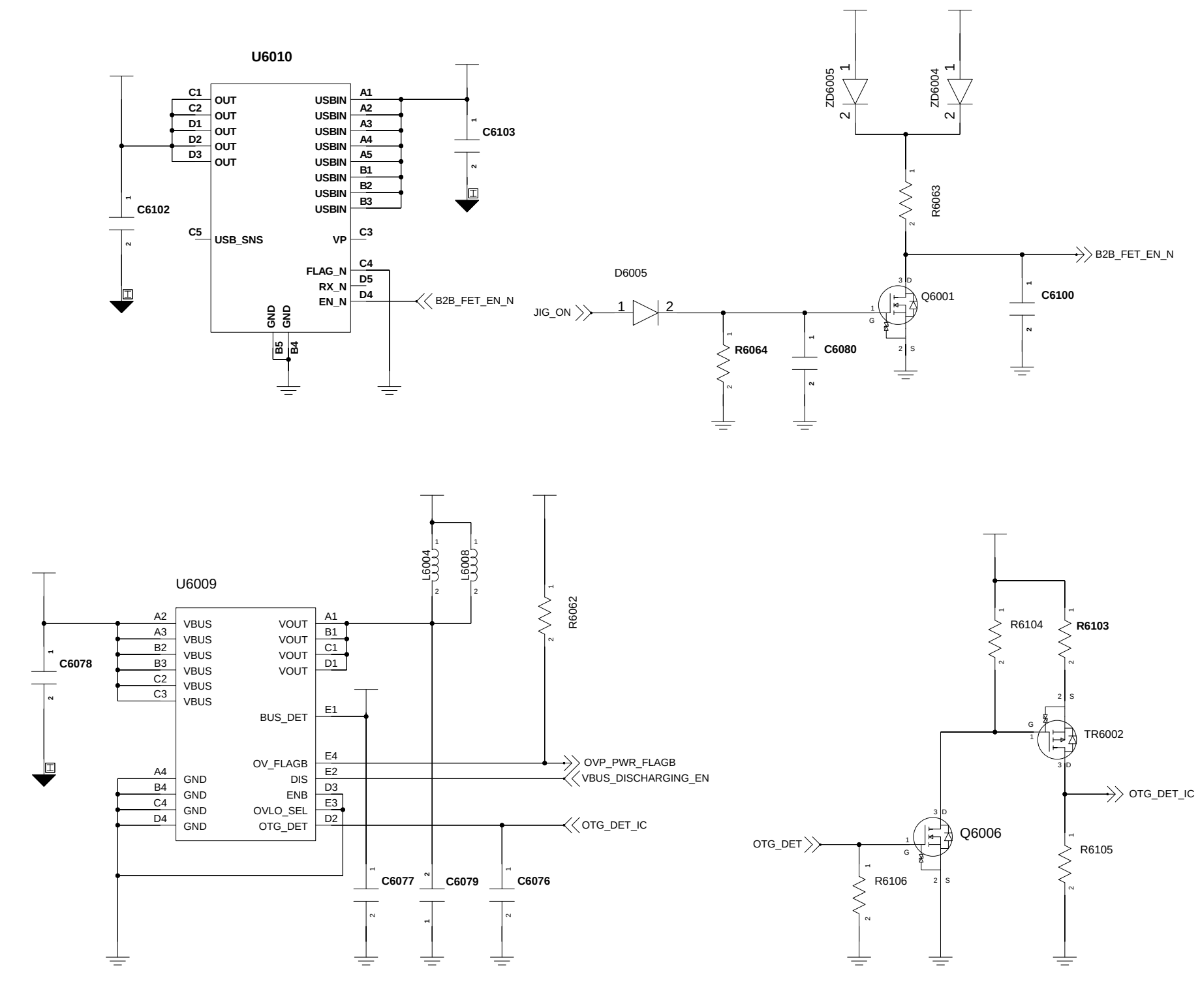
TH6004
LRPST_THM
TH6003
R6073
TH6001
R6074
BLANKET_THM
CHG_THM
C6028
TH6000

THERMISTERS



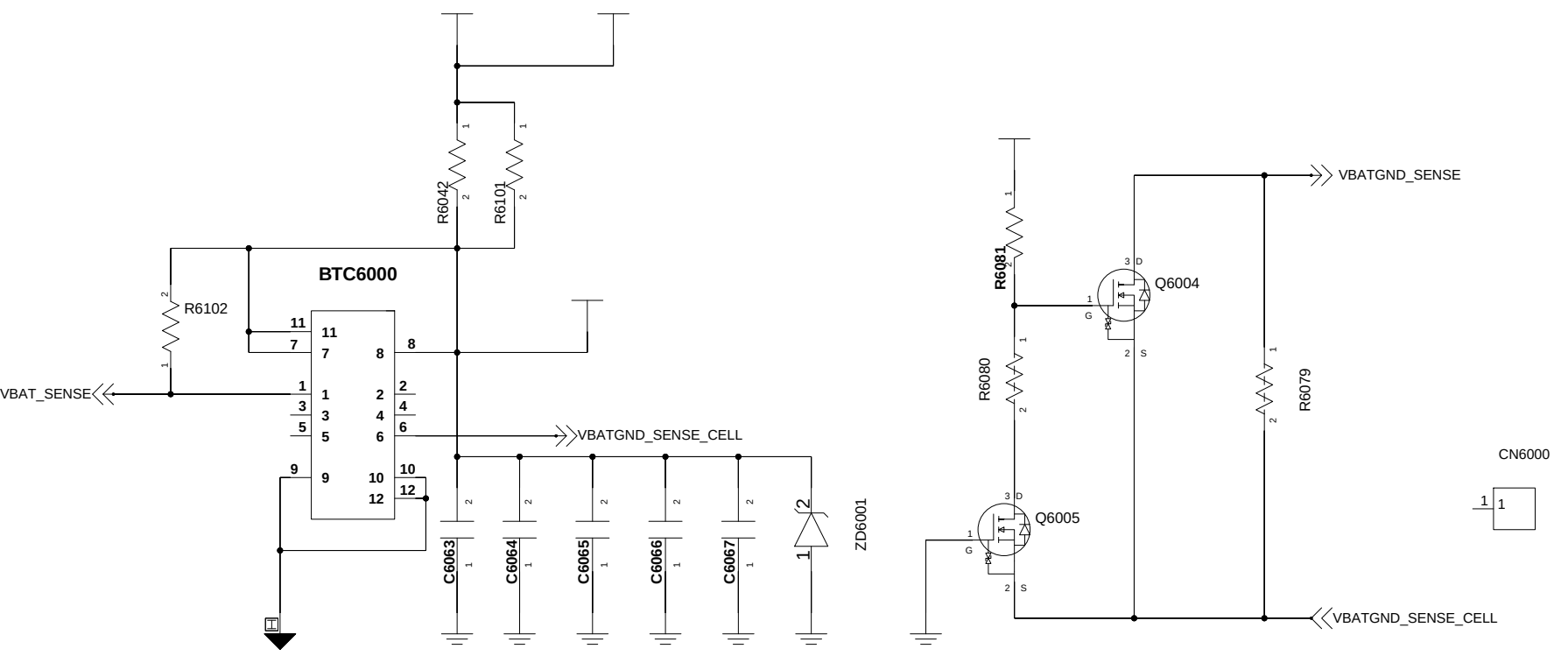
U6002
D-PAD
C6036
C6037
C6038
C6039
VL
C6040
C6045
C6046
C6047
C6048
C6049
C6050
C6051
C6052
VBUS
MIDVBUS
C2P
C2N
C1P
C1N
VOUT
PGND
CDRVH
CDRVL
SDA
SCL
NINT
NEN
FPWM
AGND
BATSNSP
BATSNSN
CSNSP
CSNSN
TEST1
TEST2
NC
THEM
R6026
R6027
R6028
C6053
DIRECT_SDA_1P8
DIRECT_SCL_1P8
DC_IRQ_N
VBAT_SENSE
VBATGND_SENSE
R6029
TH6002

Direct Charging



U6010
OUT
USBIN
USB_SNS
VP
FLAG_N
RX_N
EN_N
GND
C6102
C6103
B2B_FET_EN_N
ZD6005
ZD6004
R6063
B2B_FET_EN_N
D6005
JIG_ON
R6064
C6080
Q6001
C6100
U6009
VBUS
VOUT
BUS_DET
OV_FLAGB
DIS
ENB
OVLO_SEL
OTG_DET
C6078
L6004
L6008
R6062
C6077
C6079
C6076
OVP_PWR_FLAGB
VBUS_DISCHARGING_EN
OTG_DET_IC
R6104
R6103
TR6002
OTG_DET
Q6006
R6106
R6105

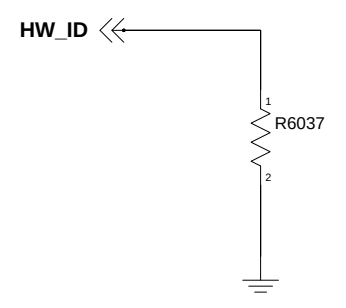
OVP



R6042
R6101
R6102
BTC6000
VBAT_SENSE
VBATGND_SENSE_CELL
C6063
C6064
C6065
C6066
C6067
ZD6001
R6081
Q6004
R6080
Q6005
R6079
VBATGND_SENSE
CN6000

Battery Con

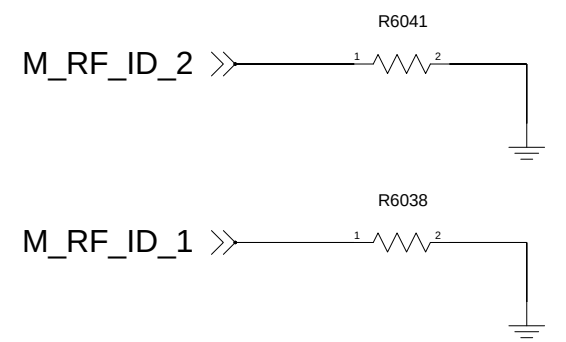
HW ID



R6037

System

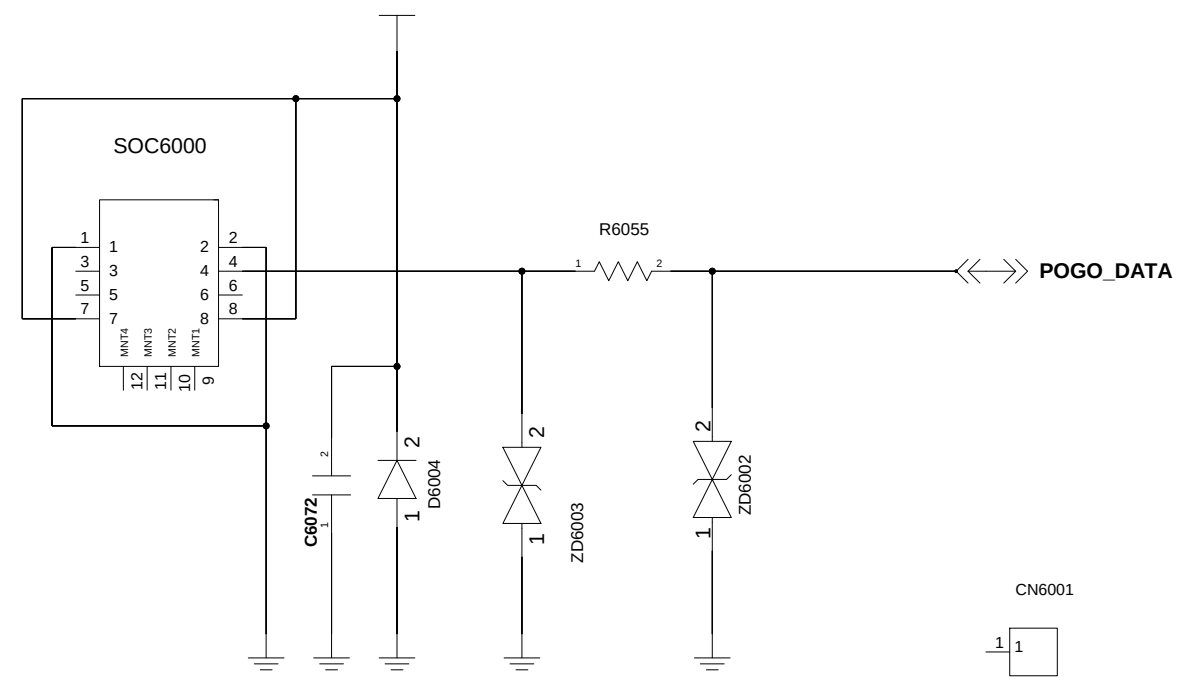
M_RF_ID



M_RF_ID_2
R6041
M_RF_ID_1
R6038

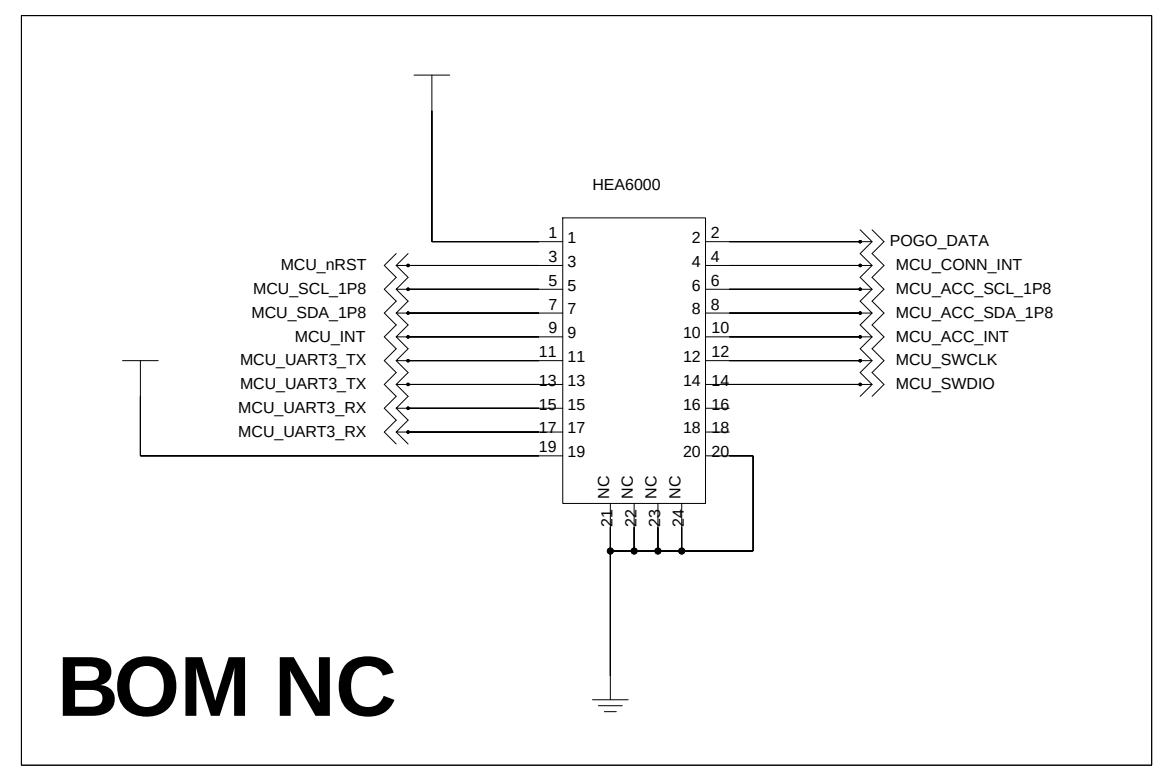
Check R11011 for M_RF_ID_0

POGO



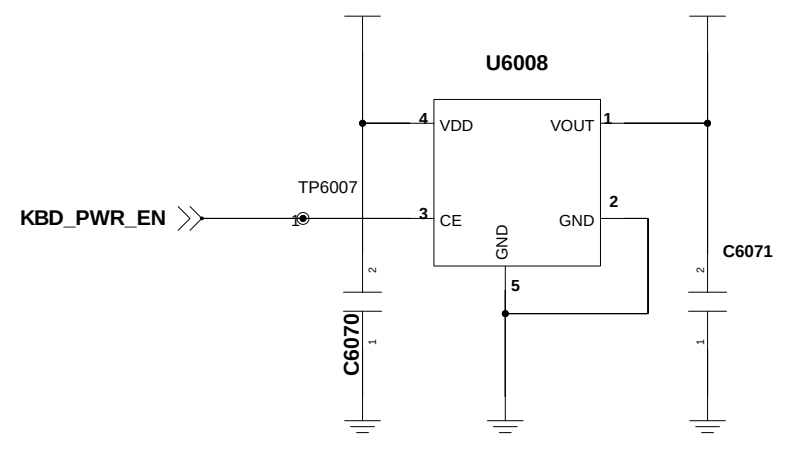
SOC6000
R6055
POGO_DATA
C6072
D6004
ZD6003
ZD6002
CN6001

POGO BtoB Con.

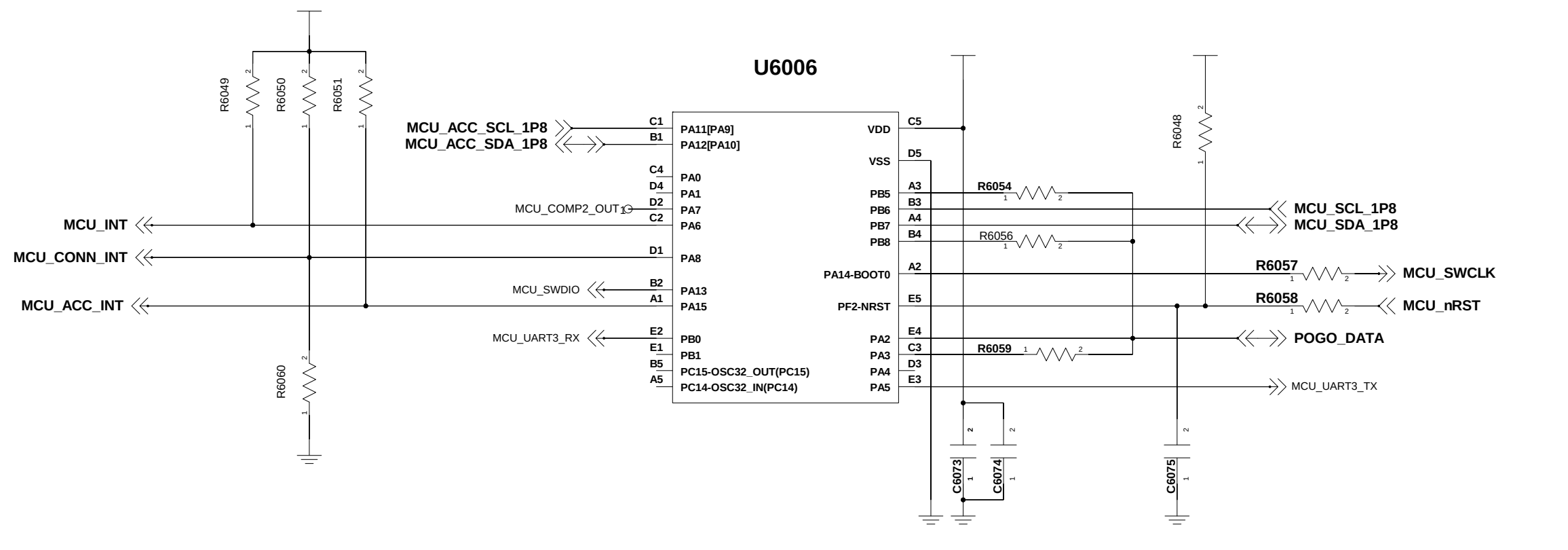


HEA6000
MCU_nRST
MCU_SCL_1P8
MCU_SDA_1P8
MCU_INT
MCU_UART3_TX
MCU_UART3_TX
MCU_UART3_RX
MCU_UART3_RX
POGO_DATA
MCU_CONN_INT
MCU_ACC_SCL_1P8
MCU_ACC_SDA_1P8
MCU_ACC_INT
MCU_SWCLK
MCU_SWDIO
NC
BOM NC

POGO Debug



U6008
VDD
VOUT
CE
GND
TP6007
KBD_PWR_EN
C6070
C6071

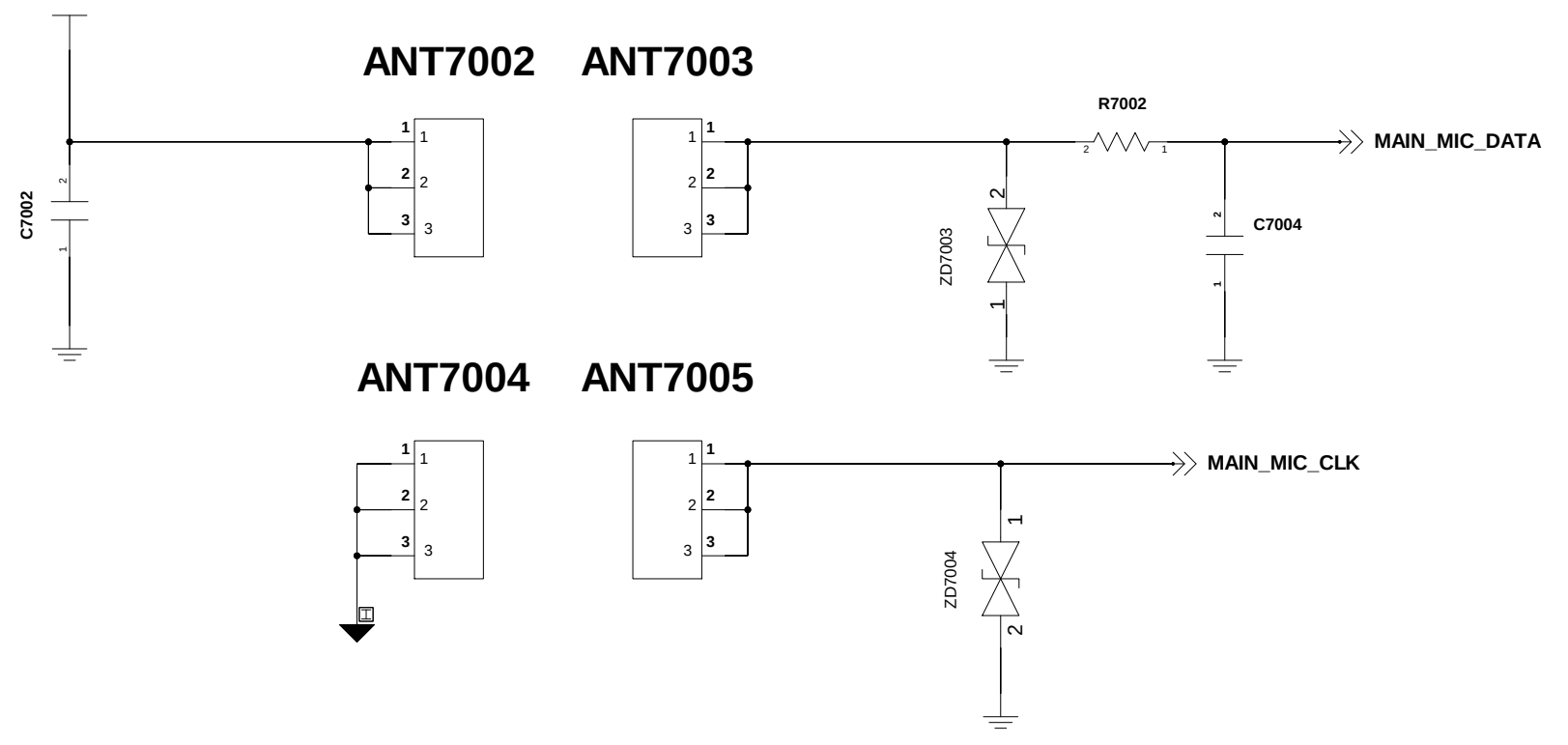


U6006
R6049
R6050
R6051
R6060
MCU_ACC_SCL_1P8
MCU_ACC_SDA_1P8
MCU_COMP2_OUT
MCU_INT
MCU_CONN_INT
MCU_SWDIO
MCU_ACC_INT
MCU_UART3_RX
PA11[PA9]
PA12[PA10]
PA0
PA1
PA7
PA6
PA8
PA13
PA15
PB0
PB1
PC15-OSC32_OUT(PC15)
PC14-OSC32_IN(PC14)
VDD
VSS
PB5
PB6
PB7
PB8
PA14-BOOT0
PF2-NRST
PA2
PA3
PA4
PA5
R6048
R6054
R6056
R6057
R6058
R6059
MCU_SCL_1P8
MCU_SDA_1P8
MCU_SWCLK
MCU_nRST
POGO_DATA
MCU_UART3_TX
C6073
C6074
C6075

POGO MCU

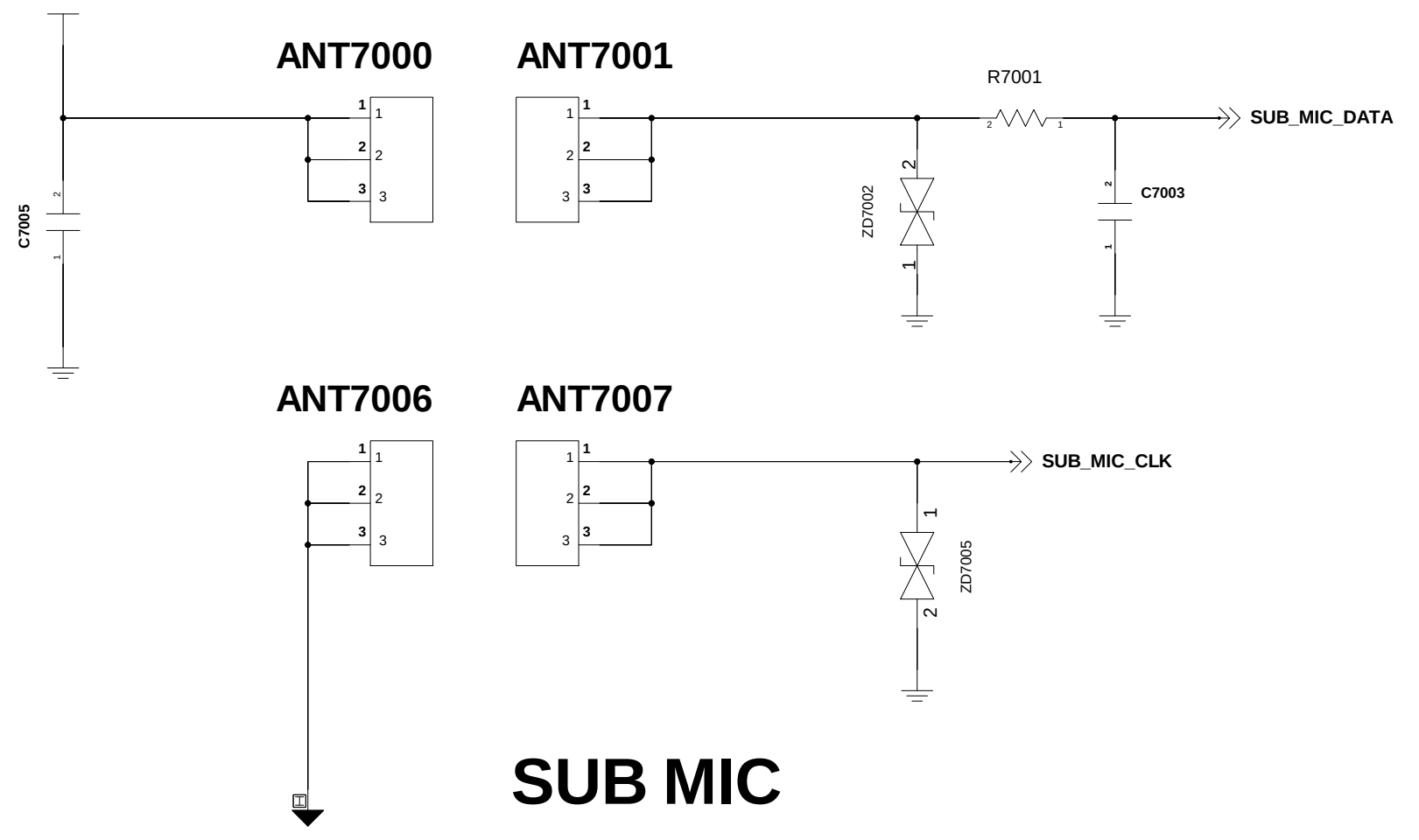
TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 6 of 15
Date

AUDIO



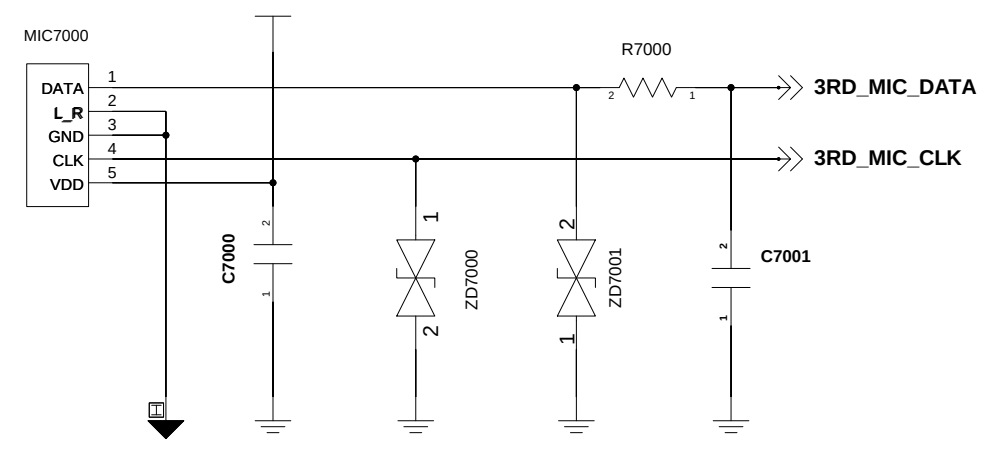
C7002
ANT7002
ANT7003
ANT7004
ANT7005
R7002
MAIN_MIC_DATA
ZD7003
C7004
MAIN_MIC_CLK
ZD7004

MAIN MIC



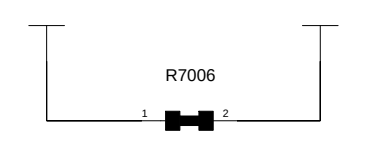
C7005
ANT7000
ANT7001
ANT7006
ANT7007
R7001
SUB_MIC_DATA
ZD7002
C7003
SUB_MIC_CLK
ZD7005

SUB MIC

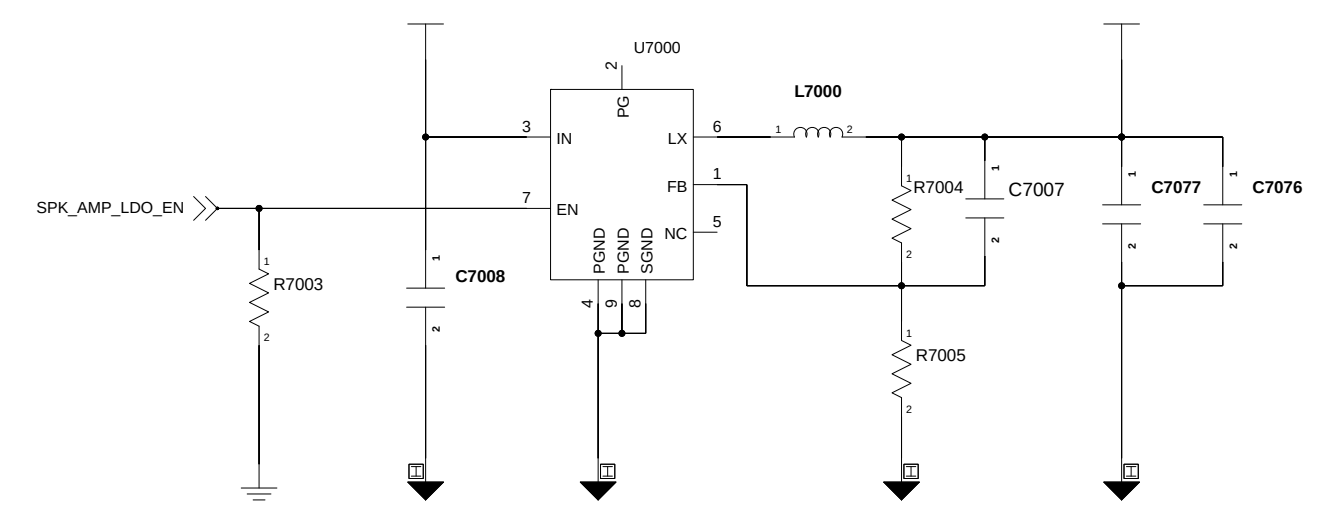


MIC7000
DATA
L_R
GND
CLK
VDD
R7000
3RD_MIC_DATA
3RD_MIC_CLK
C7000
ZD7000
ZD7001
C7001

3rd MIC

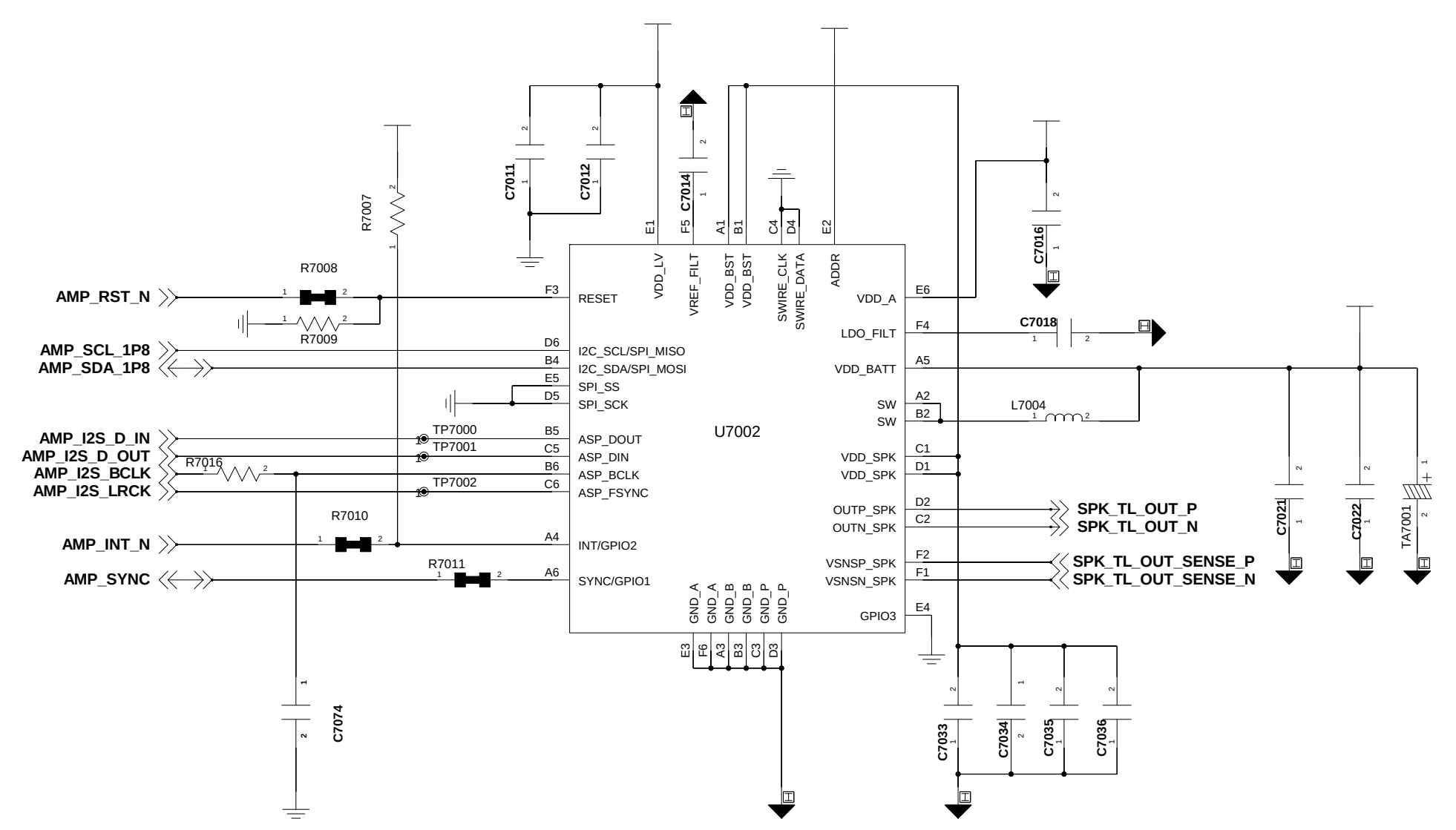


Place on array dummy
(SPK AMP SMD F/T)



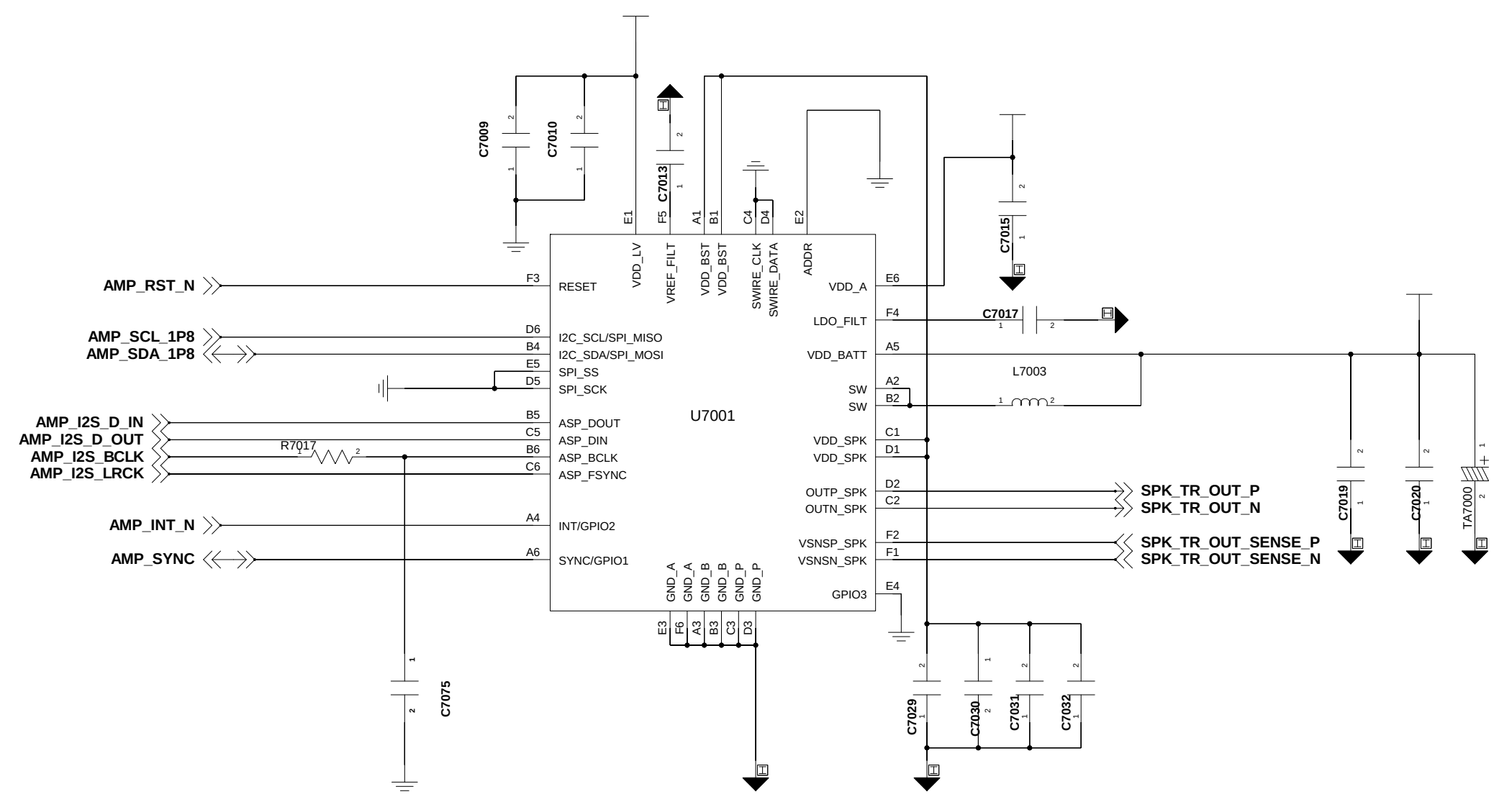
U7000
SPK_AMP_LDO_EN
IN
EN
PG
LX
FB
NC
PGND
PGND
SGND
L7000
R7003
C7008
R7004
C7007
R7005
C7077
C7076

AUDIO DCDC



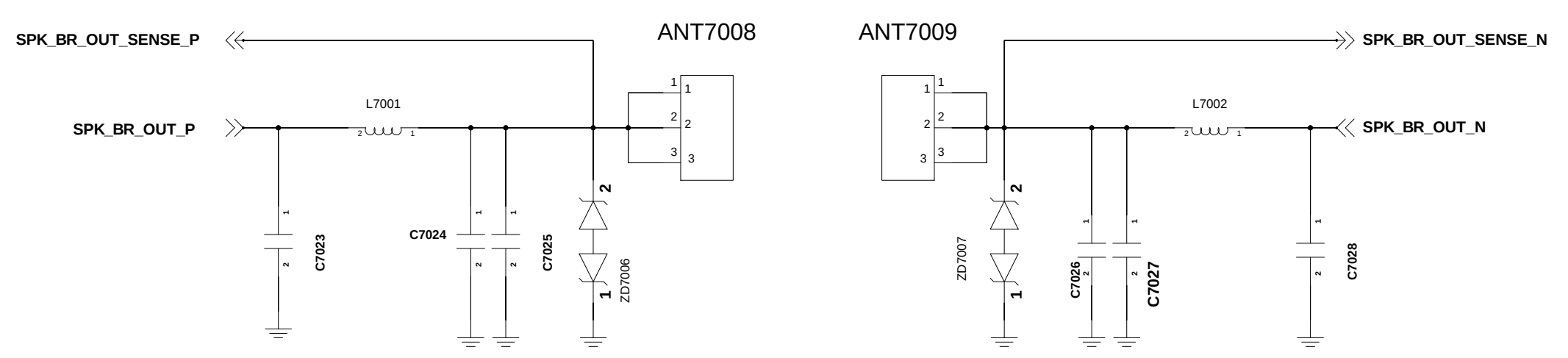
R7007
C7011
C7012
C7014
C7016
R7008
R7009
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
R7016
R7010
R7011
TP7000
TP7001
TP7002
U7002
RESET
I2C_SCL/SPI_MISO
I2C_SDA/SPI_MOSI
SPI_SS
SPI_SCK
ASP_DOUT
ASP_DIN
ASP_BCLK
ASP_FSYNC
INT/GPIO2
SYNC/GPIO1
VDD_LV
VREF_FILT
VDD_BST
VDD_BST
SWIRE_CLK
SWIRE_DATA
ADDR
VDD_A
LDO_FILT
VDD_BATT
SW
SW
VDD_SPK
VDD_SPK
OUTP_SPK
OUTN_SPK
VSNSP_SPK
VSNSN_SPK
GPIO3
GND_A
GND_A
GND_B
GND_B
GND_P
GND_P
C7018
L7004
SPK_TL_OUT_P
SPK_TL_OUT_N
SPK_TL_OUT_SENSE_P
SPK_TL_OUT_SENSE_N
C7021
C7022
TA7001
C7074
C7033
C7034
C7035
C7036

SPK TOP LEFT AMP



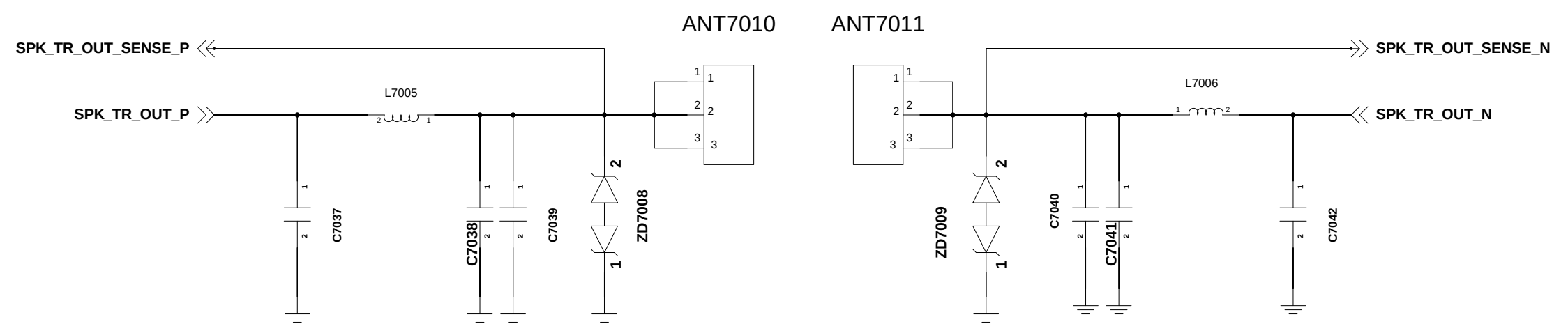
C7009
C7010
C7013
C7015
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
R7017
U7001
RESET
I2C_SCL/SPI_MISO
I2C_SDA/SPI_MOSI
SPI_SS
SPI_SCK
ASP_DOUT
ASP_DIN
ASP_BCLK
ASP_FSYNC
INT/GPIO2
SYNC/GPIO1
VDD_A
LDO_FILT
VDD_BATT
SW
SW
VDD_SPK
VDD_SPK
OUTP_SPK
OUTN_SPK
VSNSP_SPK
VSNSN_SPK
GPIO3
C7017
L7003
SPK_TR_OUT_P
SPK_TR_OUT_N
SPK_TR_OUT_SENSE_P
SPK_TR_OUT_SENSE_N
C7019
C7020
TA7000
C7075
C7029
C7030
C7031
C7032

SPK TOP RIGHT AMP



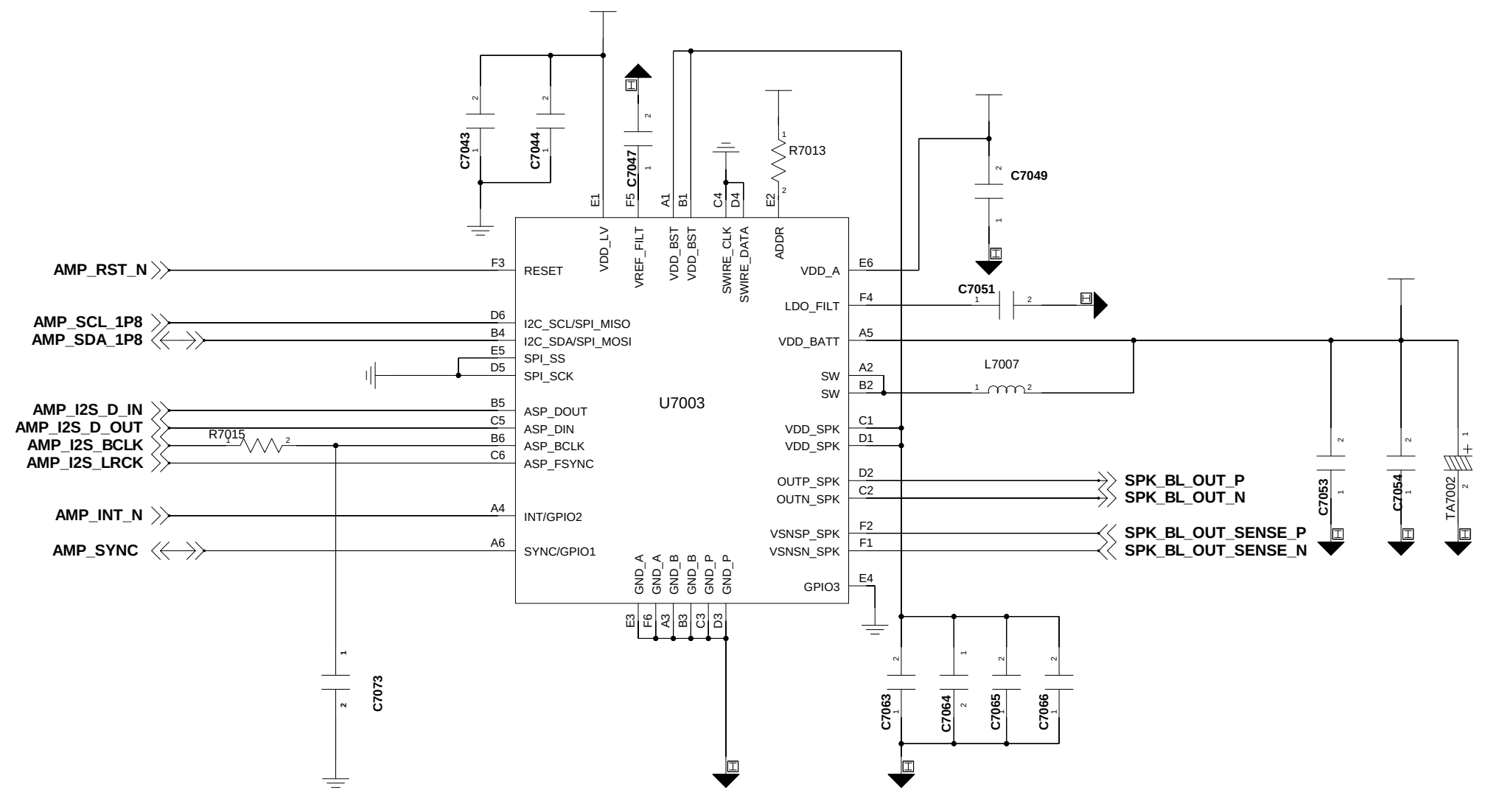
SPK_BR_OUT_SENSE_P
SPK_BR_OUT_P
ANT7008
ANT7009
L7001
L7002
C7023
C7024
C7025
ZD7006
ZD7007
C7026
C7027
C7028
SPK_BR_OUT_SENSE_N
SPK_BR_OUT_N

SPK BR CON



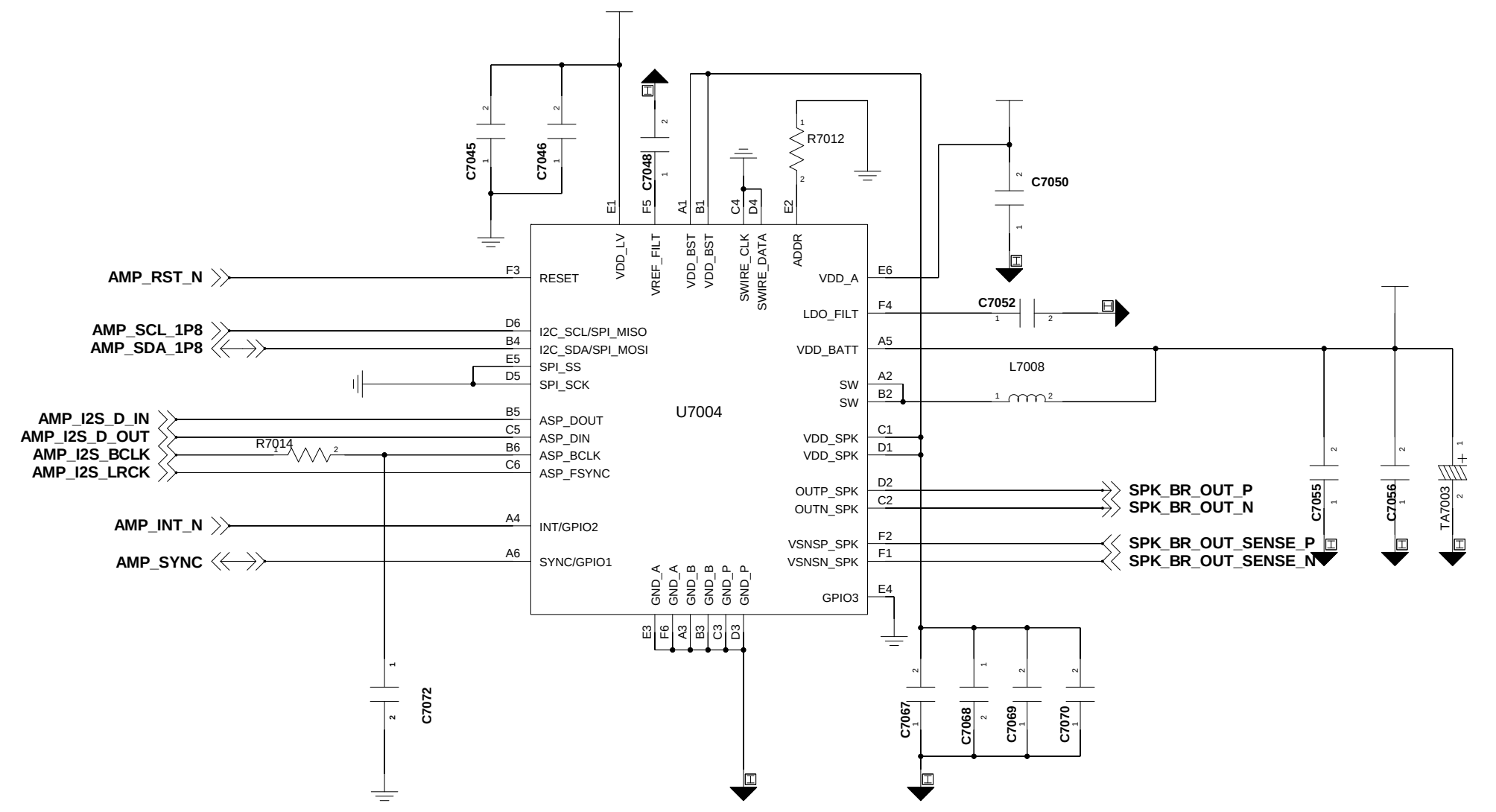
SPK_TR_OUT_SENSE_P
SPK_TR_OUT_P
ANT7010
ANT7011
L7005
L7006
C7037
C7038
C7039
ZD7008
ZD7009
C7040
C7041
C7042
SPK_TR_OUT_SENSE_N
SPK_TR_OUT_N

SPK TR CON



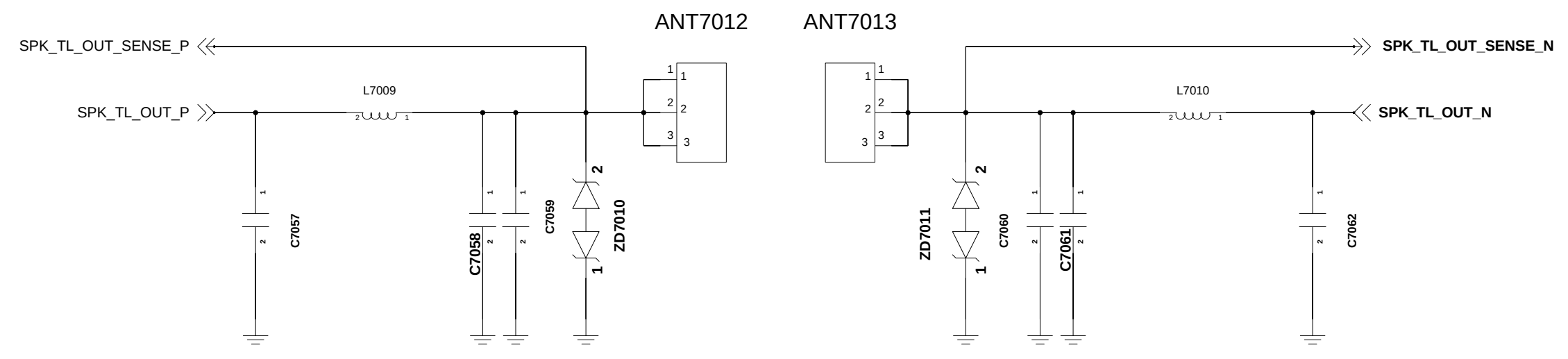
C7043
C7044
C7047
R7013
C7049
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
R7015
U7003
C7051
L7007
SPK_BL_OUT_P
SPK_BL_OUT_N
SPK_BL_OUT_SENSE_P
SPK_BL_OUT_SENSE_N
C7053
C7054
TA7002
C7073
C7063
C7064
C7065
C7066

SPK BOTTOM LEFT AMP



C7045
C7046
C7048
R7012
C7050
AMP_RST_N
AMP_SCL_1P8
AMP_SDA_1P8
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
AMP_I2S_LRCK
AMP_INT_N
AMP_SYNC
R7014
U7004
C7052
L7008
SPK_BR_OUT_P
SPK_BR_OUT_N
SPK_BR_OUT_SENSE_P
SPK_BR_OUT_SENSE_N
C7055
C7056
TA7003
C7072
C7067
C7068
C7069
C7070

SPK BOTTOM RIGHT AMP

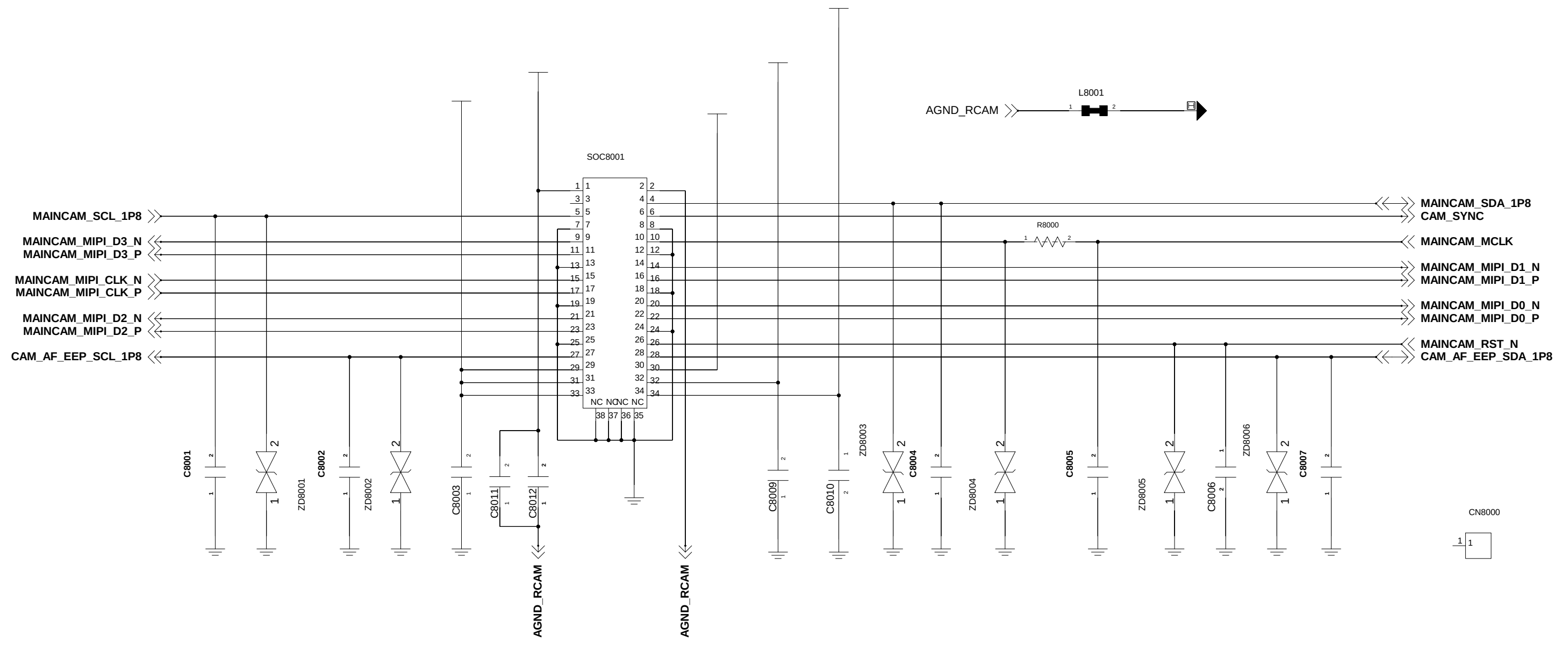


SPK_TL_OUT_SENSE_P
SPK_TL_OUT_P
ANT7012
ANT7013
L7009
L7010
C7057
C7058
C7059
ZD7010
ZD7011
C7060
C7061
C7062
SPK_TL_OUT_SENSE_N
SPK_TL_OUT_N

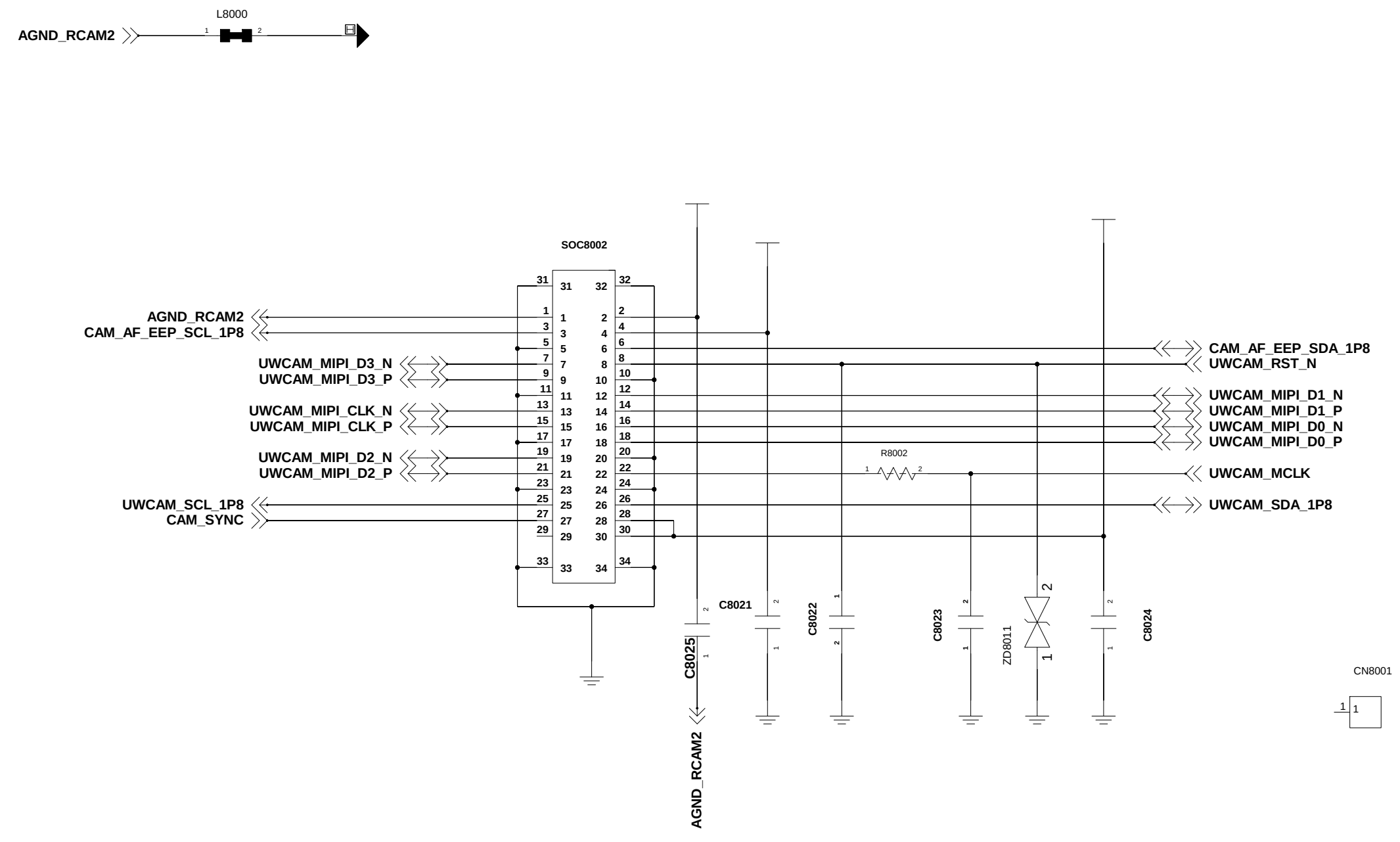
SPK TL CON

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 7 of 15
Date

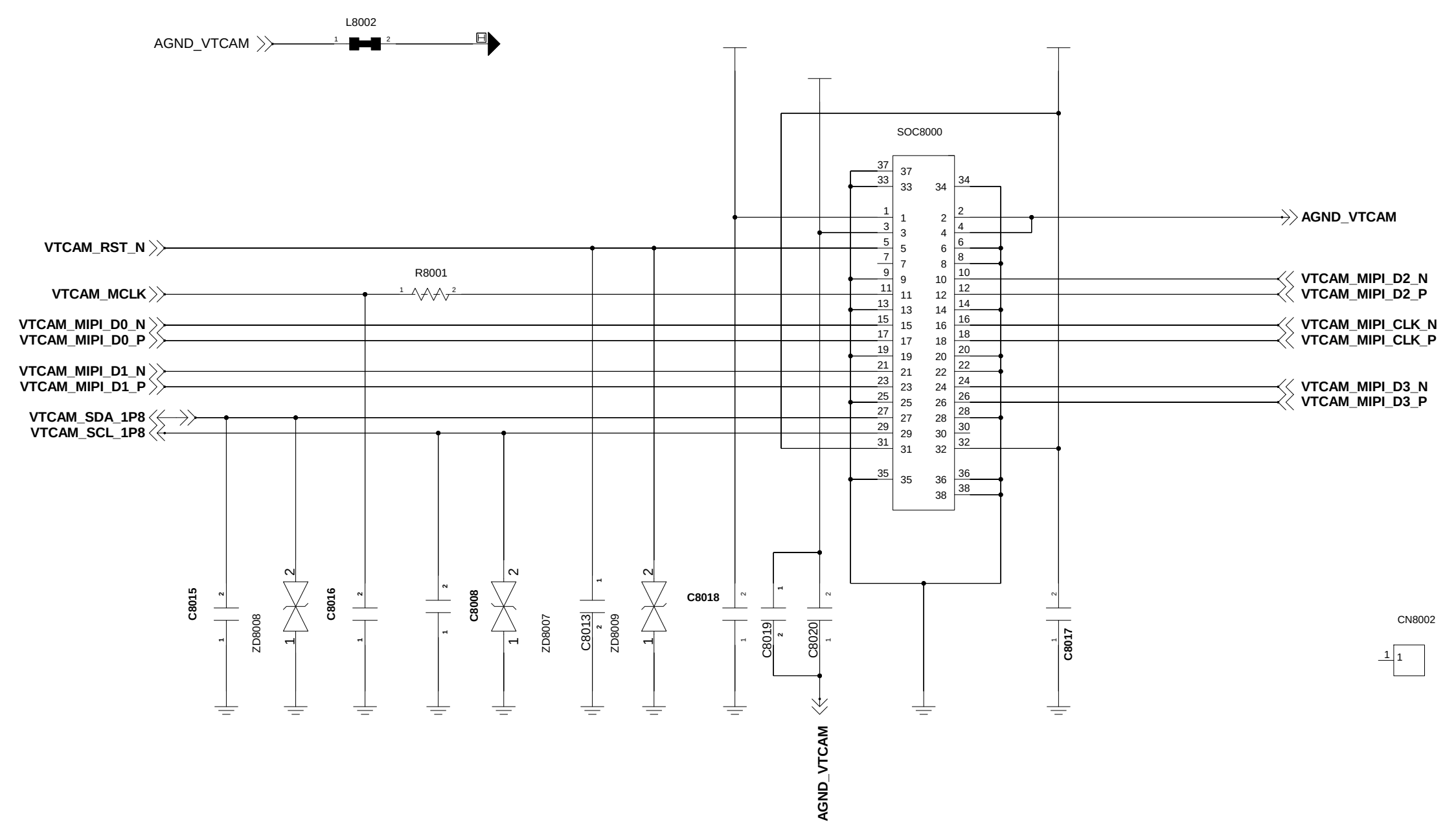
CAMERA


AGND_RCAM
L8001
SOC8001
MAINCAM_SCL_1P8
MAINCAM_MIPI_D3_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_CLK_N
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D2_P
CAM_AF_EEP_SCL_1P8
MAINCAM_SDA_1P8
CAM_SYNC
MAINCAM_MCLK
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_D0_P
MAINCAM_RST_N
CAM_AF_EEP_SDA_1P8
R8000
C8001
ZD8001
C8002
ZD8002
C8003
C8011
C8012
AGND_RCAM
AGND_RCAM
C8009
C8010
ZD8003
C8004
ZD8004
C8005
ZD8005
C8006
ZD8006
C8007
CN8000

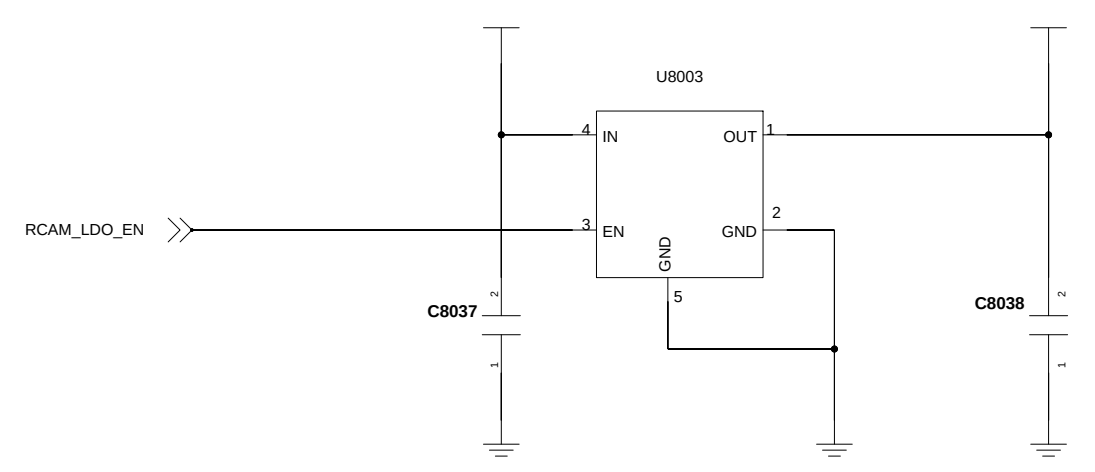
13M_CAM (RCAM)

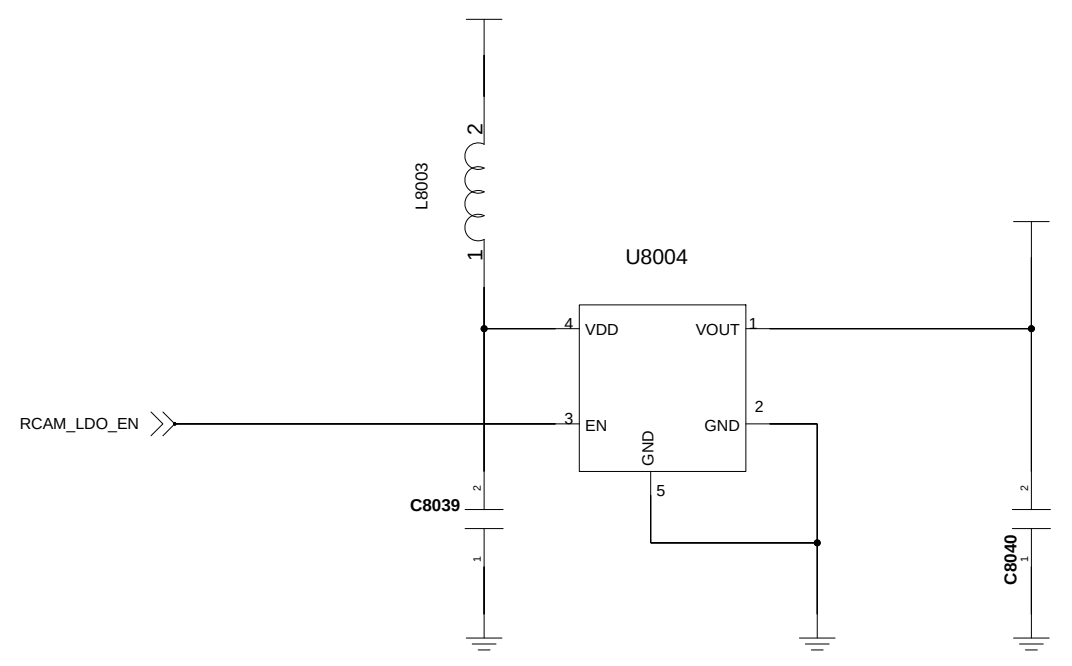

AGND_RCAM2
L8000
SOC8002
AGND_RCAM2
CAM_AF_EEP_SCL_1P8
UWCAM_MIPI_D3_N
UWCAM_MIPI_D3_P
UWCAM_MIPI_CLK_N
UWCAM_MIPI_CLK_P
UWCAM_MIPI_D2_N
UWCAM_MIPI_D2_P
UWCAM_SCL_1P8
CAM_SYNC
CAM_AF_EEP_SDA_1P8
UWCAM_RST_N
UWCAM_MIPI_D1_N
UWCAM_MIPI_D1_P
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
UWCAM_MCLK
UWCAM_SDA_1P8
R8002
C8025
C8021
C8022
C8023
ZD8011
C8024
AGND_RCAM2
CN8001

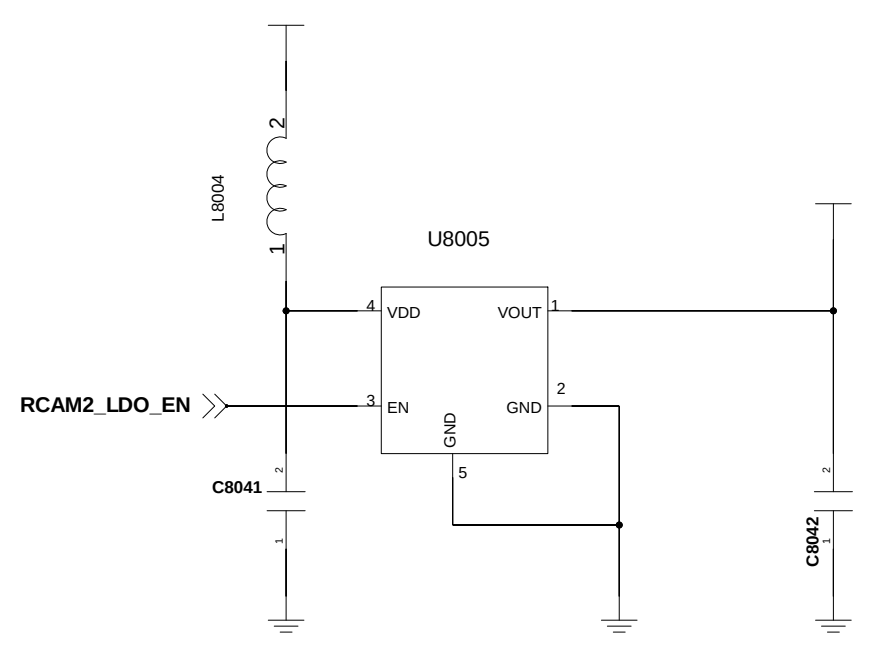
8M_UW CAM (RCAM2)

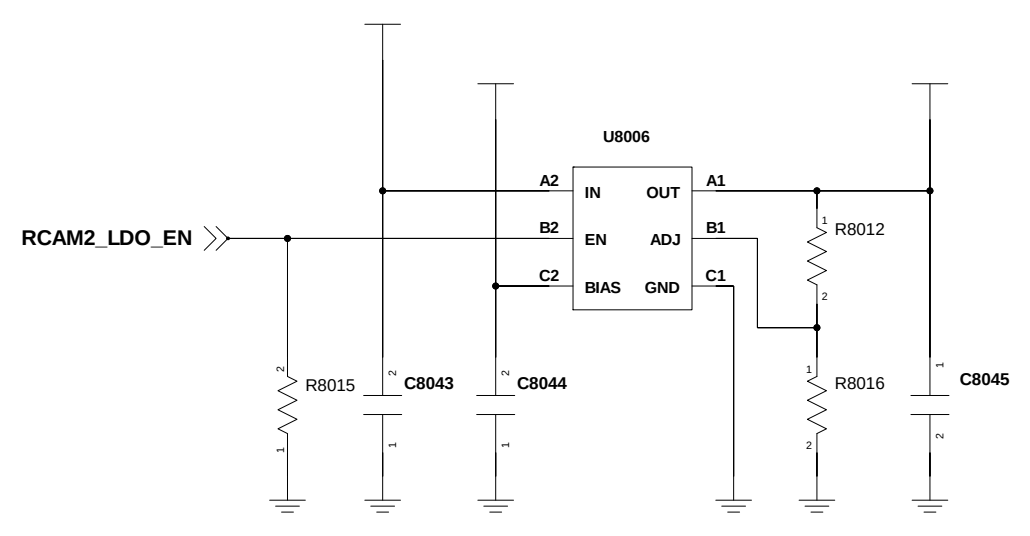

AGND_VTCAM
L8002
SOC8000
VTCAM_RST_N
VTCAM_MCLK
R8001
VTCAM_MIPI_D0_N
VTCAM_MIPI_D0_P
VTCAM_MIPI_D1_N
VTCAM_MIPI_D1_P
VTCAM_SDA_1P8
VTCAM_SCL_1P8
AGND_VTCAM
VTCAM_MIPI_D2_N
VTCAM_MIPI_D2_P
VTCAM_MIPI_CLK_N
VTCAM_MIPI_CLK_P
VTCAM_MIPI_D3_N
VTCAM_MIPI_D3_P
C8015
ZD8008
C8016
C8008
ZD8007
C8013
ZD8009
C8018
C8019
C8020
C8017
AGND_VTCAM
CN8002

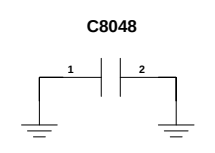
12M CAM (FCAM)


U8003
RCAM_LDO_EN
IN
OUT
EN
GND
GND
C8037
C8038

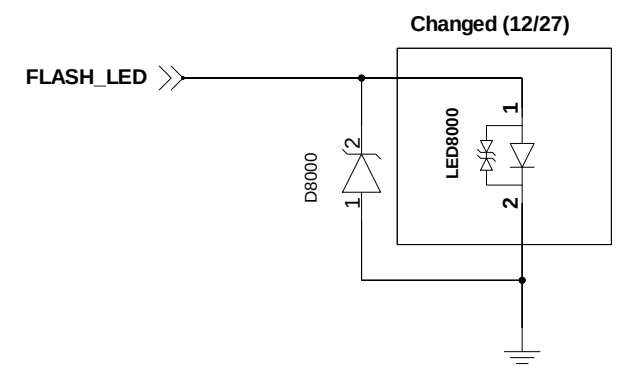

L8003
U8004
VDD
VOUT
RCAM_LDO_EN
EN
GND
GND
C8039
C8040


L8004
U8005
VDD
VOUT
RCAM2_LDO_EN
EN
GND
GND
C8041
C8042

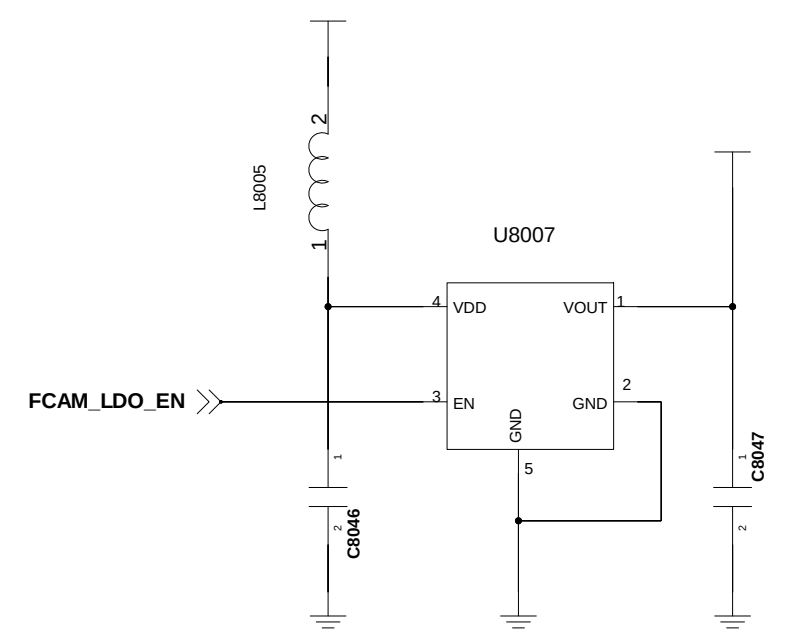

U8006
RCAM2_LDO_EN
A2
IN
OUT
A1
B2
EN
ADJ
B1
C2
BIAS
GND
C1
R8012
R8015
C8043
C8044
R8016
C8045



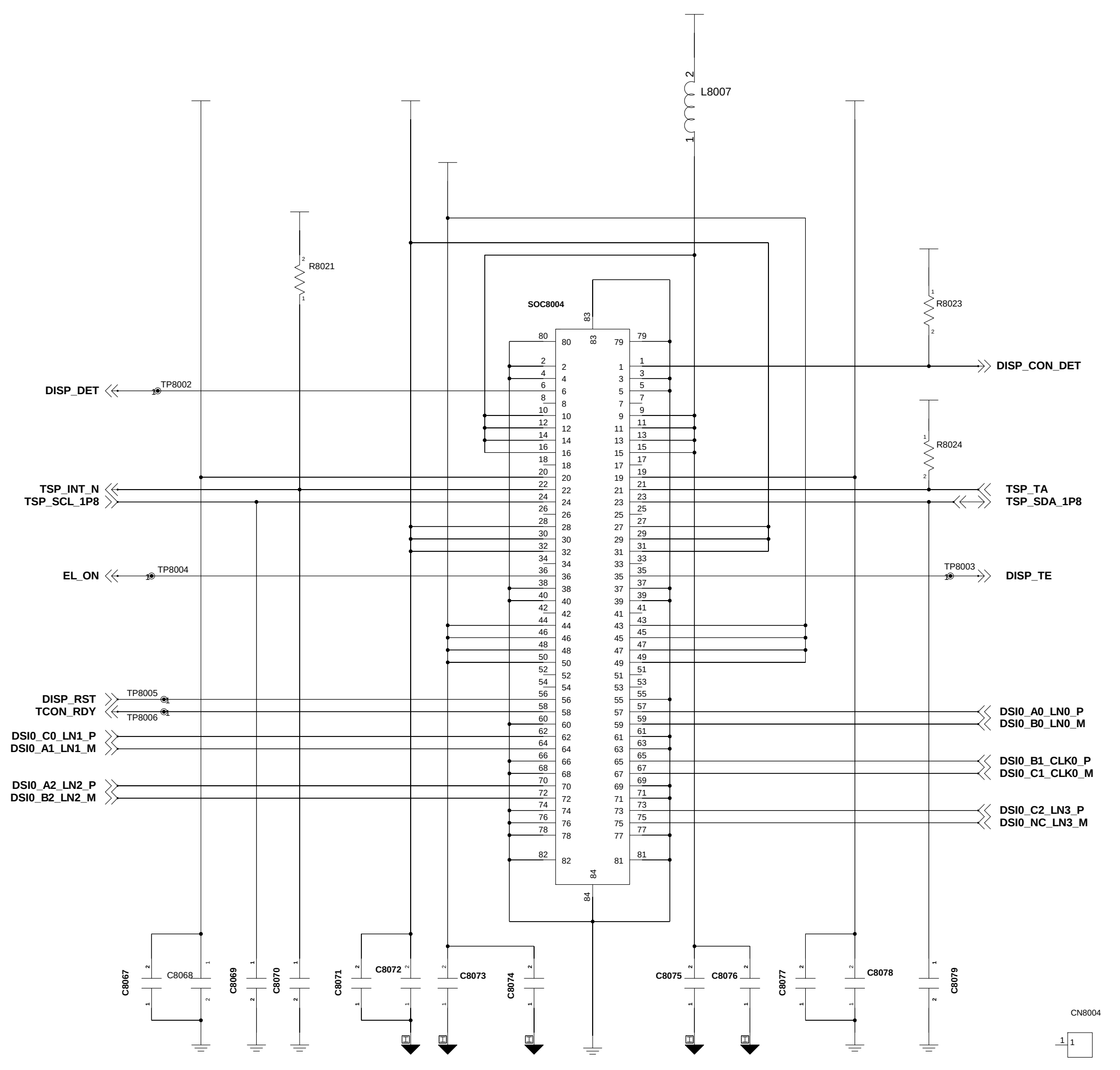
Dummy


Changed (12/27)
FLASH_LED
D8000
LED8000

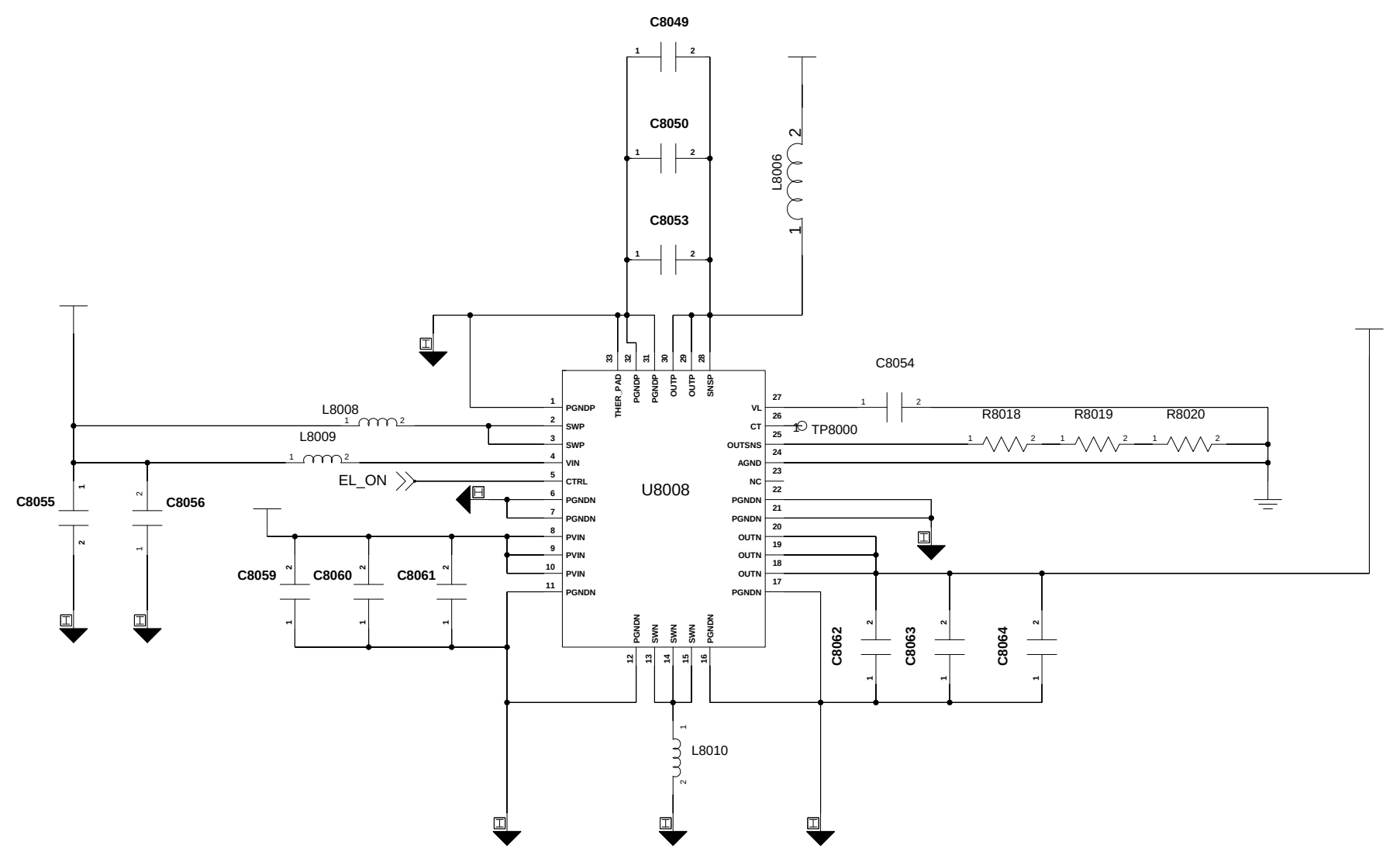
FLASH LED


L8005
U8007
VDD
VOUT
FCAM_LDO_EN
EN
GND
GND
C8046
C8047

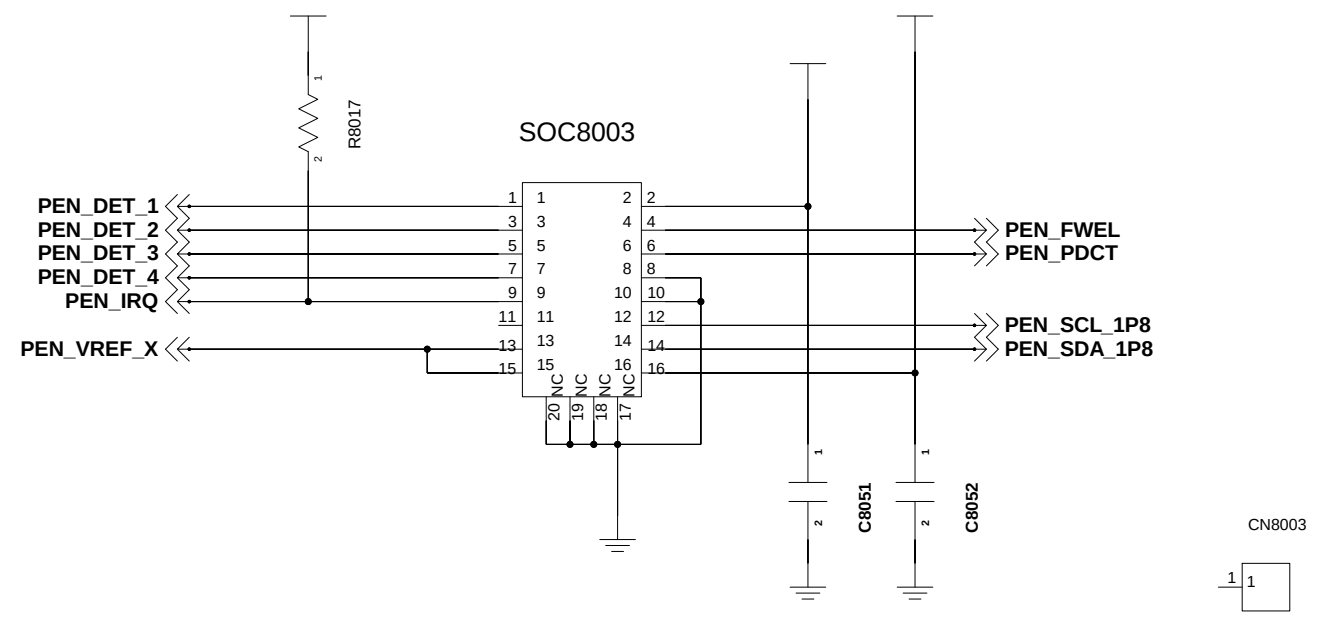
DISPLAY


L8007
R8021
SOC8004
R8023
R8024
DISP_DET
TP8002
DISP_CON_DET
TSP_INT_N
TSP_SCL_1P8
TSP_TA
TSP_SDA_1P8
EL_ON
TP8004
TP8003
DISP_TE
DISP_RST
TCON_RDY
TP8005
TP8006
DSI0_C0_LN1_P
DSI0_A1_LN1_M
DSI0_A2_LN2_P
DSI0_B2_LN2_M
DSI0_A0_LN0_P
DSI0_B0_LN0_M
DSI0_B1_CLK0_P
DSI0_C1_CLK0_M
DSI0_C2_LN3_P
DSI0_NC_LN3_M
C8067
C8068
C8069
C8070
C8071
C8072
C8073
C8074
C8075
C8076
C8077
C8078
C8079
CN8004

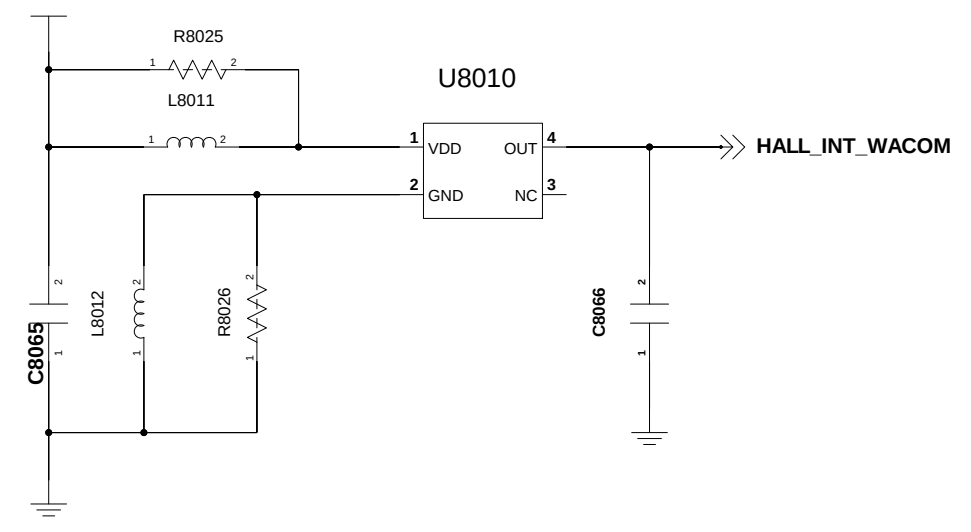
OCTA Con.


C8049
C8050
C8053
L8006
C8054
L8008
L8009
EL_ON
U8008
TP8000
R8018
R8019
R8020
C8055
C8056
C8059
C8060
C8061
C8062
C8063
C8064
L8010
PGNDP
SWP
VIN
CTRL
PGNDN
PVIN
VL
CT
OUTSNS
AGND
NC
OUTN

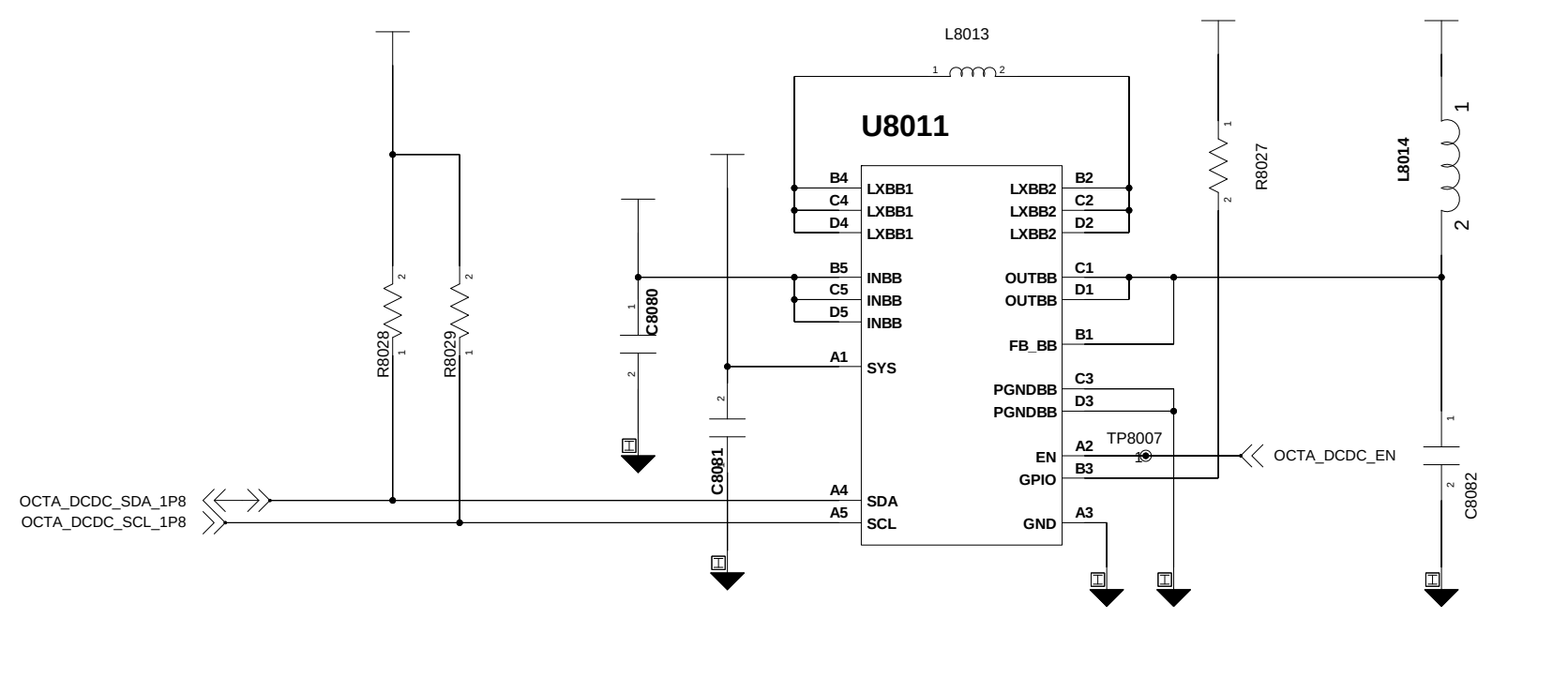
OCTA POWER


R8017
SOC8003
PEN_DET_1
PEN_DET_2
PEN_DET_3
PEN_DET_4
PEN_IRQ
PEN_VREF_X
PEN_FWEL
PEN_PDCT
PEN_SCL_1P8
PEN_SDA_1P8
C8051
C8052
CN8003

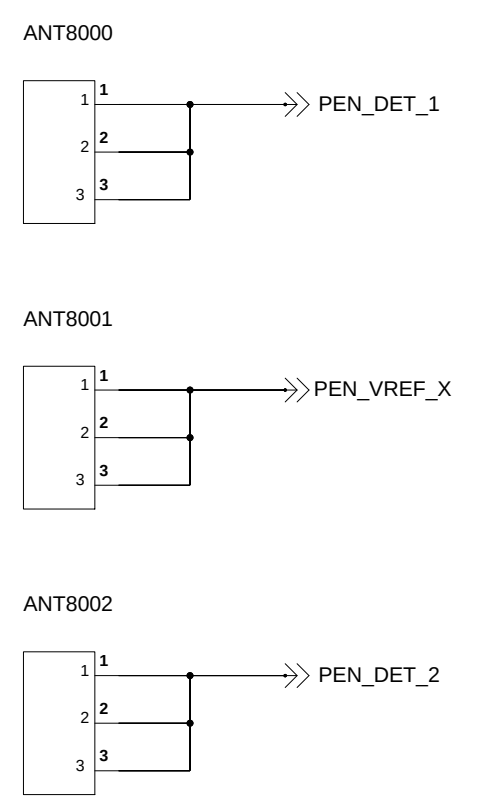
WACOM CONN
(MAIN <-> OCTA / WACOM)


R8025
L8011
U8010
VDD
OUT
GND
NC
HALL_INT_WACOM
C8065
L8012
R8026
C8066

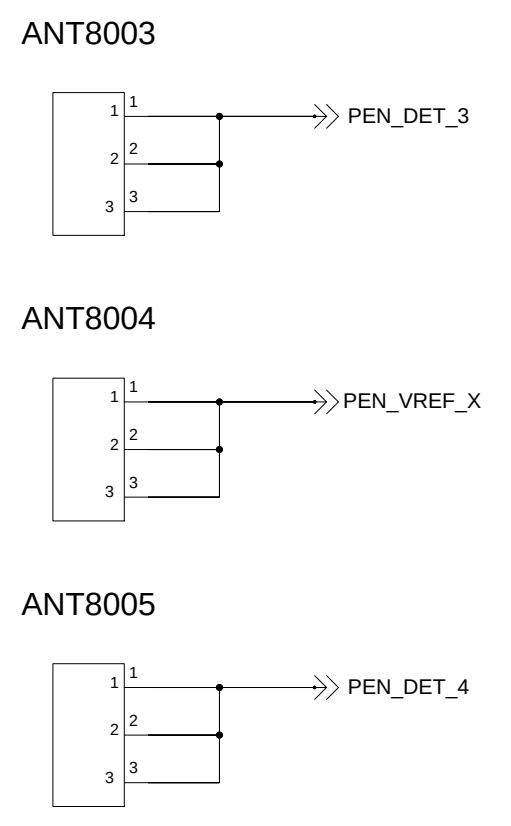
WACOM HALL_IC


L8013
U8011
LXBB1
LXBB2
INBB
OUTBB
FB_BB
SYS
PGNDBB
EN
GPIO
SDA
SCL
GND
R8027
L8014
R8028
R8029
C8080
C8081
TP8007
OCTA_DCDC_EN
OCTA_DCDC_SDA_1P8
OCTA_DCDC_SCL_1P8
C8082

TCON BUCK BOOSTER


ANT8000
PEN_DET_1
ANT8001
PEN_VREF_X
ANT8002
PEN_DET_2

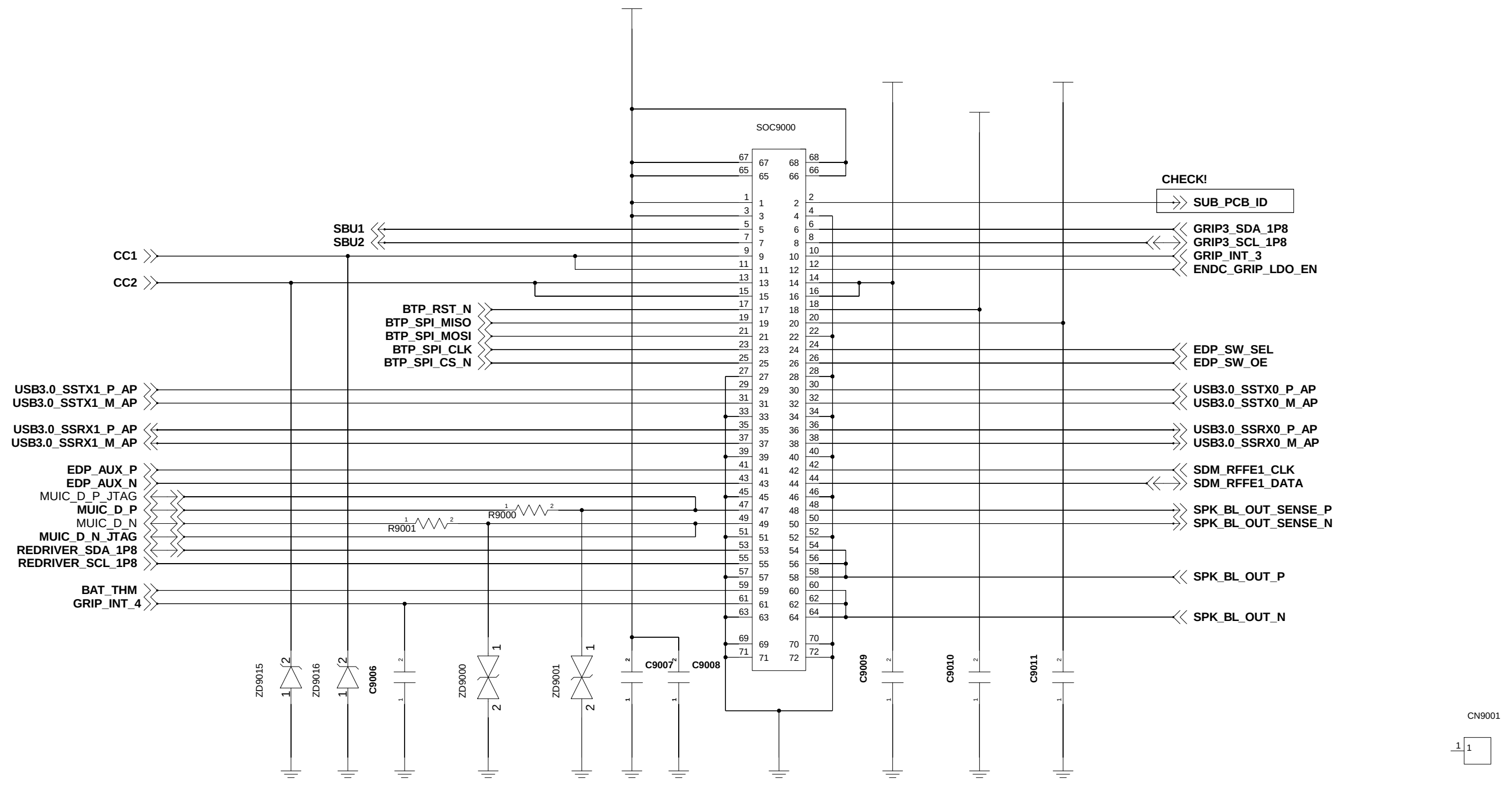
PEN CHG C/T


ANT8003
PEN_DET_3
ANT8004
PEN_VREF_X
ANT8005
PEN_DET_4

PEN CHG C/T_2

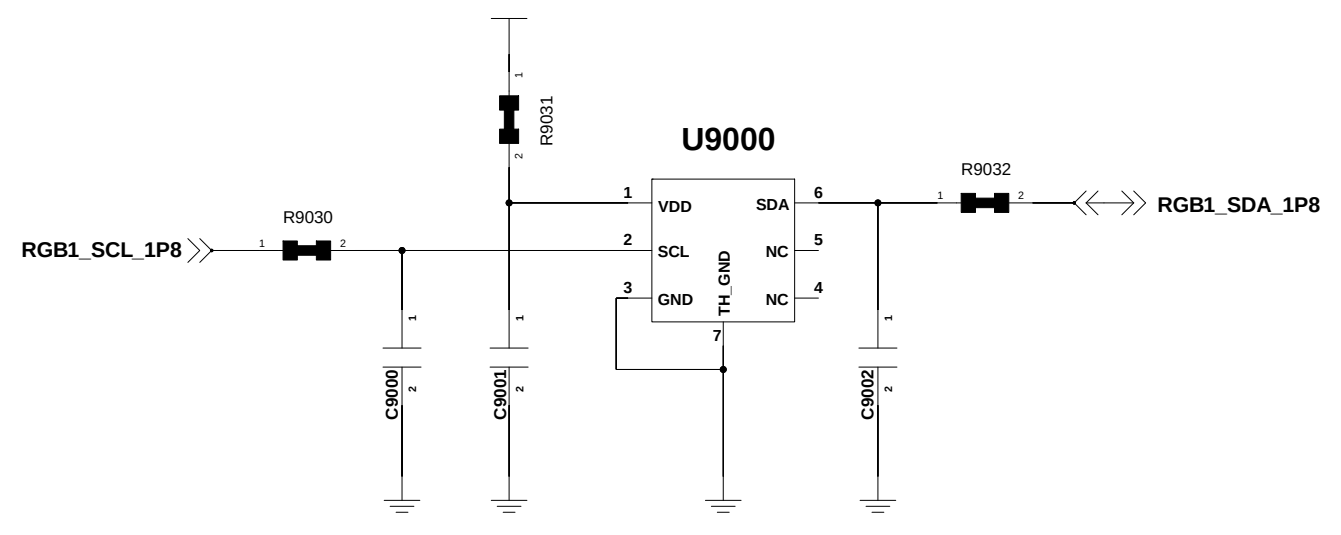
TITLE TabS7P
SIZE A1 DWG NO REV
DRAWN BY User Name
SHEET 8 of 15 Date

SENSOR / SD/SIM / KEY / I/F



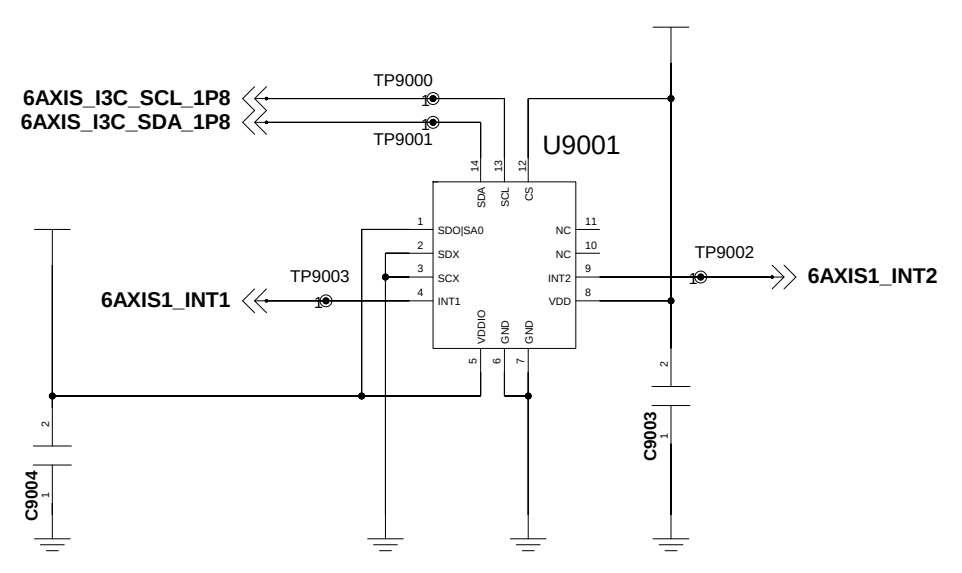
SOC9000
CHECK!
SUB_PCB_ID
SBU1
SBU2
CC1
CC2
GRIP3_SDA_1P8
GRIP3_SCL_1P8
GRIP_INT_3
ENDC_GRIP_LDO_EN
BTP_RST_N
BTP_SPI_MISO
BTP_SPI_MOSI
BTP_SPI_CLK
BTP_SPI_CS_N
EDP_SW_SEL
EDP_SW_OE
USB3.0_SSTX1_P_AP
USB3.0_SSTX1_M_AP
USB3.0_SSTX0_P_AP
USB3.0_SSTX0_M_AP
USB3.0_SSRX1_P_AP
USB3.0_SSRX1_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX0_M_AP
EDP_AUX_P
EDP_AUX_N
SDM_RFFE1_CLK
SDM_RFFE1_DATA
MUIC_D_P_JTAG
MUIC_D_P
MUIC_D_N
MUIC_D_N_JTAG
SPK_BL_OUT_SENSE_P
SPK_BL_OUT_SENSE_N
R9000
R9001
REDRIVER_SDA_1P8
REDRIVER_SCL_1P8
SPK_BL_OUT_P
BAT_THM
GRIP_INT_4
SPK_BL_OUT_N
ZD9015
ZD9016
C9006
ZD9000
ZD9001
C9007
C9008
C9009
C9010
C9011
CN9001

I/F CONN
(MAIN <-> USB)



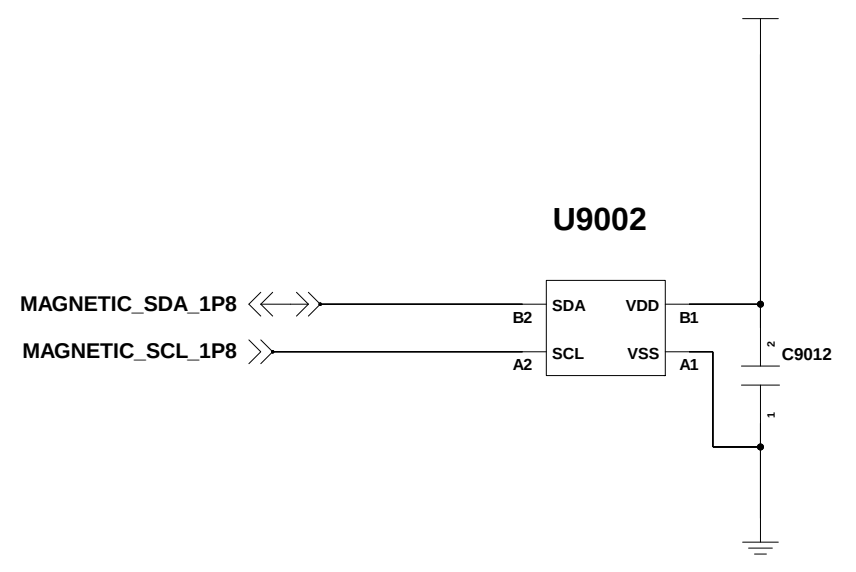
U9000
R9031
R9030
R9032
RGB1_SCL_1P8
RGB1_SDA_1P8
VDD
SCL
GND
SDA
NC
NC
TH_GND
C9000
C9001
C9002

RGB Sensor 1



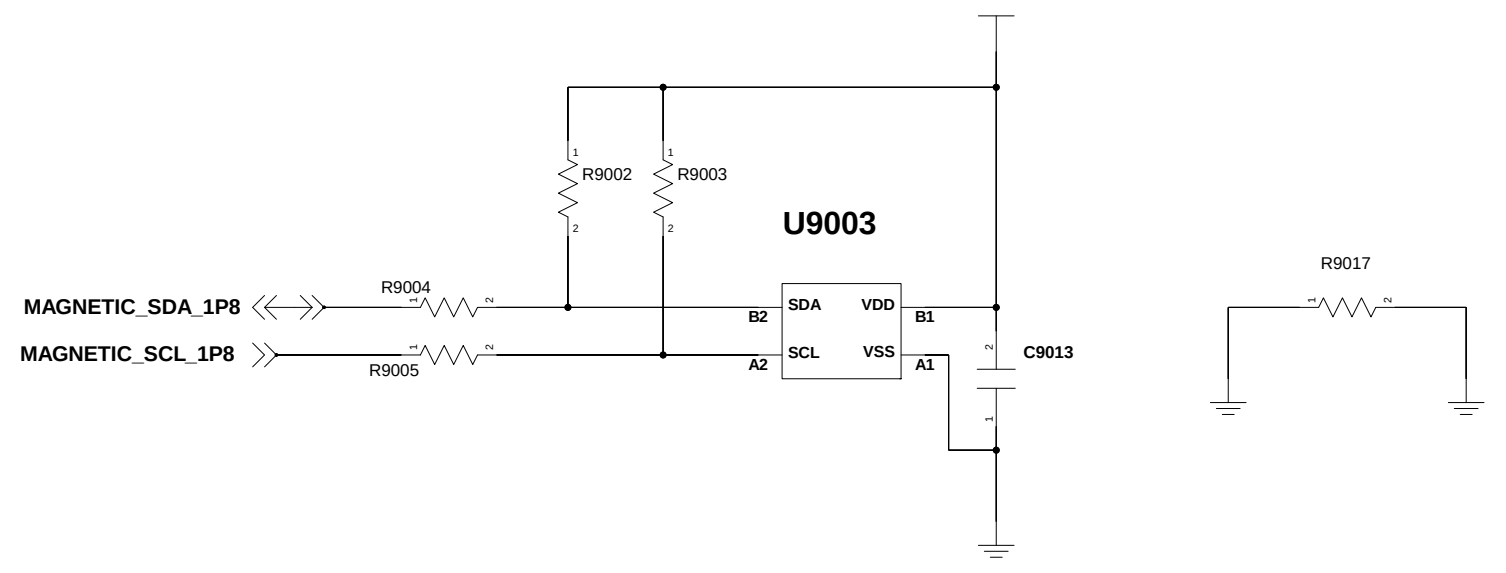
6AXIS_I3C_SCL_1P8
6AXIS_I3C_SDA_1P8
TP9000
TP9001
U9001
TP9003
6AXIS1_INT1
TP9002
6AXIS1_INT2
SDO/SA0
SDX
SCX
INT1
NC
NC
INT2
VDD
SDA
SCL
CS
VDDIO
GND
GND
C9004
C9003

6 Axis (GYRO,ACCEL)



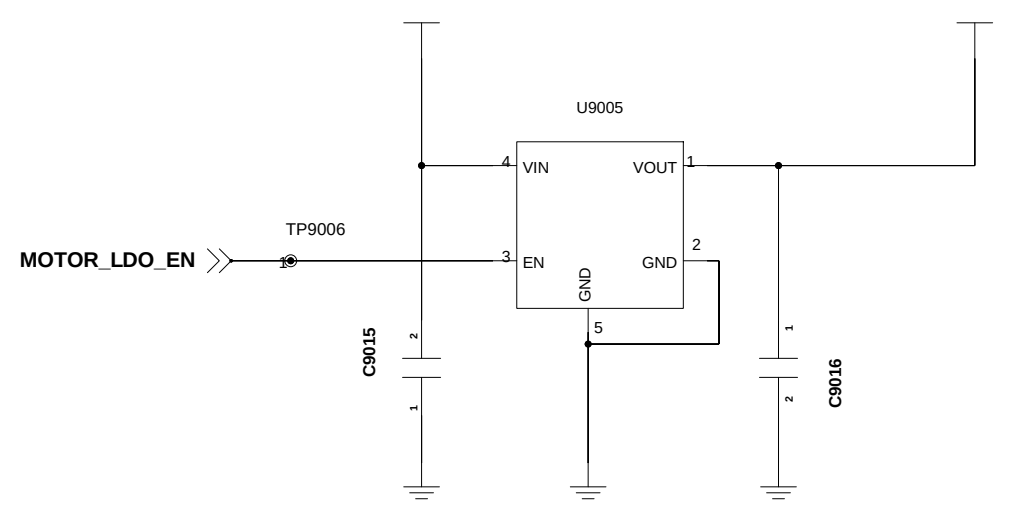
U9002
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
SDA
SCL
VDD
VSS
B2
A2
B1
A1
C9012

Magnetic SENSOR



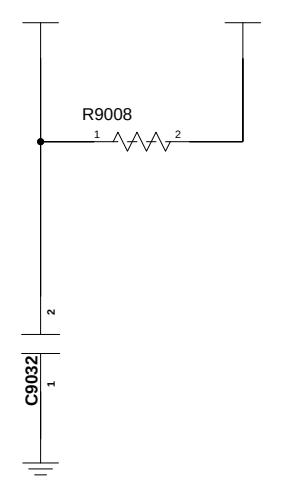
R9002
R9003
U9003
R9004
R9005
MAGNETIC_SDA_1P8
MAGNETIC_SCL_1P8
SDA
SCL
VDD
VSS
B2
A2
B1
A1
C9013
R9017

Magnetic SENSOR Option



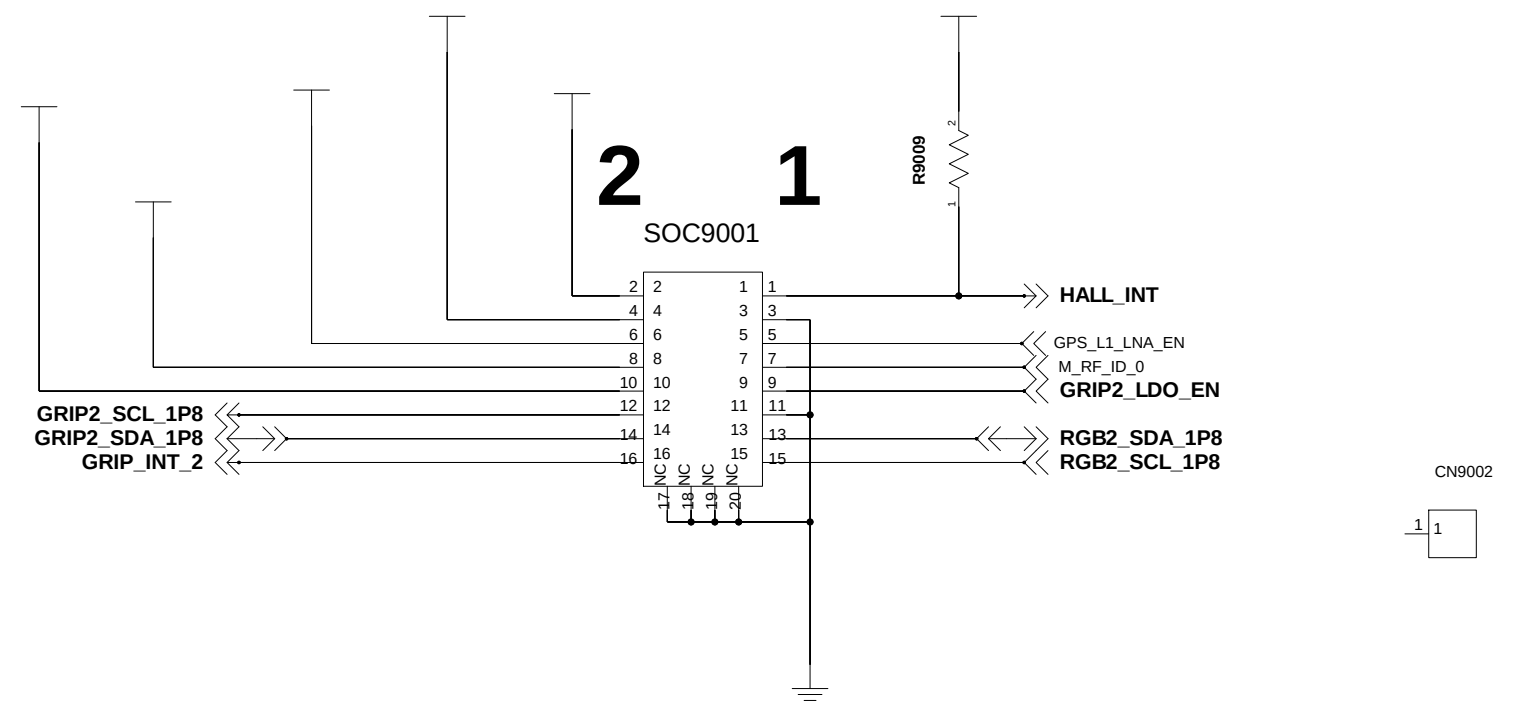
U9005
TP9006
MOTOR_LDO_EN
VIN
EN
GND
VOUT
GND
C9015
C9016

MOTOR LDO



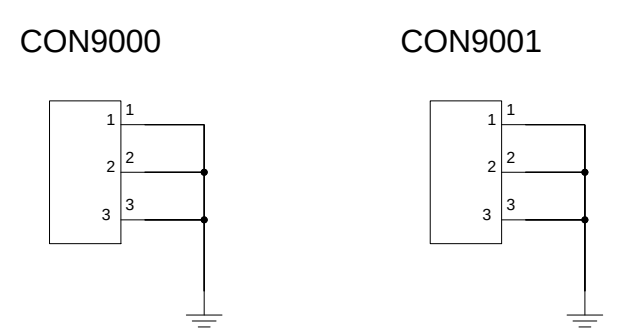
R9008
C9032

FOD



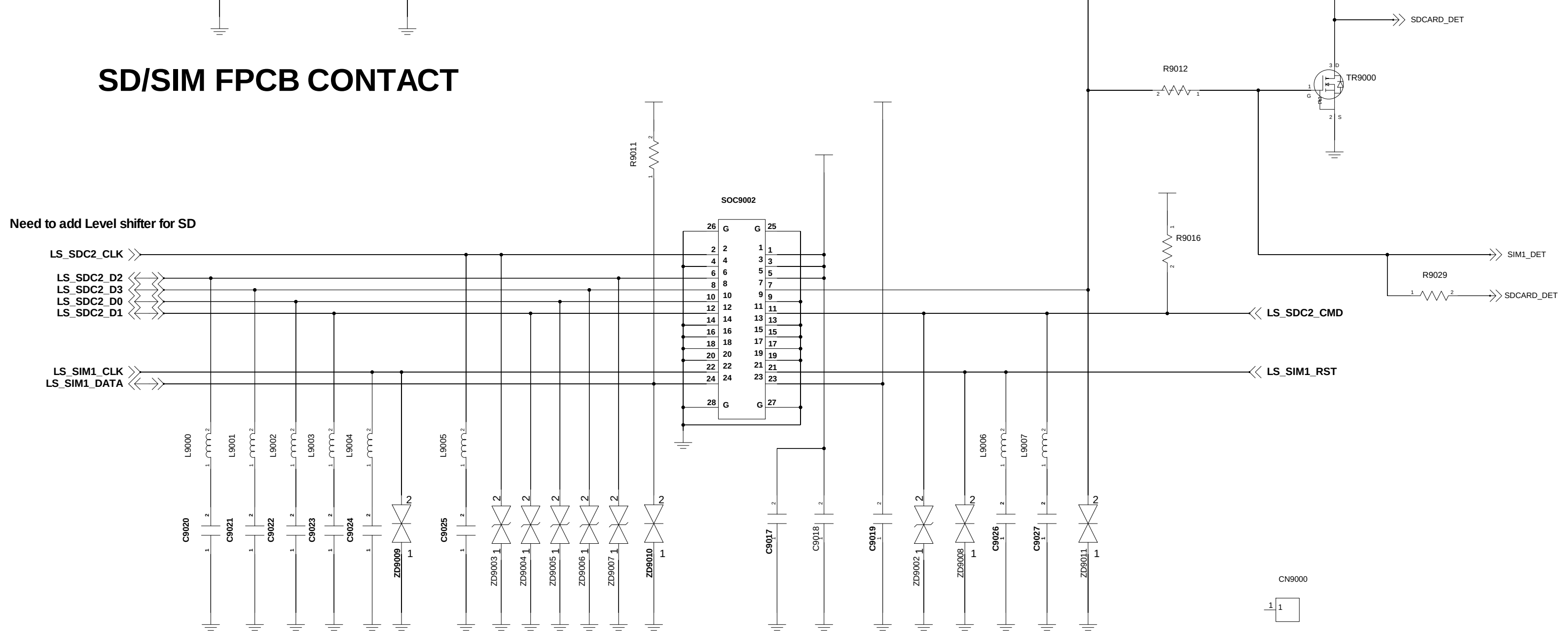
2
1
SOC9001
R9009
HALL_INT
GPS_L1_LNA_EN
M_RF_ID_0
GRIP2_LDO_EN
RGB2_SDA_1P8
RGB2_SCL_1P8
GRIP2_SCL_1P8
GRIP2_SDA_1P8
GRIP_INT_2
NC
CN9002

SUB CONN
(MAIN <-> ANT SUB)



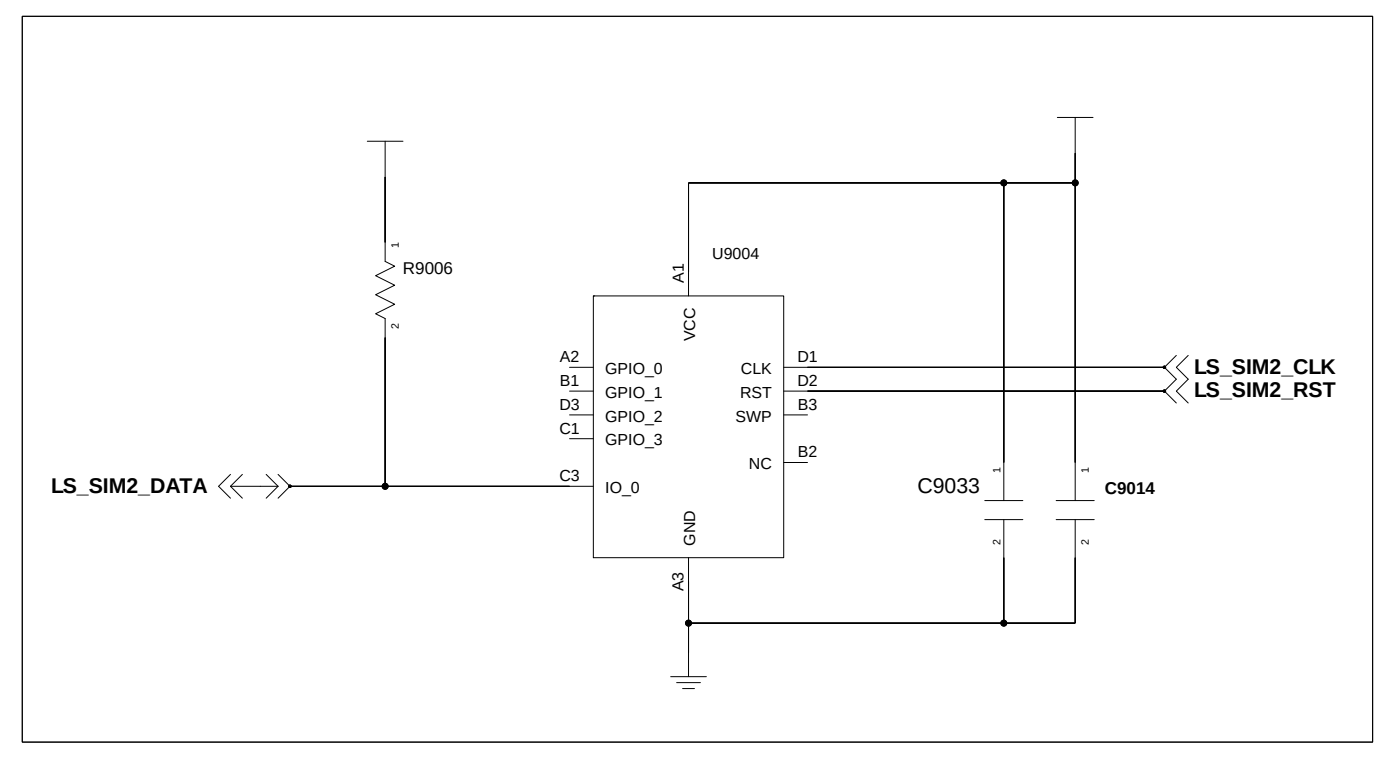
CON9000
CON9001

SD/SIM FPCB CONTACT



Need to add Level shifter for SD
LS_SDC2_CLK
LS_SDC2_D2
LS_SDC2_D3
LS_SDC2_D0
LS_SDC2_D1
LS_SIM1_CLK
LS_SIM1_DATA
R9011
SOC9002
G
R9010
R9027
SDCARD_DET
R9012
TR9000
R9016
SIM1_DET
R9029
SDCARD_DET
LS_SDC2_CMD
LS_SIM1_RST
L9000
L9001
L9002
L9003
L9004
L9005
L9006
L9007
C9020
C9021
C9022
C9023
C9024
ZD9009
C9025
ZD9003
ZD9004
ZD9005
ZD9006
ZD9007
ZD9010
C9017
C9018
C9019
ZD9002
ZD9008
C9026
C9027
ZD9011
CN9000

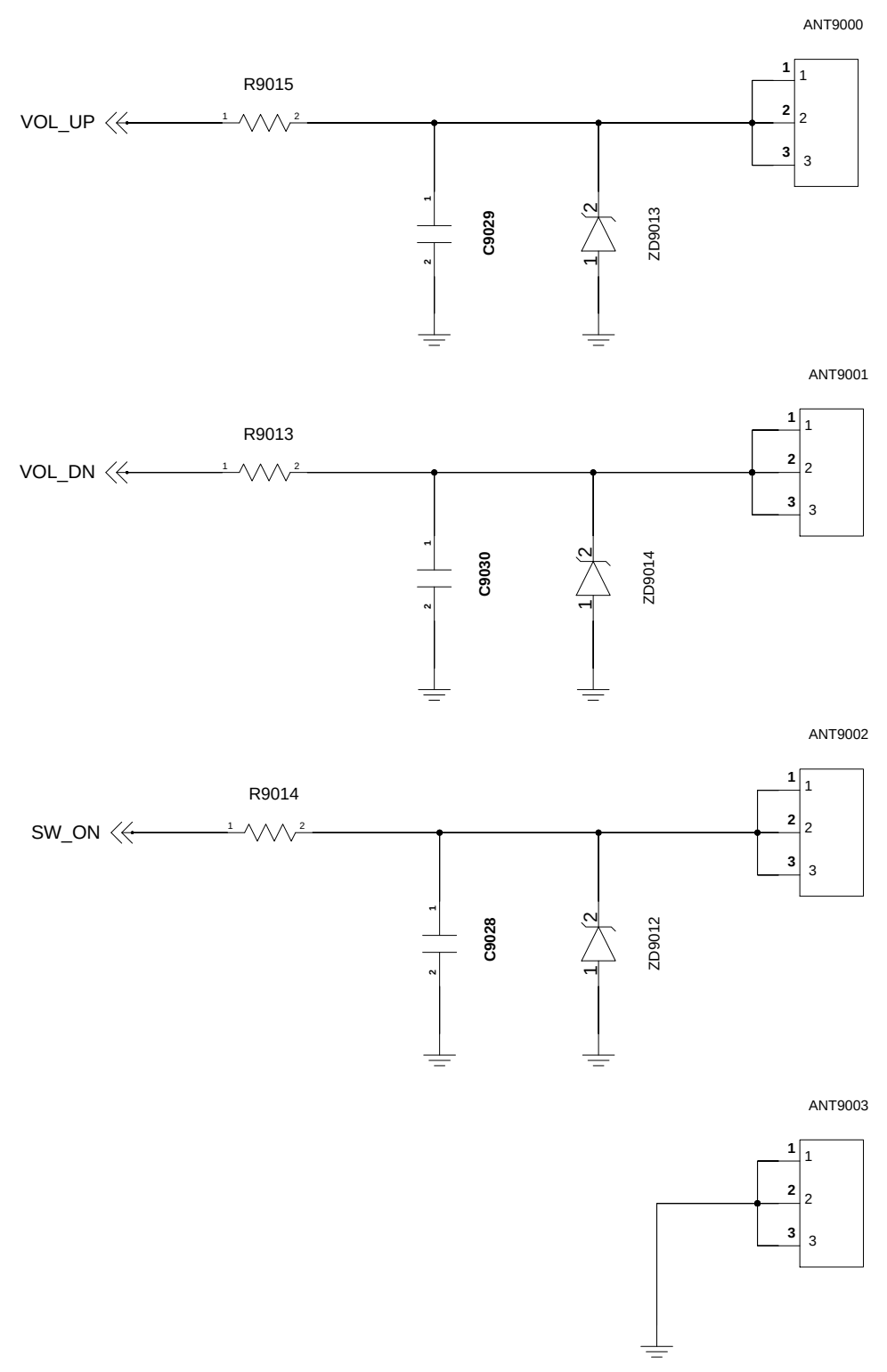
SIM/SD SOCKET Conn.



R9006
U9004
VCC
GPIO_0
GPIO_1
GPIO_2
GPIO_3
IO_0
GND
CLK
RST
SWP
NC
LS_SIM2_DATA
LS_SIM2_CLK
LS_SIM2_RST
C9033
C9014

eSIM

WIFI : delete



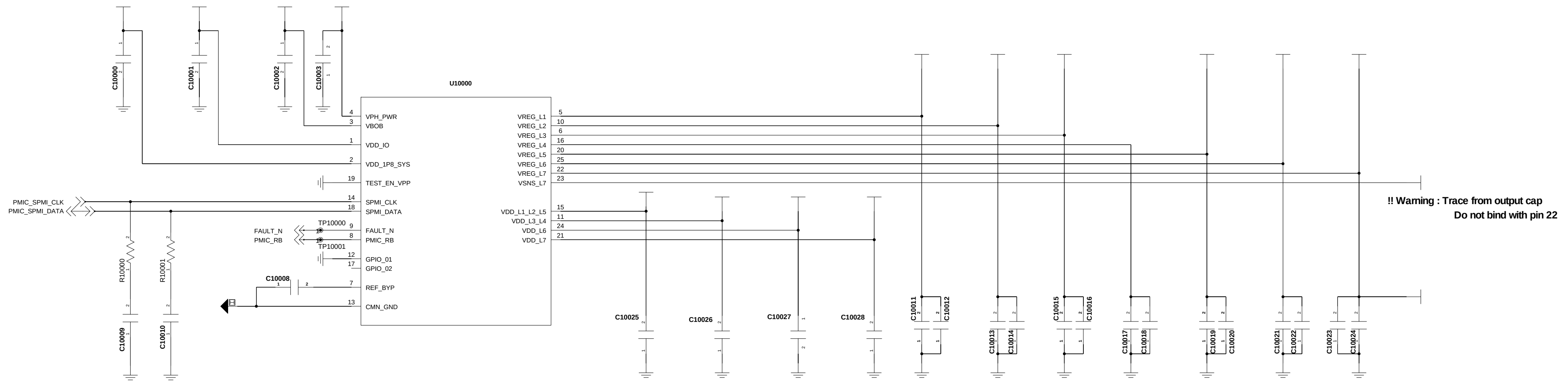
ANT9000
R9015
VOL_UP
C9029
ZD9013
ANT9001
R9013
VOL_DN
C9030
ZD9014
ANT9002
R9014
SW_ON
C9028
ZD9012
ANT9003

Key Contact

TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 9 of 15
Date

Power

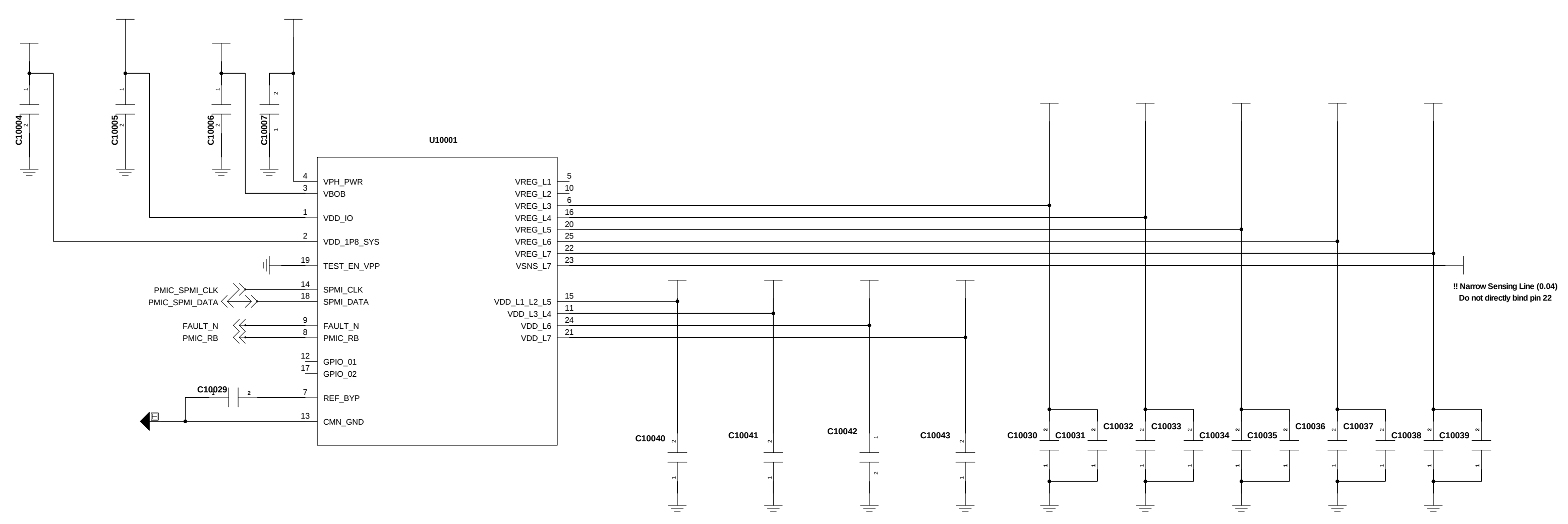
RF PMIC


U10000
C10000
C10001
C10002
C10003
VPH_PWR
VBOB
VDD_IO
VDD_1P8_SYS
TEST_EN_VPP
SPMI_CLK
SPMI_DATA
FAULT_N
PMIC_RB
GPIO_01
GPIO_02
REF_BYP
CMN_GND
VREG_L1
VREG_L2
VREG_L3
VREG_L4
VREG_L5
VREG_L6
VREG_L7
VSNS_L7
VDD_L1_L2_L5
VDD_L3_L4
VDD_L6
VDD_L7
PMIC_SPMI_CLK
PMIC_SPMI_DATA
FAULT_N
PMIC_RB
TP10000
TP10001
R10000
R10001
C10008
C10009
C10010
C10025
C10026
C10027
C10028
C10011
C10012
C10013
C10014
C10015
C10016
C10017
C10018
C10019
C10020
C10021
C10022
C10023
C10024
!! Warning : Trace from output cap
Do not bind with pin 22

PMR735D #0

SUB6 : Delete

WIFI : Delete

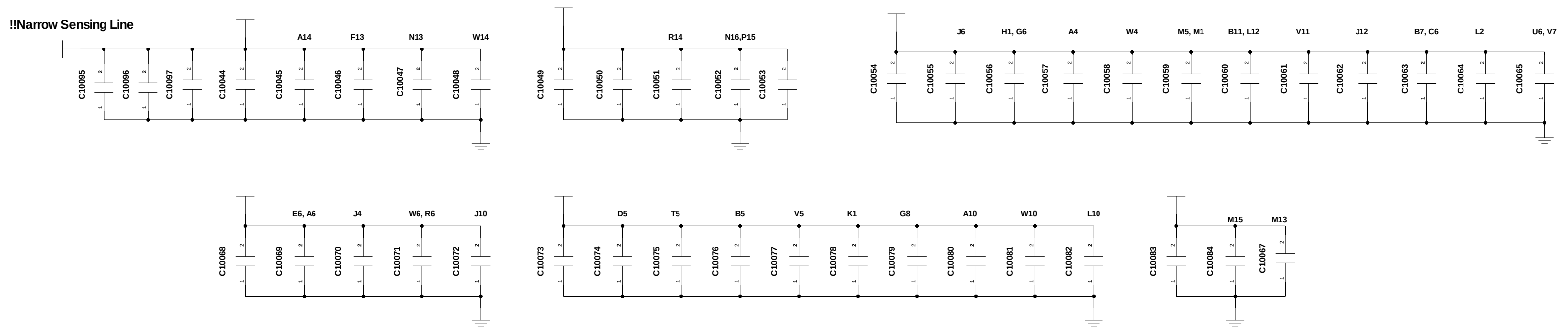

U10001
C10004
C10005
C10006
C10007
VPH_PWR
VBOB
VDD_IO
VDD_1P8_SYS
TEST_EN_VPP
SPMI_CLK
SPMI_DATA
FAULT_N
PMIC_RB
GPIO_01
GPIO_02
REF_BYP
CMN_GND
PMIC_SPMI_CLK
PMIC_SPMI_DATA
FAULT_N
PMIC_RB
C10029
VREG_L1
VREG_L2
VREG_L3
VREG_L4
VREG_L5
VREG_L6
VREG_L7
VSNS_L7
VDD_L1_L2_L5
VDD_L3_L4
VDD_L6
VDD_L7
!! Narrow Sensing Line (0.04)
Do not directly bind pin 22
C10040
C10041
C10042
C10043
C10030
C10031
C10032
C10033
C10034
C10035
C10036
C10037
C10038
C10039

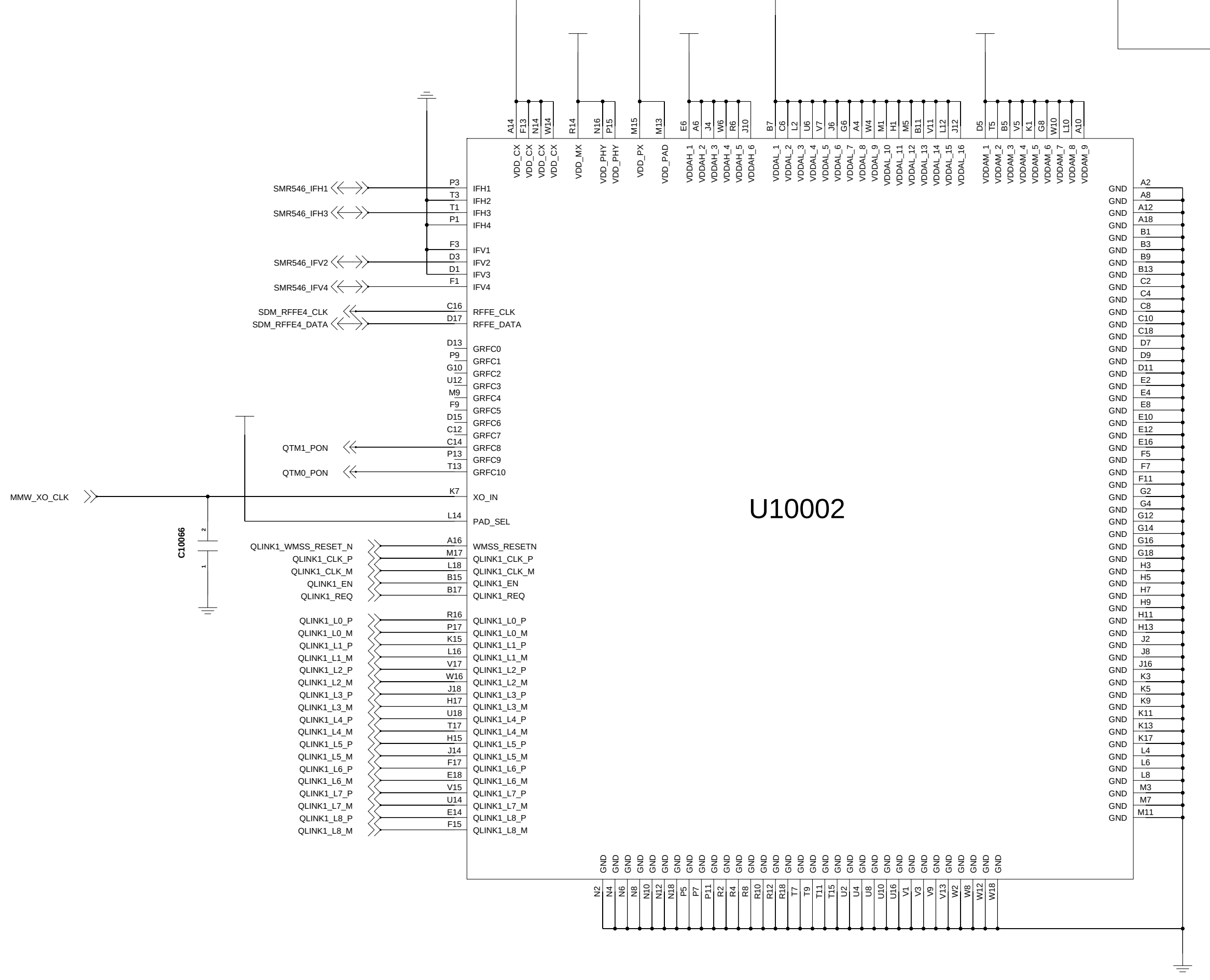
PMR735D #1

5G IFIC (SMR546)

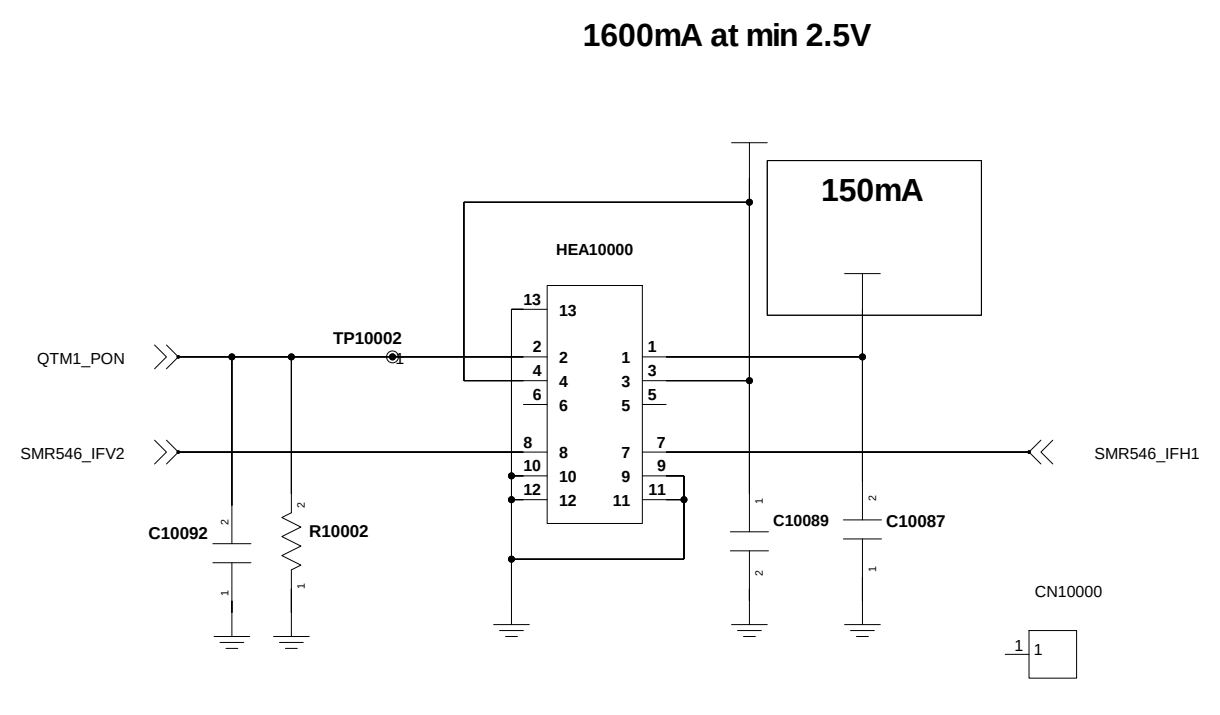
SUB6 : Delete

WIFI : Delete


!!Narrow Sensing Line
A14
F13
N13
W14
C10095
C10096
C10097
C10044
C10045
C10046
C10047
C10048
R14
N16,P15
C10049
C10050
C10051
C10052
C10053
J6
H1, G6
A4
W4
M5, M1
B11, L12
V11
J12
B7, C6
L2
U6, V7
C10054
C10055
C10056
C10057
C10058
C10059
C10060
C10061
C10062
C10063
C10064
C10065
E6, A6
J4
W6, R6
J10
C10068
C10069
C10070
C10071
C10072
D5
T5
B5
V5
K1
G8
A10
W10
L10
C10073
C10074
C10075
C10076
C10077
C10078
C10079
C10080
C10081
C10082
M15
M13
C10083
C10084
C10067


U10002
SMR546_IFH1
SMR546_IFH3
SMR546_IFV2
SMR546_IFV4
SDM_RFFE4_CLK
SDM_RFFE4_DATA
QTM1_PON
QTM0_PON
MMW_XO_CLK
C10066
QLINK1_WMSS_RESET_N
QLINK1_CLK_P
QLINK1_CLK_M
QLINK1_EN
QLINK1_REQ
QLINK1_L0_P
QLINK1_L0_M
QLINK1_L1_P
QLINK1_L1_M
QLINK1_L2_P
QLINK1_L2_M
QLINK1_L3_P
QLINK1_L3_M
QLINK1_L4_P
QLINK1_L4_M
QLINK1_L5_P
QLINK1_L5_M
QLINK1_L6_P
QLINK1_L6_M
QLINK1_L7_P
QLINK1_L7_M
QLINK1_L8_P
QLINK1_L8_M
IFH1
IFH2
IFH3
IFH4
IFV1
IFV2
IFV3
IFV4
RFFE_CLK
RFFE_DATA
GRFC0
GRFC1
GRFC2
GRFC3
GRFC4
GRFC5
GRFC6
GRFC7
GRFC8
GRFC9
GRFC10
XO_IN
PAD_SEL
WMSS_RESETN
VDD_CX
VDD_MX
VDD_PHY
VDD_PX
VDD_PAD
VDDAH_1
VDDAH_6
VDDAL_1
VDDAL_16
VDDAM_1
VDDAM_9
GND

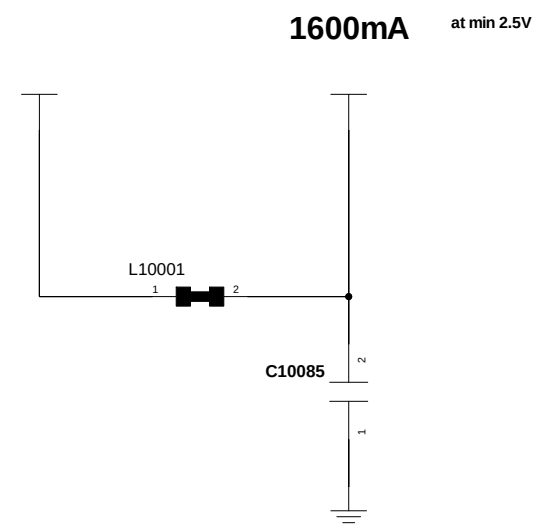
1600mA at min 2.5V


150mA
HEA10000
TP10002
QTM1_PON
SMR546_IFV2
SMR546_IFH1
C10092
R10002
C10089
C10087
CN10000

Need to check connector P/N and pin map

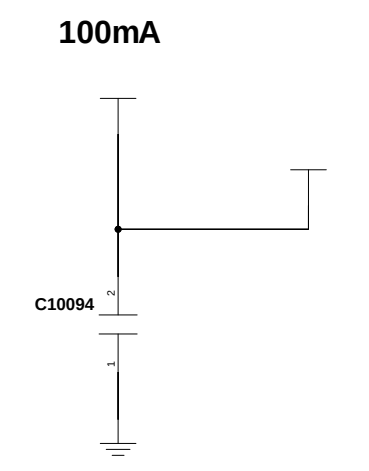
mmW_ANT_CONN (BACK)

1600mA at min 2.5V

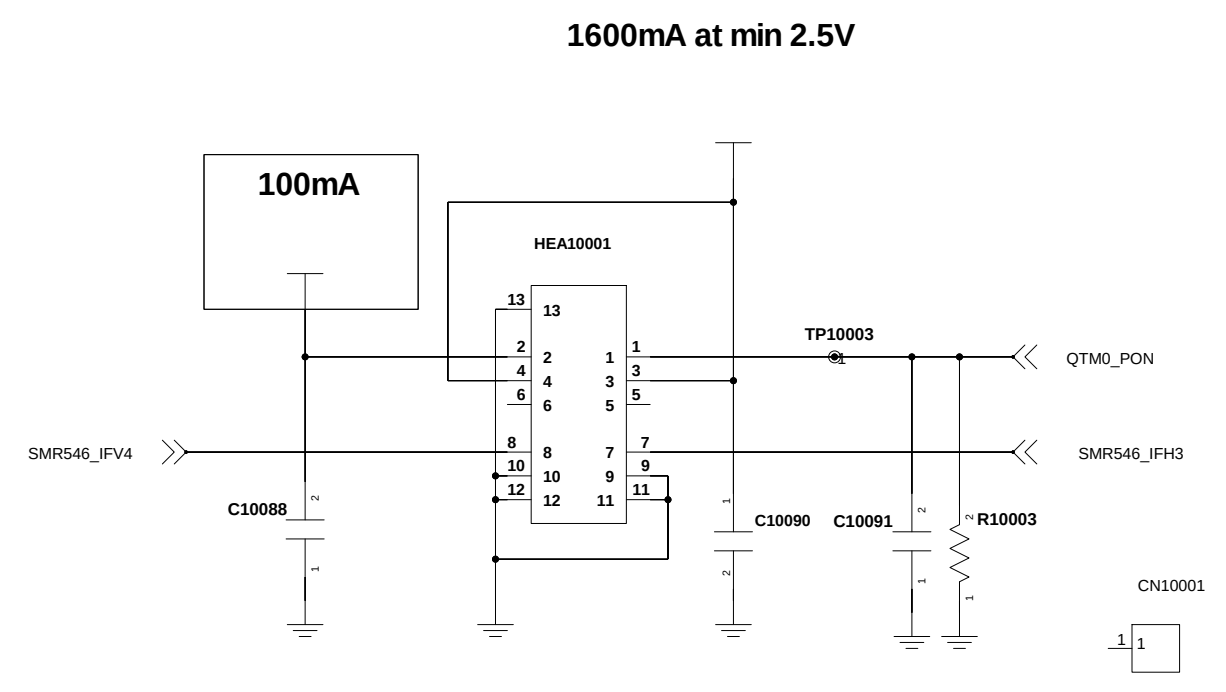

L10001
C10085

LOAD SWITCH

100mA



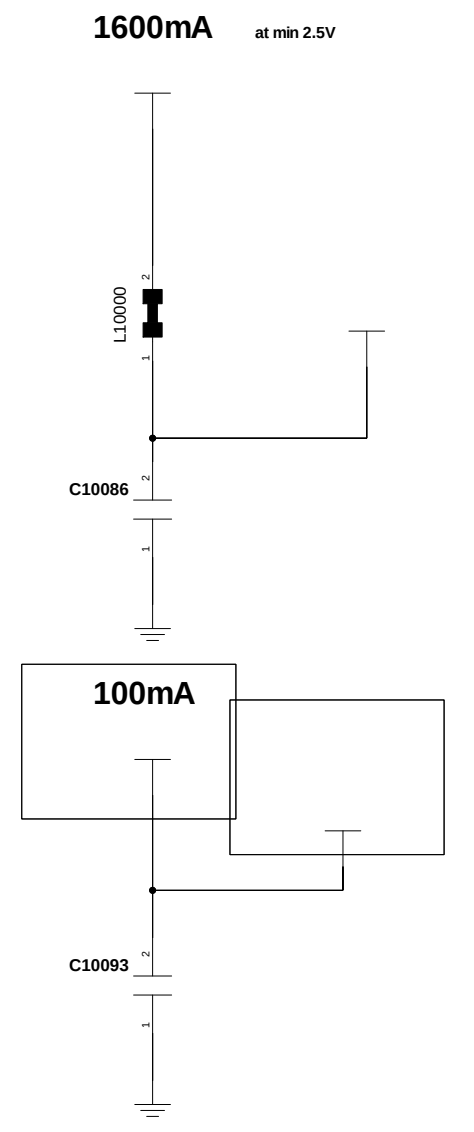
1600mA at min 2.5V


100mA
HEA10001
TP10003
QTM0_PON
SMR546_IFV4
SMR546_IFH3
C10088
C10090
C10091
R10003
CN10001

Need to check connector P/N and pin map

mmW_ANT_CONN (FRONT)

1600mA at min 2.5V


L10000
C10086
100mA
C10093

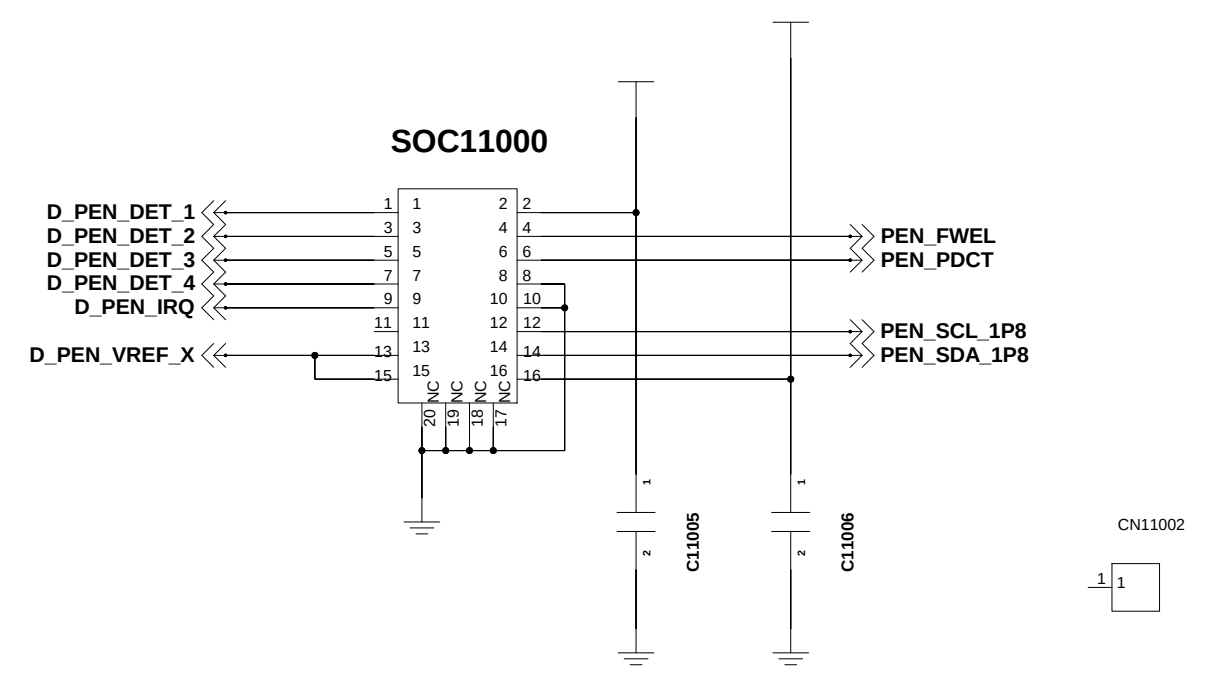
Radiation

TITLE	TabS7P	
SIZE A1	DWG NO PDM NUMBER	REV Rev
DRAWN BY User Name	SHEET 10 of 15	Date Drawing Date

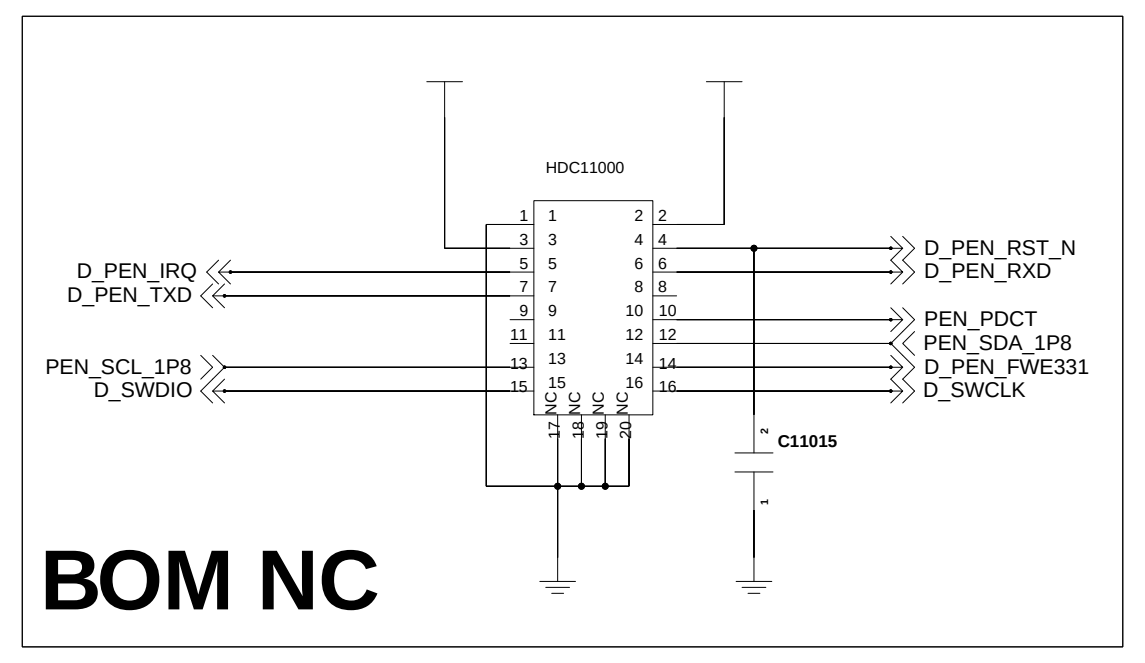
WACOM PCB (Ext.)

< SMD F/T >

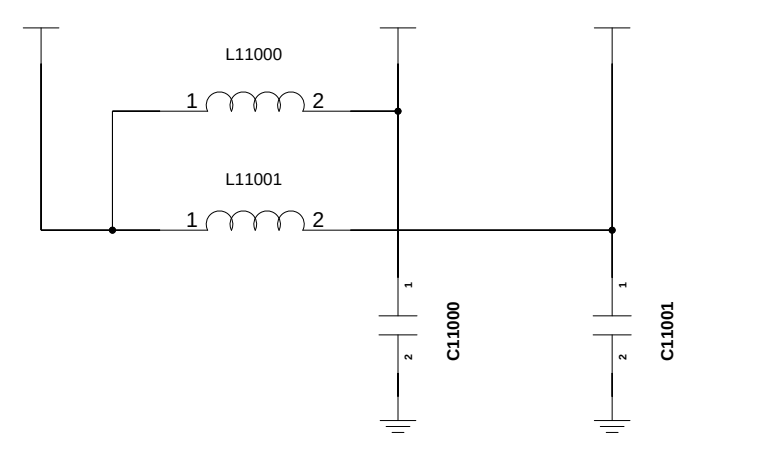
1. VREG_L13B_3P3
2. VREG_L15B_1P8
3. PEN_SCL_1P8
4. PEN_SDA_1P8
5. PEN_FWEL
6. PEN_PDCT
7. GND

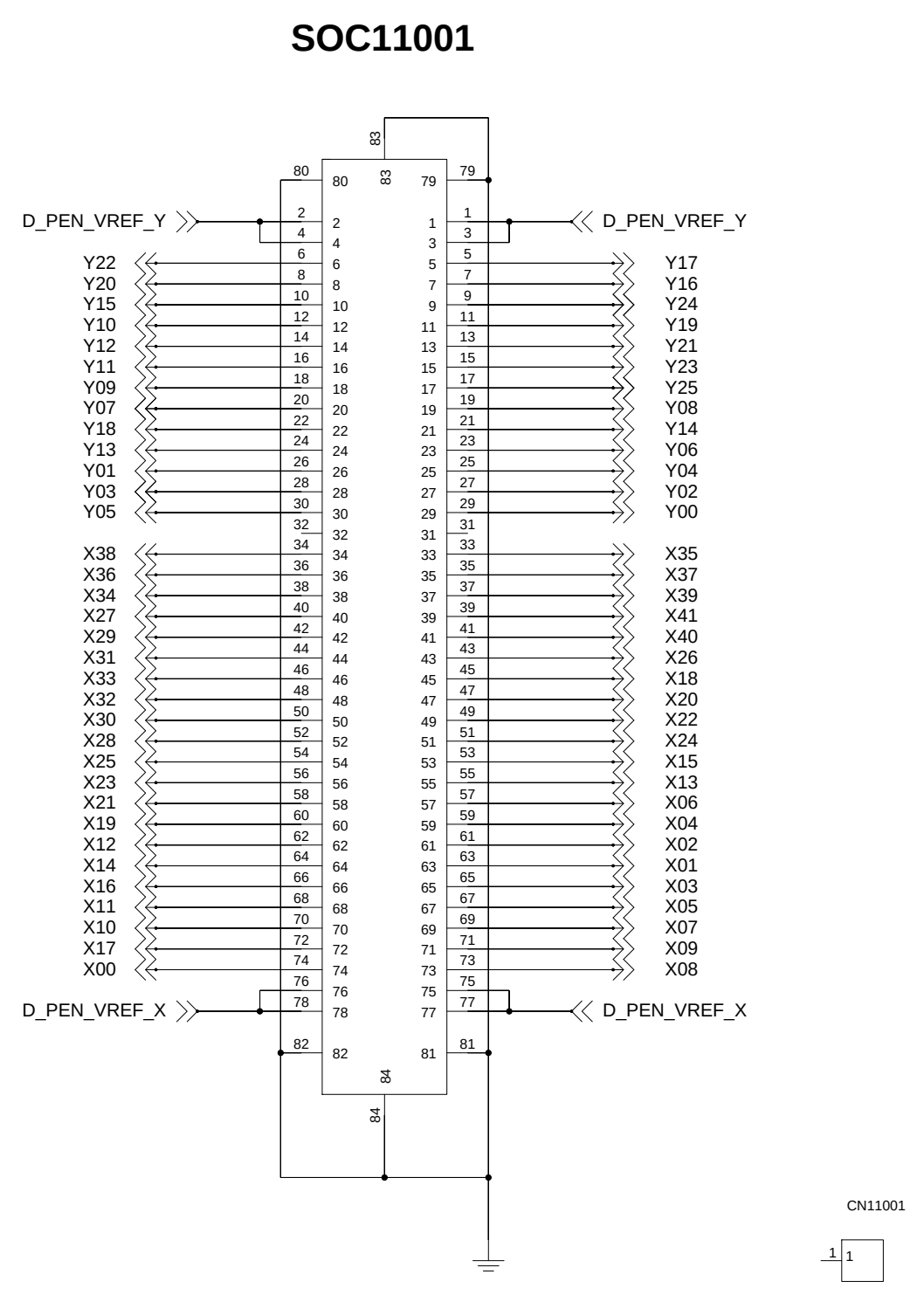

SOC11000
D_PEN_DET_1
D_PEN_DET_2
D_PEN_DET_3
D_PEN_DET_4
D_PEN_IRQ
D_PEN_VREF_X
PEN_FWEL
PEN_PDCT
PEN_SCL_1P8
PEN_SDA_1P8
C11005
C11006
CN11002

WACOM I/F CON.

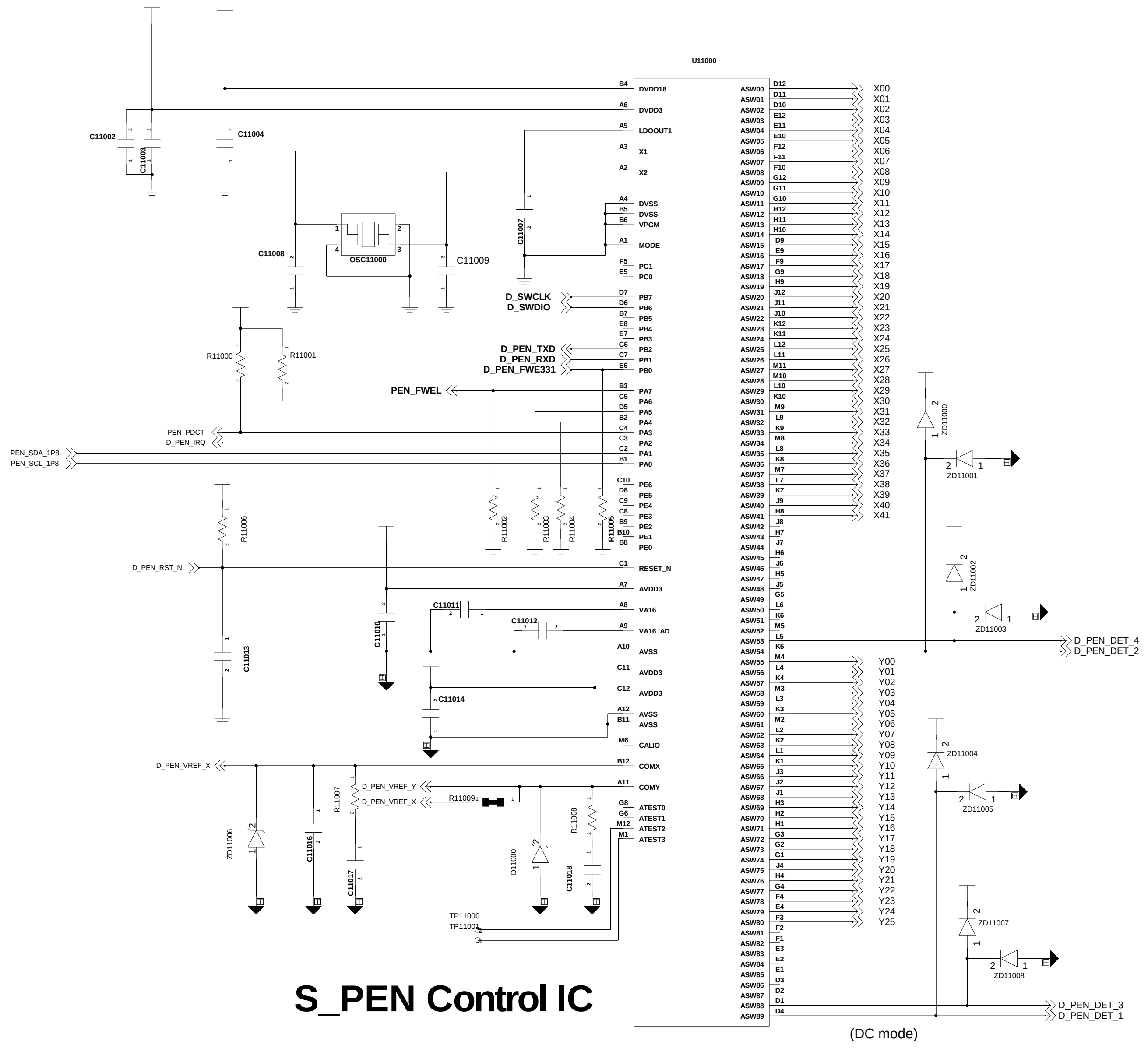

HDC11000
D_PEN_IRQ
D_PEN_TXD
PEN_SCL_1P8
D_SWDIO
D_PEN_RST_N
D_PEN_RXD
PEN_PDCT
PEN_SDA_1P8
D_PEN_FWE331
D_SWCLK
C11015
BOM NC

DEBUGER


L11000
L11001
C11000
C11001

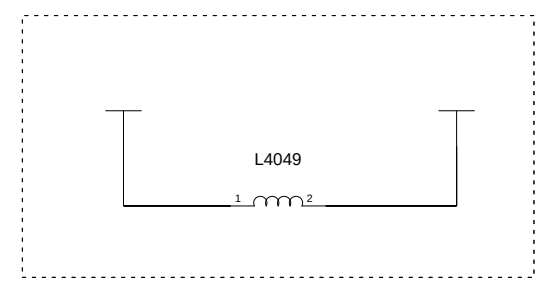

SOC11001
D_PEN_VREF_Y
Y22 Y20 Y15 Y10 Y12 Y11 Y09 Y07 Y18 Y13 Y01 Y03 Y05
X38 X36 X34 X27 X29 X31 X33 X32 X30 X28 X25 X23 X21 X19 X12 X14 X16 X11 X10 X17 X00
Y17 Y16 Y24 Y19 Y21 Y23 Y25 Y08 Y14 Y06 Y04 Y02 Y00
X35 X37 X39 X41 X40 X26 X18 X20 X22 X24 X15 X13 X06 X04 X02 X01 X03 X05 X07 X09 X08
D_PEN_VREF_X
CN11001

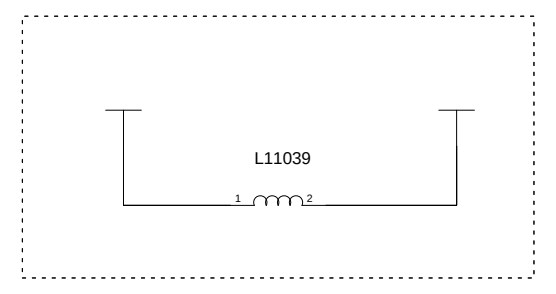
78pin S-PEN Connector

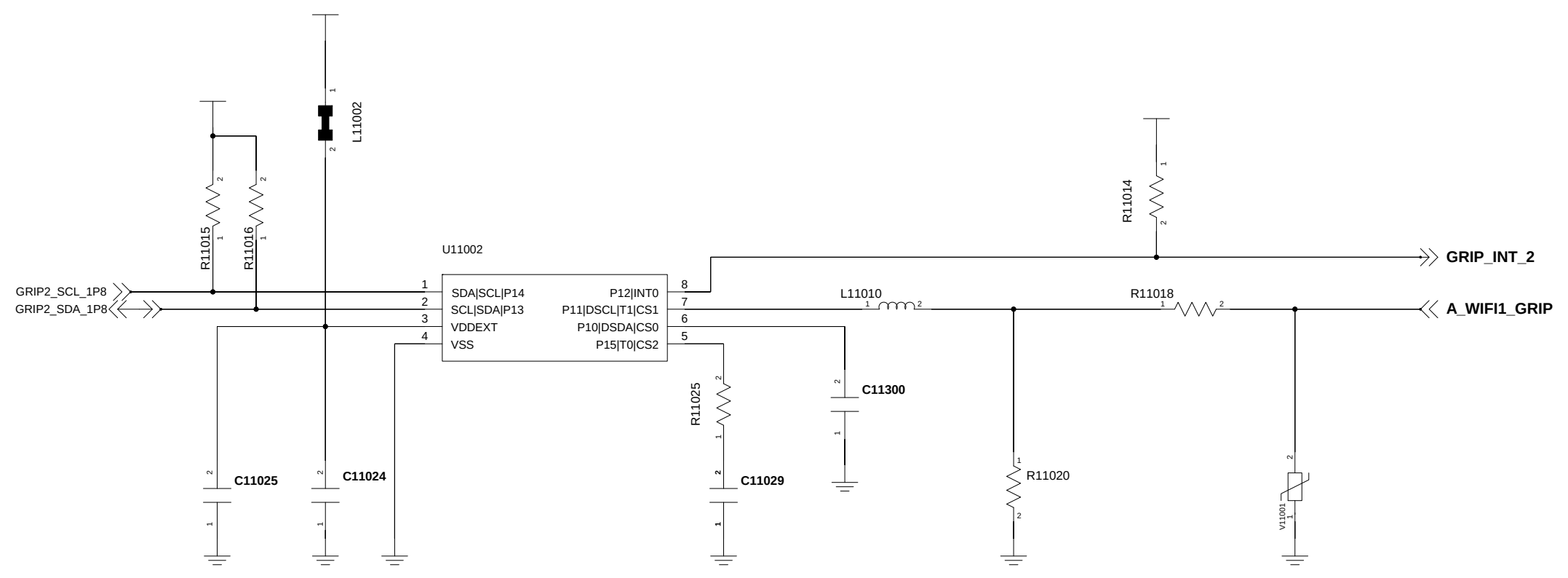

U11000
C11002
C11003
C11004
C11008
OSC11000
C11009
C11007
D_SWCLK
D_SWDIO
R11000
R11001
D_PEN_TXD
D_PEN_RXD
D_PEN_FWE331
PEN_FWEL
PEN_PDCT
D_PEN_IRQ
PEN_SDA_1P8
PEN_SCL_1P8
R11002
R11003
R11004
R11005
R11006
D_PEN_RST_N
C11010
C11011
C11012
C11013
C11014
D_PEN_VREF_X
D_PEN_VREF_Y
R11007
R11009
R11008
ZD11006
C11016
C11017
D11000
C11018
TP11000
TP11001
DVDD18 DVDD3 LDOOUT1 X1 X2 DVSS DVSS VPGM MODE PC1 PC0 PB7 PB6 PB5 PB4 PB3 PB2 PB1 PB0 PA7 PA6 PA5 PA4 PA3 PA2 PA1 PA0 PE6 PE5 PE4 PE3 PE2 PE1 PE0 RESET_N AVDD3 VA16 VA16_AD AVSS AVDD3 AVDD3 AVSS AVSS CALIO COMX COMY ATEST0 ATEST1 ATEST2 ATEST3
X00 X01 X02 X03 X04 X05 X06 X07 X08 X09 X10 X11 X12 X13 X14 X15 X16 X17 X18 X19 X20 X21 X22 X23 X24 X25 X26 X27 X28 X29 X30 X31 X32 X33 X34 X35 X36 X37 X38 X39 X40 X41
Y00 Y01 Y02 Y03 Y04 Y05 Y06 Y07 Y08 Y09 Y10 Y11 Y12 Y13 Y14 Y15 Y16 Y17 Y18 Y19 Y20 Y21 Y22 Y23 Y24 Y25
ZD11000
ZD11001
ZD11002
ZD11003
ZD11004
ZD11005
ZD11007
ZD11008
D_PEN_DET_4
D_PEN_DET_2
D_PEN_DET_3
D_PEN_DET_1
(DC mode)

S_PEN Control IC

ANT SUB PCB (Ext.)



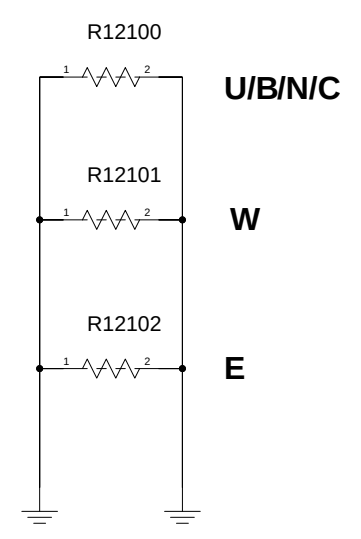



U11002
GRIP2_SCL_1P8
GRIP2_SDA_1P8
SDA|SCL|P14
SCL|SDA|P13
VDDEXT
VSS
P12|INT0
P11|DSCL|T1|CS1
P10|DSDA|CS0
P15|T0|CS2
R11015
R11016
L11002
C11025
C11024
R11025
C11029
C11300
L11010
R11014
R11018
R11020
GRIP_INT_2
A_WIFI1_GRIP

WIFI_GRIP_SENSOR

ANT PCB REGION

ANT PCB	SHORT
X818U (U), X816B/N/C (B/N/C)	R12100
X810 (W)	R12101
ETC (E)	R12102

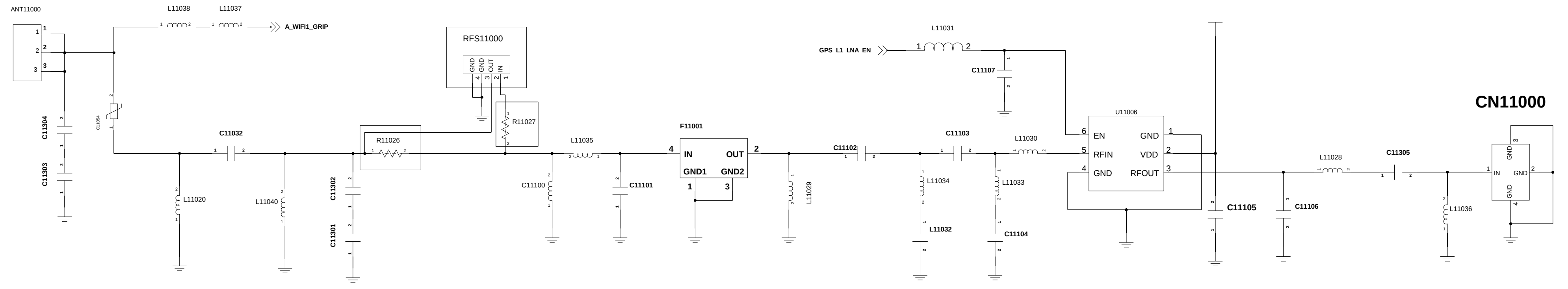

R12100
U/B/N/C
R12101
W
R12102
E

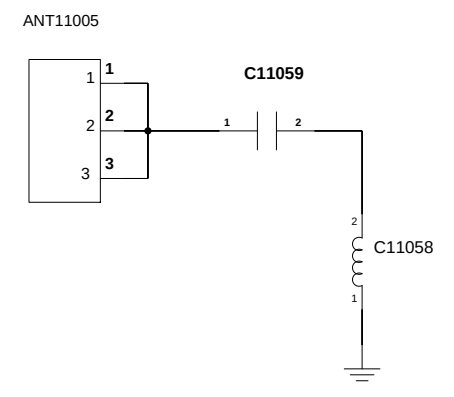
< SMD F/T >

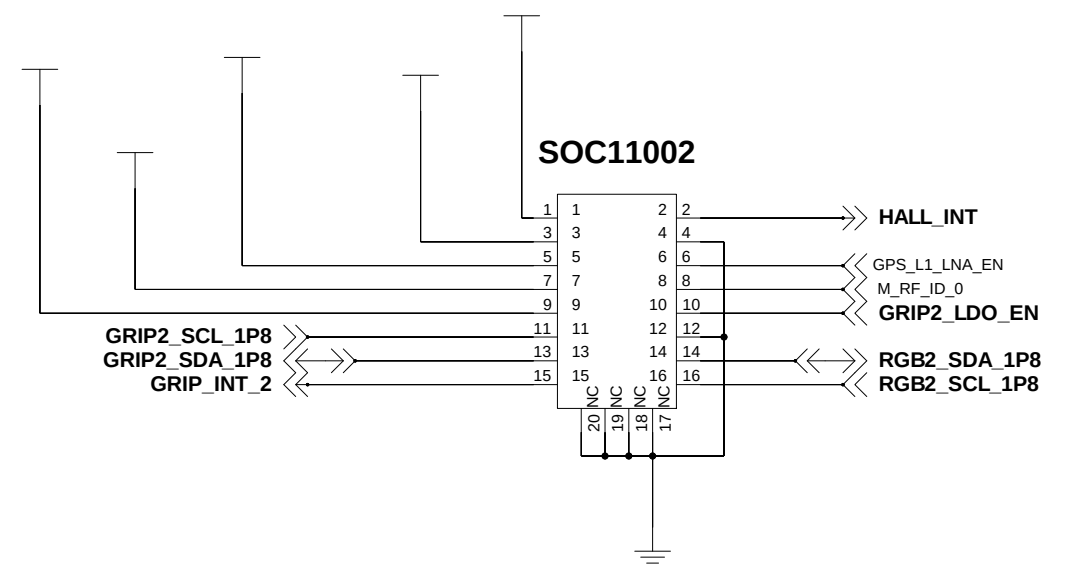
1. VDD_GRIP2_3P0_LDO
2. GRIP2_SCL_1P8
3. GRIP2_SDA_1P8
4. GRIP_INT_2
5. SDM_RFFE1_CLK
6. SDM_RFFE1_DATA
7. VREG_L15B_1P8
8. HALL_INT
9. VREG_L16B_3P0
10. RGB2_SCL_1P8
11. RGB2_SDA_1P8
12. GND

GPS ANT (GPS_L1)

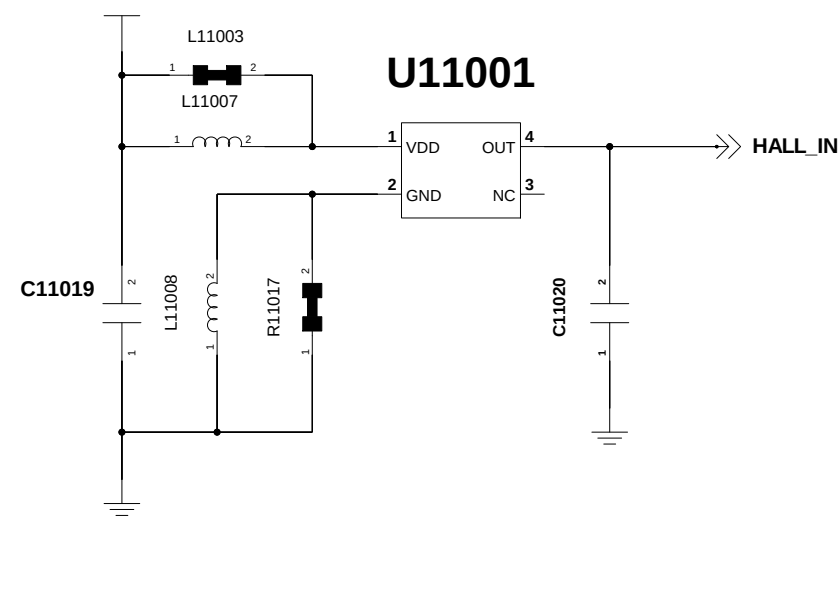
GPS_L1


ANT11000
L11038
L11037
A_WIFI1_GRIP
C11304
C11303
C11054
C11032
L11020
L11040
C11302
C11301
R11026
RFS11000
GND GND OUT IN
R11027
C11100
L11035
C11101
F11001
IN OUT GND1 GND2
L11029
GPS_L1_LNA_EN
L11031
C11107
C11102
C11103
L11034
L11032
L11030
L11033
C11104
U11006
EN RFIN GND GND VDD RFOUT
C11105
C11106
L11028
C11305
L11036
CN11000
IN GND GND GND

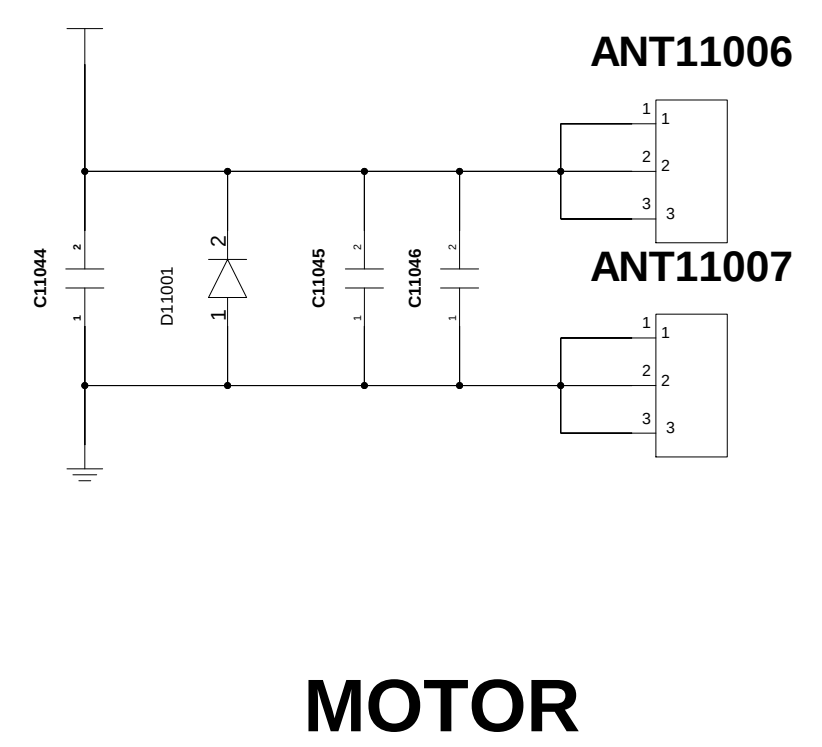

ANT11005
C11059
C11058


SOC11002
GRIP2_SCL_1P8
GRIP2_SDA_1P8
GRIP_INT_2
HALL_INT
GPS_L1_LNA_EN
M_RF_ID_0
GRIP2_LDO_EN
RGB2_SDA_1P8
RGB2_SCL_1P8
CN11003

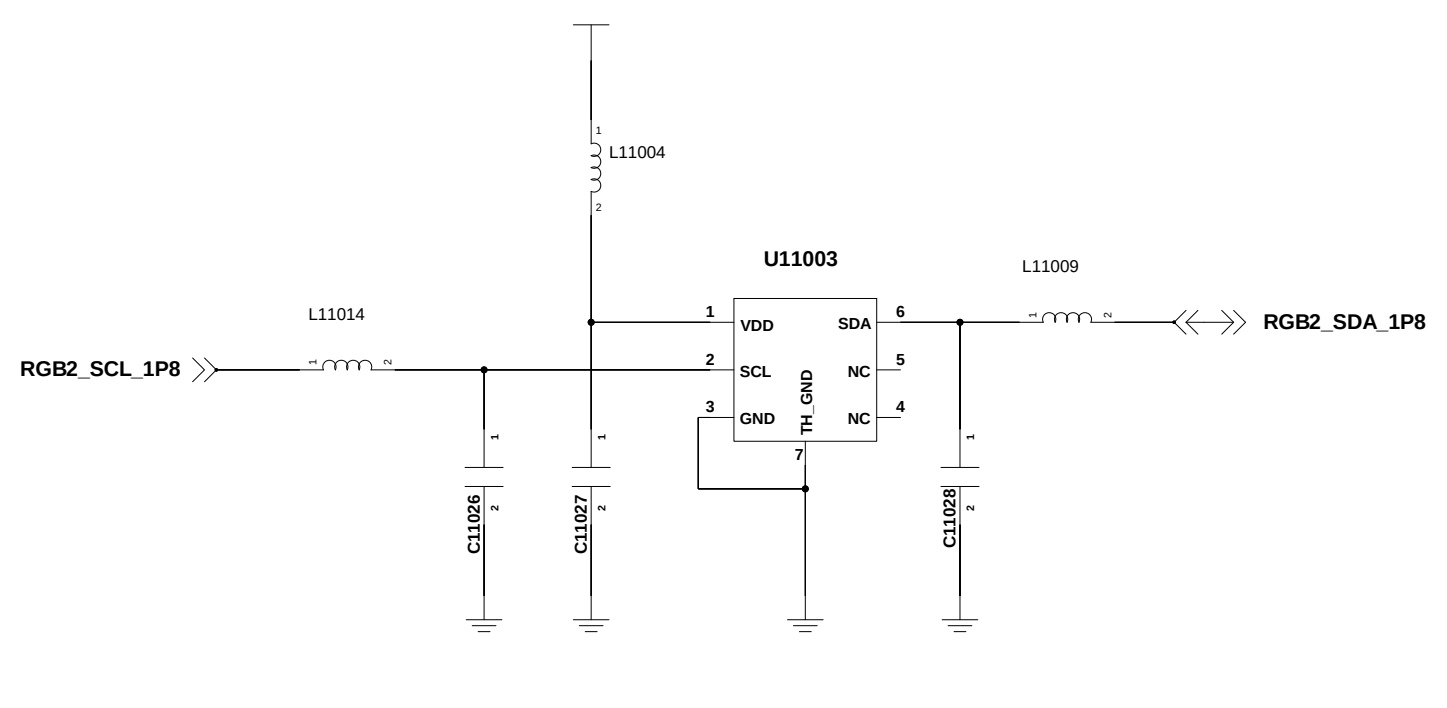
SUB CONN (ANT SUB <-> MAIN)


L11003
L11007
U11001
VDD OUT GND NC
HALL_INT
C11019
L11008
R11017
C11020

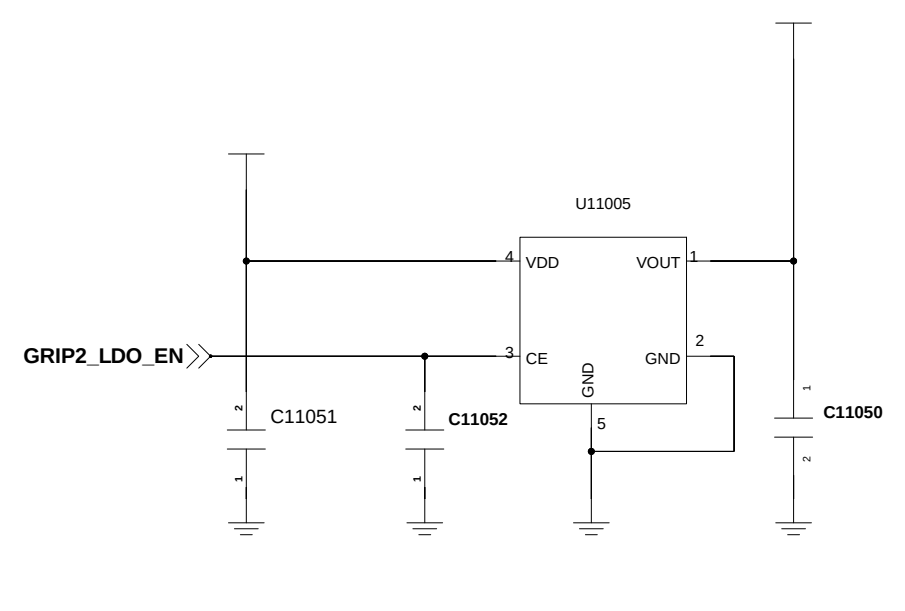
COVER_DET HALL_IC


ANT11006
ANT11007
C11044
D11001
C11045
C11046

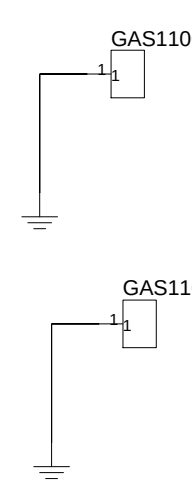
MOTOR


L11004
U11003
VDD SCL GND SDA NC NC TH_GND
RGB2_SCL_1P8
L11014
C11026
C11027
C11028
L11009
RGB2_SDA_1P8

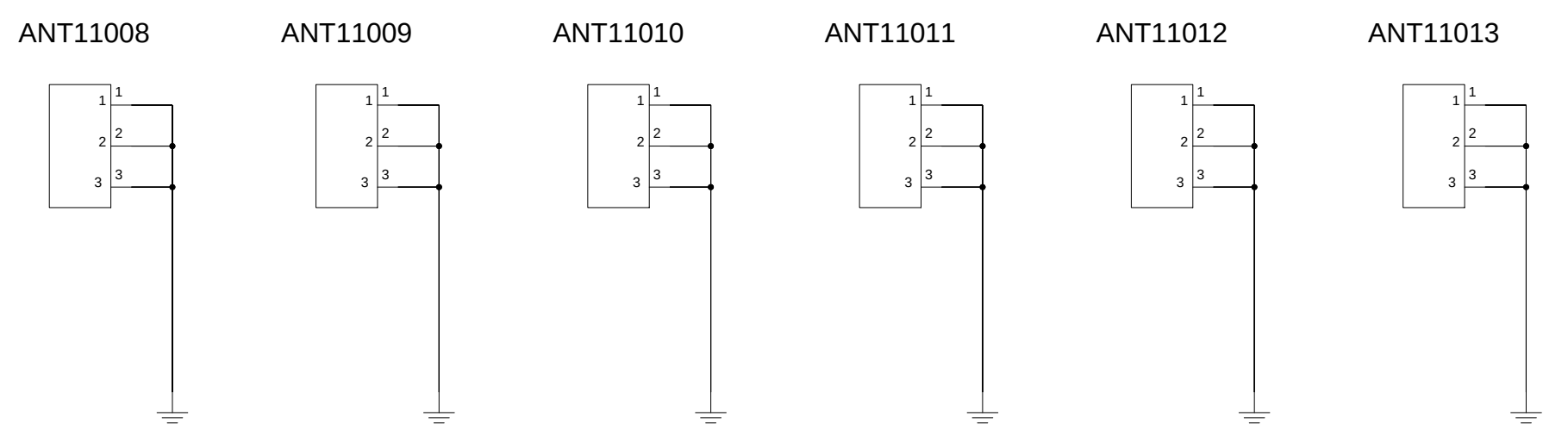
RGB Sensor 2


U11005
VDD VOUT CE GND GND
GRIP2_LDO_EN
C11051
C11052
C11050

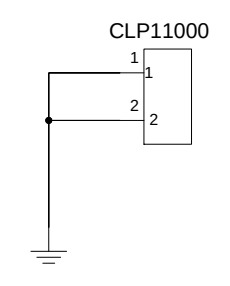
GRIP2 LDO


GAS11000
GAS11001

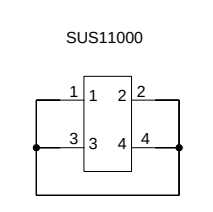
FRONT CONTACT (TOP)


ANT11008
ANT11009
ANT11010
ANT11011
ANT11012
ANT11013

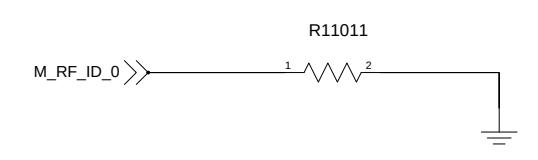
REAR CONTACT (BOT)



Cable CLIP

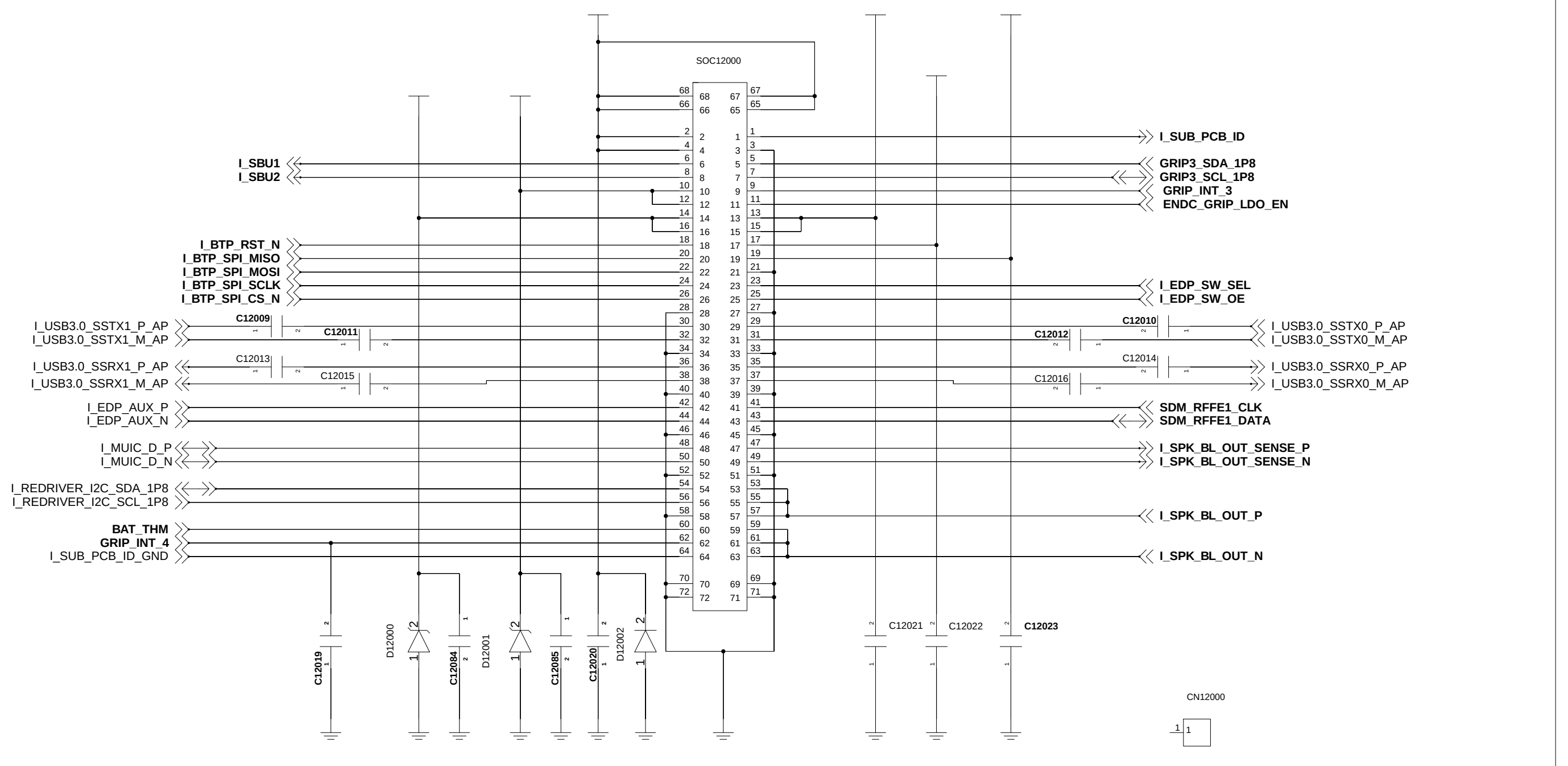


SCREW SUS (0.5mm)

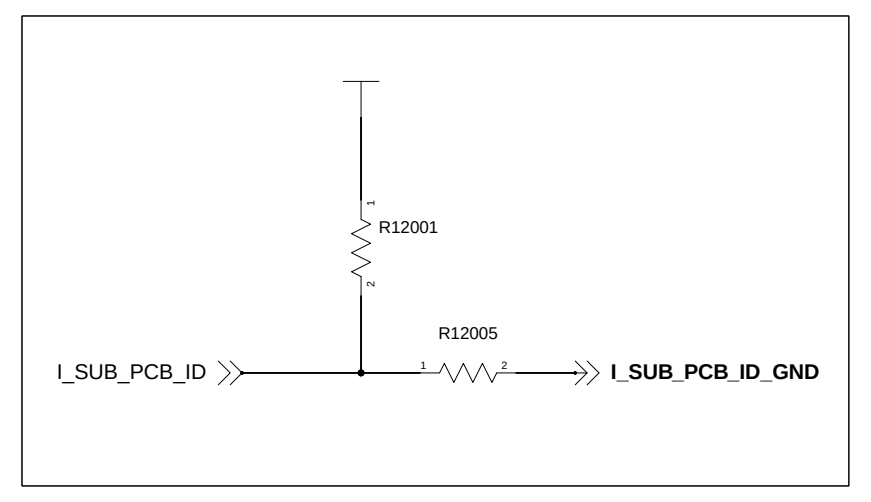

M_RF_ID_0
R11011

TITLE TabS7P
SIZE A1
DRAWN BY User Name
SHEET 11 of 15

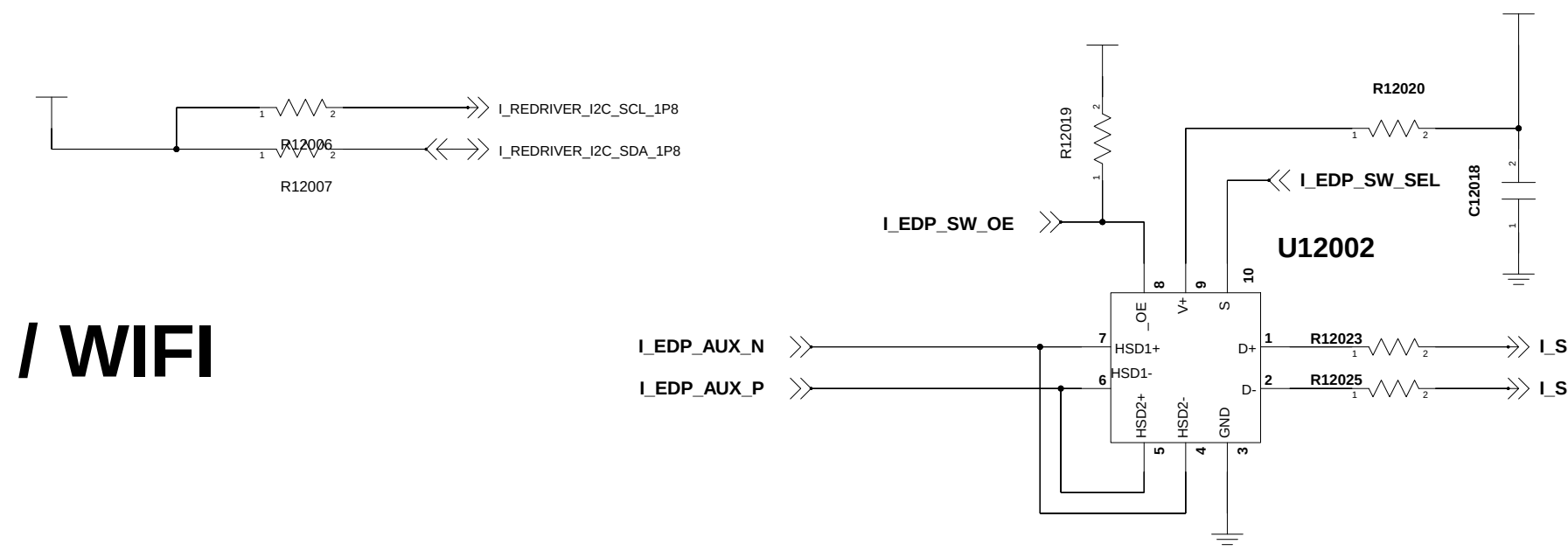
USB PCB (Ext.)


SOC12000
I_SUB_PCB_ID
I_SBU1
I_SBU2
GRIP3_SDA_1P8
GRIP3_SCL_1P8
GRIP_INT_3
ENDC_GRIP_LDO_EN
I_BTP_RST_N
I_BTP_SPI_MISO
I_BTP_SPI_MOSI
I_BTP_SPI_SCLK
I_BTP_SPI_CS_N
I_EDP_SW_SEL
I_EDP_SW_OE
I_USB3.0_SSTX1_P_AP
I_USB3.0_SSTX1_M_AP
I_USB3.0_SSRX1_P_AP
I_USB3.0_SSRX1_M_AP
I_USB3.0_SSTX0_P_AP
I_USB3.0_SSTX0_M_AP
I_USB3.0_SSRX0_P_AP
I_USB3.0_SSRX0_M_AP
C12009
C12011
C12013
C12015
C12010
C12012
C12014
C12016
I_EDP_AUX_P
I_EDP_AUX_N
SDM_RFFE1_CLK
SDM_RFFE1_DATA
I_MUIC_D_P
I_MUIC_D_N
I_SPK_BL_OUT_SENSE_P
I_SPK_BL_OUT_SENSE_N
I_REDRIVER_I2C_SDA_1P8
I_REDRIVER_I2C_SCL_1P8
I_SPK_BL_OUT_P
BAT_THM
GRIP_INT_4
I_SUB_PCB_ID_GND
I_SPK_BL_OUT_N
C12019
D12000
C12084
D12001
C12085
C12020
D12002
C12021
C12022
C12023
CN12000

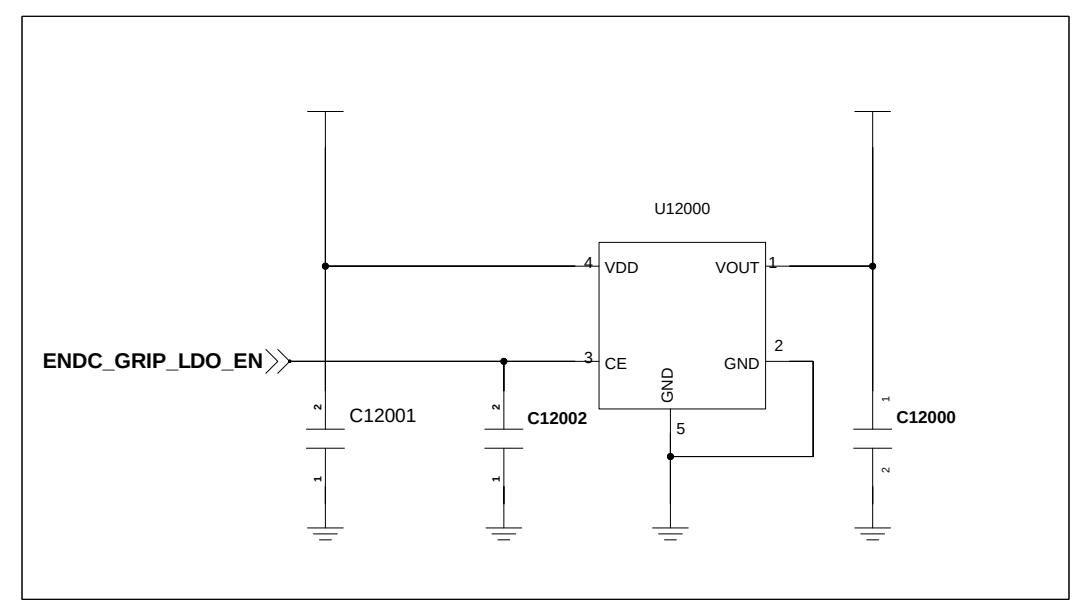
I/F CONN (USB)


R12001
R12005
I_SUB_PCB_ID
I_SUB_PCB_ID_GND

mmW,Sub6 / WIFI

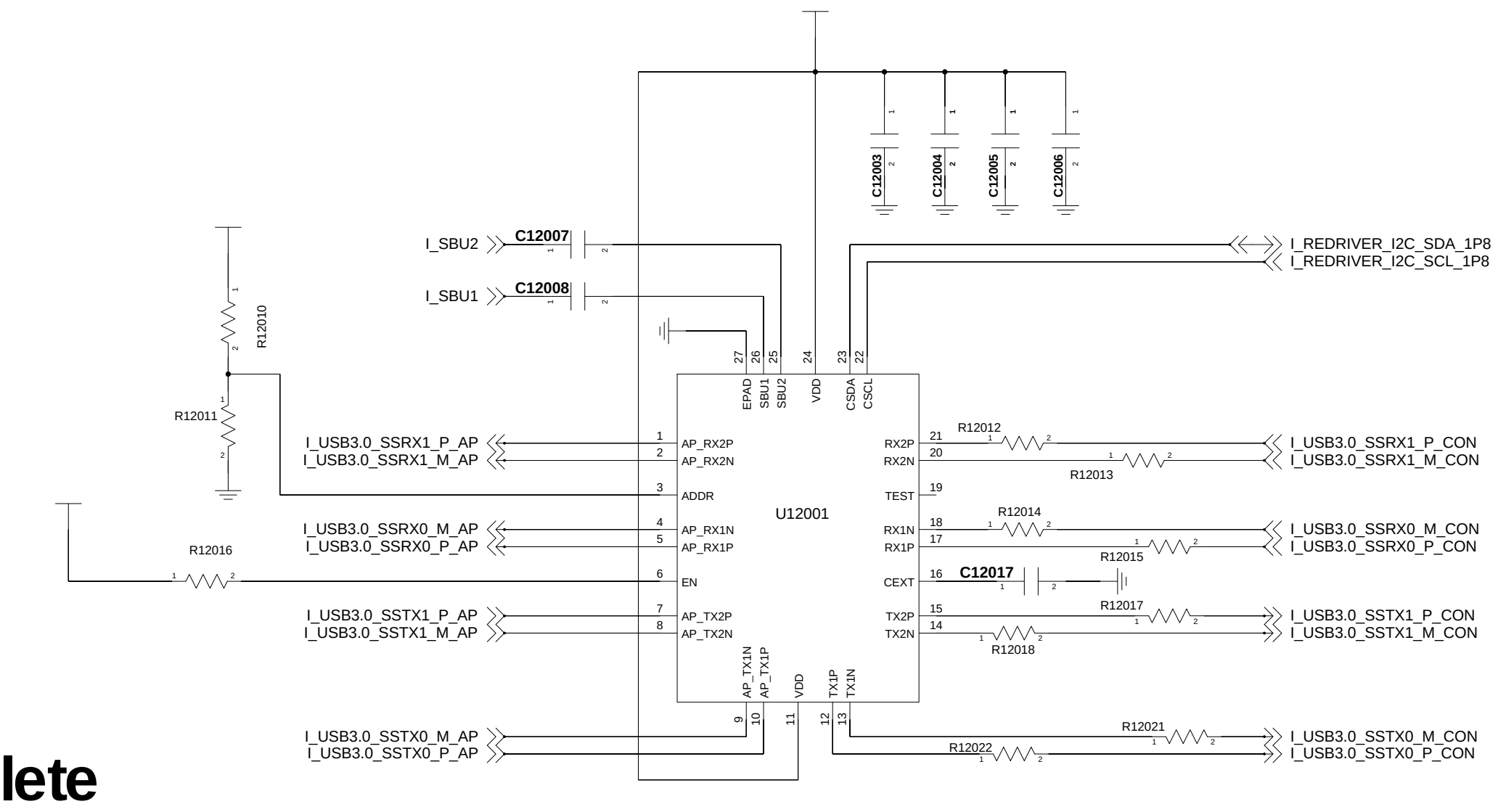

R12006
R12007
I_REDRIVER_I2C_SCL_1P8
I_REDRIVER_I2C_SDA_1P8
R12019
R12020
C12018
I_EDP_SW_OE
I_EDP_SW_SEL
U12002
I_EDP_AUX_N
I_EDP_AUX_P
HSD1+
HSD1-
HSD2+
HSD2-
OE
V+
S
D+
D-
GND
R12023
R12025
I_SBU1
I_SBU2

DP Switch

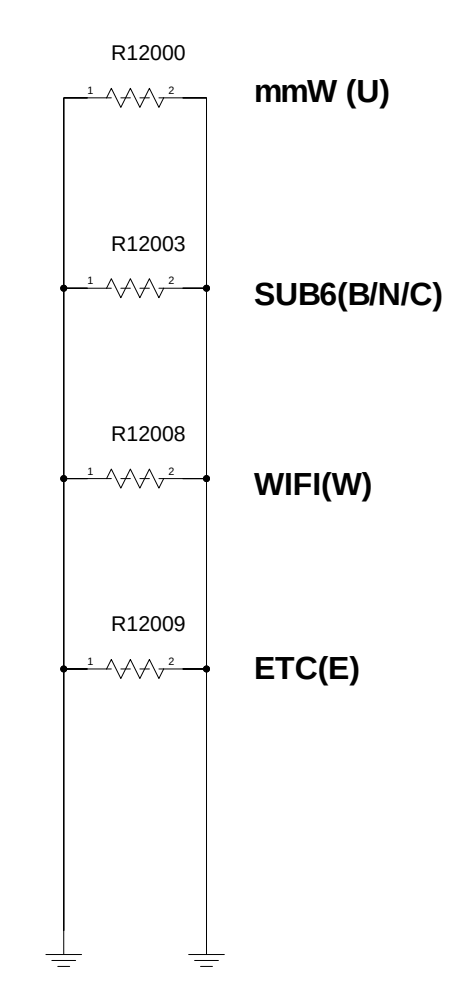

U12000
VDD
VOUT
CE
GND
ENDC_GRIP_LDO_EN
C12001
C12002
C12000

WIFI delete

GRIP3 LDO


C12003
C12004
C12005
C12006
I_SBU2
C12007
I_SBU1
C12008
I_REDRIVER_I2C_SDA_1P8
I_REDRIVER_I2C_SCL_1P8
R12010
R12011
R12016
U12001
EPAD
SBU1
SBU2
VDD
CSDA
CSCL
AP_RX2P
AP_RX2N
ADDR
AP_RX1N
AP_RX1P
EN
AP_TX2P
AP_TX2N
AP_TX1N
AP_TX1P
TX1P
TX1N
RX2P
RX2N
TEST
RX1N
RX1P
CEXT
TX2P
TX2N
I_USB3.0_SSRX1_P_AP
I_USB3.0_SSRX1_M_AP
I_USB3.0_SSRX0_M_AP
I_USB3.0_SSRX0_P_AP
I_USB3.0_SSTX1_P_AP
I_USB3.0_SSTX1_M_AP
I_USB3.0_SSTX0_M_AP
I_USB3.0_SSTX0_P_AP
R12012
R12013
R12014
R12015
C12017
R12017
R12018
R12021
R12022
I_USB3.0_SSRX1_P_CON
I_USB3.0_SSRX1_M_CON
I_USB3.0_SSRX0_M_CON
I_USB3.0_SSRX0_P_CON
I_USB3.0_SSTX1_P_CON
I_USB3.0_SSTX1_M_CON
I_USB3.0_SSTX0_M_CON
I_USB3.0_SSTX0_P_CON

REDRIVER & DP

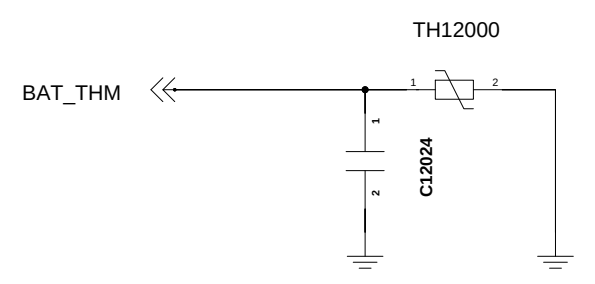

R12000
mmW (U)
R12003
SUB6(B/N/C)
R12008
WIFI(W)
R12009
ETC(E)

USB PCB REGION

USB PCB	SHORT
mmW(U)	R12000
SUB6(B/N/C)	R12003
WIFI(W)	R12008
ETC(E)	R12009

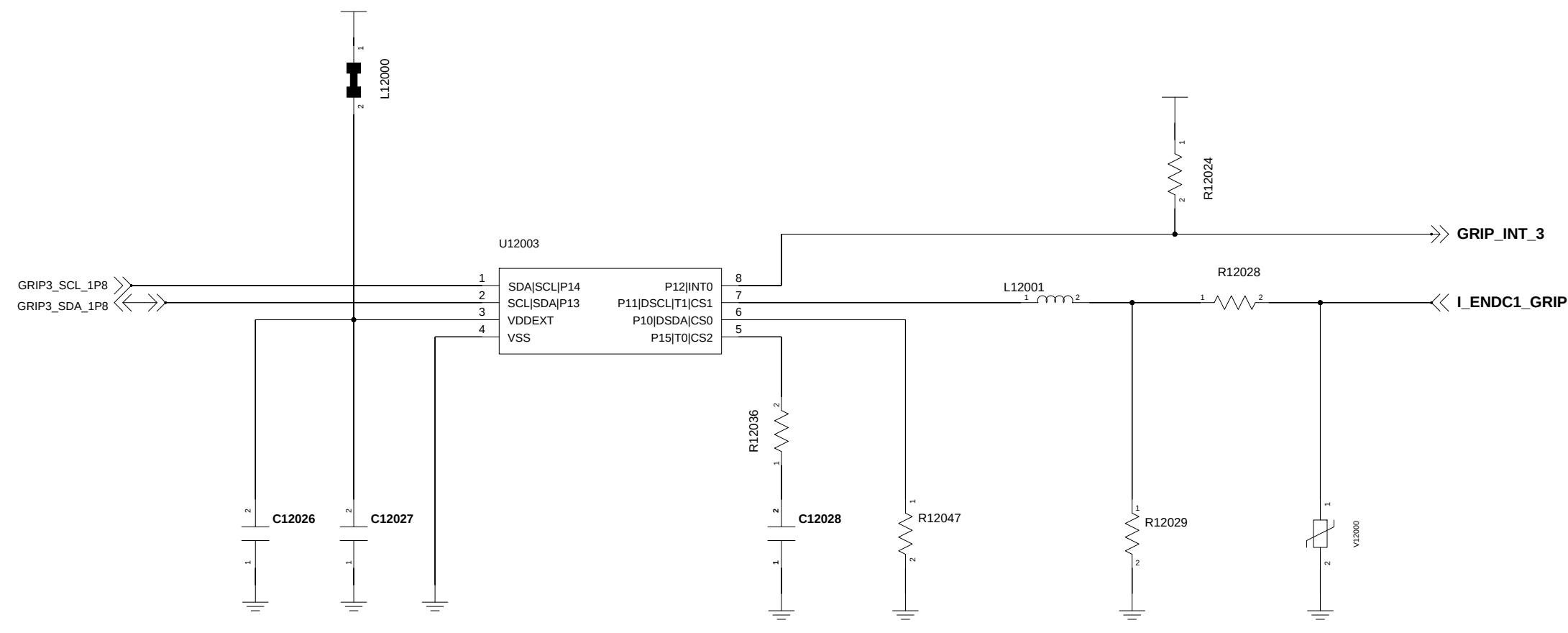
< USB PCB - SMD F/T >

1. VREG_L15B_1P8
2. BAT_THM
3. SDM_RFFE1_CLK
4. SDM_RFFE1_DATA
5. GND
6. VREG_BOB1_3P3
7. GRIP3_SCL_1P8
8. GRIP3_SDA_1P8
9. ENDC_GRIP_LDO_EN

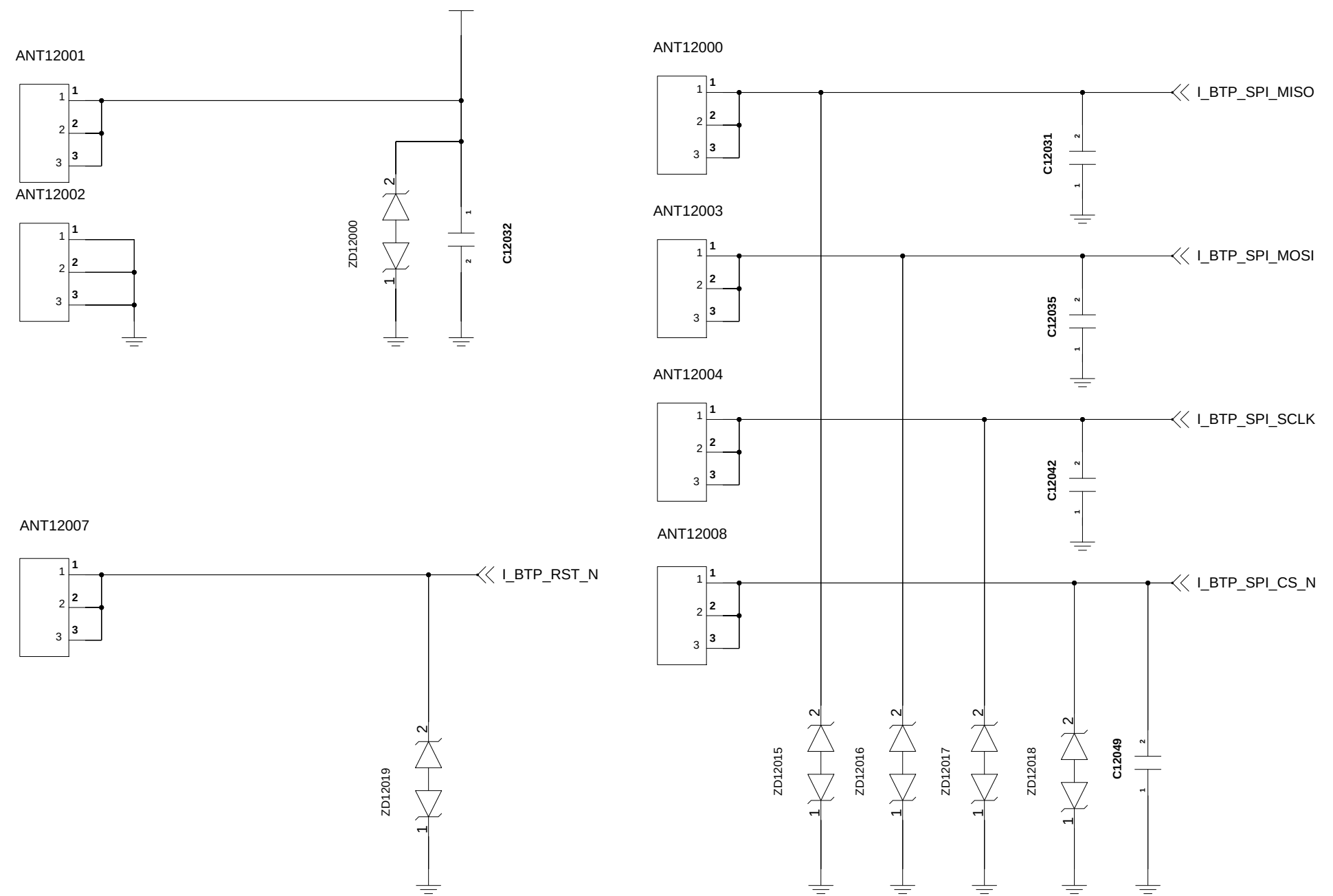

TH12000
BAT_THM
C12024

BAT THM

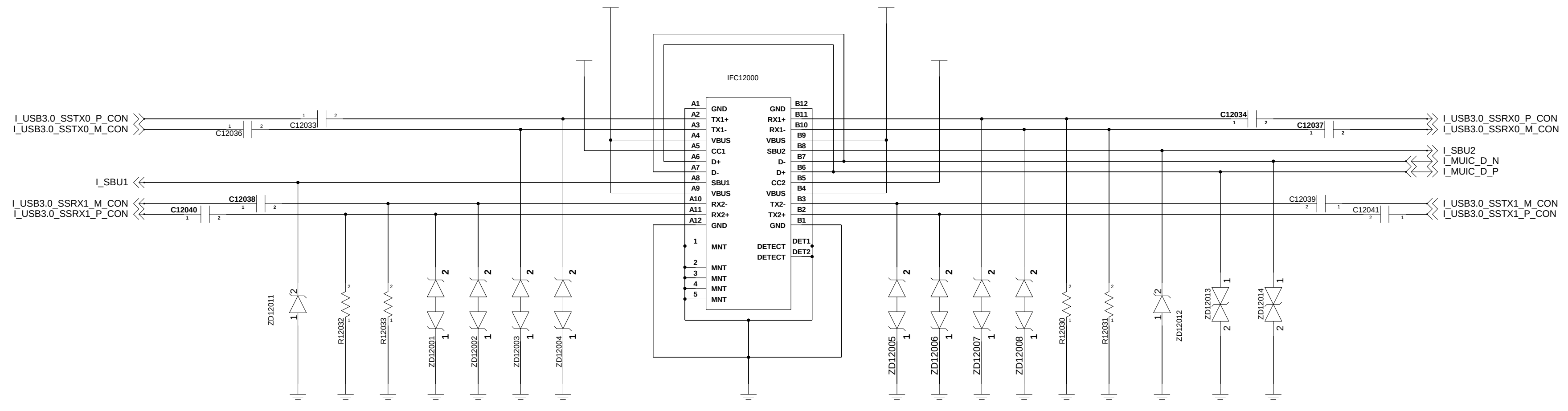
WIFI : Delete


L12000
R12024
U12003
GRIP3_SCL_1P8
GRIP3_SDA_1P8
SDA|SCL|P14
SCL|SDA|P13
VDDEXT
VSS
P12|INT0
P11|DSCL|T1|CS1
P10|DSDA|CS0
P15|T0|CS2
GRIP_INT_3
L12001
R12028
I_ENDC1_GRIP
R12036
C12026
C12027
C12028
R12047
R12029
V12000

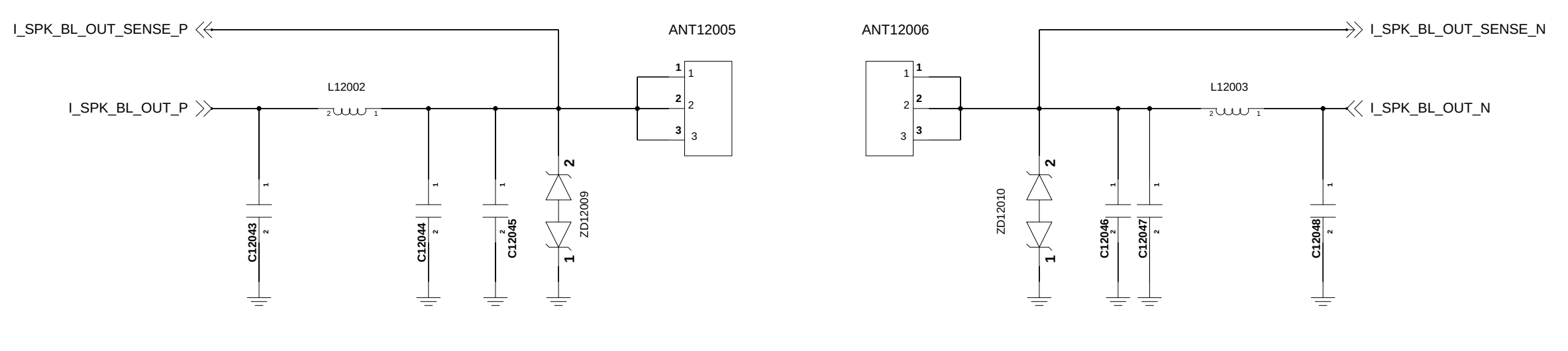
ENDC_GRIP_SENSOR


ANT12001
ANT12002
ZD12000
C12032
ANT12000
ANT12003
ANT12004
ANT12008
I_BTP_SPI_MISO
I_BTP_SPI_MOSI
I_BTP_SPI_SCLK
I_BTP_SPI_CS_N
C12031
C12035
C12042
ANT12007
I_BTP_RST_N
ZD12019
ZD12015
ZD12016
ZD12017
ZD12018
C12049

FOD CON.

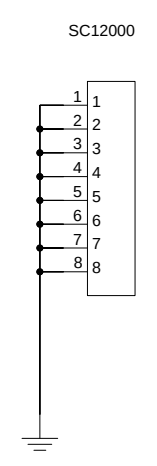

IFC12000
I_USB3.0_SSTX0_P_CON
I_USB3.0_SSTX0_M_CON
C12036
C12033
I_SBU1
I_USB3.0_SSRX1_M_CON
I_USB3.0_SSRX1_P_CON
C12038
C12040
GND
TX1+
TX1-
VBUS
CC1
D+
D-
SBU1
RX2-
RX2+
MNT
DETECT
RX1+
RX1-
SBU2
CC2
TX2-
TX2+
DET1
DET2
C12034
C12037
I_USB3.0_SSRX0_P_CON
I_USB3.0_SSRX0_M_CON
I_SBU2
I_MUIC_D_N
I_MUIC_D_P
C12039
C12041
I_USB3.0_SSTX1_M_CON
I_USB3.0_SSTX1_P_CON
ZD12011
R12032
R12033
ZD12001
ZD12002
ZD12003
ZD12004
ZD12005
ZD12006
ZD12007
ZD12008
R12030
R12031
ZD12012
ZD12013
ZD12014

Type-C Conn.


I_SPK_BL_OUT_SENSE_P
I_SPK_BL_OUT_P
L12002
ANT12005
ANT12006
I_SPK_BL_OUT_SENSE_N
I_SPK_BL_OUT_N
L12003
C12043
C12044
C12045
ZD12009
ZD12010
C12046
C12047
C12048

SPK BL CON.

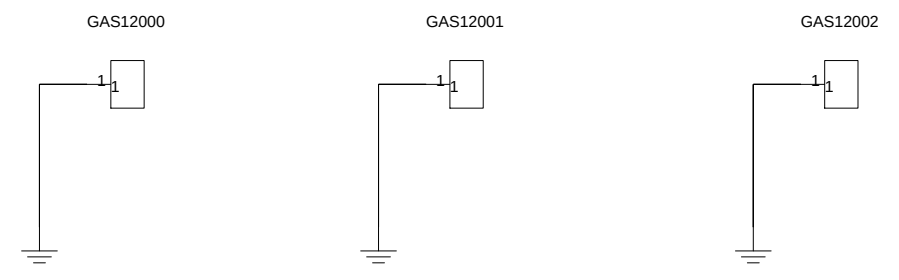
SHIELD CAN



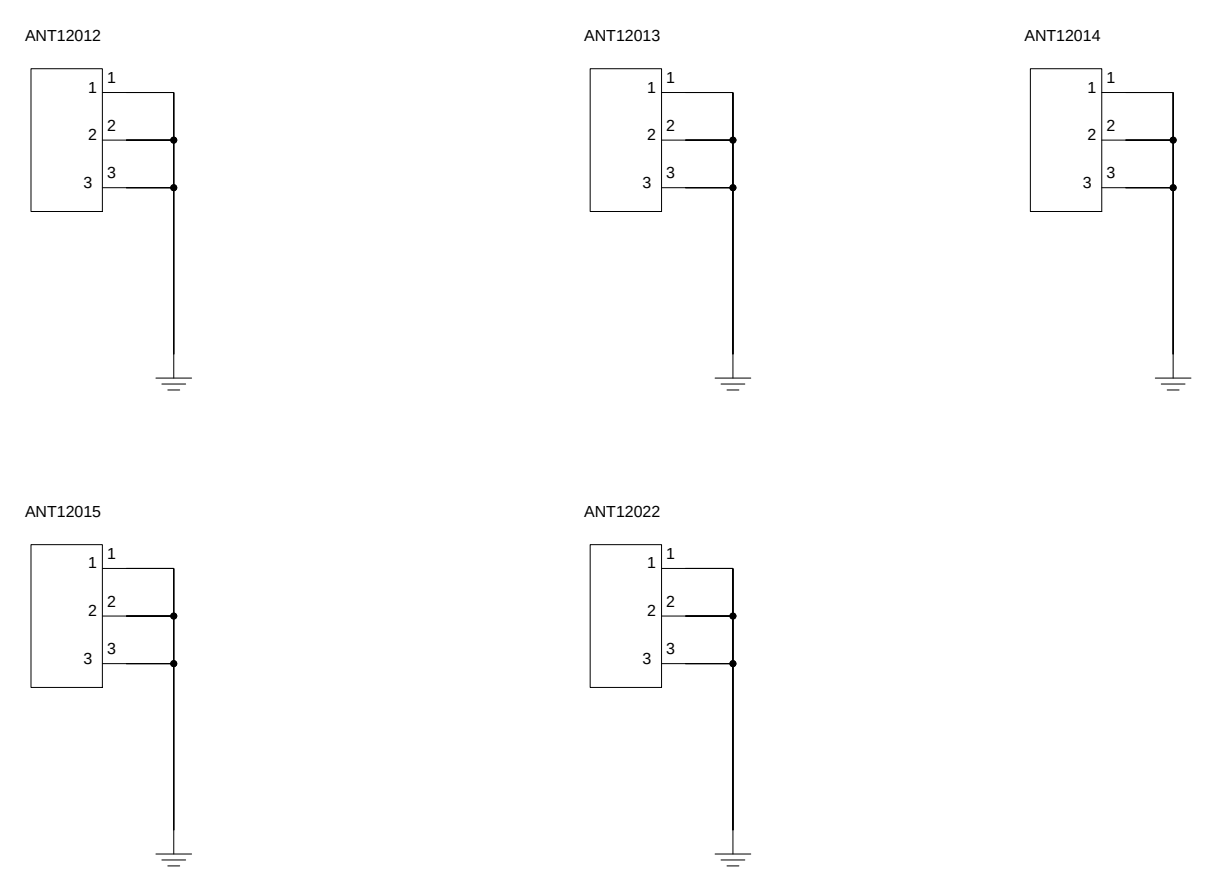
Re-Driver

< BOTTOM S/C >

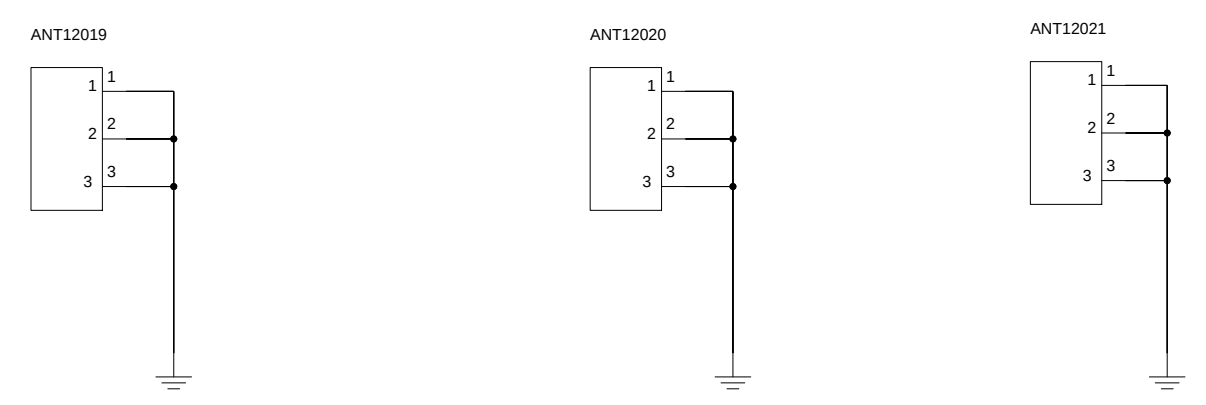
BOTHHOLE/CONTACT


GAS12000
GAS12001
GAS12002

Front CONTACT (JSM 1.3T)

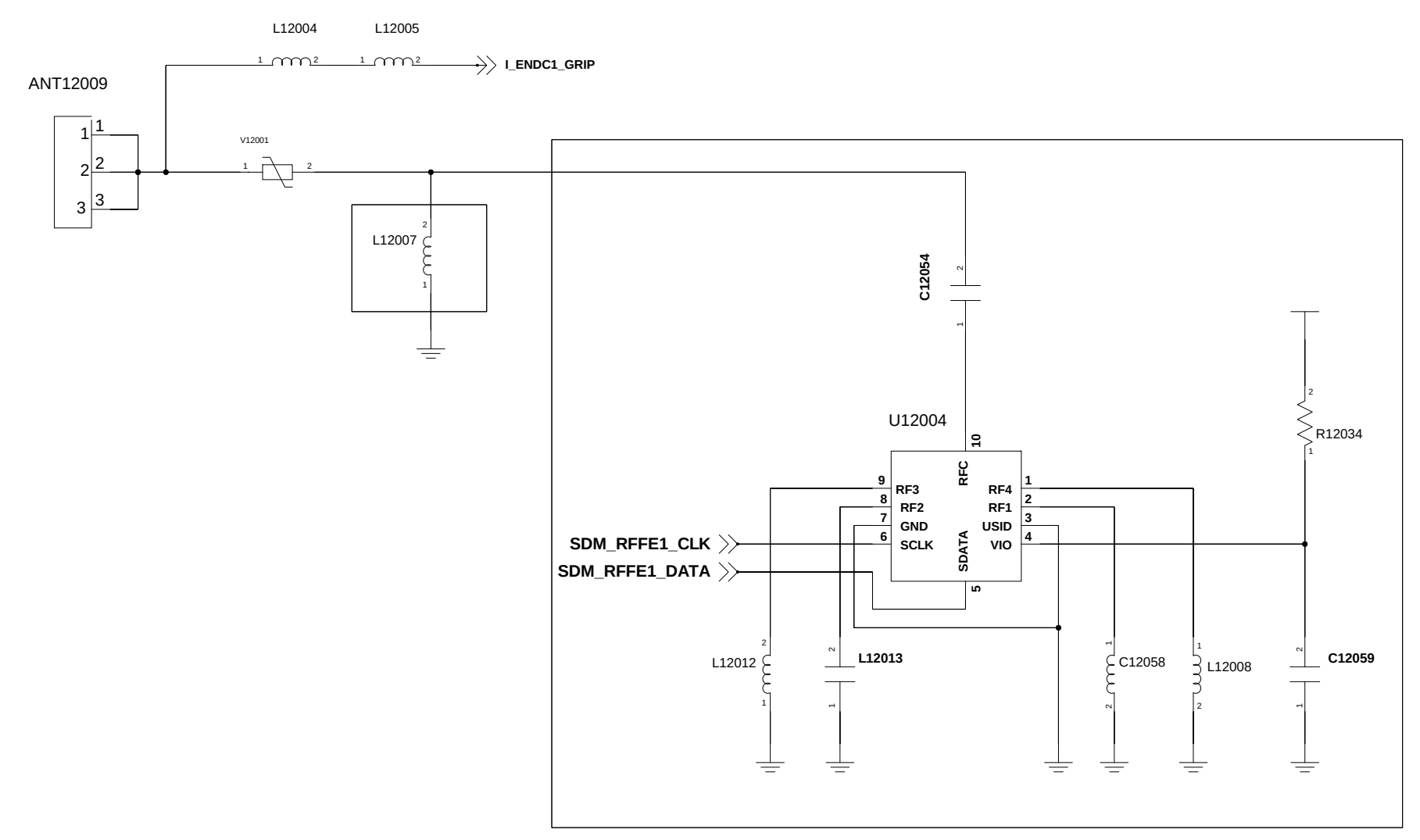

ANT12012
ANT12013
ANT12014
ANT12015
ANT12022

REAR CONTACT (1.5mm)


ANT12019
ANT12020
ANT12021

REAR CONTACT (1.3mm)

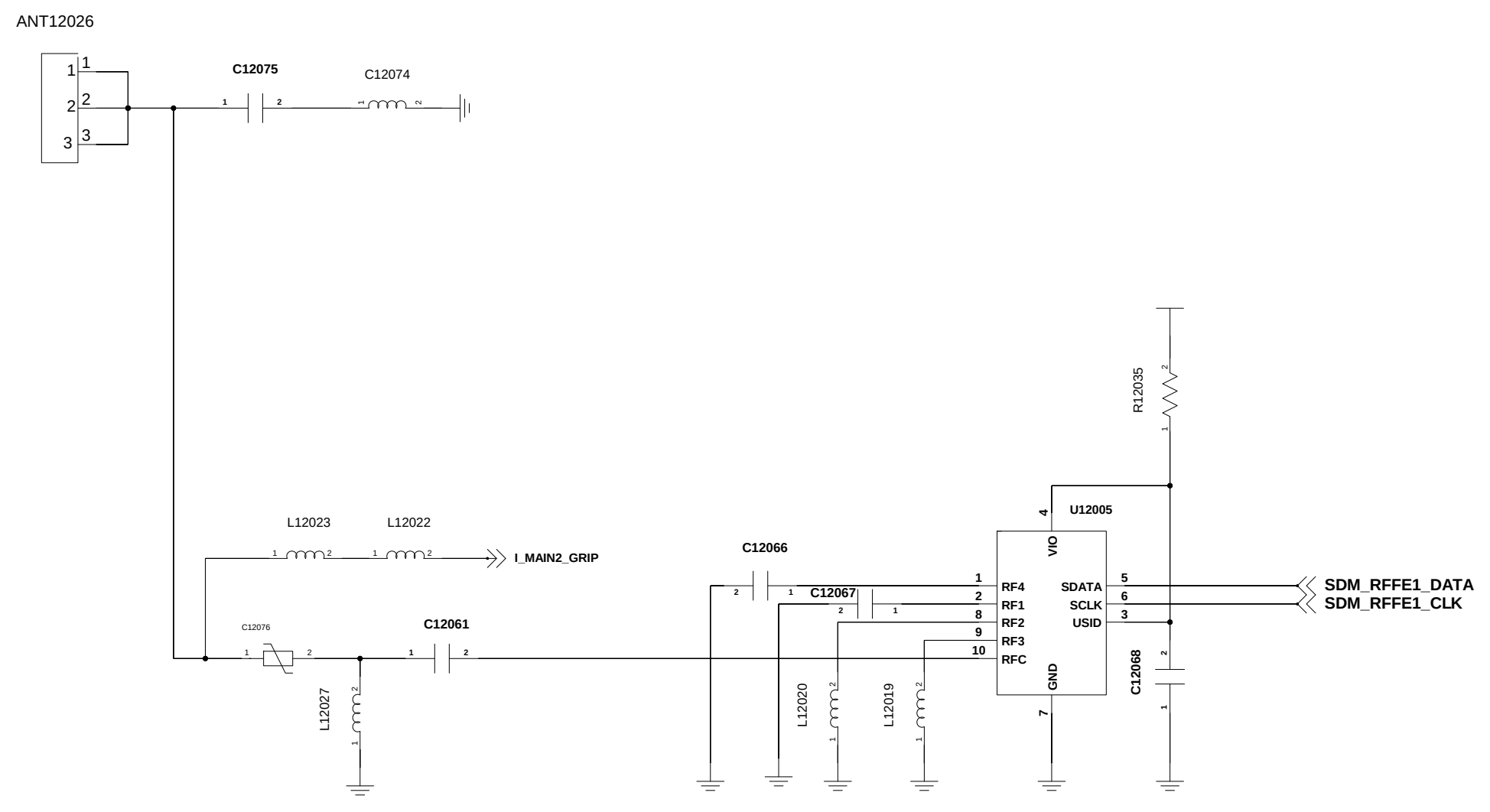
WIFI : Delete


ANT12009
L12004
L12005
I_ENDC1_GRIP
V12001
L12007
C12054
U12004
RF3
RF2
GND
SCLK
SDATA
RFC
RF4
RF1
USID
VIO
R12034
SDM_RFFE1_CLK
SDM_RFFE1_DATA
L12012
L12013
C12058
L12008
C12059

Sub6 delete

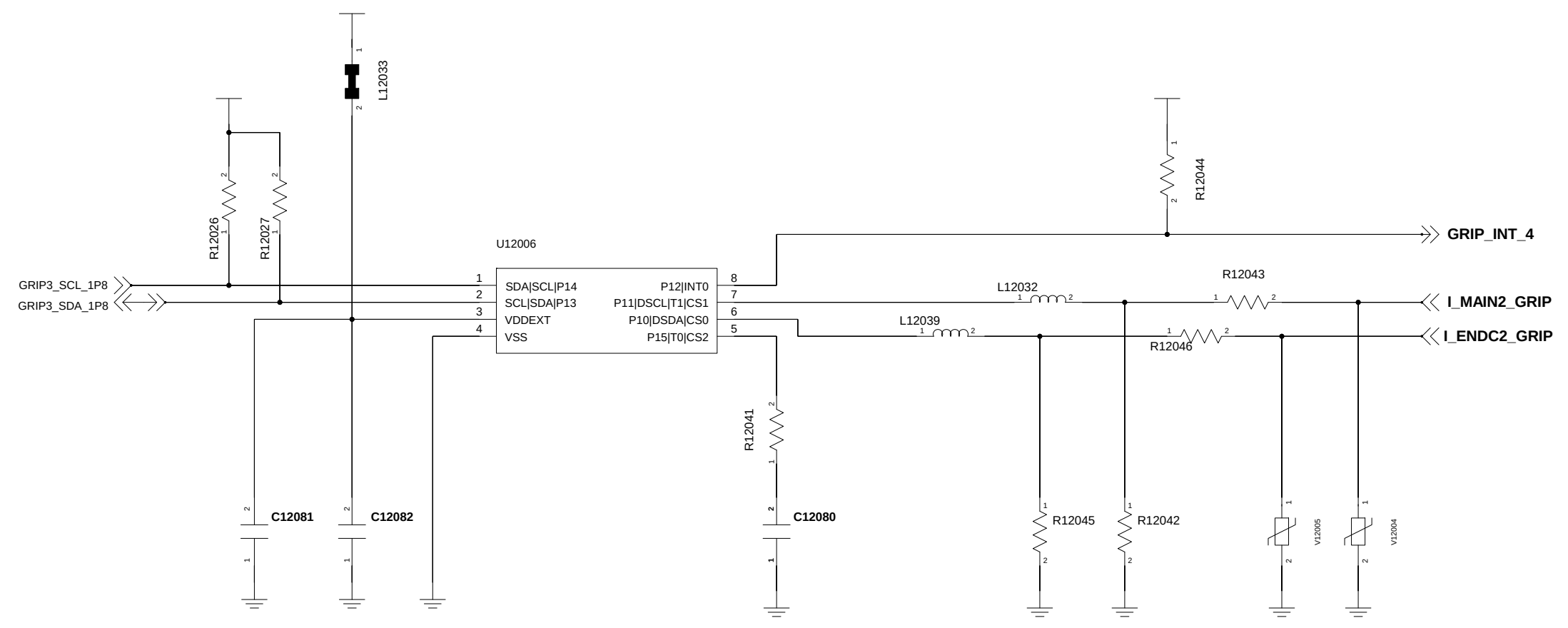
SUB1 ANT

(DRX_L/M/HB, ENDC_L/M/HB, N41 SRS)

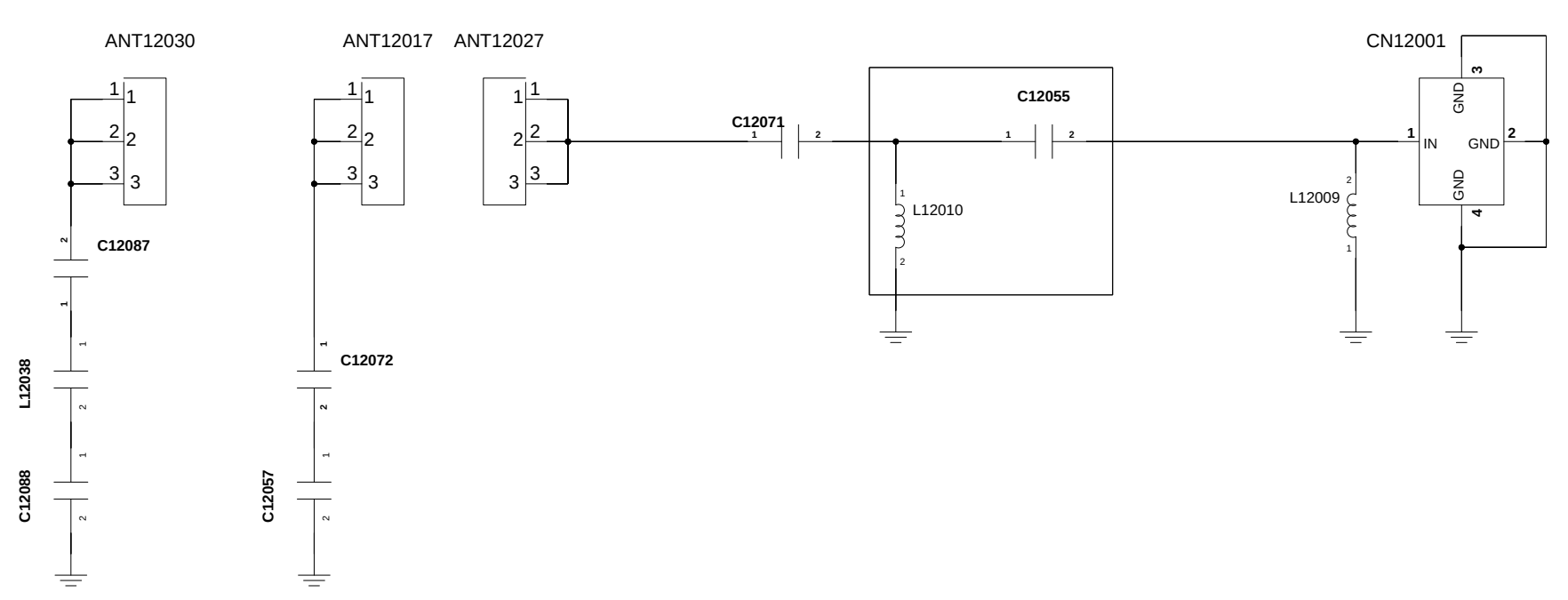

ANT12026
C12075
C12074
L12023
L12022
I_MAIN2_GRIP
C12076
C12061
L12027
C12066
C12067
L12020
L12019
R12035
U12005
VIO
RF4
RF1
RF2
RF3
RFC
SDATA
SCLK
USID
GND
C12068
SDM_RFFE1_DATA
SDM_RFFE1_CLK

MAIN2 ANT XGND SW / GRIP

(N77, N78, B48 TRX)

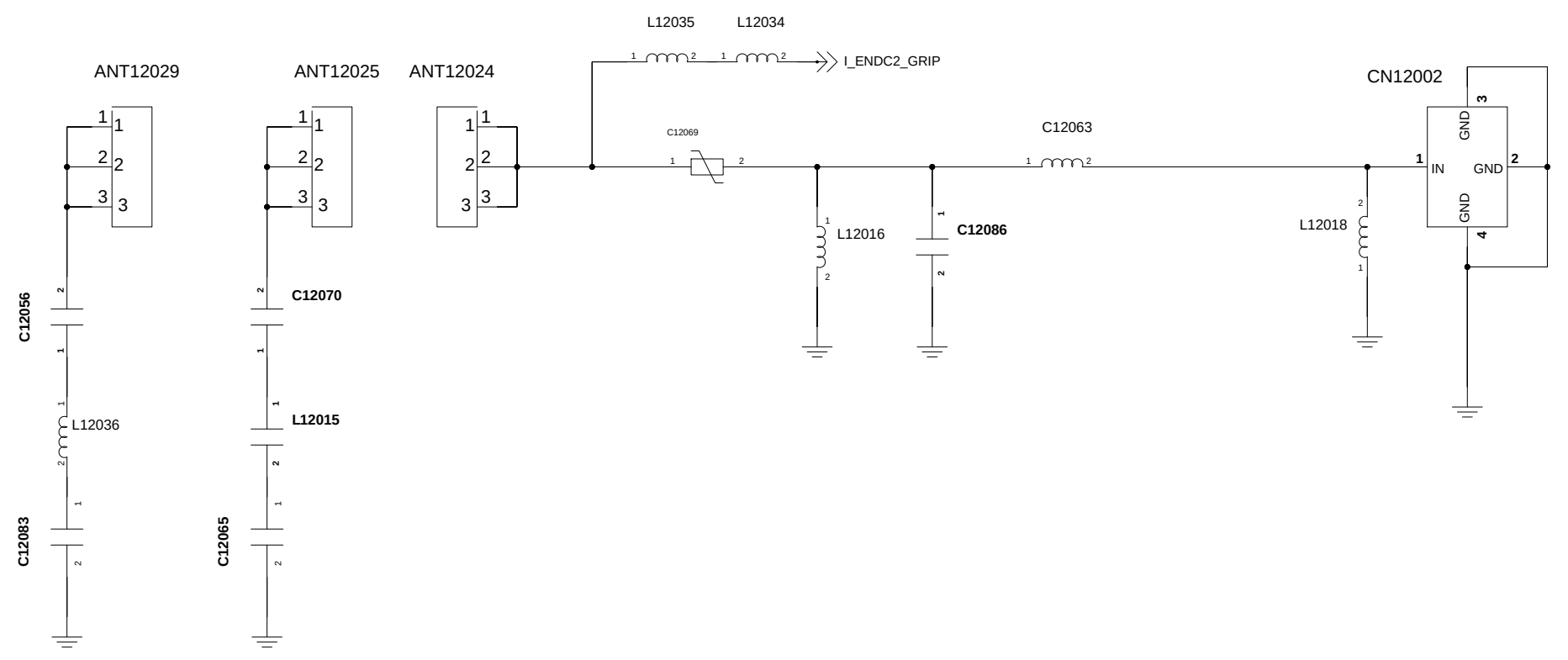

L12033
R12026
R12027
U12006
GRIP3_SCL_1P8
GRIP3_SDA_1P8
SDA|SCL|P14
SCL|SDA|P13
VDDEXT
VSS
P12|INT0
P11|DSCL|T1|CS1
P10|DSDA|CS0
P15|T0|CS2
R12044
GRIP_INT_4
L12032
R12043
I_MAIN2_GRIP
L12039
R12046
I_ENDC2_GRIP
R12041
C12081
C12082
C12080
R12045
R12042
V12005
V12004

MAIN2_GRIP_SENSOR


ANT12030
ANT12017
ANT12027
C12087
L12038
C12088
C12072
C12057
C12071
C12055
L12010
CN12001
L12009
IN
GND

SUB2 ANT

(DRX_MIMO_M/HB, DRX_n77 / n78, N41 SRS)


ANT12029
ANT12025
ANT12024
L12035
L12034
I_ENDC2_GRIP
C12069
L12016
C12086
C12063
L12018
CN12002
IN
GND
C12056
L12036
C12083
C12070
L12015
C12065

SUB4 ANT

(PRX_MIMO_MB / HB, DRX_MIMO_n77 / n78)

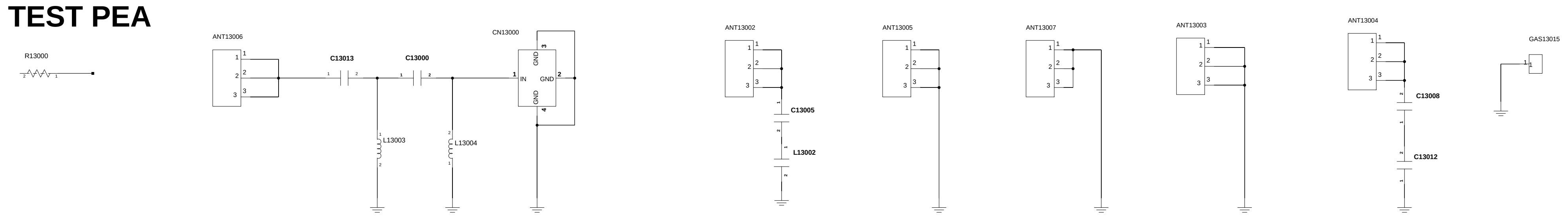
L12028
CLP12000
CLP12002

Cable CLIP

TITLE	TabS7P
SIZE	A1
DWG NO	
REV	
DRAWN BY	User Name
SHEET	12 of 15
Date	

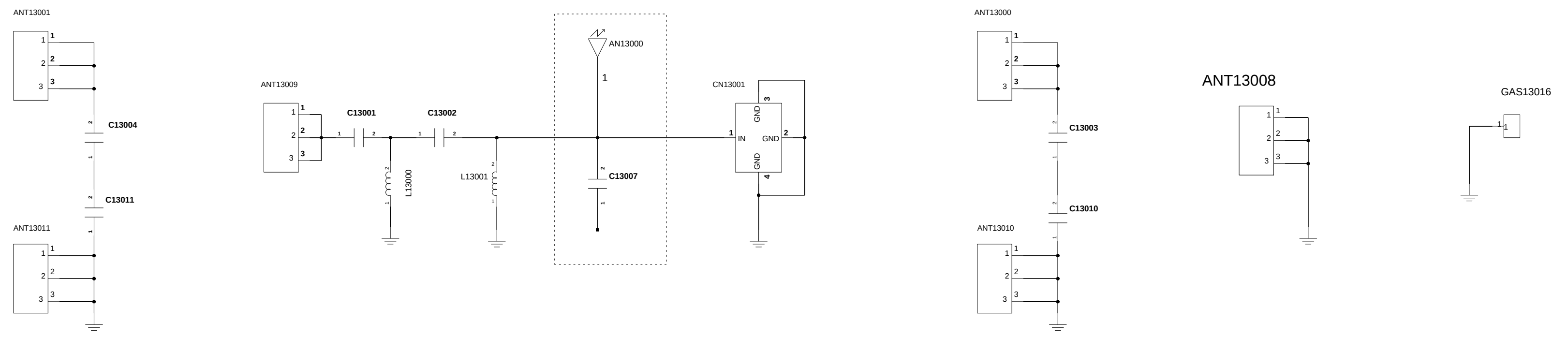
WIFI TL PCB (Ext.)

5G TEST PEA

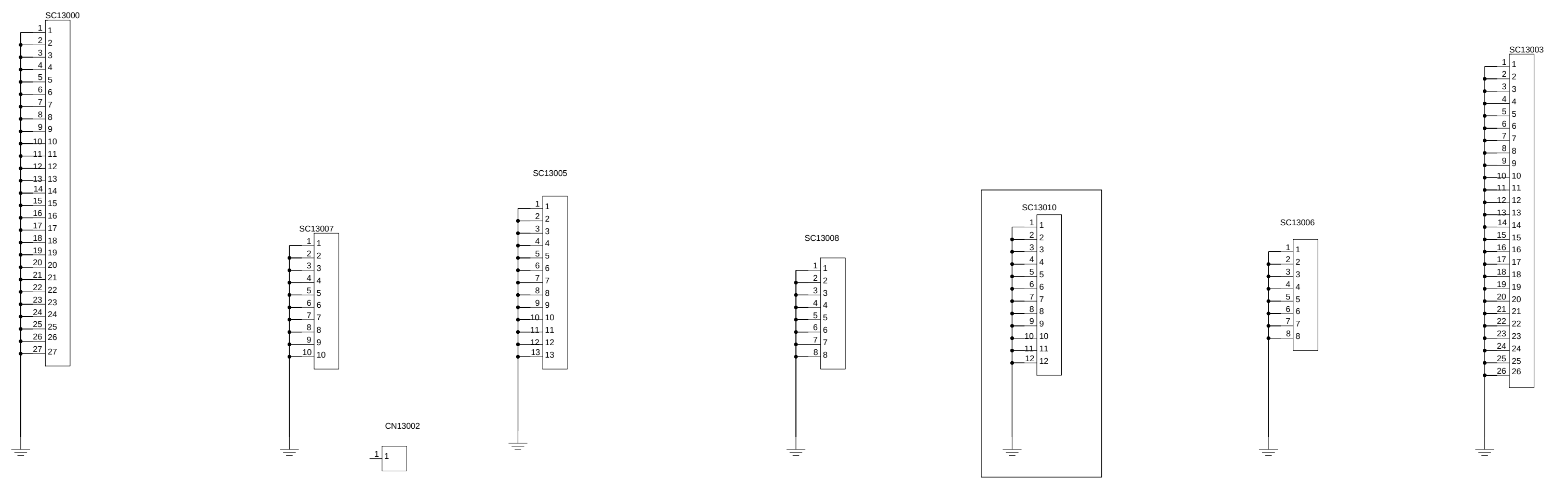

R13000
ANT13006
C13013
C13000
CN13000
IN
GND
GND
GND
L13003
L13004
ANT13002
C13005
L13002
ANT13005
ANT13007
ANT13003
ANT13004
C13008
C13012
GAS13015

WIFI TR PCB (Ext.)

PEA ANT


ANT13001
C13004
C13011
ANT13011
ANT13009
C13001
C13002
L13000
L13001
AN13000
C13007
CN13001
IN
GND
GND
GND
ANT13000
C13003
C13010
ANT13010
ANT13008
GAS13016

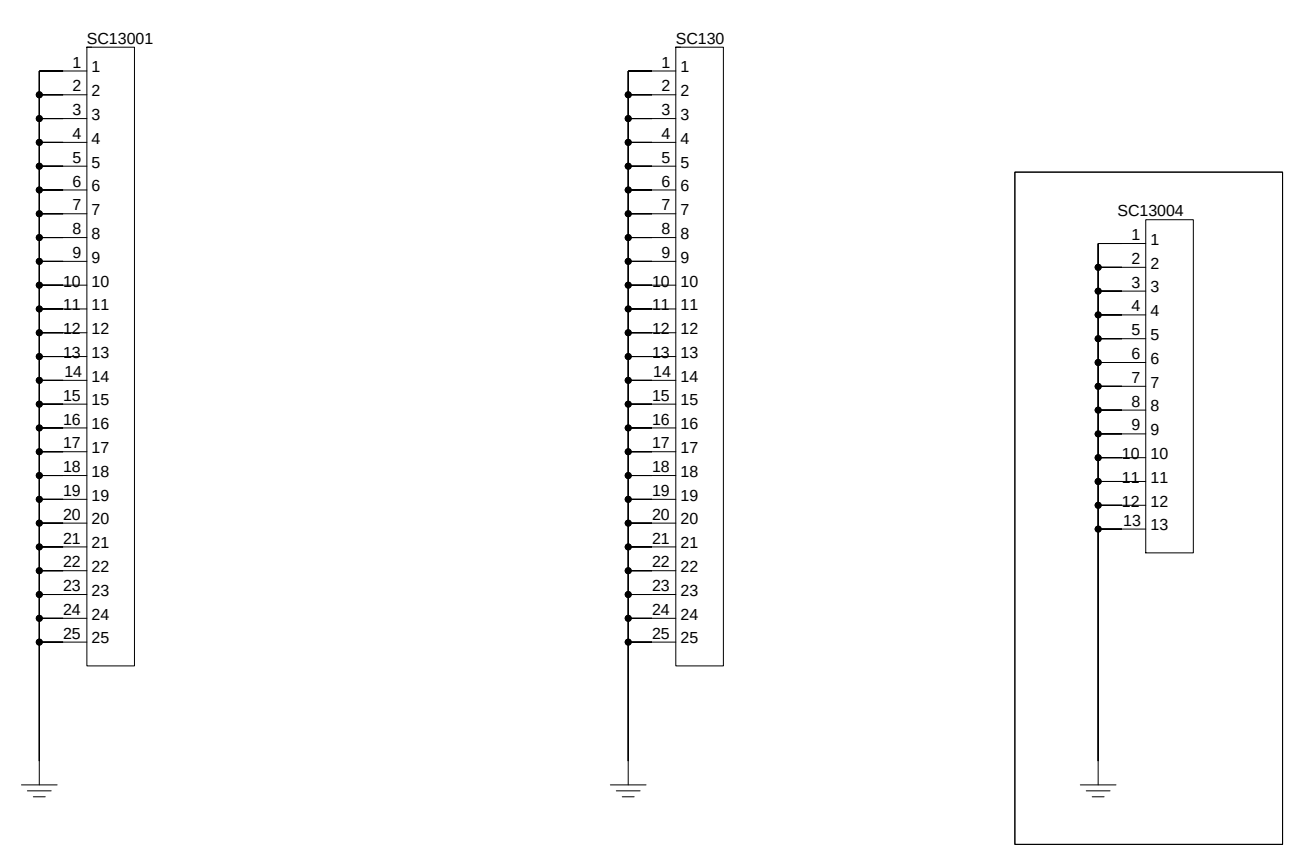
CLIP / SMR / SUS / SHIELDCAN for Main PCB


SC13000
SC13007
CN13002
SC13005
SC13008
SC13010
SC13006
SC13003

RF1 **OSC** **SPK** **UFS** **5G** **OCTA** **OVP**

5G
mmW:ADD
SUB6, WIFI:OPEN

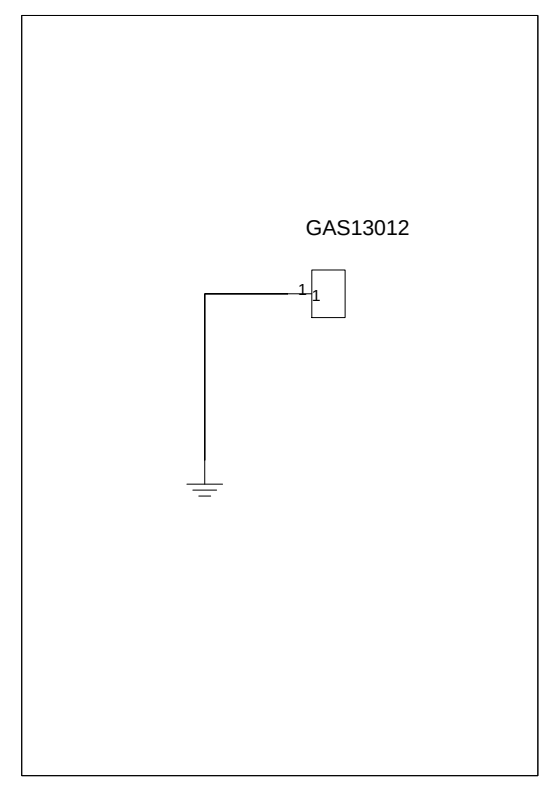
< TOP S/C >


SC13001
SC130
SC13004

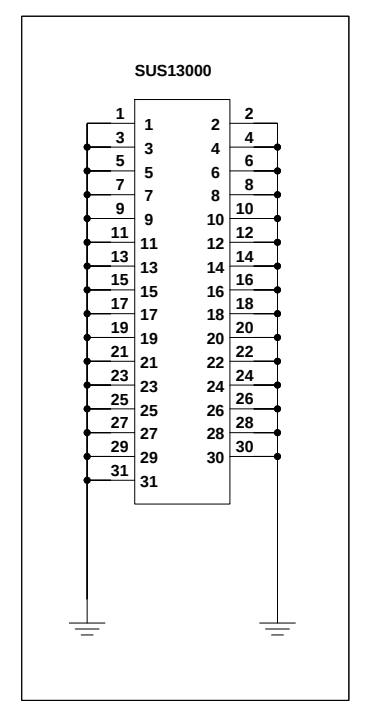
WIFI **AP** **RF2**

RF2
mmW, SUB6:ADD
WIFI:OPEN

< BOTTOM S/C >

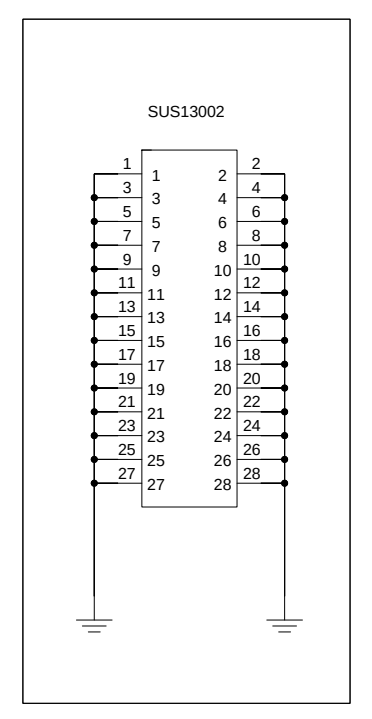


need for CAM



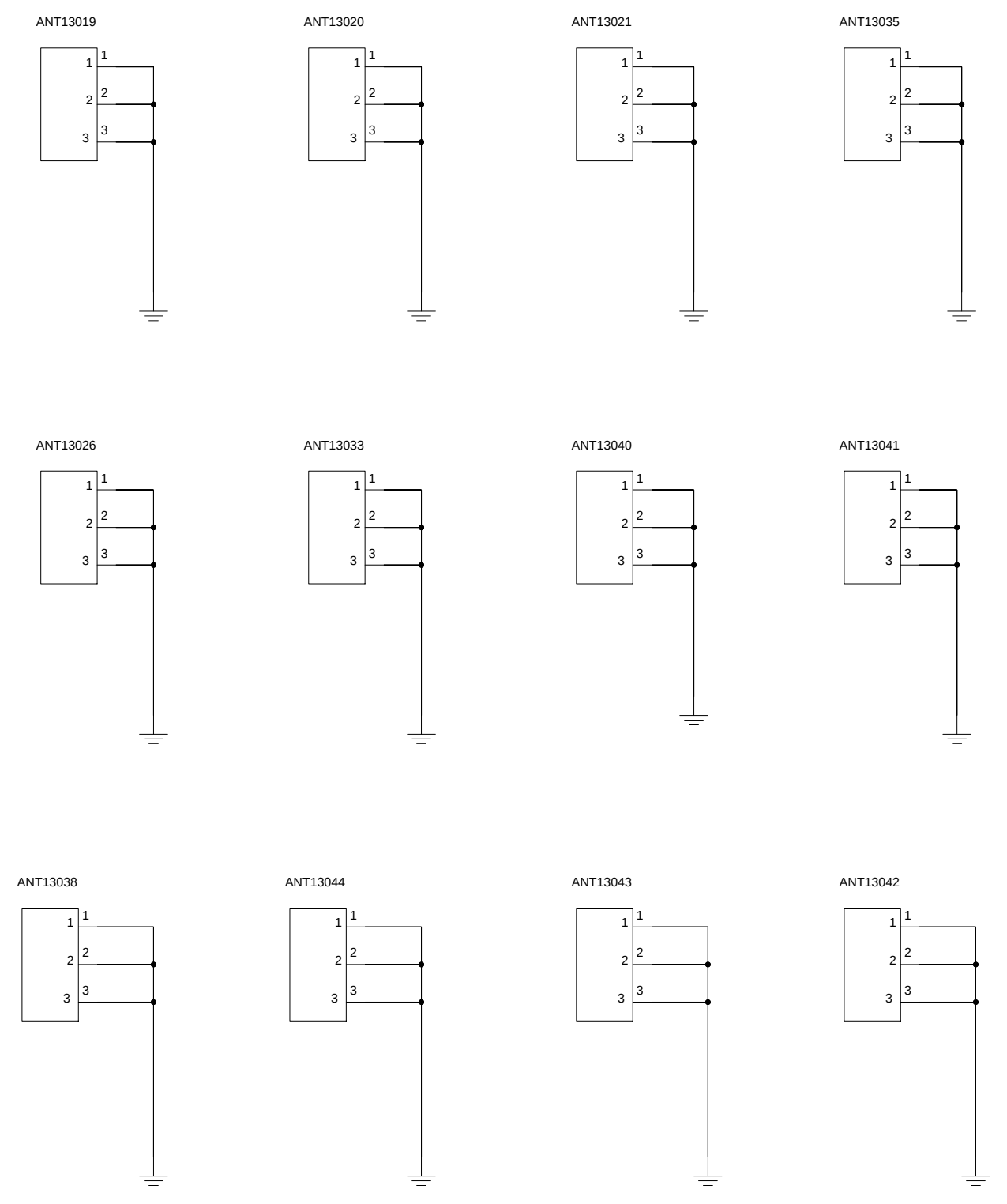
mmWAVE SUS
mmW:ADD
SUB6, WIFI:OPEN

< TOP SUS >

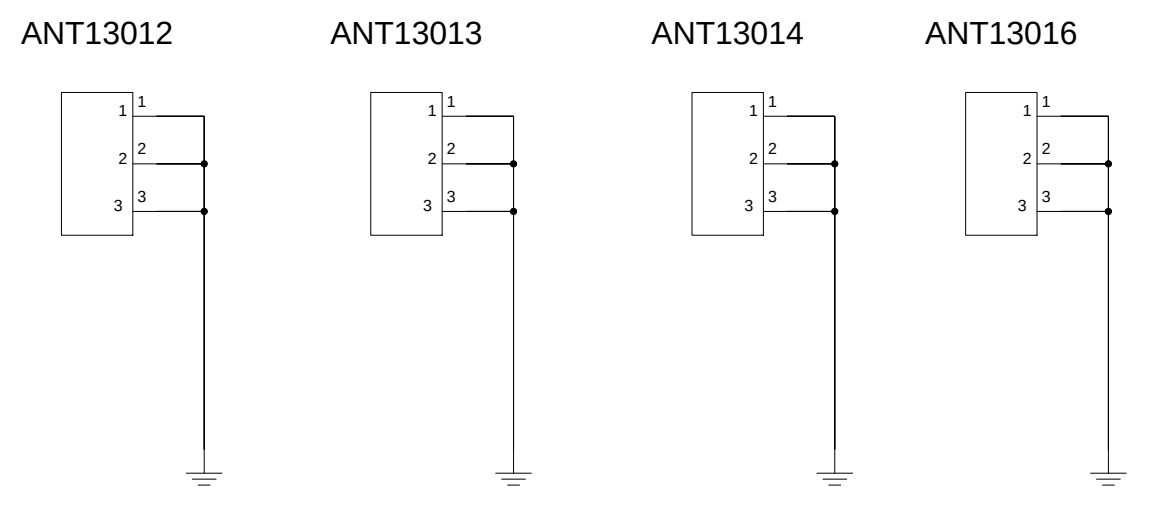


PEN-DECO SUS
mmW:OPEN
SUB6, WIFI:ADD

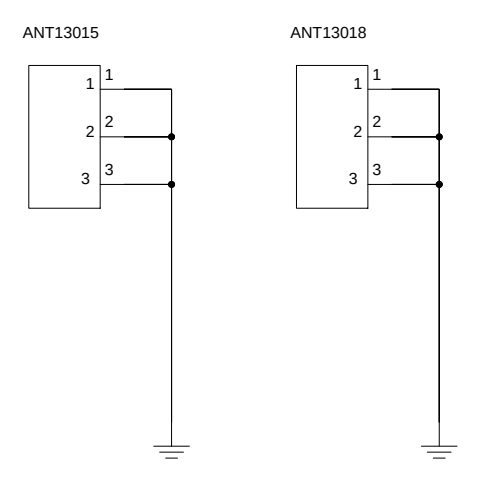
< BOT SUS >


ANT13019
ANT13020
ANT13021
ANT13035
ANT13026
ANT13033
ANT13040
ANT13041
ANT13038
ANT13044
ANT13043
ANT13042

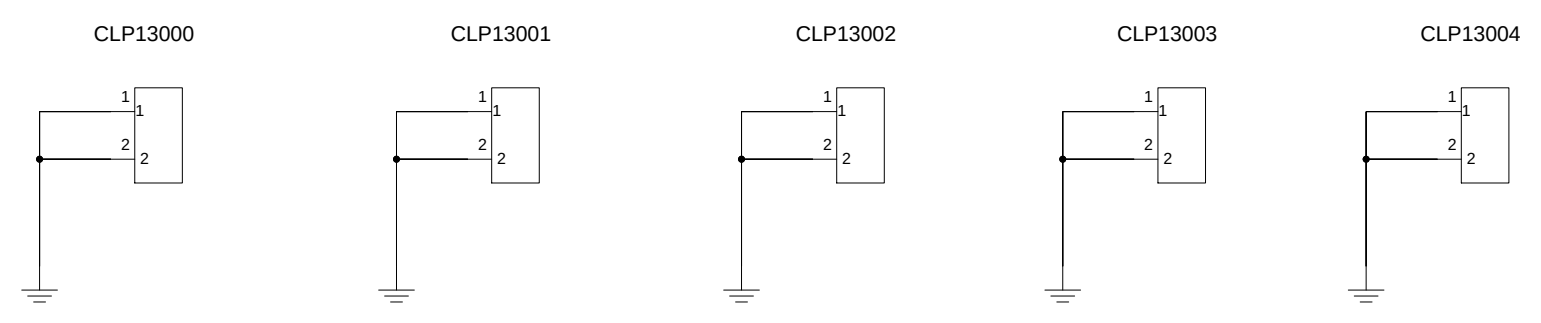
REAR CONTACT (1.5T)


ANT13012
ANT13013
ANT13014
ANT13016

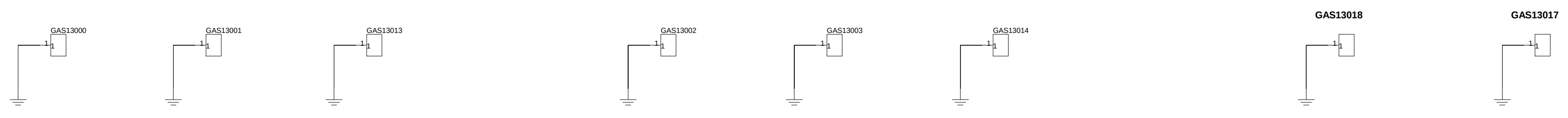
REAR CONTACT (1.1T)


ANT13015
ANT13018

REAR CONTACT (1.5T)

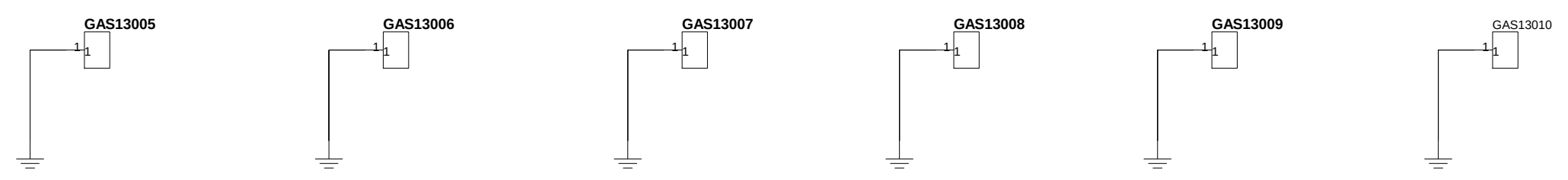

CLP13000
CLP13001
CLP13002
CLP13003
CLP13004

Cable CLIP (1.1T)


GAS13000
GAS13001
GAS13013
GAS13002
GAS13003
GAS13014
GAS13018
GAS13017

REAR CONTACT (SMR 1.4T) **REAR CONTACT (SMR 1.1T-OPEN)** **REAR CONTACT (SMR 0.8T)**

need for Core PCB


GAS13005
GAS13006
GAS13007
GAS13008
GAS13009
GAS13010

Front CONTACT (2.2T)

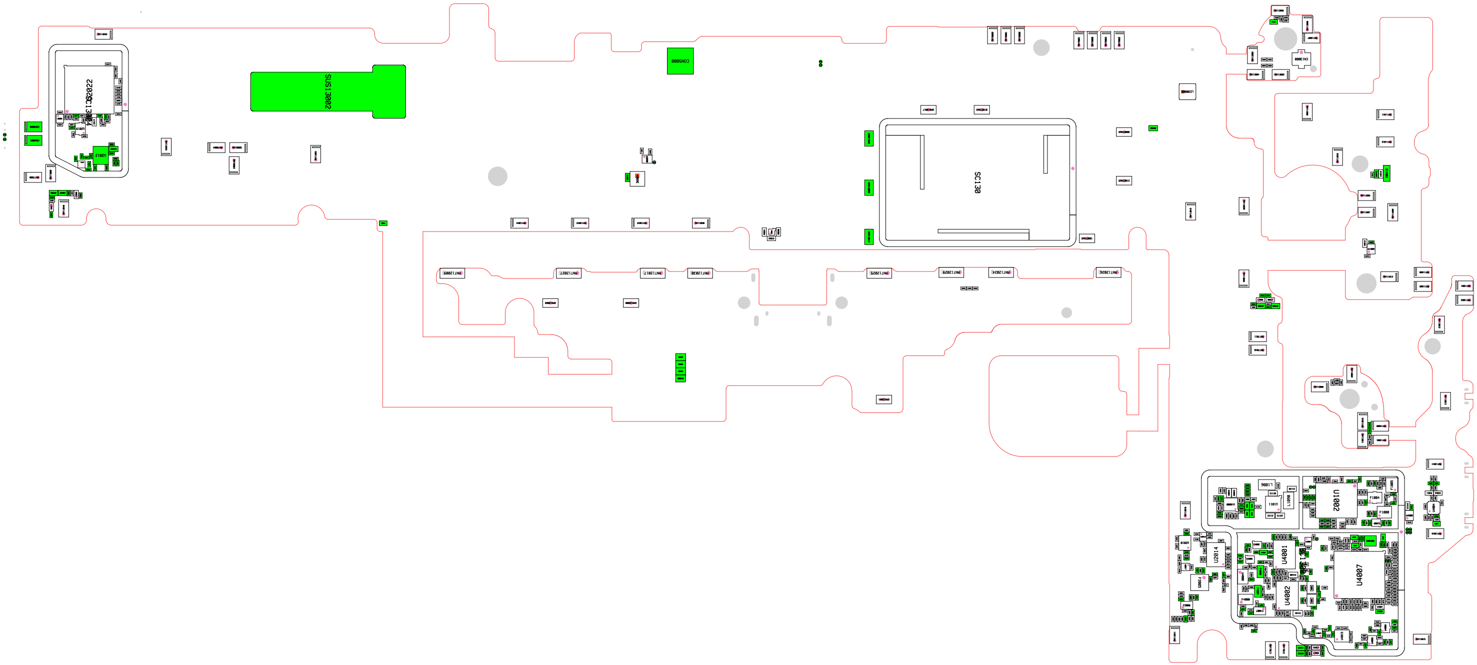
TITLE TabS7P
SIZE A1
DWG NO
REV
DRAWN BY User Name
SHEET 13 of 15
Date

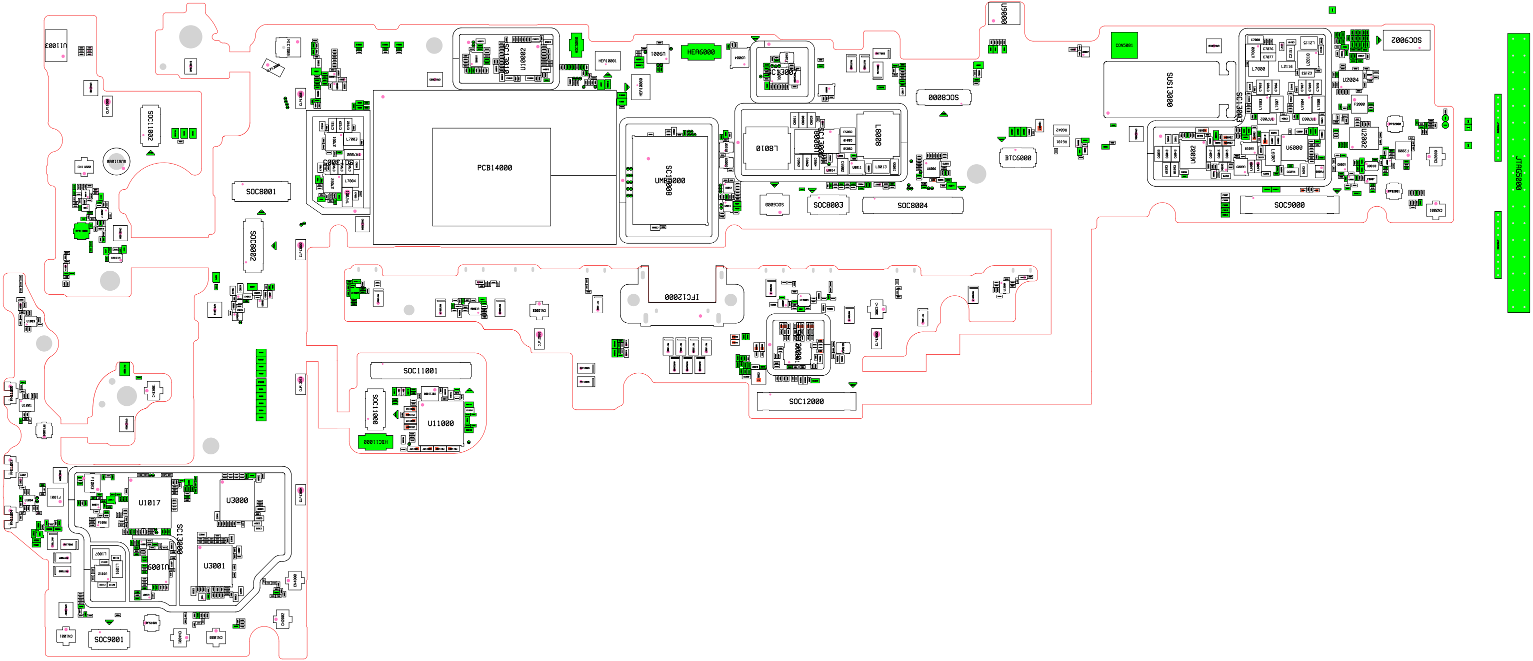
Sencondary PCB PAD (Main PCB)

PCB14000
(1/2)
PCIE0_TX1_M_WLAN
PCIE0_TX0_M_WLAN
PCIE0_REFCLK_M_WLAN
PCIE0_RX1_M_WLAN
PCIE0_RX0_M_WLAN
CC_SCL_1P8
CC_SDA_1P8
6AXIS1_INT2
6AXIS1_INT1
GRIP2_SCL_1P8
BT_UART_RXD
BT_UART_TXD
PCIE0_TX1_P_WLAN
PCIE0_TX0_P_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_RX1_P_WLAN
PCIE0_RX0_P_WLAN
REDRIVER_SDA_1P8
UWCAM_SDA_1P8
CAM_AF_EEP_SDA_1P8
GRIP2_SDA_1P8
WCN_SW_CTRL
BT_EN
WCN_SW_CTRL_WL_CX
UW_VTCAM_SDA_1P8
PMIC_SPMI_DATA
PMIC_SPMI_CLK
UWCAM_MIPI_D3_P
UWCAM_MIPI_D3_N
SDC2_CLK
REDRIVER_SCL_1P8
UWCAM_SCL_1P8
MAINCAM_SDA_1P8
MAINCAM_RST_N
BT_UART_RTS
BT_UART_CTS
CC_DIR
M_RF_ID_1
BAT_ID
SLEEP_CLK
WLAN_BT_SLIMBUS_CLK
WLAN_BT_SLIMBUS_DATA
UWCAM_MIPI_D2_P
UWCAM_MIPI_D2_N
MAINCAM_SCL_1P8
GRIP_INT_3
CAM_AF_EEP_SCL_1P8
MCU_SCL_1P8
UW_VTCAM_SCL_1P8
HALL_INT
CABLE_DET_GPS
M_RF_ID_2
M_RF_ID_0
3RD_MIC_CLK
3RD_MIC_DATA
UWCAM_MIPI_D1_P
UWCAM_MIPI_D1_N
UW_VTCAM_MCLK
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
VTCAM_MIPI_CLK_N
VTCAM_MIPI_D0_N
VTCAM_MIPI_D1_N
VTCAM_MIPI_D2_N
VTCAM_MIPI_D3_N
SDC2_D1
SDC2_CMD
SDC2_D2
CABLE_DET_MAIN2
PEN_IRQ
ground
DSI0_NC_LN3_M
DSI0_C2_LN3_P
DSI0_A2_LN2_P
FAULT_N
DSI0_A1_LN1_M
DSI0_B1_CLK0_P
DSI0_B2_LN2_M
DSI0_C0_LN1_P
DSI0_C1_CLK0_M
DSI0_A0_LN0_P
DSI0_B0_LN0_M
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_CLK_N
PMIC_RB
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D3_N
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_D2_P
MAINCAM_MCLK
UWCAM_MIPI_CLK_P
UWCAM_MIPI_CLK_N
MAIN_MIC_DATA
MAIN_MIC_CLK
PON_RESET_N
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
PCIE0_RST_N
PCIE0_WAKE_N
CABLE_DET_WIFI0
MCU_SWCLK
MCU_SDA_1P8
WLAN_EN
SECURE_NVM_SCL_1P8
VTCAM_RST_N
SDC2_D3
SDC2_D0
VTCAM_MCLK
VTCAM_MIPI_D3_P
VTCAM_MIPI_D2_P
VTCAM_MIPI_D1_P
VTCAM_MIPI_D0_P
VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D0_P
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D1_P
UWCAM_MCLK
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
SUB_MIC_CLK
SUB_MIC_DATA
PMK_CLK6_CXO
PCIE0_CLK_REQ_N
KBD_PWR_EN
MCU_nRST
DC_IRQ_N
SECURE_NVM_SDA_1P8
UW_VTCAM_RST_N
UWCAM_RST_N
PCB14000
(2/2)
QLINK2_EN
QLINK2_REQ
QLINK2_WMSS_RESET_N
FCAM_LDO_EN
JTAG_TDI
QLINK2_L1_P
QLINK2_L3_P
JTAG_TMS
JTAG_TDO
QLINK2_L1_M
QLINK2_L3_M
JTAG_TCK
RCAM2_LDO_EN
SPK_AMP_LDO_EN
JTAG_RST_OUT_N
QLINK2_L0_M
QLINK2_L2_P
KBD_DCDC_SCL_1P8
RCAM_LDO_EN
JTAG_TRST_N
MOTOR_LDO_EN
QLINK2_L0_P
QLINK2_L2_M
KBD_DCDC_SDA_1P8
GRIP3_SCL_1P8
OCTA_DCDC_SCL_1P8
AMP_INT_N
MCU_CONN_INT
MCU_INT
DIRECT_SDA_1P8
GRIP3_SDA_1P8
OCTA_DCDC_SDA_1P8
QLINK2_L4_P
QLINK2_CLK_M
DIRECT_SCL_1P8
AMP_SDA_1P8
AMP_SCL_1P8
QLINK2_L4_M
QLINK2_CLK_P
QLINK1_EN
QLINK1_REQ
QLINK1_WMSS_RESET_N
QLINK0_EN
QLINK0_REQ
QLINK0_WMSS_RESET_N
QLINK1_L0_M
QLINK1_L0_P
QLINK1_L2_M
QLINK1_L2_P
QLINK0_L3_P
QLINK0_L0_P
QLINK0_L3_M
QLINK0_L0_M
QLINK1_L4_P
QLINK1_L4_M
QLINK1_L7_P
QLINK1_L7_M
QLINK0_L1_P
QLINK0_CLK_M
QLINK1_L1_P
QLINK1_L1_M
QLINK1_CLK_P
QLINK1_CLK_M
QLINK0_L1_M
QLINK0_CLK_P
QLINK1_L5_M
QLINK1_L5_P
QLINK1_L3_M
QLINK1_L3_P
QLINK0_L2_P
QLINK0_L4_P
QLINK0_L2_M
QLINK0_L4_M
QLINK1_L8_P
QLINK1_L8_M
QLINK1_L6_P
QLINK1_L6_M
SIM1_EN
BLANKET_THM
SDM_RFFE0_CLK
SDM_RFFE0_DATA
AMP_I2S_LRCK
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
DISP_RST
DISP_DET
SIM2_EN
FLASH_LED
FLASH_LED
SDM_RFFE1_CLK
SDM_RFFE1_DATA
OCTA_DCDC_EN
VTCAM_SDA_1P8
VTCAM_SCL_1P8
FLASH_LED
WCN_HOST_SOL_OUT
GRIP_INT_2
HALL_INT_WACOM
SDM_RFFE4_DATA
SDM_RFFE4_CLK
BAT_THM
CHG_THM
SUB_CON_DET
WCN_WLAN_SOL_IN
SUB_PCB_ID
SIM1_DET
SIM2_DET
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
RGB1_SCL_1P8
GRIP1_LDO_EN
VOL_UP
CHG_DET
SDCARD_DET
RGB1_SDA_1P8
RGB2_SCL_1P8
MAGNETIC_SCL_1P8
MAGNETIC_SDA_1P8
BTP_RST_N
BTP_SPI_CS_N
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_MISO
GRIP_INT_1
EDP_AUX_N
USB3.0_SSTX1_M_AP
USB3.0_SSTX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX1_P_AP
eUSB_AP_D_P
UW_VTCAM_LDO_EN
EDP_AUX_P
USB3.0_SSTX1_P_AP
USB3.0_SSTX0_P_AP
USB3.0_SSRX0_M_AP
USB3.0_SSRX1_M_AP
eUSB_AP_D_N
UFS0_TX0_P
UFS0_TX1_P
UFS0_RX0_P
UFS0_RX1_P
UFS0_RESET_N
UFS0_TX0_M
UFS0_TX1_M
UFS0_RX0_M
UFS0_RX1_M
CABLE_DET_WIFI1
SIM1_DATA_1P2
SIM1_CLK_1P2
BTP_LDO_EN
SM_PS_HOLD
WLAN_COEX_SDXUART_RX
TSP_SCL_1P8
GPS_L1_LNA_EN
LAA_AS_EN
LAA_EN_0
TSP_SDA_1P8
WLAN_COEX_SDXUART_TX
PEN_LDO_EN
SIM2_DATA_1P2
SIM2_CLK_1P2
ENDC_GRIP_LDO_EN
TSP_INT_N
MCU_ACC_INT
eUSB_SCL_1P8
PEN_SDA_1P8
HWID_LRPST_THM
GRIP2_LDO_EN
AP_UART_TXD
AP_UART_RXD
eUSB_SDA_1P8
OTG_DET
VBUS_DISCHARGING_EN
PEN_SCL_1P8
CAM_IO_LDO_EN
TYPE_C_INT
AMP_RST_N
AP_USB_BOOT
PEN_PDCT
SIM1_RST_1P2
SIM2_RST_1P2
TCON_RDY
PEN_FWEL
DISP_TE
CABLE_DET_SUB1
EDP_SW_OE
CABLE_DET_SUB2
CABLE_DET_SUB4
IF_PMIC_INTB
CABLE_DET_SUB3
GRIP1_SDA_1P8
DISP_CON_DET
HWID_LRPST_THM_SEL
OVP_PWR_FLAGB
EDP_SW_SEL
GRIP1_SCL_1P8
eUSB_RPTR_RST_N
RGB2_SDA_1P8
6AXIS_I3C_SCL_1P8
6AXIS_I3C_SDA_1P8
GRIP_INT_4

TITLE	TabS7P	
SIZE A1	DWG NO PDM NUMBER	REV Rev
SHEET 14 of 15	Date Drawing Date	

DRAWN BY User Name


SUS13002
SC2022
SC13004
U2812
U2819
CON5000
SC130
ANT12009
ANT12027
ANT12017
ANT12038
ANT12025
ANT12029
ANT12024
ANT12026
U1002
L1006
L1090
U1811
F1004
F1008
F1005
U4001
U4002
U4007
U2014
U2018
F2005
F2006
F4000
U4015


U11003
SOC11002
SUS11000
CN11000
MIC7000
CLP13000
CLP11000
CLP13001
CLP13002
CLP13003
CLP13004
SOC8001
SOC8002
SC13005
U7001
U7002
L7003
L7004
TA7000
TA7001
C7831
C7829
C7832
PCB14000
SC13010
U10002
HDC5000
HEA10001
HEA10000
U9001
HEA6000
U9004
SC13007
U6012
U9000
UME5000
SC13008
L8010
SC13006
L8008
L8013
U8011
U6006
SOC8000
SOC6000
SOC8003
SOC8004
BTC6000
R6042
R6101
CON5001
SUS13000
SC13003
U7000
L7000
L2115
L2116
C2151
C2153
U2018
C7876
C7877
C7088
U7003
U7004
L7007
L7008
TA7002
TA7003
U6002
U6000
L6007
U6010
U6009
C6094
SOC9002
U2004
U2002
F2000
F2007
F2008
U8018
RFS2000
RFS2001
CN2000
CN2001
SOC9000
JTAG5000
IFC12000
CN12001
CN12002
CLP12000
CLP12002
SC12000
U12002
U12003
SOC12000
SOC11001
SOC11000
U11000
HDC11000
CN13001
U1001
U1003
U1004
RFS1000
F1001
F1003
F1006
U1017
U3000
U3001
SC13000
U1009
U1012
L1007
L1091
CN4000
CN2002
CN1001
CN1000
CN4001
RFS1001
SOC9001

TAB_S9_CORE PCB_REV0.5 (S9, S9+, S9 Ultra)

- CORE PCB TYPE -

2023.05.26

sheet01 : AP_SM8550

sheet02 : PMIC

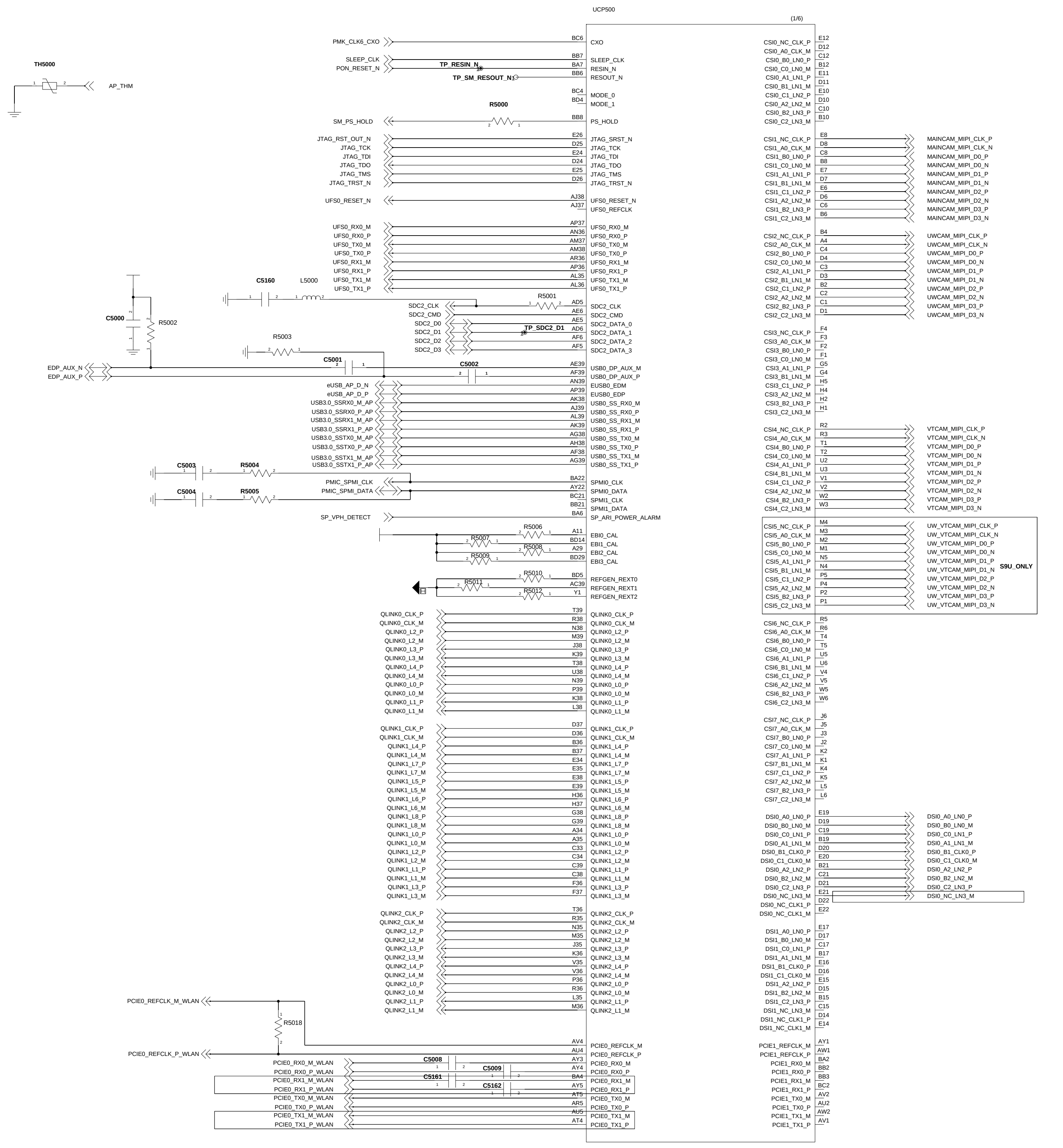
sheet03 : CORE PCB

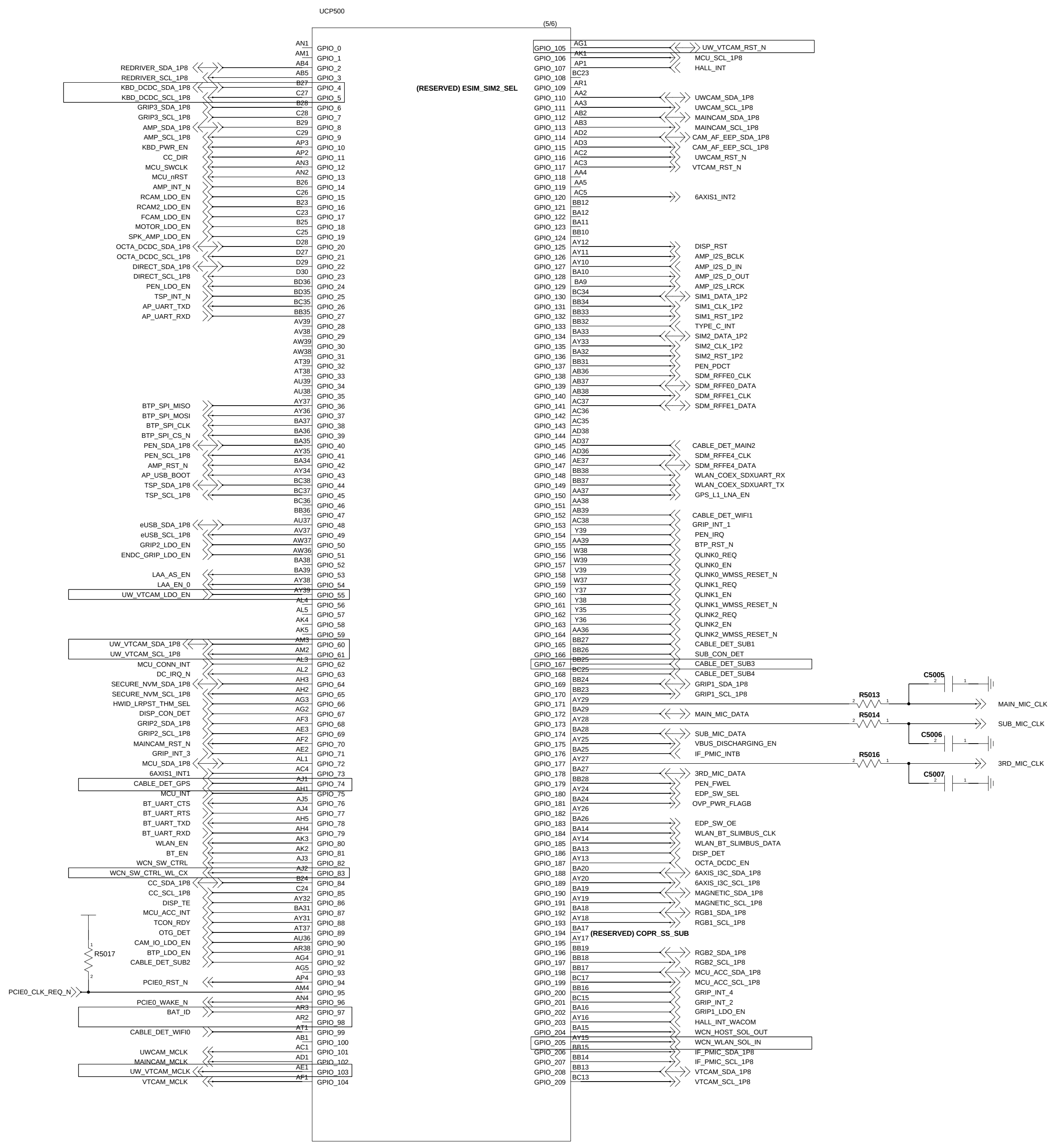
TITLE	TabS9_CORE			
SIZE A1	DWG NO	PDM NUMBER	REV Rev	
DRAWN BY User Name	SHEET 1 of 4		Date	Drawing Date

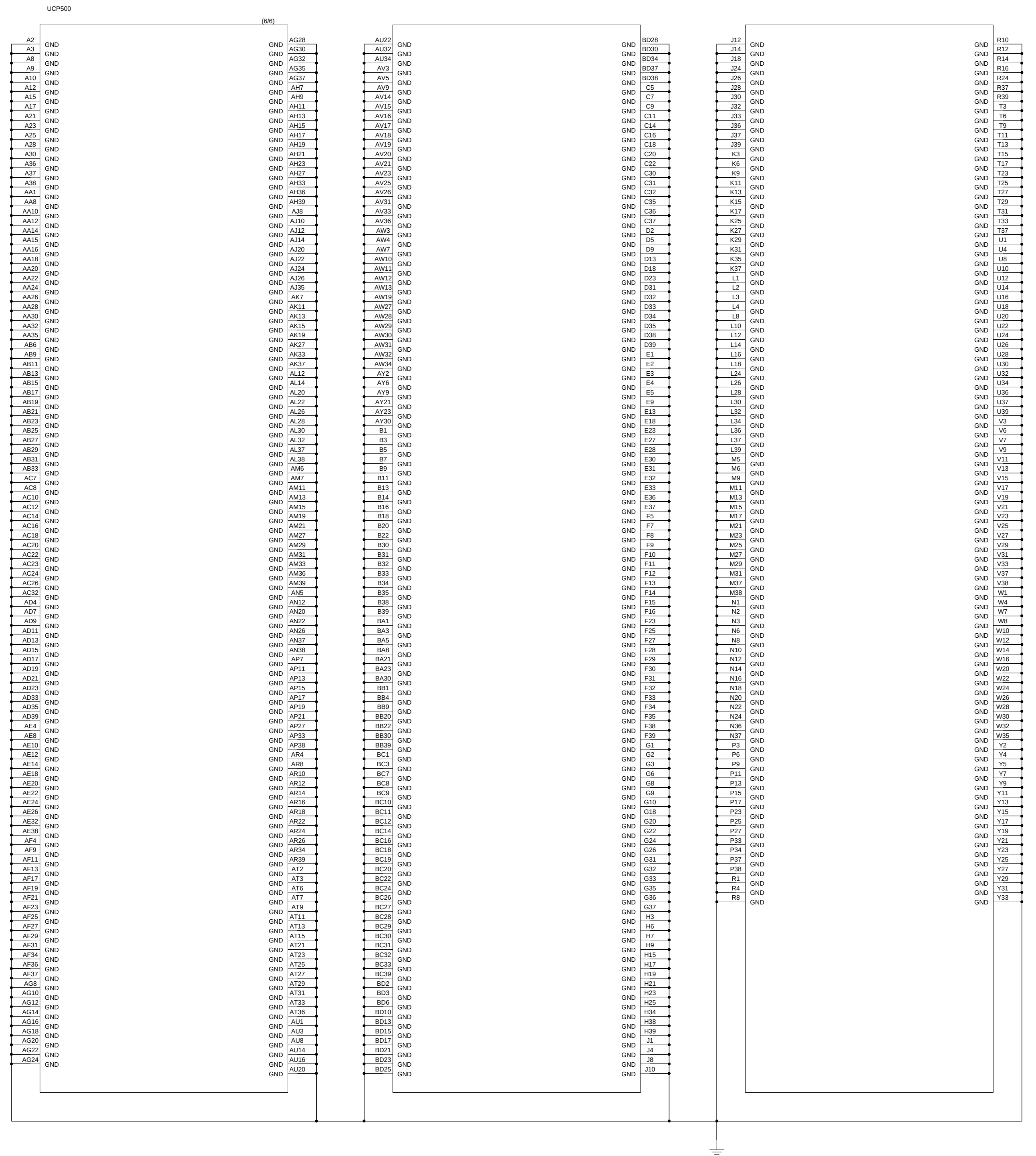
SM8550 (INTERFACE/GPIO) (Core PCB)

mmWAVE, SUB6 : SM8550

System

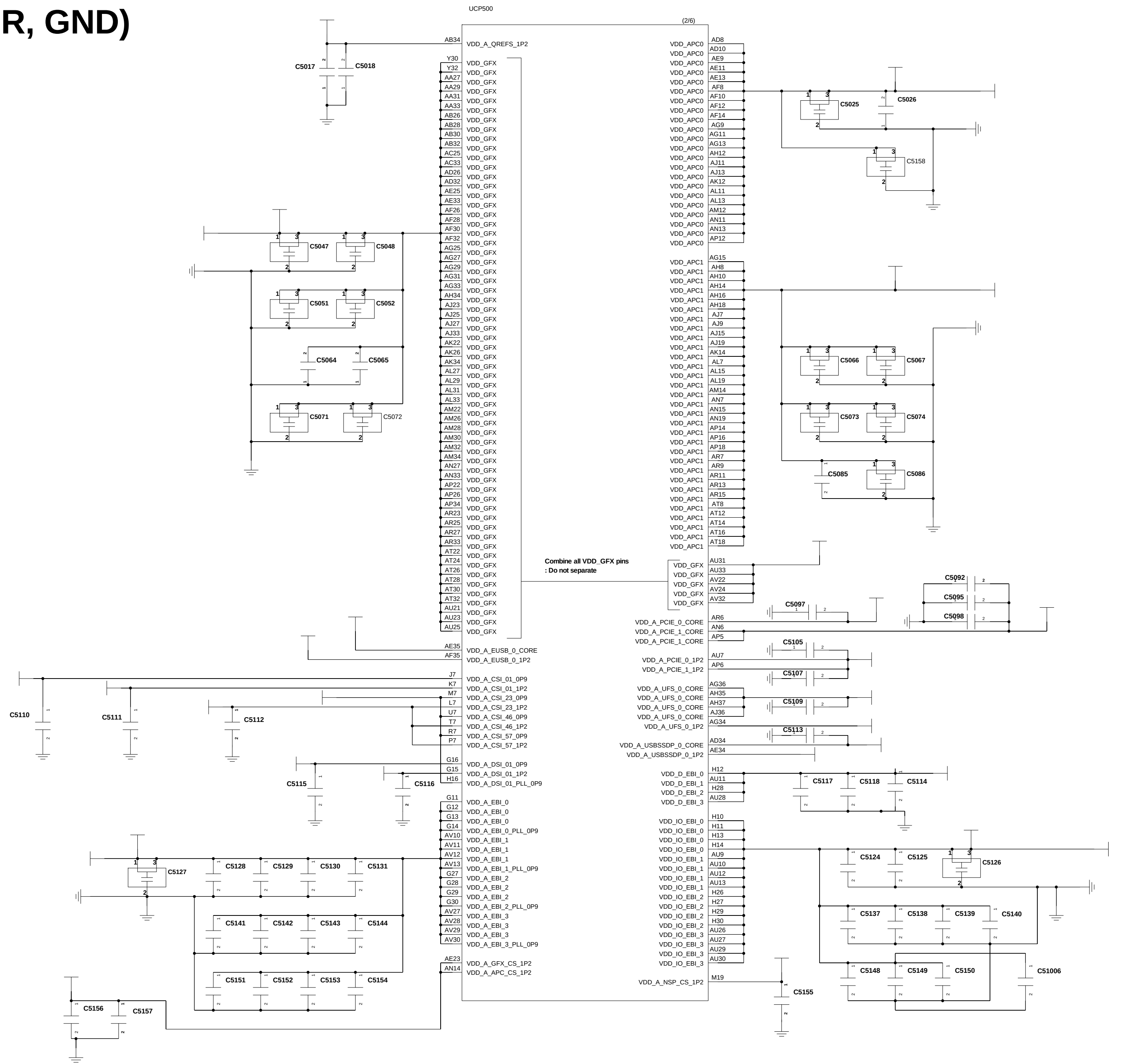

UCP500
(1/6)
TH5000
AP_THM
PMK_CLK6_CXO
SLEEP_CLK
PON_RESET_N
TP_RESIN_N
TP_SM_RESOUT_N
R5000
SM_PS_HOLD
JTAG_RST_OUT_N
JTAG_TCK
JTAG_TDI
JTAG_TDO
JTAG_TMS
JTAG_TRST_N
UFS0_RESET_N
C5160
L5000
C5000
R5002
R5001
R5003
TP_SDC2_D1
C5001
C5002
EDP_AUX_N
EDP_AUX_P
C5003
R5004
C5004
R5005
PMIC_SPMI_CLK
PMIC_SPMI_DATA
SP_VPH_DETECT
SP_ARI_POWER_ALARM
R5006
R5007
R5008
R5009
R5010
R5011
R5012
MAINCAM_MIPI_CLK_P
UWCAM_MIPI_CLK_P
VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_CLK_P
S9U_ONLY
DSI0_NC_LN3_M
PCIE0_REFCLK_M_WLAN
R5018
PCIE0_REFCLK_P_WLAN
C5008
C5009
C5161
C5162

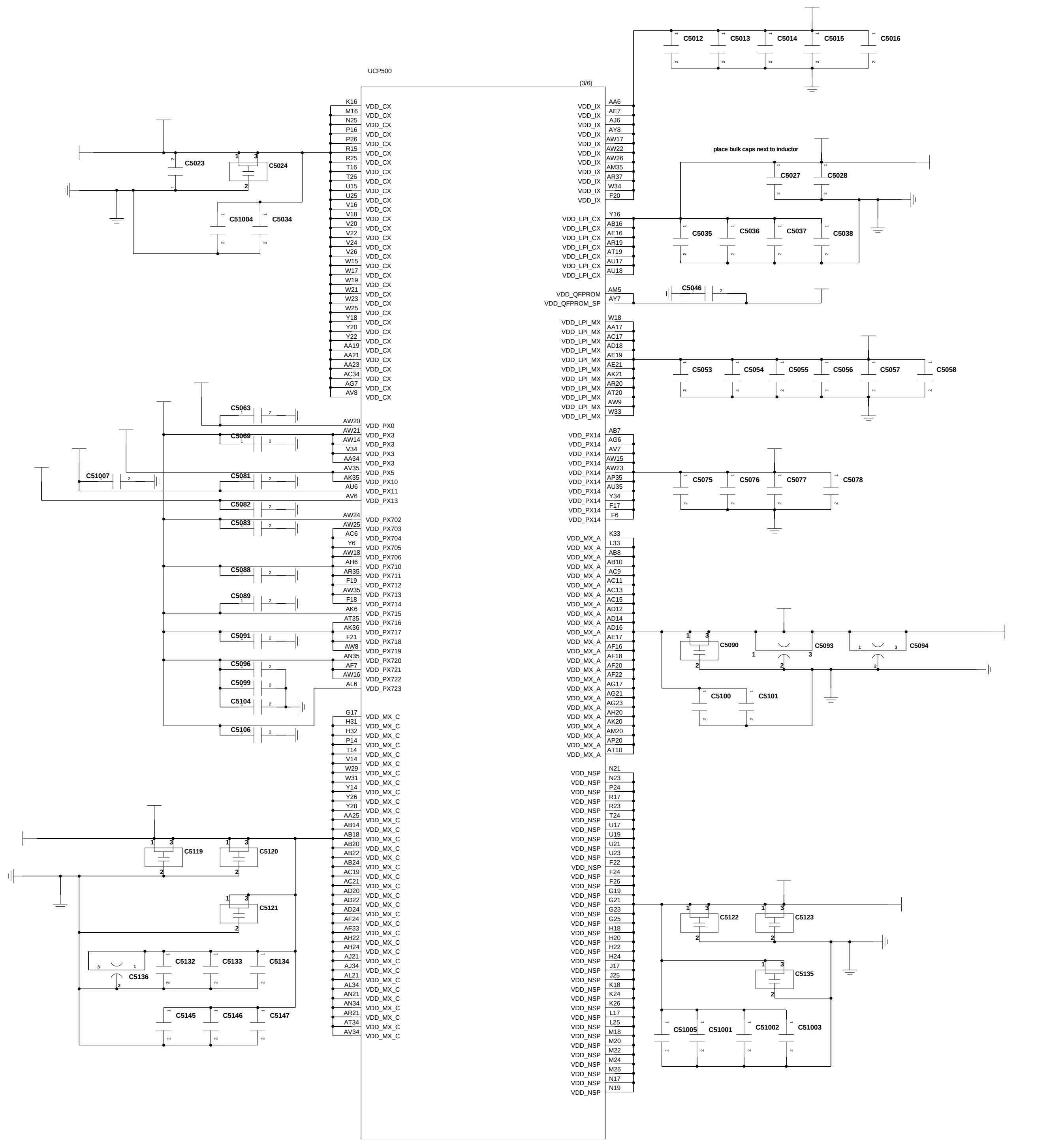

UCP500
(5/6)
(RESERVED) ESIM_SIM2_SEL
(RESERVED) COPR_SS_SUB
REDRIVER_SDA_1P8
KBD_DCDC_SDA_1P8
UW_VTCAM_RST_N
MCU_SCL_1P8
HALL_INT
R5013
R5014
R5016
C5005
C5006
C5007
MAIN_MIC_CLK
SUB_MIC_CLK
3RD_MIC_CLK
R5017
PCIE0_CLK_REQ_N
VTCAM_SCL_1P8

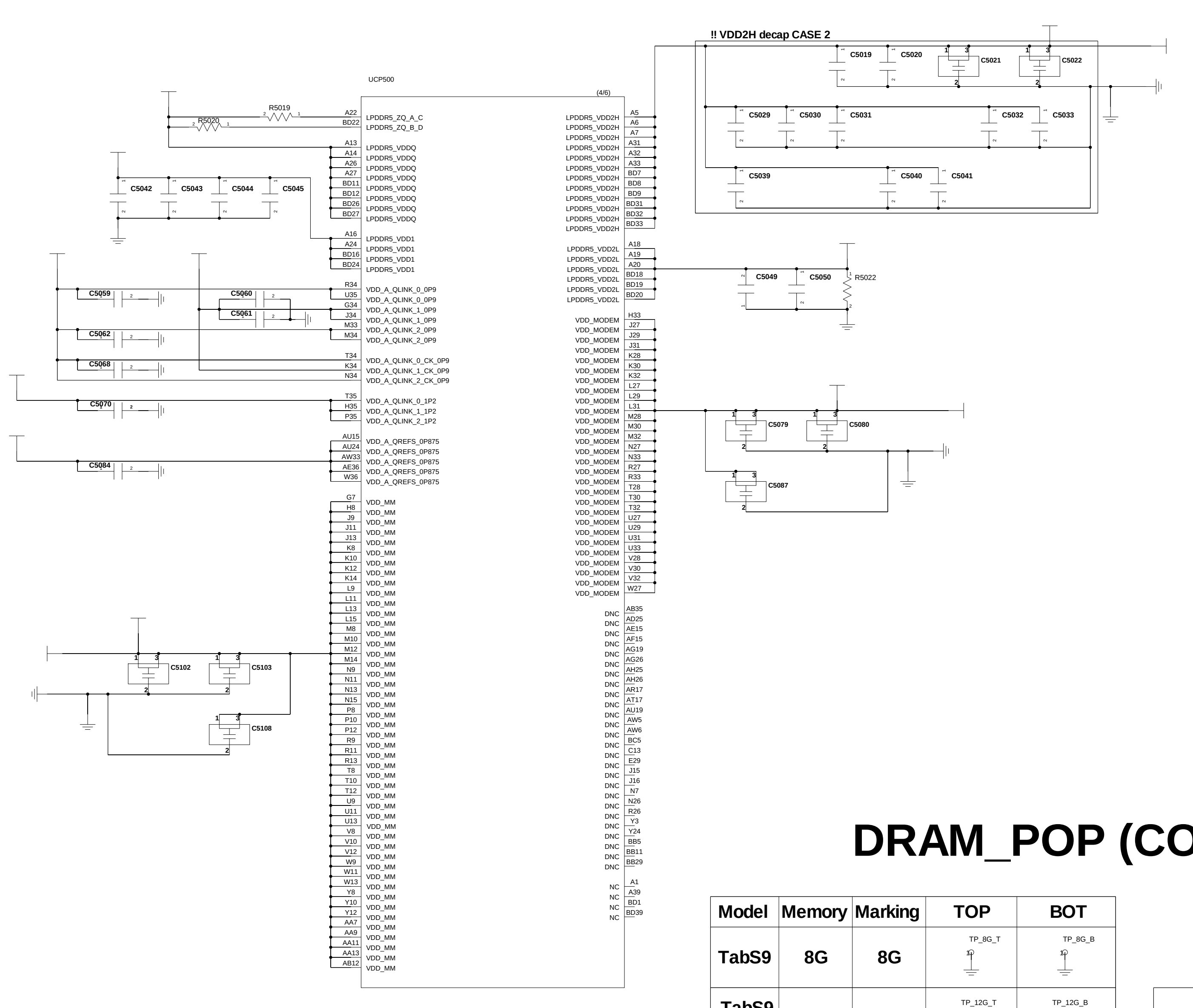

UCP500
(6/6)
GND

SM8550

(POWER, GND)


UCP500
(2/6)
VDD_A_QREFS_1P2
VDD_GFX
VDD_APC0
VDD_APC1
Combine all VDD_GFX pins
: Do not separate
VDD_A_PCIE_0_CORE
VDD_A_EUSB_0_CORE
VDD_A_CSI_01_0P9
VDD_A_DSI_01_0P9
VDD_A_EBI_0
VDD_D_EBI_0
VDD_IO_EBI_0
VDD_A_GFX_CS_1P2
VDD_A_APC_CS_1P2
VDD_A_NSP_CS_1P2
C5017
C5018
C5025
C5026
C5158
C5047
C5048
C5051
C5052
C5064
C5065
C5071
C5072
C5066
C5067
C5073
C5074
C5085
C5086
C5092
C5095
C5097
C5098
C5105
C5107
C5109
C5113
C5110
C5111
C5112
C5115
C5116
C5117
C5118
C5114
C5127
C5128
C5129
C5130
C5131
C5141
C5142
C5143
C5144
C5151
C5152
C5153
C5154
C5124
C5125
C5126
C5137
C5138
C5139
C5140
C5148
C5149
C5150
C51006
C5155
C5156
C5157


UCP500
(3/6)
VDD_CX
VDD_IX
VDD_LPI_CX
VDD_QFPROM
VDD_QFPROM_SP
VDD_LPI_MX
VDD_PX0
VDD_PX3
VDD_PX14
VDD_MX_A
VDD_MX_C
VDD_NSP
place bulk caps next to inductor
C5012
C5013
C5014
C5015
C5016
C5023
C5024
C51004
C5034
C5027
C5028
C5035
C5036
C5037
C5038
C5046
C5053
C5054
C5055
C5056
C5057
C5058
C5063
C5069
C51007
C5081
C5082
C5083
C5088
C5089
C5091
C5096
C5099
C5104
C5106
C5075
C5076
C5077
C5078
C5090
C5093
C5094
C5100
C5101
C5119
C5120
C5121
C5132
C5133
C5134
C5136
C5145
C5146
C5147
C5122
C5123
C5135
C51005
C51001
C51002
C51003


UCP500
(4/6)
!! VDD2H decap CASE 2
R5019
R5020
LPDDR5_ZQ_A_C
LPDDR5_ZQ_B_D
LPDDR5_VDDQ
LPDDR5_VDD1
LPDDR5_VDD2H
LPDDR5_VDD2L
VDD_A_QLINK_0_0P9
VDD_A_QLINK_0_CK_0P9
VDD_A_QLINK_0_1P2
VDD_A_QREFS_0P875
VDD_MM
VDD_MODEM
DNC
NC
R5022
C5019
C5020
C5021
C5022
C5029
C5030
C5031
C5032
C5033
C5039
C5040
C5041
C5042
C5043
C5044
C5045
C5049
C5050
C5059
C5060
C5061
C5062
C5068
C5070
C5084
C5079
C5080
C5087
C5102
C5103
C5108

DRAM_POP (CORE)

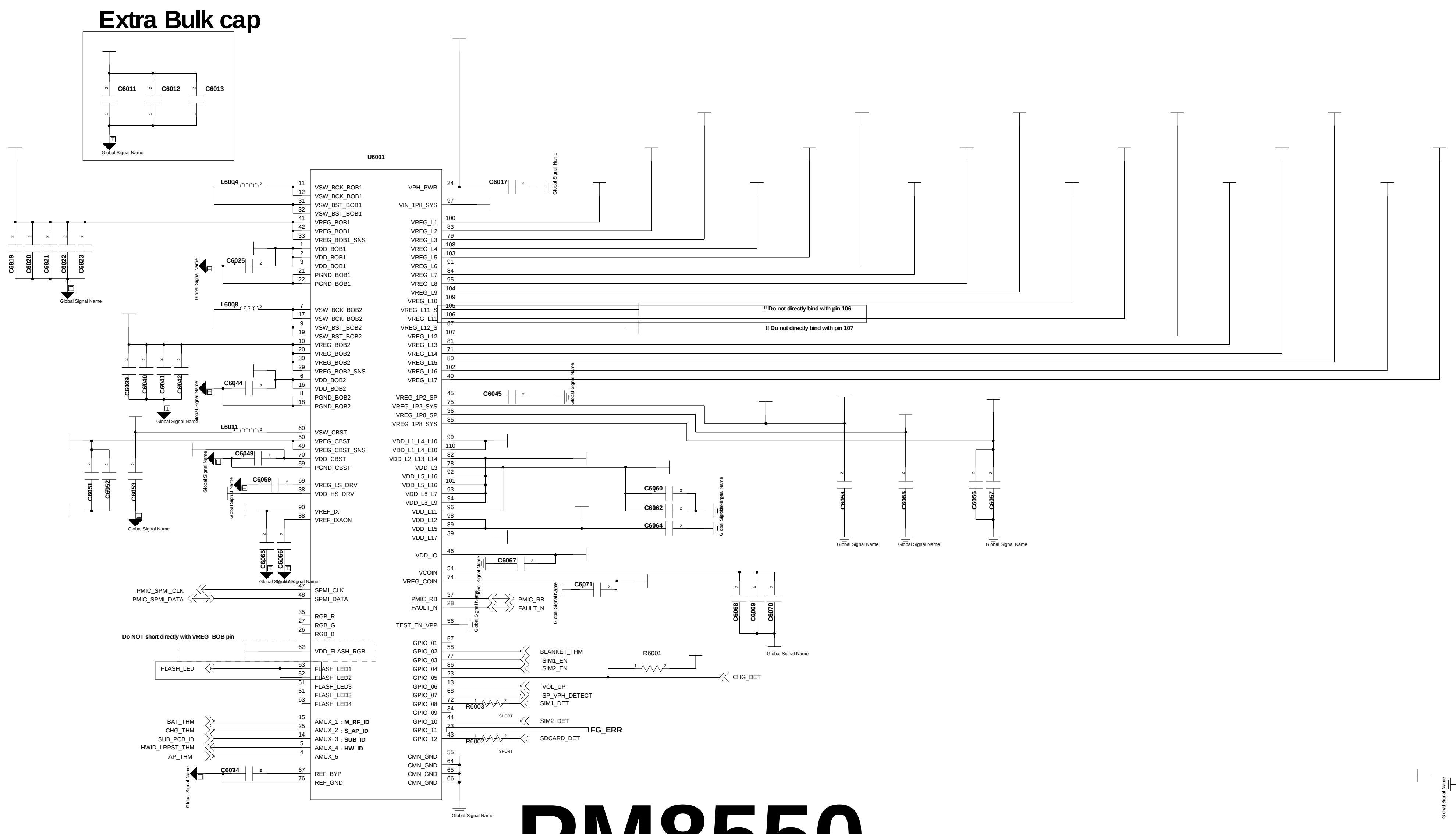
Model	Memory	Marking	TOP	BOT
TabS9	8G	8G	TP_8G_T	TP_8G_B
TabS9 TabS9P	12G	12G	TP_12G_T	TP_12G_B
TabS9U	12G	12U	TP_12U_T	TP_12U_B
TabS9U	16G	16U	TP_16U_T	TP_16U_B

8G	
12G	
16G	

TITLE TabS9_CORE
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 2 of 4
Date Drawing Date

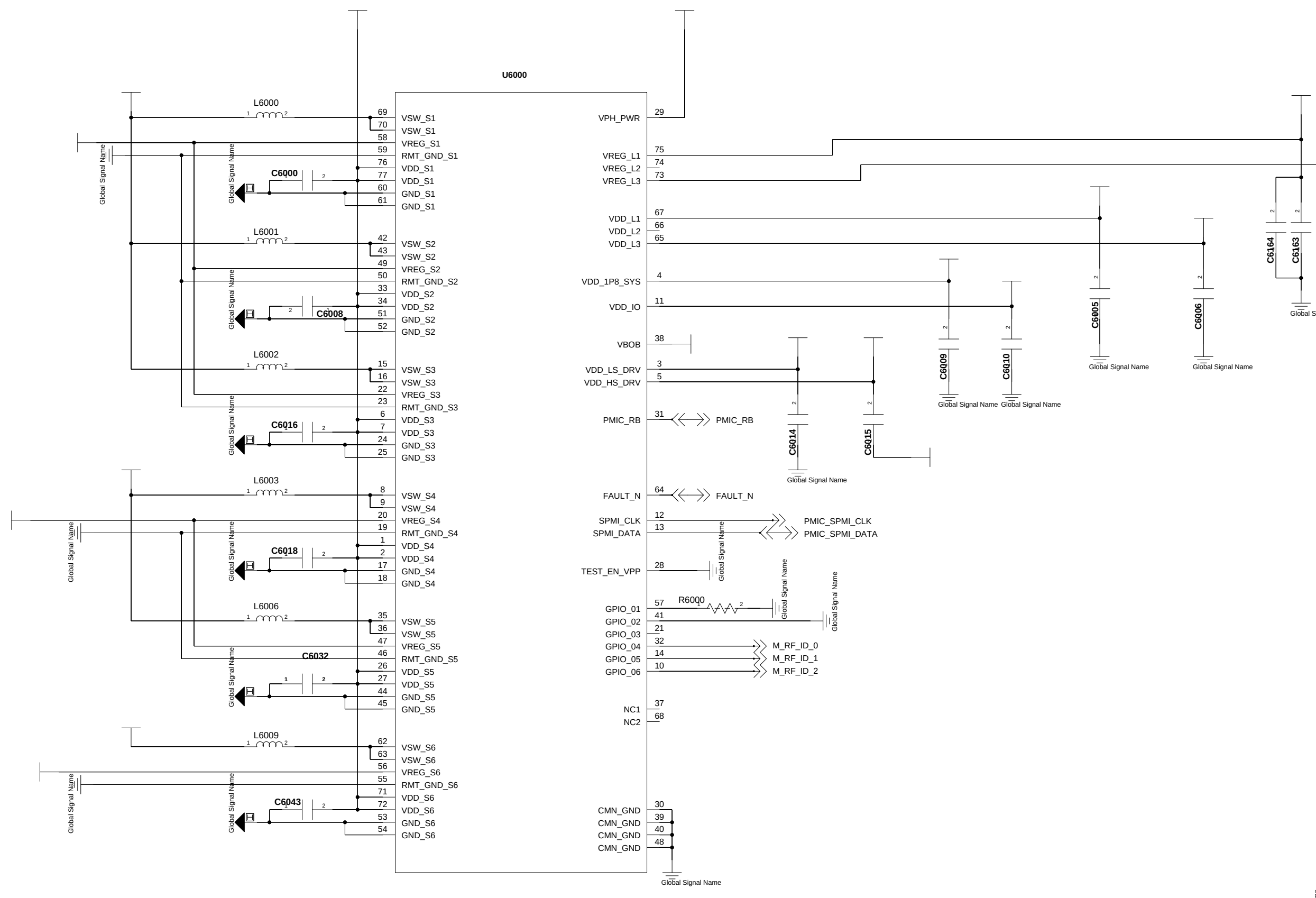
AP PMIC set (Core PCB)

PMIC (PM8550/8550VS/8550VE)



Extra Bulk cap
C6011
C6012
C6013
U6001
L6004
L6008
L6011
C6019
C6020
C6021
C6022
C6023
C6025
C6039
C6040
C6041
C6042
C6044
C6049
C6051
C6052
C6053
C6059
C6065
C6066
C6017
C6045
C6054
C6055
C6056
C6057
C6060
C6062
C6064
C6067
C6071
C6068
C6069
C6070
C6074
R6001
R6002
R6003
VPH_PWR
VIN_1P8_SYS
!! Do not directly bind with pin 106
!! Do not directly bind with pin 107
PMIC_SPMI_CLK
PMIC_SPMI_DATA
PMIC_RB
FAULT_N
Do NOT short directly with VREG_BOB pin
FLASH_LED
BAT_THM
CHG_THM
SUB_PCB_ID
HWID_LRPST_THM
AP_THM
M_RF_ID
S_AP_ID
SUB_ID
HW_ID
BLANKET_THM
SIM1_EN
SIM2_EN
CHG_DET
VOL_UP
SP_VPH_DETECT
SIM1_DET
SIM2_DET
FG_ERR
SDCARD_DET
SHORT
Global Signal Name

PM8550



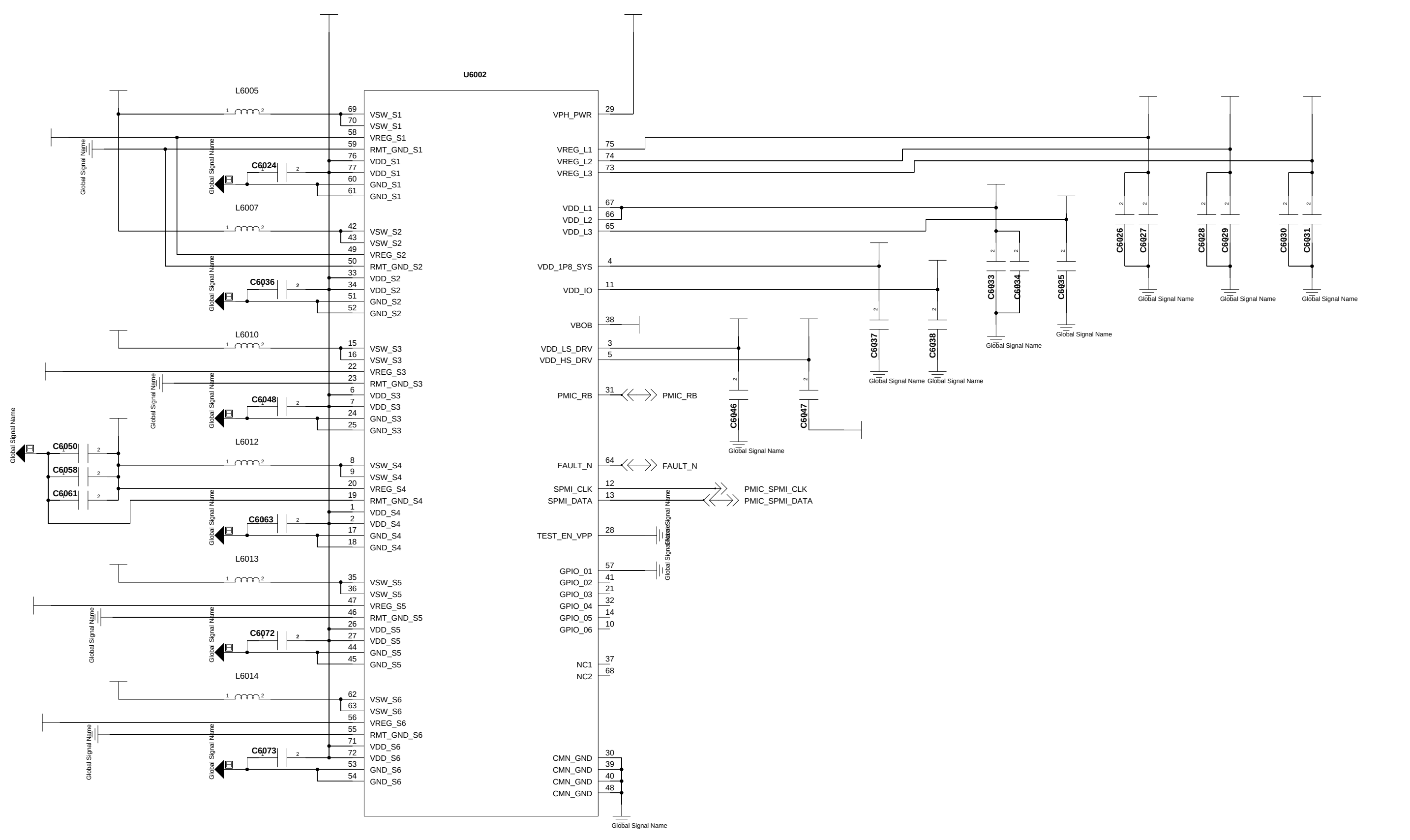
U6000
L6000
L6001
L6002
L6003
L6006
L6009
C6000
C6008
C6016
C6018
C6032
C6043
C6005
C6006
C6009
C6010
C6014
C6015
R6000
VPH_PWR
PMIC_RB
FAULT_N
PMIC_SPMI_CLK
PMIC_SPMI_DATA
M_RF_ID_0
M_RF_ID_1
M_RF_ID_2
Global Signal Name

PM8550VS_C

!! CAUTION !!

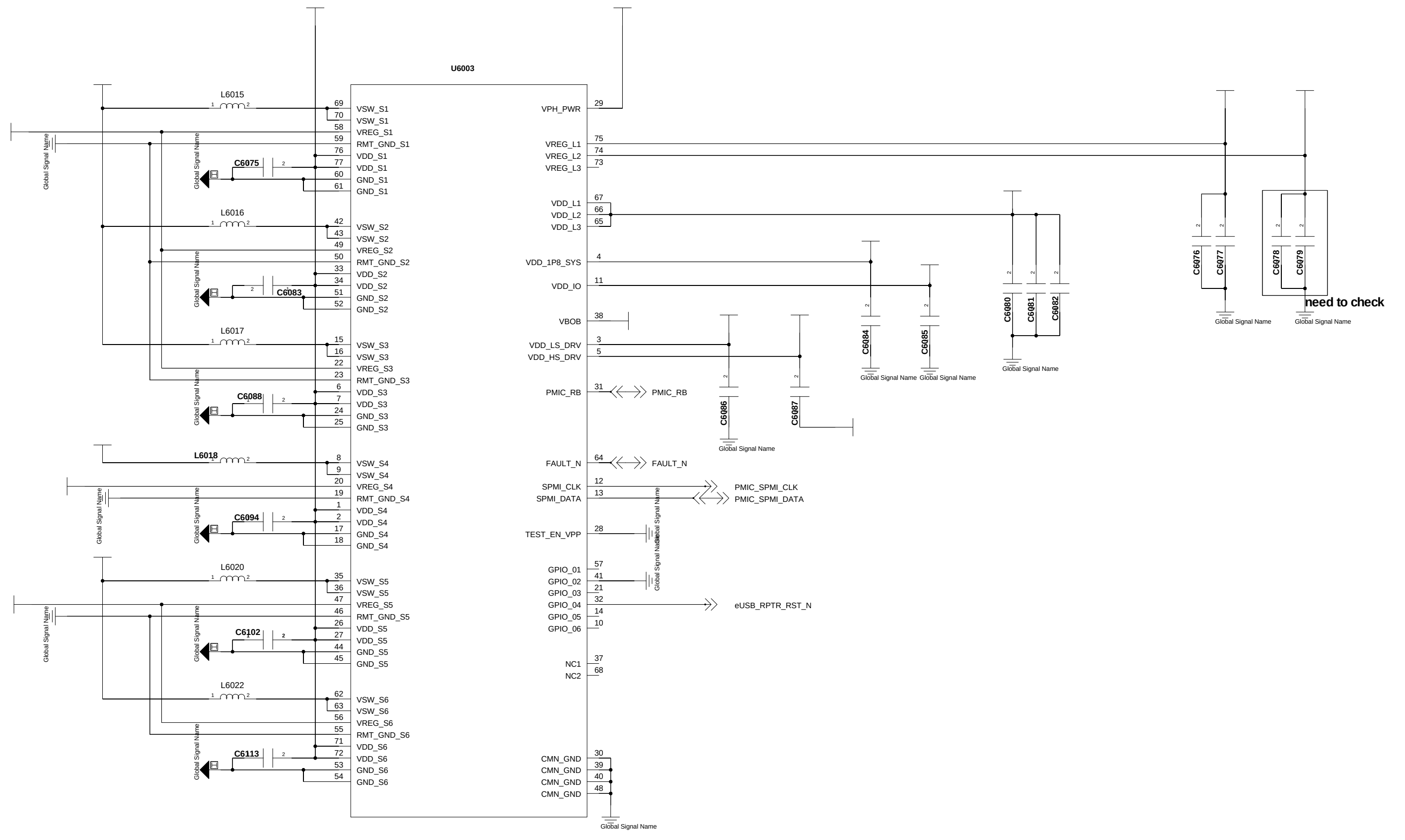
Place Cin 10uF Bulk Caps at corner of PM8550VS and PM8550VE as close as possible

- **PM8550VS : VDD_S1, S3, S4, S6**
- **PM8550VE : VDD_S1, S4, S5, S8**



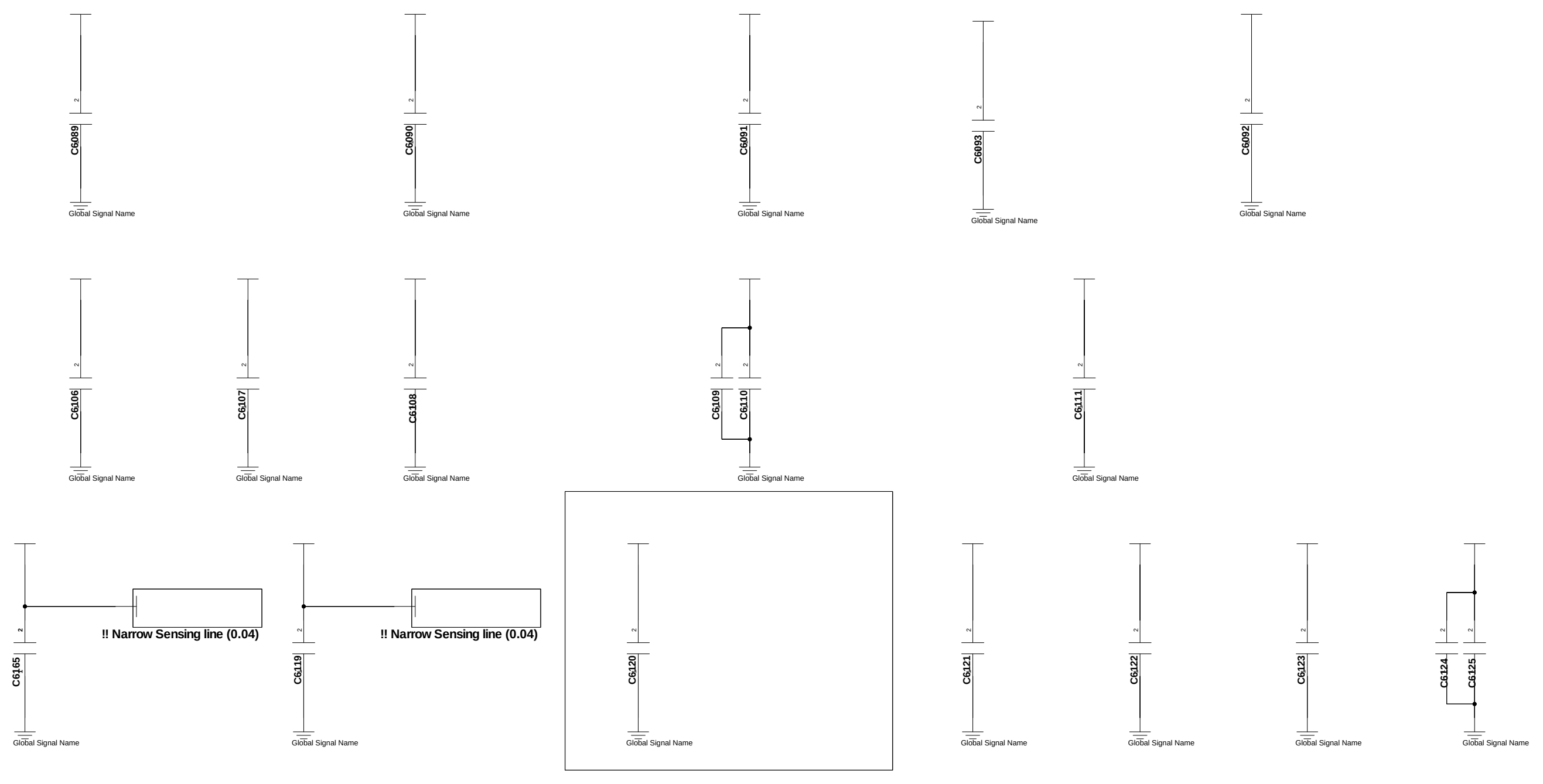
C6164
C6163
C6003
C6004
U6002
L6005
L6007
L6010
L6012
L6013
L6014
C6024
C6036
C6048
C6050
C6058
C6061
C6063
C6072
C6073
C6026
C6027
C6028
C6029
C6030
C6031
C6033
C6034
C6035
C6037
C6038
C6046
C6047
VPH_PWR
PMIC_RB
FAULT_N
PMIC_SPMI_CLK
PMIC_SPMI_DATA
Global Signal Name

PM8550VS_E



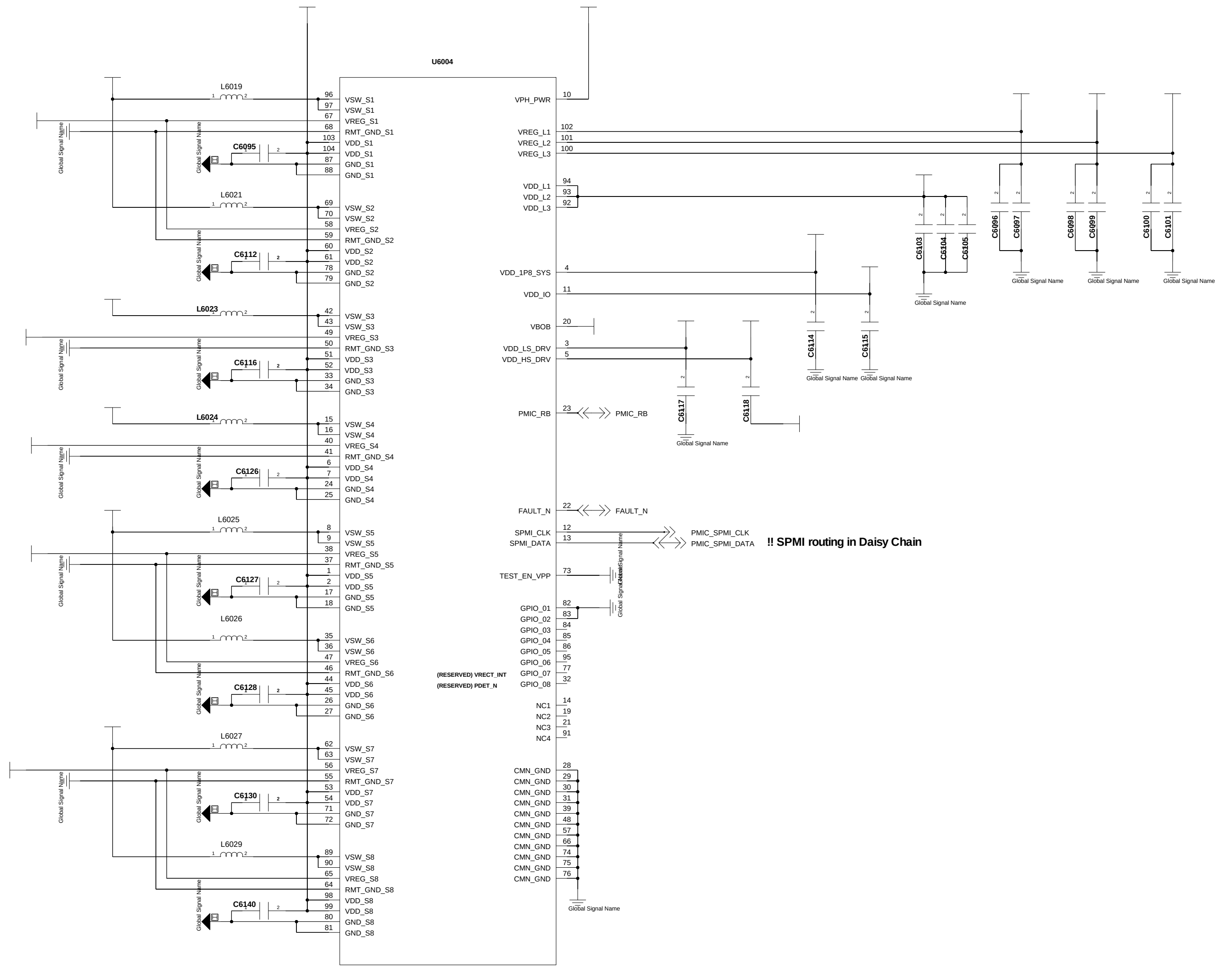
U6003
L6015
L6016
L6017
L6018
L6020
L6022
C6075
C6083
C6088
C6094
C6102
C6113
C6076
C6077
C6078
C6079
C6080
C6081
C6082
C6084
C6085
C6086
C6087
need to check
VPH_PWR
PMIC_RB
FAULT_N
PMIC_SPMI_CLK
PMIC_SPMI_DATA
eUSB_RPTR_RST_N
Global Signal Name

PM8550VS_D



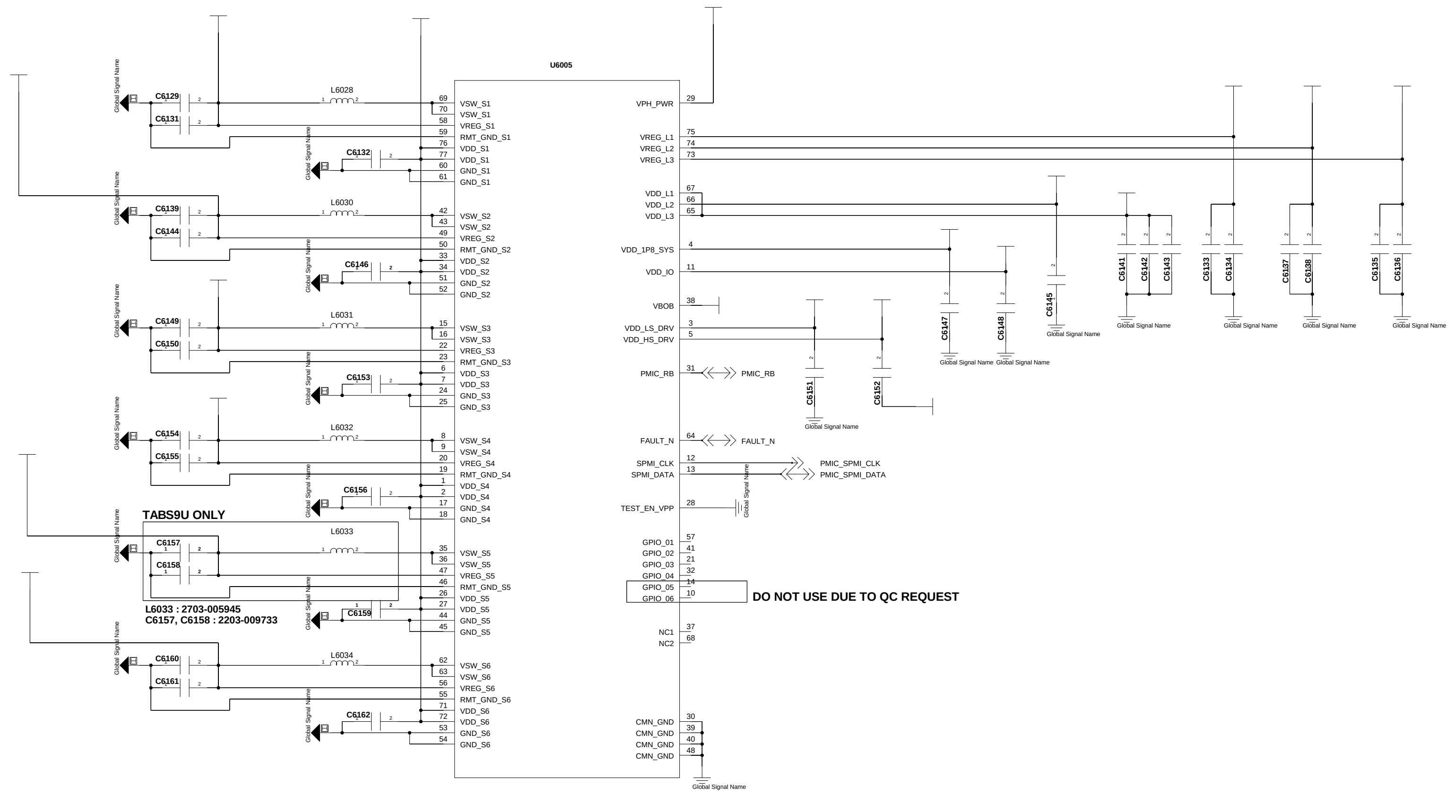
C6089
C6090
C6091
C6093
C6092
C6106
C6107
C6108
C6109
C6110
C6111
C6165
!! Narrow Sensing line (0.04)
C6119
!! Narrow Sensing line (0.04)
C6120
C6121
C6122
C6123
C6124
C6125
Global Signal Name

PM8550 LDO OUT CAP



U6004
L6019
L6021
L6023
L6024
L6025
L6026
L6027
L6029
C6095
C6112
C6116
C6126
C6127
C6128
C6130
C6140
C6096
C6097
C6098
C6099
C6100
C6101
C6103
C6104
C6105
C6114
C6115
C6117
C6118
C6135
C6136
VPH_PWR
PMIC_RB
FAULT_N
PMIC_SPMI_CLK
PMIC_SPMI_DATA
!! SPMI routing in Daisy Chain
(RESERVED) VRECT_INT
(RESERVED) PDET_N
Global Signal Name

PM8550VE_F



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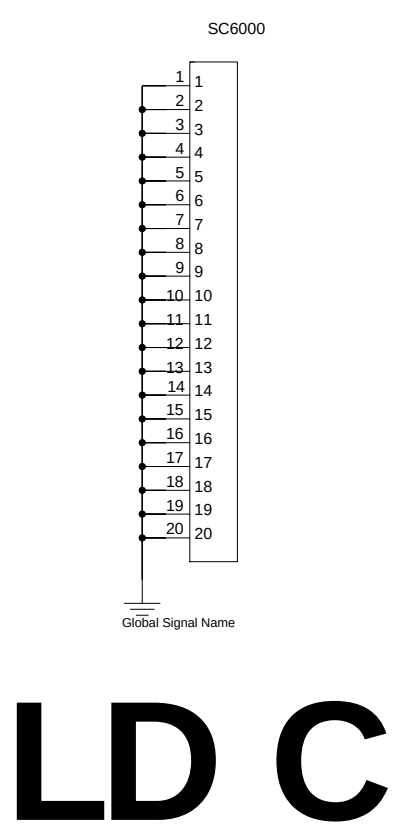
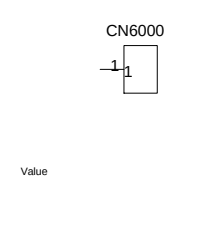
C6161
C6162
TABS9U ONLY
L6033 : 2703-005945
C6157, C6158 : 2203-009733
C6141
C6142
C6143
C6133
C6134
C6137
C6138
C6135
C6136
C6145
C6147
C6148
C6151
C6152
VPH_PWR
PMIC_RB
FAULT_N
PMIC_SPMI_CLK
PMIC_SPMI_DATA
DO NOT USE DUE TO QC REQUEST
Global Signal Name

PM8550VS_G



SC6000
Global Signal Name



CN6000
Value

SHIELD CAN

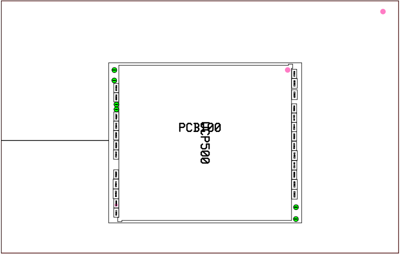
2D BARCORD

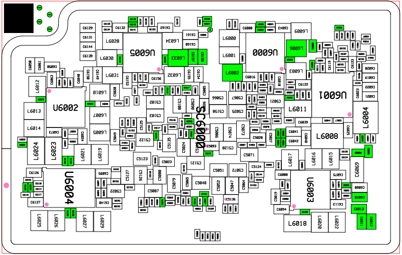
TITLE TabS9_CORE
SIZE A1 DWG NO PDM NUMBER REV Rev
DRAWN BY User Name
SHEET 3 of 4 Date Drawing Date

COREPCB PAD(Core PCB)

PCB100
(1/2)
PCIE0_TX1_M_WLAN
PCIE0_TX0_M_WLAN
PCIE0_REFCLK_M_WLAN
PCIE0_RX1_M_WLAN
PCIE0_RX0_M_WLAN
CC_SCL_1P8
CC_SDA_1P8
6AXIS1_INT2
6AXIS1_INT1
GRIP2_SCL_1P8
BT_UART_RXD
BT_UART_TXD
Global Signal Name
PCIE0_TX1_P_WLAN
PCIE0_TX0_P_WLAN
PCIE0_REFCLK_P_WLAN
PCIE0_RX1_P_WLAN
PCIE0_RX0_P_WLAN
REDRIVER_SDA_1P8
UWCAM_SDA_1P8
CAM_AF_EEP_SDA_1P8
GRIP2_SDA_1P8
WCN_SW_CTRL
BT_EN
WCN_SW_CTRL_WL_CX
UW_VTCAM_SDA_1P8
PMIC_SPMI_DATA
PMIC_SPMI_CLK
UWCAM_MIPI_D3_P
UWCAM_MIPI_D3_N
SDC2_CLK
REDRIVER_SCL_1P8
UWCAM_SCL_1P8
MAINCAM_SDA_1P8
MAINCAM_RST_N
BT_UART_RTS
BT_UART_CTS
CC_DIR
M_RF_ID_1
BAT_ID
SLEEP_CLK
WLAN_BT_SLIMBUS_CLK
WLAN_BT_SLIMBUS_DATA
UWCAM_MIPI_D2_P
UWCAM_MIPI_D2_N
MAINCAM_SCL_1P8
GRIP_INT_3
CAM_AF_EEP_SCL_1P8
MCU_SCL_1P8
UW_VTCAM_SCL_1P8
HALL_INT
CABLE_DET_GPS
M_RF_ID_2
M_RF_ID_0
3RD_MIC_CLK
3RD_MIC_DATA
UWCAM_MIPI_D1_P
UWCAM_MIPI_D1_N
UW_VTCAM_MCLK
UW_VTCAM_MIPI_D1_N
UW_VTCAM_MIPI_CLK_N
UW_VTCAM_MIPI_D0_N
UW_VTCAM_MIPI_D3_P
UW_VTCAM_MIPI_D3_N
VTCAM_MIPI_CLK_N
VTCAM_MIPI_D0_N
VTCAM_MIPI_D1_N
VTCAM_MIPI_D2_N
VTCAM_MIPI_D3_N
SDC2_D1
SDC2_CMD
SDC2_D2
Global Signal Name
CABLE_DET_MAIN2
PEN_IRQ
ground
DSI0_NC_LN3_M
DSI0_C2_LN3_P
DSI0_A2_LN2_P
FAULT_N
DSI0_A1_LN1_M
DSI0_B1_CLK0_P
DSI0_B2_LN2_M
DSI0_C0_LN1_P
DSI0_C1_CLK0_M
DSI0_A0_LN0_P
DSI0_B0_LN0_M
MAINCAM_MIPI_D0_P
MAINCAM_MIPI_D0_N
MAINCAM_MIPI_CLK_N
PMIC_RB
MAINCAM_MIPI_D1_N
MAINCAM_MIPI_D1_P
MAINCAM_MIPI_CLK_P
MAINCAM_MIPI_D3_N
MAINCAM_MIPI_D2_N
MAINCAM_MIPI_D3_P
MAINCAM_MIPI_D2_P
MAINCAM_MCLK
UWCAM_MIPI_CLK_P
UWCAM_MIPI_CLK_N
MAIN_MIC_DATA
MAIN_MIC_CLK
PON_RESET_N
IF_PMIC_SDA_1P8
IF_PMIC_SCL_1P8
PCIE0_RST_N
PCIE0_WAKE_N
CABLE_DET_WIFI0
MCU_SWCLK
MCU_SDA_1P8
WLAN_EN
SECURE_NVM_SCL_1P8
VTCAM_RST_N
SDC2_D3
SDC2_D0
VTCAM_MCLK
VTCAM_MIPI_D3_P
VTCAM_MIPI_D2_P
VTCAM_MIPI_D1_P
VTCAM_MIPI_D0_P
VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D2_P
UW_VTCAM_MIPI_D2_N
UW_VTCAM_MIPI_D0_P
UW_VTCAM_MIPI_CLK_P
UW_VTCAM_MIPI_D1_P
UWCAM_MCLK
UWCAM_MIPI_D0_N
UWCAM_MIPI_D0_P
SUB_MIC_CLK
SUB_MIC_DATA
PMK_CLK6_CXO
PCIE0_CLK_REQ_N
KBD_PWR_EN
MCU_NRST
DC_IRQ_N
SECURE_NVM_SDA_1P8
UW_VTCAM_RST_N
UWCAM_RST_N
Global Signal Name
PCB100
(2/2)
QLINK2_EN
QLINK2_REQ
QLINK2_WMSS_RESET_N
FCAM_LDO_EN
JTAG_TDI
QLINK2_L1_P
QLINK2_L3_P
JTAG_TMS
JTAG_TDO
QLINK2_L1_M
QLINK2_L3_M
JTAG_TCK
RCAM2_LDO_EN
SPK_AMP_LDO_EN
JTAG_RST_OUT_N
QLINK2_L0_M
QLINK2_L2_P
KBD_DCDC_SCL_1P8
RCAM_LDO_EN
JTAG_TRST_N
MOTOR_LDO_EN
QLINK2_L0_P
QLINK2_L2_M
KBD_DCDC_SDA_1P8
GRIP3_SCL_1P8
OCTA_DCDC_SCL_1P8
AMP_INT_N
MCU_CONN_INT
MCU_INT
DIRECT_SDA_1P8
GRIP3_SDA_1P8
OCTA_DCDC_SDA_1P8
QLINK2_L4_P
QLINK2_CLK_M
DIRECT_SCL_1P8
AMP_SDA_1P8
AMP_SCL_1P8
QLINK2_L4_M
QLINK2_CLK_P
QLINK1_EN
QLINK1_REQ
QLINK1_WMSS_RESET_N
QLINK0_EN
QLINK0_REQ
QLINK0_WMSS_RESET_N
QLINK1_L0_M
QLINK1_L0_P
QLINK1_L2_M
QLINK1_L2_P
QLINK0_L3_P
QLINK0_L0_P
QLINK0_L3_M
QLINK0_L0_M
QLINK1_L4_P
QLINK1_L4_M
QLINK1_L7_P
QLINK1_L7_M
QLINK0_L1_P
QLINK0_CLK_M
QLINK1_L1_P
QLINK1_L1_M
QLINK1_CLK_P
QLINK1_CLK_M
QLINK0_L1_M
QLINK0_CLK_P
Global Signal Name
QLINK1_L5_M
QLINK1_L5_P
QLINK1_L3_M
QLINK1_L3_P
QLINK0_L2_P
QLINK0_L4_P
QLINK0_L2_M
QLINK0_L4_M
QLINK1_L8_P
QLINK1_L8_M
QLINK1_L6_P
QLINK1_L6_M
SIM1_EN
BLANKET_THM
SDM_RFFE0_CLK
SDM_RFFE0_DATA
AMP_I2S_LRCK
AMP_I2S_D_IN
AMP_I2S_D_OUT
AMP_I2S_BCLK
DISP_RST
DISP_DET
SIM2_EN
FLASH_LED
FLASH_LED
SDM_RFFE1_CLK
SDM_RFFE1_DATA
OCTA_DCDC_EN
VTCAM_SDA_1P8
VTCAM_SCL_1P8
FLASH_LED
WCN_HOST_SOL_OUT
GRIP_INT_2
HALL_INT_WACOM
SDM_RFFE4_DATA
SDM_RFFE4_CLK
BAT_THM
CHG_THM
SUB_CON_DET
WCN_WLAN_SOL_IN
SUB_PCB_ID
SIM1_DET
SIM2_DET
MCU_ACC_SDA_1P8
MCU_ACC_SCL_1P8
RGB1_SCL_1P8
GRIP1_LDO_EN
VOL_UP
CHG_DET
SDCARD_DET
RGB1_SDA_1P8
RGB2_SCL_1P8
MAGNETIC_SCL_1P8
MAGNETIC_SDA_1P8
Global Signal Name
BTP_RST_N
BTP_SPI_CS_N
BTP_SPI_CLK
BTP_SPI_MOSI
BTP_SPI_MISO
GRIP_INT_1
EDP_AUX_N
USB3.0_SSTX1_M_AP
USB3.0_SSTX0_M_AP
USB3.0_SSRX0_P_AP
USB3.0_SSRX1_P_AP
eUSB_AP_D_P
UW_VTCAM_LDO_EN
EDP_AUX_P
USB3.0_SSTX1_P_AP
USB3.0_SSTX0_P_AP
USB3.0_SSRX0_M_AP
USB3.0_SSRX1_M_AP
eUSB_AP_D_N
Global Signal Name
UFS0_TX0_P
UFS0_TX1_P
UFS0_RX0_P
UFS0_RX1_P
UFS0_RESET_N
UFS0_TX0_M
UFS0_TX1_M
UFS0_RX0_M
UFS0_RX1_M
CABLE_DET_WIFI1
SIM1_DATA_1P2
SIM1_CLK_1P2
BTP_LDO_EN
SM_PS_HOLD
WLAN_COEX_SDXUART_RX
TSP_SCL_1P8
GPS_L1_LNA_EN
LAA_AS_EN
LAA_EN_0
TSP_SDA_1P8
WLAN_COEX_SDXUART_TX
PEN_LDO_EN
SIM2_DATA_1P2
SIM2_CLK_1P2
ENDC_GRIP_LDO_EN
TSP_INT_N
MCU_ACC_INT
eUSB_SCL_1P8
PEN_SDA_1P8
HWID_LRPST_THM
GRIP2_LDO_EN
AP_UART_TXD
AP_UART_RXD
eUSB_SDA_1P8
OTG_DET
VBUS_DISCHARGING_EN
PEN_SCL_1P8
CAM_IO_LDO_EN
TYPE_C_INT
AMP_RST_N
AP_USB_BOOT
PEN_PDCT
SIM1_RST_1P2
SIM2_RST_1P2
TCON_RDY
PEN_FWEL
DISP_TE
CABLE_DET_SUB1
EDP_SW_OE
CABLE_DET_SUB2
CABLE_DET_SUB4
IF_PMIC_INTB
CABLE_DET_SUB3
GRIP1_SDA_1P8
DISP_CON_DET
HWID_LRPST_THM_SEL
OVP_PWR_FLAGB
EDP_SW_SEL
GRIP1_SCL_1P8
eUSB_RPTR_RST_N
RGB2_SDA_1P8
6AXIS_I3C_SCL_1P8
6AXIS_I3C_SDA_1P8
GRIP_INT_4
Global Signal Name

TITLE TabS9_CORE
SIZE A1
DWG NO PDM NUMBER
REV Rev
DRAWN BY User Name
SHEET 4 of 4
Date Drawing Date


PCB100
CP500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